

JSEP

Volume 48 • Supplement • May 2026

North American Society for the Psychology of Sport and Physical Activity

Annual Conference

June 18–20, 2026

Montréal, Québec

Contents

Keynote Speakers	S1
Symposia	S5
Free Communications: ORAL and Posters	S8
<i>Motor Development Abstracts</i>	S8
<i>Motor Learning and Control Abstracts</i>	S16
<i>Sport and Exercise Psychology</i>	S46

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Keynote Speakers

NASPSPA Keynote Lecture

The legacy of NASPSPA to sport psychology

Glyn C. Roberts, Norwegian School of Sports Sciences

Research in sport psychology dates to the 19th Century, with the most cited articles being Tripplet (1897) and Fitz (1895). Research within laboratories began with Wundt (1879) in Leipzig, Diem (1921) in Berlin, and Griffith (1925) in Illinois. However, it was not until the early 60s that sport psychology reemerged after the World War. The first world congress of sport psychology (ISSP) met in 1965, two American scientists invited ISSP to meet in Washington in 1968 and invented the name of the organization North American Society of Sport Psychology and Physical Activity. A name we retain to this day! NASPSPA was formally recognized as a national organization in 1967. After Allerton Park conference in 1973, NASPSPA created 3 sections, sport psychology, motor learning, and motor development. NASPSPA grew in numbers and influence during the 70s and early 80s and led to other national organizations following suit (e.g., SCAPPS, 1969; FEPSAC, 1969) and by the end of the 80s we had multiple societies (e.g., BASES 1984; AASP, 1986; ASPASP, 1989). NASPSPA saw growth in motor learning and control and with the popularity of health and exercise psychology in the 90s and 00s, thrived. However, following the creation of AASP in 1986, it drew many sport psychology people away from NASPSPA. The 90s witnessed a diminished presence of sport psychologists and with the rise of the entrepreneurial university, sport psychology vanished from most Research 1 universities in the US based on its inability to establish external funding. Ironically, during the 90s and 00s sport psychology grew in Europe and on the Pacific Rim. The journal records of publishing authors during this period reflect this trend and the center of gravity of research in sport psychology was clearly in Europe. More recently, the growing dismay with the way AASP focused more on applied and professional issues led to an increasing number of sport psychologists returning to NASPSPA. NASPSPA celebrates 60 years this year. The talk concludes with some reflections on the future of sport psychology within NASPSPA.

Keynote Speaker: Motor Development

Crossroads are made for encounters: Motor development at the heart of interdisciplinary convergence

Caterina Pesce, Michigan State University

The essence of motor development research, praxis and policy is inherently boundary-crossing. A meaningful step into interdisciplinarity of motor development with pediatric exercise science and physiology was made almost two decades ago, when the Stodden model paved the way for motor development to enter the public health arena as a key factor for positive trajectories of physical health. A further interdisciplinary encounter was made in the last decade with developmental exercise psychology and neuroscience to encompass mental (cognitive and socio-emotional) health outcomes of motor development. In particular, motor development intersected with a line of physical activity and cognition research that, beyond the dosage of physical activity, focused on its qualitative characteristics such as coordinative and cognitive task complexity, and on the intertwined development of movement and cognition. This holistic view on the role of

motor development and its increasingly ecological and contextualized understanding and assessment, along with the growing interest for physical activity pedagogies that embrace exploration and discovery is bringing motor development to a most intriguing crossroad. I propose that it is where motor development meets creativity. In the light of advancements in the neuroscience of creativity, and in research on motor creativity and the relation between physical activity, cognition and creativity, I suggest that the time is mature to broaden the scope (definition, assessment and promotion) of motor competence to encompass motor creativity. Beyond the fundamental, foundational and functional characteristics of motor skills in motor developmental research, creativity adds the unique feature of originality of movement actions that emerges from both deliberate, cognitively demanding and unintentional, sensorimotor processes. Finally, I discuss implications and perspectives for practice that can lead from interdisciplinary research on motor development to transdisciplinary innovation in holistic development promotion with motor competence at its core.

Keynote Speaker: Motor Learning and Control

From possibility to action: Affordance actualization as a biomarker of health, risk, and recovery

Christopher Connaboy, Rosalind Franklin University

Health and performance are commonly evaluated via isolated biomechanical, physiological, or cognitive metrics. Yet humans function as integrated systems acting within complex environments. This keynote advances a unifying perspective grounded in ecological dynamics: health and behavior can be understood as the capacity to perceive, select, and actualize affordances under changing constraints. I define affordance actualization as the lawful transformation of perceived action possibilities into functional behavior and propose it as a system-level biomarker of readiness, risk, and recovery. Data from NASA analog missions and simulated military operational stress studies investigating stress, fatigue, isolation, and sleep disruption demonstrate that performance degradation is often expressed first as subtle changes in the ability to detect and act upon emerging opportunities for action. Rather than reflecting isolated cognitive deficits, these changes represent shifts in the stability of the organism–environment system. Parallel findings emerge in the study of mild traumatic brain injury (mTBI). In individuals recovering from concussion, traditional clinical measures frequently fail to capture persistent disruptions in sensorimotor regulation and adaptive decision-making. Experimental work examining sensorimotor and neurocognitive behavior following mTBI suggests that impaired performance reflects a diminished capacity to effectively actualize affordances within dynamic task contexts. Recovery, therefore, is not simply the absence of symptoms, but the restoration of adaptive perception–action coupling. Across these domains, affordance actualization provides a unifying framework for understanding how stress, neurological disruption, and recovery influence functional behavior. By reframing health as the dynamic capacity to convert possibility into action, this perspective offers a pathway toward ecologically grounded biomarkers capable of detecting risk earlier and guiding recovery strategies in operational and clinical settings.

Keynote Speaker: Sport and Exercise Psychology

Beyond the hype: Engineering precision physical activity interventions with artificial intelligence

David Conroy, University of Michigan

Despite decades of research, the efficacy of physical activity interventions has plateaued: intervention effects are not increasing and long-term adherence to behavior change remains elusive. This stagnation stems in part from the difficulty of tailoring interventions to individuals with unique learning histories and the limitations of static theories that fail to capture the dynamic person and context interactions underlying behavior. One of the most promising paths forward involves transitioning from “one-size-fits-all” intervention approaches to precision behavioral interventions enabled by wearable technology and artificial intelligence. To bridge the gap between theoretical promise and clinical reality, I highlight advances from distinct experimental frameworks that evaluate the potential of scalable artificial intelligence approaches to physical activity promotion: dynamic dosing and precision content generation. Control optimization trial results demonstrate how model predictive control algorithms can adaptively “dose” digital interventions by selecting intervention content and message timing to maintain a target physical activity level. Factorial experiments to optimize generative artificial intelligence prompts illustrate how motivational target engagement and the perceived relevance of messages can be improved to address fundamental challenges in the science of behavior change and engagement. Integrating dynamic dosing and precision content generation ensures that each individual receives the right support at the right time. These emerging approaches require theoretical evolution and a paradigm shift in how we train the next generation of physical activity behavior change experts.

Senior Lecturers

Motor Development

From mentee to mentor: Advancing motor competence research and developing the next generation of scientists

Leah Robinson, University of Michigan

Research in motor development increasingly recognizes the foundational role of motor competence in shaping children’s physical activity participation, health trajectories, and developmental well-being. In this talk, I reflect on my journey from mentee to mentor and discuss how mentorship, interdisciplinary collaboration, and sustained research programs can advance both scientific discovery and the training of future scholars. I will describe how early mentorship experiences shaped the development of my research program focused on children’s motor competence and physical activity. This work led to the creation of mastery-climate interventions, including the Children’s Health Activity Motor Program, designed to improve motor competence during early childhood. Findings from this research demonstrate how developmentally appropriate learning environments can enhance children’s motor skill development, physical activity engagement, and overall health. Central to this lecture is the recognition that scientific progress is closely linked to mentorship. Drawing from my experiences as a trainee and mentor, I will discuss approaches to supporting graduate students, postdoctoral fellows, and early-career investigators. Ultimately, this lecture highlights how the transition from mentee to mentor is not only a career milestone but also a responsibility to advance the science and develop the next generation of scholars.

Motor Learning and Control

Chasing the (grant) money vs. scientific passion: Can they co-exist?

Christopher Rhea, Old Dominion University

Careers often take circuitous routes, but the trajectory must start somewhere. I was fortunate to be connected with NASPSPA during my graduate education, attending my first conference in Savannah, GA in 2003. The ensuing twenty-plus years of connections to NASPSPA have led to a series of perturbations that nudged me into challenging, exciting, frustrating, and rewarding spaces. Along the way, I have attempted to balance the desire (and sometimes, need) to fund my research program while maintaining my identity as a focused scientist. An early example of this trajectory evolved when I served as the NASPSPA Student Representative and put together a student-focused presentation at the 2007 conference titled, *Grant money: Where is it and how can I get some if it?* I recruited presenters at different career stages (a student, an early career faculty member, and a senior faculty member) who had each been successful at obtaining grant funding to support their research. The goal was to provide NASPSPA student members with potential funding pathways to support their budding scientific agendas. I must note that putting together this session was not purely altruistic—I was taking notes just like everyone else while contemplating my forthcoming scientific journey. As I progressed from my doctoral training through my postdoc training and into my tenure-track faculty position, those early lessons—along with other breadcrumbs picked up along the path—led me to discover creative ways to align my research interests with funding opportunities, ultimately landing in scientific spaces I hadn’t previously imagined. The career pathway that naturally emerged may appear circuitous, but the central node tying everything together was my identity as a scientist whose research lies at the intersection of neuromotor control and advanced technology to address human health challenges. My goal in this presentation is not to juxtapose the pursuit of grant money with the pursuit of scientific passion, but to show that flexible thinking can put them into a synergistic relationship.

Sport and Exercise Psychology

Lessons from LiFEsports and life: My journey advancing quality and access in youth sport

Dawn Anderson-Butcher, The Ohio State University

Being selected as a Senior Lecturer is a tremendous honor and has encouraged me to reflect on the experiences that shaped my career. My passion for sport and exercise psychology began with what soccer provided in my own youth: a protective environment that supported my development and helped buffer me from various risks. These early experiences inspired my commitment to designing sport programs that support young people, particularly those facing social vulnerabilities. Along the way, mentors across kinesiology, social work, psychology, and education helped me connect this work to broader social issues, including behavioral mental health, school failure, and violence prevention. As a scholar working at the intersection of sport psychology and social work, I have remained committed to advancing interdisciplinary approaches to youth development. Much of my career has unfolded through *LiFEsports*, our university-wide initiative at The Ohio State University. *LiFEsports* uses sport, recreation, and mentorship to help youth build confidence, belonging, and essential life skills. Complementary efforts such as *Coach Beyond* prepare coaches and caring adults to go “beyond the Xs and Os” to support mental health and well-being. For nearly two decades, countless students, colleagues, and community partners have contributed to our research, teaching, and outreach agenda. Together, we have advanced the knowledge base on the value of youth sport and sport-based positive youth development, investigating which types of sport experiences are most effective, for whom, and under what conditions. I am proud of our collective contributions. In this presentation, I will highlight key findings from our work over the past 20 years and outline future directions for strengthening youth sport through innovative

research and design strategies. I will also discuss how sport can be leveraged to address broader social challenges facing young people today. Throughout the talk, I will share lessons learned to inform and support others as they chart their own paths in this evolving and impactful field. My hope is that this work continues to inspire new approaches to youth development through sport and encourages others to imagine what is possible when research, practice, and community come together.

Outstanding Student Paper Award Recipients

Motor Learning and Control

Effect of spatial and temporal error tolerance modulation in haptic guidance on motor learning

Lola Charbonnier, Université de Poitiers; Arnaud Decatoire, Université de Poitiers; Yannick Blandin, Université de Poitiers; Antoine Eon, Université de Poitiers; Pierre Laguillaumie, Université de Poitiers; Cécile Scotto, Université de Poitiers

Haptic guidance in rehabilitation is a promising tool for promoting motor relearning, notably with devices that allow both trajectory (kinematic) and dynamic (kinetic) learning. However, many adjustable parameters in these devices produce variable learning effects, and the key drivers remain unclear. We therefore conducted two distinct experiments, targeting two motor learning parameters: spatial error tolerance, which modulates the relative contribution of kinetics and kinematics, and temporal error tolerance, which modulates the discrete or continuous nature of the task. Both experiments involved learning a complex arm movement using a robotic device with admittance control: the robot moves only on a prescribed trajectory when the required torque is reached. In the first experiment, spatial error tolerance was manipulated by varying the tolerated participant's torque error margin. On Day 1, participants completed a five-trial pre-test without visual feedback, followed by 6 blocks of 15 acquisition trials with continuous and terminal visual feedback. The error margin during the acquisition phase was kept constant at a low (2 Nm) or high level (12 Nm), or gradually increased (2 to 12 Nm) or decreased (12 to 2 Nm) across blocks. On Day 2, a five-trial post-test without feedback assessed learning. In the second experiment, only temporal error tolerance was manipulated by varying robot stiffness, corresponding to the time to stop after being out of margin. Stiffness during acquisition was kept constant at a low (200 ms) or high level (1000 ms), or gradually increased (1000 ms to 200 ms) or decreased (200 ms to 1000 ms). Results revealed a marked effect of error margin on motor learning. Progressive or regressive variations of error margin both led to better learning between pre- and post-tests. Pilot data related to the temporal error suggested different learning trends across groups during acquisition, but limited effects on learning changes between pre- and post-tests. Our results show that varying spatial error tolerance enhances learning by engaging both kinetic and kinematic contribution, whereas temporal error tolerance influences learning strategies without affecting retention. Spatial error tolerance appears to be a key parameter, whereas temporal error tolerance has a limited impact on a complex task.

Sport and Exercise Psychology

Strategic adaptations to deceptive actions in sports: The anticipation-response-inhibition-accuracy trade-off

Carolyn Wickemeyer, Paderborn University; Matthias Weigelt, Paderborn University; Iris Gildenpenning, Paderborn University

This study investigated the impact of local (i.e., trial-to-trial sequence effects) and global (i.e., experiment-wide fake-frequency effects) context information on response inhibition for the pump fake in basketball, using a basketball-specific anticipation-response-inhibition task. A total of 29 participants (10 female, 19 male; 21.8 years) took part in the study. Like in the response-inhibition paradigm by Slater-Hammel (1960), the ARI task consisted of two trial types: In the go-trials, a basketball player performed a jump shot and participants were asked to respond at the exact point in time when the ball leaves the attacker's fingertips. In the stop-trials, a simulated pump fake was presented by pausing the video at a variable point in time before the ball left the attacker's fingertips and then rewinding the video until the ball was back at hip height. In the case of a stop-trial, participants were instructed to inhibit their response. The pump-fake frequency varied across three blocks, with either 25 %, 50 %, or 75 % pump fakes per block. The stop-signal interval was adjusted by a staircase-tracking algorithm, where the stop-signal interval either decreased or increased when participants could or could not inhibit the response. Participants' response-inhibition ability was determined by the point-of-no-return (PNR), which marks the point in time when a (prepared) defensive action can no longer be inhibited, and participants' response-precision performance was quantified using the constant error (CE). The results revealed a new psychological effect, which we term anticipation-response-inhibition-accuracy trade-off. Similar to the speed-accuracy trade-off discovered by Paul Fitts (1954), in the present study, participants either traded off response inhibition for response precision or (vice versa) traded off response precision for response inhibition. This new effect describes the strategic response adaptation to local and global context information and provides a new dimension to anticipatory action control. It has important implications for understanding how individuals balance response precision and response inhibition in dynamic and context-dependent environments.

Rethinking the physical self: A dynamic systems perspective

Kelsey Sick, Western University; Nilam Ram, Stanford University; Eva Pila, Western University

In sport and exercise psychology literatures, the physical self is often positioned as a central determinant of sport and exercise experiences and behavior. Traditionally, physical self-concept has been situated within hierarchical models of the self where domain-specific self-perceptions are organized beneath a relatively stable global evaluative construct. These models typically emphasize fixed structure and assume largely unidirectional influence across levels. In contrast, dynamic systems perspectives conceptualize self-concept as an interdependent system of cognitive, affective, and social processes that continuously influence one another over time. The present study applied a dynamic systems approach to examine how physical self-concept, global self-esteem, and physical self-related evaluations (i.e., appearance- and ability-related self-conscious emotions and social comparisons) dynamically interact across time and at different temporal scales, allowing for bidirectional and multidirectional influences rather than a fixed hierarchical structure. Young women ($N = 148$; $M_{\text{age}} = 20.47$ years, $SD = 2.04$) completed seven ecological momentary assessments per day across seven days. Multilevel vector autoregression network models were used to estimate temporal, contemporaneous, and between-person networks. Temporal analyses supported a non-hierarchical organization of constructs in which physical self-concept at one moment predicted subsequent changes in global self-esteem, whereas global self-esteem predicted later changes in physical self-conscious emotions. Physical self-concept did not directly predict subsequent changes in self-conscious emotions or social comparisons. Contemporaneous networks indicated that appearance- and ability-related self-conscious emotions formed tightly coupled, domain-specific clusters, with

selective cross-domain associations. Between-person networks captured stable individual differences in overall levels of social comparison and self-conscious emotions. Together, these findings challenge traditional hierarchical perspectives by illustrating that physical self-concept and global self-esteem are not only structurally distinct but also temporally organized in daily life. Rather than operating through a single directional pathway, these constructs appear to play distinct, time-specific roles within a broader

self-regulatory system. Conceptualizing the physical self as a dynamic system provides new insight into how momentary self-evaluations are organized and linked across time, and has important implications for sport and exercise experiences and behaviors. Funding source: KS was supported by SSHRC Canadian Graduate Scholarship–Doctoral (CGS–D; 2021–2026) and a CGS–Michael Smith Foreign Study Supplement (2023).

Symposia

Sport and Exercise Psychology

Coping and wellbeing in sport: How athletes navigate adversity in their athletic careers

Patti C. Parker, Thompson Rivers University (Organizer); Mariya Yukhy-menko-Lescroart, California State University, Fresno (Discussant)

Introduction to symposium

Patti C. Parker, Thompson Rivers University

Athletes at all levels of sport face adversity at some point in their athletic careers—adversity such as performance setbacks, injury, competition demands, and even political and systemic burdens. Today, athletes are reporting concerning levels of anxiety and depression, psychological distress, and burnout. Unfortunately, these challenges are associated with increased injury risk, poorer rehabilitation adherence, and less effective coping. With four presentations, our symposium offers novel research that generates theoretical and applied insights for athletes navigating adversity. Presentation 1 draws on attribution theory to investigate how athletes' attributions for sport setbacks can promote adaptive coping. Their study unveils that certain attribution qualities promote setback adaptability and affective wellbeing five weeks later, but not for athletes with high perfectionistic striving levels. Presentation 2 uses a narrative constructionist lens to understand how Olympic and Paralympic athletes engage in advocacy efforts as part of coping and wellbeing in elite sport. The findings highlight the significance of advocacy engagement for athlete wellbeing, but also the importance for socially privileged individuals to share in this work. Presentation 3 examines the experiences of elite Ukrainian athletes training, competing, and succeeding during wartime using a strength-based qualitative approach. Findings highlight how athletes adapt to war-related disruptions, sustain high-level performance, and use sport as a key resource for coping and wellbeing under extreme adversity. Finally, Presentation 4 involves club sport athletes and examines their mental health help-seeking—with a focus on athletic trainers. Notably, only 20% of athletes reported they would seek help from athletic trainers; however, comfort and other-efficacy in the athletic trainer predicted help-seeking intentions. Altogether, these presentations contribute to, and extend, what we know about how athletes cope and support their well-being when facing adversity. The studies range in methodological scope and consider athletes participating at various levels in sport. This symposium advances our understanding of how setback attributions, advocacy engagement, resilience under extreme adversity, and mental health help-seeking are integral to athlete wellbeing and coping in sport. Finally, these outputs have applied implications, such as sharing responsibility in advocacy engagement, developing interventions to support adaptive coping and strengthening athlete-trainer relationships.

Attributions for sport setbacks and coping: The moderating role of perfectionistic striving

Patti C. Parker, Thompson Rivers University; Amber D. Mosewich, University of Alberta

Setbacks are common occurrences in high-performance sport that are unique, contextually rich, and influenced by individual differences and cognitive processes. Attributions, or how setbacks are explained, can

influence how athletes psychologically adapt to those experiences. According to attribution theory, explaining setbacks in internal, unstable, and controllable ways (e.g., poor strategy) can be more adaptive for dealing with negative outcomes since it can produce hope, expectancies for success, and responsibility. In addition, athletes varying in perfectionistic striving—having heightened personal standards and a pursuit of excellence—may endorse attributional tendencies that can help or harm setback coping. Thus, we test three research questions in the present study: (a) Does perfectionistic striving predict athletes' attributional qualities of setbacks? (b) Do attributional qualities influence subsequent setback adaptability? (c) Does perfectionistic striving moderate the influence of attributional qualities on setback adaptability? 130 Canadian and U.S. competitive athletes ($M_{\text{age}} = 26.87$ years; 60 women, 67 men, 2 non-binary, 2 did not report) completed an initial survey measuring attribution qualities and perfectionistic striving, and a second survey 5-weeks later measuring positive and negative affect and adaptability. We employed multiple and moderation regression analysis controlling for Time 1 levels of outcomes. Our findings revealed that perfectionistic striving was not related to attributional qualities (locus, stability, controllability; $ps > .05$). In the moderation analysis, setback attributions viewed as internal, unstable, and personally controllable each significantly predicted cognitive adaptability ($\beta_s = .23, .24, .24$, respectively). Attributions seen as personally controllable also predicted higher positive affect and lower negative affect ($\beta_s = .08, -.28$). Finally, perfectionistic striving moderated the relationship between some attribution qualities and cognitive adaptability, where attributions perceived as internal, as well as personally controllable, predicted better adaptability for those with low or average perfectionistic striving ($\beta_{\text{int}} = -.20, -.23$). Perfectionistic striving also moderated the relationship between attributions seen as personally controllable and positive affect ($\beta_{\text{int}} = -.04$), and between attributions seen as unstable and negative affect ($\beta_{\text{int}} = -.29$; all $ps < .05$). Our findings imply that for athletes with lower perfectionistic striving, the way setbacks are explained (e.g., in controllable ways) can foster setback adaptability and affective well-being.

“By giving, I get so much back”: Advocacy as a wellbeing strategy in Canadian elite sport

Regan S. I. Thompson, University of British Columbia; Erica V. Bennett, University of British Columbia; Lisa R. Trainor, Swansea University; Peter R. E. Crocker, University of British Columbia; Andrea Bundon, University of British Columbia

Elite athletes navigate highly pressurized and public environments, resulting in unique performance and organizational stressors that can significantly impact psychological wellbeing. At the same time, entrenched injustices related to gender, (dis)ability, and related axes of oppression and power hierarchies within broader sport culture, create conditions where advocacy becomes an expected response to inequity rather than a deliberate strategy for improved and sustained psychological wellbeing. We examined how Canadian Olympic and Paralympic athletes diverse in gender and (dis)ability engaged in advocacy during their athletic careers, and how these efforts interwove with coping and wellbeing strategies while navigating the elite sport system. Adopting a narrative constructionist lens, we conducted life-history interviews with 14 elite athletes (8 Paralympic; 6 Olympic; 8 women, 6 men; $M_{\text{age}} = 32$ years) at three time-points (42 interviews). We generated two themes through a narrative thematic analysis illustrating the nuanced relationship between elite athlete

advocacy and psychological wellbeing. The first theme, Taking Action/ Taking Care, explored how athletes intentionally chose to engage in advocacy within and beyond their sport context, knowing that such efforts would positively impact their psychological wellbeing. Athletes told stories of fulfillment, greater purpose, and collective healing through advocacy, which helped them navigate and assign meaning to sport-related stressors and injustices. However, in the second theme, Assigned to Advocate, it is highlighted that those who shared stories of advocacy were predominantly athletes who identified as women and/or disabled; notably, individuals positioned outside privileged norms, bodies, and identities in sport. While advocacy generated influence and visibility, this was emotionally laborious and reflected broader societal narratives that position (and expect) women and marginalized groups to engage in care-oriented, collective work. These findings highlight how athletes can actively choose to engage in advocacy work as a pathway for wellbeing. However, our findings also suggest that gendered and ableist expectations continue to unevenly distribute the labour of advocacy, placing disproportionate emotional demands on marginalized athletes. Perhaps, then, the sustainability of advocacy-as-wellbeing is strengthened when athletes with greater social privilege also share this work, amplifying both social justice efforts in the elite sport context as well as collective wellbeing.

Sustaining training and competitive success during war: Experiences of elite Ukrainian athletes

Mariya Yukhymenko-Lescroart, California State University, Fresno; Jeffrey Martin, Wayne State University; Olena Voiedilova, T. H. Shevchenko National University

Since Russia's full-scale invasion of Ukraine in 2022, elite Ukrainian athletes have faced unprecedented disruptions to their training environments, competition schedules, and personal lives. Daily air raid alarms, relocation, facility closures, travel barriers, and ongoing threats to personal and family safety have fundamentally altered the conditions under which these athletes train and compete. Despite these challenges, many Ukrainian athletes have continued to perform and succeed at national and international levels. However, limited research has examined how elite athletes sustain performance, including training and competition, under conditions of prolonged armed conflict. Guided by strength-based frameworks, this study explored how elite Ukrainian athletes navigated war-related disruptions while continuing to pursue high-level sport. Semi-structured interviews were conducted with five elite Ukrainian athletes – two men and three women – representing biathlon, kickboxing, and judo, who were selected based on their sport status as national team members, national champions, World Cup and World Championship medalists, and/or prospective 2028 Olympians. Interviews explored athletes' training conditions, emotional responses to wartime disruptions, coping strategies, sources of resilience, perceived losses and gains, and the role of sport during the war. Interviews were conducted via Zoom in Ukrainian by a Ukrainian-speaking researcher. Recordings were transcribed in Ukrainian, verified for accuracy, translated into English by a bilingual researcher, and carefully reviewed. Data were analyzed using reflective thematic analysis and a descriptive phenomenological approach. Findings revealed significant disruptions to training and competition, including interruptions caused by air raids, logistical challenges, and displacement. Athletes reported elevated stress, anxiety, and grief related to personal and collective losses. Despite these challenges, participants demonstrated strong athletic identities and adaptive resilience. Sport emerged as a powerful source of emotional refuge, structure, and national representation, helping athletes maintain motivation and a sense of purpose. Participants also identified unexpected opportunities, including international training partnerships and increased global visibility. This study contributes novel insights into athlete transitions under extreme adversity, highlighting sport

as a critical anchor for coping – identity, resilience, and purpose – during wartime and proposing ways for sport psychologists to support athletes during periods of extreme adversity and traumatic life events.

Mental health help-seeking in club sports: A focus on athletic trainers

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The National Athletic Trainers Association has identified athletic trainers as a primary source of mental health help for athletes. Yet, little is known about the perceived role of athletic trainers by athletes and why athletes may or may not seek help from them. While NCAA athletes have generally received more research attention on mental health and help-seeking, it is estimated that more than two million university students participate in club sport each year. The aim of this study was to examine (1) who club sport athletes seek help from with a focus on athletic trainers and (2) the extent to which club sport athlete self-efficacy, other-efficacy in the athletic trainer, and comfort with the athletic trainer predicts intentions to seek help from the athletic trainer. Club sport athletes ($N=363$; 49.9% female) from 24 teams at the same university in the United States participated in a survey, which included measures of likelihood to seek help from 14 individuals for mental health support, mental health referral self-efficacy, mental health referral other-efficacy in the athletic trainer, and comfort with the athletic trainer. Results indicated athletic trainers were the third least likely individual a club athlete would go to for mental health support. Only 20.11% of participants were likely to seek help from their athletic trainer compared to up to 47% of NCAA athletes reported from other studies. Multiple regression results indicated that comfort with the athletic trainer ($\beta = .44, p < .001$) and mental health referral other-efficacy in the athletic trainer ($\beta = .19, p < .001$) were unique predictors of seeking help from the athletic trainer, while mental health referral self-efficacy was not. In other words, the more confident and comfortable the athlete was with their athletic trainer, the more likely they were to go to their athletic trainer for help with a mental health concern. Club sport athletes are under-utilizing athletic trainers but those with more positive perceptions of their trainers will lean on this relationship as a mental health resource. How training environments influence relationships with athletic trainers are discussed alongside suggestions for helping athletes navigate adversity.

Interdisciplinary

The road to excellence: A multi-continent endeavor of understanding expertise

Markus Raab, German Sport University Cologne (Organizer); Nicola Hodges, University of British Columbia, Vancouver (Discussant)

A 'whole of sport' approach? Optimizing sport development pathway creation using the FTEM and 3DAD

Clare MacMahon, La Trobe University

This presentation will outline a common approach to talent development in Australia that incorporates the FTEM (Foundation, Talent, Elite, Mastery) and has been used to delineate pathways in multiple sports (e.g., swimming, basketball) and multiple roles (i.e., athletes, referees/umpires). The FTEM framework (Weissensteiner, 2017) and the Three-Dimensional Athlete Development (3DAD) model used together provide a system-level blueprint for sport organisations, clubs, schools, coaches, and administrators. Key principles of this approach are the non-linear multidimensional consideration of progression, the identification of core capabilities to support resourcing development, and the application to both high-performance and participation sport. In addition to examples from different

contexts, data from work exploring women's sport and progression towards professionalisation will also be explored to provide the perspective of sport expertise considerations within the Australian context.

Multidimensional diagnostics and interventions in elite sports individualization

Karen Zentgraf, Goethe University

Expertise in elite sports has been studied from single disciplinary perspectives such as genetics or cognition, but multidisciplinary approaches are still rare. In a big-scale project on individualization, we suggest that aiming at significant single variables from single disciplines to better understand individual elite performance in group studies is misleading. In the project called 'in:prove', we apply multidimensional diagnostics integrating physiology, cognition, psychosocial health, and motor testing with tailored interventional recommendations via fast-and-frugal trees focusing on prospective performance development. The theoretical background for our approach is Ullén's gene-environment-interaction model. In this talk, I will present German squad athletes' data from a differential and multifactorial perspective, integrating performance-relevant variables and some of their interactions. I will discuss the gain in knowledge on expertise in sports from this approach. Nevertheless, there is a wide gap between 'significant' variables in distinguishing elite from near-elite athletes and their individual predictive value for a specific phenotype. This problem will be exemplified with in:prove motor and genetic data and placed in an

expertise perspective. This needs to be discussed to foster theoretical as well as applied expertise research.

No-size-fits-all: Challenges to the creation of evidence-informed frameworks for developing sport expertise

Joseph Baker, University of Toronto

Elite athlete performance reflects a collection of often disparate variables manifesting in unique ways at specific points in time. As research in the fields of sport expertise, athlete development, and performance-prediction has developed over the past two decades, one thing is clear – there is considerable variability in how athletes develop over the extensive periods of time required to acquire the skills necessary for expertise. In this presentation, I will discuss the challenges this variability presents for researchers, policymakers and coaches trying to come with standardized approaches to athlete development and/or general frameworks that capture the complex interactions among relevant variables. On the one hand, there are general frameworks of human development and learning that have obvious implications for improving our understanding of how athletes advance and thrive (e.g., evolutionary psychology, basic needs theory). On the other hand, these general approaches seem insufficient for capturing the obvious specificity of individual performance and the unique and evolving nature of competition environments. Holistic and evidence-informed models need to clarify this tension between generalized and individualized aspects of athlete development.

Free Communications: Oral and Poster Presentations

Motor Development Abstracts

Associations between motor competence and health-related fitness in university students

Megan M. Brown, Louisiana State University; David F. Stodden, University of South Carolina; Ryan M. Hulteen, Louisiana State University

Previous research has supported a moderate to strong association between motor competence and components of health-related fitness. However, most of the existing evidence examining motor competence and health-related fitness is in youth, not young adults. The purpose of this study was to examine associations between motor competence and health-related fitness among university students aged 18 to 25 years. Seventy-one college-aged students (55 females; age = 20.31 ± 1.01) completed field-based motor competence and fitness assessments. Motor competence was assessed with three product-oriented tasks: supine-to-stand and go (time), standing long jump (distance), and throw and catch (number of clean catches). Health-related fitness was assessed using the PACER (laps), handgrip (kilograms), push-up (number completed), and plank (time) tests. Pearson's correlations with 95% confidence intervals (95% CI) were conducted for the total sample and stratified by sex between each motor outcome and each fitness outcome. Standing long jump and throw and catch were not significantly associated with any fitness outcomes. In contrast, supine-to-stand and go performance demonstrated significant negative associations with multiple fitness variables in the total sample, including PACER ($r = -.453$; 95% CI $[-.624, -.241]$), plank ($r = -.389$; 95% CI $[-.574, -.166]$), handgrip strength ($r = -.330$; 95% CI $[-.527, -.100]$), and push-ups ($r = -.272$; 95% CI $[-.479, -.036]$). Among females, supine-to-stand and go remained significantly associated with PACER performance ($r = -.461$; 95% CI $[-.652, -.215]$) and plank hold time ($r = -.284$; 95% CI $[-.517, -.012]$). No significant associations between motor competence and fitness variables were observed in males. Findings suggest that motor competence, particularly the supine-to-stand and go, is meaningfully associated with health-related fitness in college-aged females, but not males. Standing long jump and throw and catch did not significantly associate with fitness outcomes. These findings contradict previous research demonstrating significant associations between locomotor and object control skills and health-related fitness in young adults. It is possible that the choice of assessment methods for the chosen skills could, in part, explain these differences. The male sample was quite small; thus, generalizability of these results should be treated with caution.

Investigating the impact of developmental coordination disorder on children's mental well-being: A scoping review

Malak Shaath, Brock University; Ellen Wade, Brock University; Taylor Rowe, Brock University; Terrance Wade, Brock University; Matthew Kwan, Brock University

Developmental coordination disorder (DCD) is a neurodevelopmental condition affecting ~5-6% of children (Cairney et al., 2015), and is characterized by severe impairments to fine and/or gross motor skills. Importantly, while DCD has been linked to cognitive, behavioral, and

socio-emotional difficulties (Zwicker et al., 2013), little work has aimed to synthesize the research examining the relationship between DCD and mental well-being. The purpose of the current study was to conduct a scoping review to better understand the relationship between DCD and mental well-being, and to understand where gaps in the current literature may exist. A Boolean search method was employed using relevant keywords across several scholarly databases (PubMed, Web of Science, and PsycInfo). The eligibility criteria included, but was not limited to, studies that: a) focused on children with confirmed, probable, or at-risk DCD; b) focused on specific factors of mental well-being; c) focused on factors of quality of life (QOL) – was used to narrow down results. Covidence was used as a management system that extracted and recorded key study indicators. Results of the initial search included 226 studies that were imported into Covidence for screening. Of those, 68 duplicates were flagged and excluded. Following initial title and abstract screening, 96 studies were deemed to be irrelevant, and were thus excluded. Full text review was conducted for 62 studies, for which 38 did not meet the eligibility criteria. Therefore, the final number of papers included in the review was 24 studies. Overall, findings of this scoping review suggest that children with DCD were consistently found to have reported lower wellbeing, including indicators across their quality of life. The caveat, however, is that commensurate with mental health deficits, there has been much less work that has focused on mental wellbeing. More longitudinal research with consistent measures and assessments of mental wellbeing is required for this population.

Assisted cycle therapy improved grip strength in children with Down syndrome

Shannon Ringenbach, Arizona State University; Sujan Parab, Arizona State University; Bryn Gunther, Arizona State University; Andrew Kraemer, Arizona State University; Molly Bissa, Arizona State University; Tony Vi, Arizona State University; Nicholas Asa, Arizona State University; Marcos Romero, Arizona State University; Stockton Ringenbach, Arizona State University; Isabella Colin, Arizona State University; Jordan Jaslow, Arizona State University; Sarah Erramuzpe, Arizona State University; Saray Ringenbach, Arizona State University; Eqra Rehman, Arizona State University

Approximately 60% of children with Down syndrome (DS) do not meet the recommended standard aerobic activity guidelines each week. In addition, deficits in fine motor control are evident in children with DS and have a large impact on activities of daily living (e.g., eating, grooming, dressing, etc.). The primary goal of this study is to assess the effects of an 8 week Assisted Cycling Therapy (ACT) intervention in comparison to a voluntary cycling group on manual strength in children with Down syndrome. Fourteen children ($M_{\text{Chronological age}} = 10.42$ year, $M_{\text{Mental age}} = 3.83$ year) with DS participated in this study. Eleven children were in the ACT group (i.e., cycled at a 35% greater rate than their self-selected rate with the help of a motor) and three were in the Voluntary Cycling (VC) group (i.e., cycled at self-selected pace) and they all completed 30-minute cycle sessions 2x/week for 8 weeks. Pre- and post-testing of grip force with a dynamometer was used to determine changes in manual strength. Results showed improvements in grip force in both hands following ACT but not voluntary cycling. Our results are interpreted with respect to possible neuroplasticity in the motor control areas of the brain (e.g., supplementary motor area, cerebellum, basal ganglia, etc.) following ACT, which could explain why a lower limb exercise improved upper limb motor control in children with DS.

Profiles of perceived and actual motor competence and their associations with psychosocial difficulties in young children

Leesi George-Komi, University of Michigan; Stephanie A. Palmer, University of Michigan; Christian Burke, Michigan State University; Luis Torres Villela, Michigan State University; Aaron P. Wood, Fairmont State University; Nicholas D. Myers, University of Michigan; Karin A. Pfeiffer, Michigan State University; Leah E. Robinson, University of Michigan

Perceived and actual motor competence are central to children's physical activity engagement and psychosocial development, yet children's self-perceptions do not always align with their demonstrated motor skills. During early childhood, discrepancies between perceived and actual competence may represent psychosocial risk. This study examined whether profiles of perceived-actual motor competence were associated with psychosocial functioning in young children. Baseline data were drawn from 112 school-aged children ($M_{age} = 6.73 \text{ years} \pm .99$) enrolled in the CHAMP-After School Program Study. Actual motor competence (AMC) was assessed using product-based motor skill measures and combined into a standardized composite AMC index. Perceived motor competence (PMC) was assessed using skill-matched items from the Digital Scale of Perceived Motor Competence. Children were classified into four profiles using median splits of standardized AMC and PMC scores (high AMC/high PMC; high AMC/low PMC; low AMC/high PMC; low AMC/low PMC). Psychosocial functioning was assessed using parent-reported Strengths and Difficulties Questionnaire (SDQ) scores. ANOVAs examined differences in SDQ outcomes across profiles, adjusting for sex and age. Significant profile differences emerged for total, $F(3, 105) = 3.26$, $p < .05$, and externalizing, $F(3, 105) = 2.83$, $p < .05$, SDQ symptoms. Compared to children with high AMC/high PMC, children in the low AMC/high PMC profile demonstrated greater total difficulties ($M = 12.0$ vs. 8.3 ; $\beta = 5.12$, $p < .001$), as well as higher externalizing symptoms ($M = 8.0$ vs. 5.9 ; $\beta = 2.84$, $p < .01$). The low AMC/low PMC profile also exhibited greater total difficulties compared with the high AMC/high PMC group ($M = 9.4$ vs. 8.3 ; $\beta = 3.47$, $p < .01$). Sex and age emerged as significant covariates across profiles, with girls exhibiting lower total, internalizing, and externalizing difficulties than boys (p -values $< .01$), and older children showing higher total and internalizing difficulties than their younger peers (p -values $< .01$). These findings suggest that low AMC is a key correlate of psychosocial difficulties in early childhood, with children who also overestimate their motor competence potentially at a heightened risk for outwardly visible behavioral problems. Interventions that support motor skill development alongside adaptive self-perceptions may promote psychosocial well-being in early childhood. Funding source: National Institutes of Health (NHLBI- R01- NR018830).

Process or product? Predicting changes in process object control skills across an intervention from pre-and post product motor variables

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The extent to which product variables predict process variables has been a topic of debate in motor development. Given that motor competence can be measured from a product or process perspective, understanding what variables predict intervention outcomes is helpful. This study was part of a larger fundamental motor skill (FMS) intervention study with vulnerable preschoolers that evaluated the extent to which pre-and posttest product variables predicted changes in object control (OC) scores (OCSSChg) across the intervention. Participants included 64 vulnerable

preschool children ($M = 4.40 \text{ years}$, $SD = .77$). Intervention ($n = 39$) children received a 7-wk FMS intervention provided by the early childhood teacher and an expert in motor development. Both the intervention and control children received their regular preschool program. Children were pre-and posttested on the process-oriented TGMD-3 yielding OC standard score (OCSS) and locomotor standard score (LocSS). Pre-and post product motor variables were also measure including time taken to run 20m (Run), maximum distance jumped in m (MxJmp) and maximum distance thrown in m (MxThrow), along with BMI. Pre and post Run, MxJmp, and MxThrow along with BMI and sex were entered into a linear regression predicting OCSSChg scores (post OCSS minus pre OCSS). The regression model was significant ($p < .001$) predicting 39% of the variance in OCSSChg. Sex ($p = .011$), post-MxJmp ($p = .047$), and post-Run ($p = .026$) were significant predictors. Interestingly pre-product measures were not significantly predictive of OCSSChg scores. This study shows that product variables are helpful in predicting OC outcomes following intervention and suggest that we need to work on both process and product FMS instruction during FMS interventions. Funding source: NASPSA Graduate Student Research Grant.

Movement in the making: Infant walker use and developing mobility

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Existing research on baby walker use has primarily asked whether walkers shift the timing of developmental milestones, with multiple studies reporting delayed onset of sitting, crawling, and walking among infants who use walkers compared with non-users. However, this literature often relies on cross-sectional designs, retrospective parent reports, or standardized assessments, limiting insight into infants' moment-to-moment motor behavior during everyday play. Building on this work, the goal of the present case study is to examine how infants' use of different walker types—push walkers, ride-on walkers, and sit-in walkers—relates to emerging patterns of posture (e.g., upright standing) and locomotion (e.g., cruising and walking) across the first year of life. Although prior research suggests walker use may affect trunk control, balance, and gait quality, little is known about how infants actively engage with specific walker types in naturalistic contexts. To address this gap, the present study employs monthly in-home video recordings to capture spontaneous episodes of walker use, motor exploration, and caregiver involvement. Data were collected from four participants through monthly in-home visits beginning at 1 month of age and ending at independent walking, with infants recorded for 20–40 min during natural, uninterrupted play. Recordings were analyzed in Datavyu using continuous video coding with reliability checks, classifying walker-related events by walker type, motor demands, caregiver participation, and repurposing behaviors, while posture and locomotion were coded using the same onset/offset approach to align walker use with moment-to-moment developmental behavior. Observational coding revealed variability in walking onset across infants. Earlier independent walking was observed in infants who did not engage with walkers (~12months), whereas a later onset was observed in infants who engaged with walker use (~14months). Across sessions, infants typically engage with a single walker type rather than alternating between models, suggesting stable interaction patterns. Developmental changes in walker use were also observed, with younger infants primarily using walkers for fine-motor exploration and older infants increasingly using walkers for gross-motor activities. These findings suggest that walker use is associated with developmental changes in posture and locomotion, highlighting the value of longitudinal, naturalistic observation for understanding early motor development.

WHO 24-hour movement guideline adherence and motor skill performance in preschool-aged children: Evidence from the SUNRISE USA study

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The World Health Organization's (WHO) 24-hour movement guidelines for preschool-aged children recommend 180 min of total physical activity (TPA), including 60 min of moderate-to-vigorous intensity (MVPA); < 1 hr each of sedentary screen-time and restrained sitting; and 10-13 hr of quality sleep. The aim of this analysis was to examine whether there were differences in motor skill performance based on attainment of the 24-hour WHO movement guidelines. Preschool-aged children ($n = 317$, $4.1 \pm .6$ years, 50.3% boys) in the US cohort of the SUNRISE Study participated. WHO guideline attainment was based on waist-worn accelerometry (TPA, MVPA, sleep) and a parent-report questionnaire (screen-time, sitting). Children completed four product-oriented motor skill tasks including: 9-hole pegboard, one-leg standing balance, standing long jump (SLJ), and supine timed-up-and-go (STUG). Group differences were compared using independent t -tests. Only 8.4% of children met all the guidelines; while most children met the MVPA guideline (81%), two-thirds met the restrained sitting guideline (67%), and nearly half met the TPA (41%), screen-time (45%), and sleep (49%) guidelines. Overall, 30% of the 30 possible guideline-motor skill comparisons were significantly different. Specifically, children who met the TPA guideline had better balance, on both their preferred and non-preferred legs, and SLJ performance (all p -values < .05). STUG ($p = .039$) performance was better for those who met the MVPA guideline. Children who met the sleep guideline performed better on the 9-hole pegboard with their preferred and non-preferred hand ($p < .01$) and had better balance performance on their preferred and non-preferred sides ($p < .001$). Those meeting the sitting guideline performed better on the pegboard for the non-preferred side only ($p = .02$). Meeting the screen-time guideline was not related to motor skill differences. In sum, findings suggest lower body strength and stability were associated with attainment of TPA, while MVPA attainment was related to better motor coordination and agility, suggesting more PA engagement might facilitate more practice in these skills. Healthy sleeping habits reflected better fine and gross motor skills which might reflect stable household environments and opportunities to practice a range of motor tasks. Future work should explore causal relationships and potential bidirectional influences for these health factors.

Controlling and neglectful but not supportive physical activity parenting practices are associated with advantageous health-related fitness outcomes in children

Stephanie Palmer, University of Michigan; Olivia Henn, University of Michigan; Charlotte Meeker, University of Michigan; Rebecca Hasson, University of Michigan; Alison Miller, University of Michigan; Anao Zhang, Washington University St. Louis; Leah Robinson, University of Michigan

Physical activity supports the development of health-related fitness (HRF) in children, including aerobic fitness, muscular strength and endurance,

flexibility, and body composition. Physical activity parenting practices (PAPP) are associated with children's physical activity and may contribute to HRF; however, the association between PAPP and HRF has not been examined. The purpose of this study was to examine associations between supportive PAPP (autonomy-promoting and structure practices) and unsupportive PAPP (controlling/neglectful practices) and children's aerobic fitness, muscular strength, and body composition. We hypothesized that supportive PAPP would be positively associated with these fitness outcomes and that unsupportive PAPP would be negatively associated with these outcomes in children aged 5–8 years. Parents ($N = 55$; $M_{\text{age}} = 39.64$ years; 82.6% mothers) of children ($M_{\text{age}} = 6.36$ years; 55.1% female) enrolled in the Children's Health Activity Motor Program–Afterschool Program (CHAMP-ASP) completed the Physical Activity Parenting Practices Questionnaire–Short Form (PAPP-SF). Subscale scores were calculated for structure, autonomy-promoting, and controlling/neglectful practices. Children completed the 6-min walk test, grip strength assessment, and body composition measures. Linear regression analyses showed that autonomy-promoting practices are associated with higher BMI ($\beta = .32$, 95% CI: .05, .58, $p = .02$), with no associations observed for structure practices. Controlling/neglectful practices were associated with better 6-min walk performance ($\beta = .34$, 95% CI: .08, .60, $p = .01$) and lower BMI ($\beta = -.27$, 95% CI: $-.53$, .00, $p = .05$). These findings suggest that associations between PAPP and HRF may be more complex than hypothesized and highlight the need for further investigation into how specific parenting practices relate to children's fitness outcomes. Funding source: R01NR018830; K12TR004374; Rackham Graduate Student Research Award

The role of the caregiver and siblings: Does support differ with multiple children?

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The present study investigates whether patterns of caregiver support during infant play differ by birth order, with a particular focus on hands-on parental support. By conducting naturalistic, longitudinal observations in-home, we aimed to capture how caregiver behavior and support strategies evolve over time and how birth order may shape these interactions, thereby clarifying the environmental and interpersonal factors that constrain or support early motor development. A total of 20 infants (11 male) participated. Of the participants, 11 were firstborns and 9 were later-born. All parents or primary caregivers consented to monthly home visits, continuing until the infant reached the milestone of independent walking. At each visit, major gross motor milestones—sitting, hands-and-knees crawling, cruising, and walking—were assessed either through parental report or direct observation during play. We examined the types of support infants received or used during play and tested whether these patterns differed between birth-order groups across the observation period. Second, we explored whether the type of support was related to infants' progress in achieving gross motor milestones (i.e., Sitting and Cruising). To test both aims, we used a linear mixed-effects model to evaluate how age, birth order, and their interaction predicted the type of support received. As infants grew older, they spent significantly less time in external support locations. A significant interaction between age and birth order indicated that the developmental decline and reliance on external support was less pronounced for firstborn infants than for later-born infants. Time in caregiver's manual support decreased with age and neither birth order nor its interaction with age was not significant. Hold time decreased across infant age as expected, but birth and its interaction with age were not significant. During a 100-day sitting practice window, we observed an increase in the percentage of time in spatial-support and decrease in the percentage of manual caregiver support. Yet birth order and its interactions

remained significantly unchanged. Across a 100-day practice window for upright skills, spatial-support increased with age, while caregiver manual support did not vary. Birth order was not associated with effects of either spatial or manual-support for upright skill practice.

From standing to walking: A longitudinal characterization of the development of independent walking in infants

Promise Robinson, Michigan State University; Kang Gao, Michigan State University; Tianxiang Zhang, Michigan State University; Subir Biswas, Michigan State University; Mei-Hua Lee, Michigan State University

The acquisition of independent walking in infants is a pivotal milestone in early motor development, and is hypothesized to trigger cascading changes in social and cognitive development. While laboratory studies have documented early walking behaviors, less is known about how infants transition from independent standing to walking in everyday home environments. This longitudinal study examines the transition in movement activity across the transition from standing to walking. We specifically examined the role of exploration by quantifying how infants allocate their time between independent standing and walking, as well as the proportion of time spent in supported standing and walking, either with assistance from caregivers or using objects for support. Infants were followed from the onset of independent standing (10 s) until they were able to take 10 independent steps. Home visits were conducted one to two times per week, during which infants were video-recorded for up to four hours in their natural home environments. Videos were coded in Datavyu for different activities - including sitting, crawling, standing, and walking. Preliminary analyses indicate that infants exhibit diverse patterns of activity, often transitioning rapidly from one state to another—for example, from sitting to standing or from standing to walking within a few seconds. The long-term recordings also showed evidence of individual differences in infant-caregiver interaction in terms of time that infant relied on weight support from the caregiver during walking. During periods of full independent weight-bearing, there was evidence of significant exploration where time was distributed across multiple activities, not exclusively walking. These results highlight the necessity for recordings in the naturalistic home environment and the role of motor exploration in the non-linear progression toward independent walking. Funding source: NSF

Evidence for a potential sensitive period in learning to float on the back

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Sensitive periods in development refer to windows of time during which an organism shows heightened anatomical, physiological, and behavioral responsiveness to environmental influences. The existence of these periods raises important questions about the optimal timing of exposure to learning new skills. Despite their relevance to skill acquisition theory and practice, sensitive periods have received relatively little empirical attention in the context of early motor learning. Here, we describe evidence suggesting the existence of a sensitive period for learning to float on the back and discuss its implications for the development of foundational aquatic competencies. We assessed back floating skill in 93 children ($M_{\text{age}} = 14.1$ months, range = 5–25 months) before and after a 20-week Train-the-Trainer intervention designed to empower caregivers to teach foundational aquatic skills. Caregivers completed pre- and post-intervention questionnaires as well as bi-weekly surveys

tracking time spent in the water with their children. We assessed back floating skill pre- and post-intervention using an ordinal scale ranging from .5 to 5. Preliminary analyses indicated that back floating skill improved significantly following the intervention (Wilcoxon signed-rank test, $Z = -6.95$, $p < .001$). Ordinal logistic regression revealed that younger starting age predicted higher baseline float ratings ($\chi^2 = 67.69$, $p < .001$; OR $\approx .23$). Starting age also significantly predicted final float rating following intervention ($\chi^2 = 35.46$, $p < .001$; OR $\approx .53$), indicating that earlier starters were more likely to achieve higher post-intervention proficiency. However, when baseline float rating was included as a covariate, starting age no longer predicted final float rating independently, suggesting that early exposure influenced later outcomes primarily through its effect on initial skill acquisition. We discuss the implications of these findings for theories of learning and development and highlight their practical relevance for optimizing the timing of early aquatic skill instruction.

Mediating pathways between motor competence and BMI in young adults: The role of physical fitness, MVPA, and perceived motor competence

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Motor competence is a foundational component of health and development and has been associated with physical fitness, physical activity (PA) engagement, and healthy weight status. Although these relationships are established, less is known about the mechanisms explaining how motor competence influences weight status in adulthood. The purpose of this study was to examine whether physical fitness, moderate-to-vigorous PA (MVPA), and perceived motor competence (PMC) mediate the relationship between motor competence and body mass index (BMI) in young adults. A total of 223 participants ($M_{\text{age}} = 21.55 \pm 1.67$ years; 42% male) completed the Movement Assessment Battery for Children—Third Edition (standardized t -scores) to assess motor competence. Muscular strength, a component of physical fitness, was assessed using handgrip performance (kgf), PMC was assessed with one item from the Physical Activity-Related Health Competence questionnaire, and participants completed the 7-Day Physical Activity Recall and average daily MVPA was calculated. Height and weight were objectively measured; BMI was calculated (kg/m^2). Mediation analyses were run using Hayes' PROCESS macro for SPSS (Model 4) with gender as a covariate and 5,000 bootstrapped resamples to determine the indirect effects; unstandardized coefficients are reported. Three separate analyses evaluated each mediator in the relationship between motor competence and BMI. Results indicated a significant indirect effect of motor competence on BMI through handgrip [$b = .019$; 95% CI (.0036–.0446)] and the direct effect of motor competence on BMI remained significant after accounting for handgrip [$b = -.0952$, $p = .012$]. There was no significant indirect effect of motor competence on BMI through MVPA [$b = .0017$; 95% CI (–.0039–.0105)], and the direct effect of motor competence on BMI was not significant accounting for MVPA [$b = -.0699$, $p = .0771$]. There was no significant indirect effect of motor competence on BMI through PMC [$b = -.0077$; 95% CI (–.0310–.0062)], and the direct effect of motor competence on BMI was not significant accounting for PMC [$b = -.04$, $p = .25$]. Findings highlight muscular strength as a mechanism linking motor competence and BMI in young adults, whereas MVPA and PMC did not explain the association. Interventions targeting young adults may benefit from integrating motor competence with physical fitness activities to more effectively influence weight-related outcomes.

Evaluating motor proficiency using standard procedures and social narratives in elementary-aged children

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Motor competence is critical for development and provides the foundation for lifelong physical activity and health. Conducting valid and appropriate motor skill assessments is important as the data can be used to inform programming and identify motor delays. However, children may have difficulties understanding and following standardized assessment instructions. Social narratives (SN) utilize descriptors (e.g. words and illustrations) to instruct individuals about novel situations and are shown to reduce anxiety, improve procedural understanding, and offer visual support. Therefore, the purpose of this study was to compare motor performance using the Bruininks-Oseretsky Test of Motor Proficiency – 2nd Edition, Short Form (BOT-2 SF) standard assessment protocol with administration of BOT-2 SF using a social narrative in elementary-aged children. Participants were 22 children (64% males, 7.3+1.3y) from the Southeastern region of the US. The BOT-2 SF was used to assess motor proficiency (14 items, fine motor precision and integration, manual dexterity, bilateral coordination, balance, running speed/agility, upper limb coordination, and strength), and a sum of the BOT-2 SF scores was reported (max = 83). Participants completed the BOT-2-SF in two separate conditions, separated by one week: 1) the standard administration (STD), and 2) the STD accompanied by an SN. These conditions were completed in a counter-balanced order. The social narrative featured a tablet utilizing illustrations of a robot character and audio instructions that were implemented in a story-like format. Paired-samples *t*-tests with bootstrap resampling (5,000 iterations) were used to examine differences in motor proficiency between the two administration methods (STD and SN). On average, the BOT-2 SF score using STD (48.2+11.8) was similar to the SN score (49.0+12.2). The mean difference between the modalities was -.8, 95% CI [-2.828, 1.273] and was not significant [$t(21) = -.704, p = .489$]. These findings demonstrate that the addition of social narratives to motor proficiency assessments in elementary-aged children does not negatively impact their motor skill performance. Future research comparing the two administration methods could include enjoyment and anxiety measures as well as explore whether differences exist between administration methods in different populations (e.g. autism, intellectual and developmental disabilities).

Trends in parental perceptions of child motor skills following a 9-week school-based intervention

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While evidence supports the benefits of engaging in school-based movement intervention on objectively measured child health outcomes, less is known about whether parents see any differences in their children's health outcomes after engaging in an intervention. Parent perceptions may be important to consider when examining ways to boost intervention buy-in or community adoption and/or implementation of programs. Therefore, this pilot project examined whether parental perceptions of a child's motor competence changed following their children's completion of a 9-week e-health intervention implemented in school (e:MAP). Children ($N=49$) were recruited from a single preschool center that served low-income families in the Midwestern United States. Only 7 families completed both the baseline and post-intervention parent surveys. Parents were asked about their child's ball skills (score of 0-100), locomotor skills (score of 0-100), and provided a global rating of their child's skills compared to peers (5-point Likert scale). To assess changes in parental perceptions, we used

Wilcoxon Signed-Rank tests for continuous measures and the Marginal Homogeneity test for categorical data. Given the small sample size ($n=7$), exact *p*-values and effect sizes (*r*) were calculated and used in the interpretation of results. In total, 57% of parents reported improved perceptions. While statistical significance was not reached, there was a positive trend in parental perceptions of child locomotor skills ($M_{diff} = +9.86; r = .48$). There was also a positive trend for motor skill ratings, which improved from "Average" to "High" ($M = 3.57$ to 4.14). While limited by a small sample size, these pilot findings support that the eMAP may lead to positive shifts in parental perceptions and child motor skill categorization, particularly regarding locomotor skills. Future large-scale longitudinal studies are warranted to further examine these relationships and determine the impact of school-based interventions on home perceptions relating to child health. Funding source: Michigan Health Endowment Fund

Parental screen usage and preschoolers' motor skill performance

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With the increasing prevalence of screentime, opportunities that promote motor development may be reduced, potentially leading to decreased physical activity. Studies have examined how children's screentime impacts motor development, but less is known about the influence of parental screentime behaviors. Therefore, the purpose of this analysis was to examine the relationship between children's motor skill performance and parental screentime. Participants were 313 preschool-aged children ($M_{age} = 4.06 \pm .59$ years; 50.2% male) in the U.S. cohort of the international SUNRISE study. Parents completed a questionnaire assessing screentime, specifically during mealtimes, travel, playtime, walks, and bedtime routines. Motor skills included four product-oriented assessments: 9-hole pegboard, one-leg balance, supine timed-up-and-go (STUG), and standing long jump. Analyses of covariance were conducted to examine the influence of parental screentime (never, low, high) on children's motor skill performance, using child age and sex as covariates. Responses to parent screentime were relatively evenly distributed. During mealtimes, 29.4% of parents never used screens, 35.7% had low, and 28.1% had high usage (6.8% missing). During playtime, 18.2% of parents never used screens, 35.5% had low, and 37.7% had high usage (8.6% missing). During travel, 28.9% never used screens, 31.9% had low, and 33.8% had high usage (5.4% missing). During walks, 59.1% never used screens, 26.5% had low, and 7% had high usage (7.4% missing). During bedtime, 54.9% never used screens, 21.4% had low, and 15.9% had high usage (7.8% missing). Parental screen use during mealtimes was related to poorer performance on 9-hole pegboard, $p = .005$ (preferred hand), STUG ($p = .046$), and 9-hole pegboard non-preferred hand approached significance ($p = .057$). Post-hoc results indicated children performed worse on motor skills when parents had higher screentime during mealtimes compared to those with low usage (all $ps = .029$). For playtime, balance was not significantly related to parental screentime, but a trend was observed, $p = .052$. Age was a significant covariate for all analyses; sex was significant for select motor outcomes. High parental screentime during mealtimes negatively influenced fine motor and motor coordination tasks. Results

highlight that parental screentime is not detrimental in all settings, but may influence some motor skill performance.

Parent-reported motor challenges and objective performance: Agreement and discrepancies across developmental groups

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Motor difficulties are common in Autism Spectrum Disorder (ASD) and Developmental Coordination Disorder (DCD). Parent-report questionnaires are widely used to screen for motor challenges. However, it remains unclear how well parent perceptions of motor difficulties align with objectively measured motor performance across developmental groups. This study examined group differences in motor performance and parent-reported motor difficulties, as well as the relationship between parent perception and objective motor outcomes. Seventy-five children between the ages of 5-18 years (26 autistic, 23 with DCD, 26 neurotypical; $M_{\text{age}} = 10.0$, $SD_{\text{age}} = 3.3$) completed the Movement Assessment Battery for Children – Third Edition (MABC-3) motor assessment, while parents completed the MABC-3 Checklist. One-way analysis of variance were conducted to examine group differences in motor performance and parent-reported motor difficulties. Associations between checklist scores and motor assessment scores were examined using Spearman's Rho correlation across the full sample and within each group. Significant group differences were observed in both objective motor performance and parent-reported motor difficulties. Autistic children and those with DCD demonstrated significantly lower motor performance on the MABC-3 compared to neurotypical (NT) peers ($p < .001$), while parents of autistic children and those with DCD reported significantly greater motor difficulties compared to parents of NT children ($p < .001$). Across the full sample, parent-reported motor difficulties were strongly and negatively associated with objective motor performance, indicating that higher perceived difficulties were related to lower motor scores. Within-group analyses revealed significant negative associations for autistic and neurotypical children, whereas the association was weaker and not statistically significant for those with DCD. These findings suggest that parent-report checklists generally reflect objective motor performance; however, the strength of this relationship varies across developmental groups. Results highlight the value of parent-report measures for motor screening while underscoring the importance of objective assessment in clinical and research contexts. Funding source: Pearson Assessments

Assessing the accuracy of parent-reported motor competence in children

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The concordance between parental perception of child motor competence and objectively assessed child motor competence remains understudied. This study examined the agreement between parent-reported motor competence and objectively assessed child motor competence (Test of Gross Motor Development-3rd Edition [TGMD3]) in young children. A cohort of 49 children was recruited from a single preschool center that served low-income families in the Midwestern United States. Only 17 families completed the baseline parent surveys, and only these families were included in the current investigation. Parents completed a short series of questionnaires, which included three questions asking about their child's ball skills (score of 0-100), locomotor skills (score of 0-100), and a global rating of their child's skills compared to peers (5-point Likert scale). Children completed the TGMD-3. To match data scoring, TGMD percentile scores for ball skill and locomotor scores were used, as well as the final categorization of global motor competence (ranging from 1-5). Data were

analyzed using Pearson correlations, Intraclass Correlation Coefficients (ICC; two-way mixed, consistency) for agreement, and Cohen's Weighted Kappa for categorical classification. Results revealed a significant systematic "optimism bias," with parents consistently overestimating their child's proficiency (TGMD3 ball skill percentile = 22.62% vs. parent-reported ball skills = 75.62, $r = .5$; ICC = .489; TGMD3 locomotor = 20.54 vs. parent-reported locomotor skills = 75.62, $r = .23$; ICC = .181). Categorical analysis demonstrated no agreement between parent global rating and child TGMD categorization (Kappa = -.018). Additionally, 94% of children (16/17) objectively scored "Below Average" or "Impaired," yet 0% of parents correctly identified their child as "Low" or "Very Low," resulting in a 0% sensitivity rate for identifying motor delays. The findings suggest that parents typically overestimate their children's skills and are currently unable to accurately score or categorize their young children's motor competence as assessed with the TGMD3. Funding source: Michigan Health Endowment Fund: Nutrition and Healthy Lifestyles

Affective responses and energy expenditure during skill-based and continuous physical activities in youth: The utility of a Fun Scale

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Understanding the interplay between physical exertion, cognitive demand, and affective experience is essential to designing developmentally appropriate physical activity interventions. This study synthesized data from three investigations involving 92 children and adolescents ($M_{\text{age}} = 9.7 \pm 2.9$ years) to examine the relationship between energy expenditure (EE), perceived exertion (RPE), and fun during discrete motor skill practice and continuous endurance activity using a novel 10-point "Fun Scale." Participants engaged in structured sessions involving discrete object projection skills (throwing, catching, kicking, receiving) and a continuous task (PACER run), completed under both single- and dual-task conditions. EE was measured via indirect calorimetry (COSMED K4b2/K5), RPE through a validated child-adapted scale, and fun was rated using a Fun Scale. Across all sessions, EE exceeded moderate-to-vigorous physical activity thresholds (range = 4.5-11.5 METs). Despite increased RPE under higher-intensity and dual-task conditions, fun ratings remained consistently high (7.5-10). Notably, continuous tasks elicited the highest RPE values (up to 7.3), yet participants still reported moderately high enjoyment ($M_{\text{fun}} = 8.7$). In skill-based tasks, fun was positively associated with motor competence; higher-skilled participants reported significantly greater enjoyment and exhibited better cognitive performance, particularly under dual-task conditions. Although fun ratings showed a slight decline with increasing age, Fun scores remained high across all ages. Findings support the utility of the Fun Scale as a valid tool for capturing affective engagement during physical activity. Fun emerged as a resilient affective response even in the presence of increased metabolic and cognitive load, suggesting its potential as a psychological buffer. This research underscores the value of integrating enjoyable, skill-based tasks into physical education and youth interventions to promote sustained physical activity, enhance motor development, and support cognitive challenge. The Fun Scale represents a practical instrument to operationalize and monitor enjoyment, an often-overlooked but critical variable in movement science and pediatric health promotion.

Eye tracking in an end means task

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Action perception, the ability to observe, interpret, and understand others' goal-directed actions, is a foundational process supporting learning, imitation, and the development of skilled behavior. Although extensively

studied in simple, unimanual tasks, it has rarely been examined in multi-step, goal-directed actions. We present preliminary data assessing action perception using an end-means task, examining associations among age, task performance, and visual attention to a skilled performer. Sixteen typically developing children (8–18 months) participated. Children retrieved a toy from an opaque box (Task 1), a clear box (Task 2), and a clear box nested inside an opaque box (Task 3). Standardized verbal cues and assistance were provided, and tasks ended when the toy was grasped. Trained coders scored task onset and completion, hand contacts, and assistance. After task performance, children watched videos of an adult completing the same tasks while eye-gaze was recorded at 60 Hz (Tobii). Associations among age, task duration, contact duration, and proportion of attention to the skilled performer were analyzed using linear models, with effect sizes quantified as Cohen's f^2 . For Tasks 1 and 2, associations between age, task duration, contact duration, and visual attention were small-to-moderate and nonsignificant ($R^2 \leq .22$, $f^2 \leq .28$, $p > .05$). In contrast, Task 3 showed robust associations between performance and visual attention. Longer task duration and contact duration were strongly linked to greater attention ($R^2 = .52$ and $.48$; $f^2 = 1.10$ and $.92$; $p < .01$). Age showed a moderate association ($R^2 = .25$, $f^2 = .33$; $p = .05$). Visual attention to skilled action is strongly related to performance in a complex, multi-step end-means task. This pattern suggests that nested, goal-directed actions place greater demands on motor planning and action understanding, making them more sensitive indicators of developmental change. Integrating behavioral performance measures with eye tracking offers a promising approach for linking motor development with perceptual and cognitive processes. Future work will extend this framework to clinical populations, including children with cerebral palsy, to examine how action perception can support motor learning and rehabilitation. Funding source: Two National Institutes of Health grants: (PI: Jill Heathcock, Award Number: 5 R01HD107739-03) and (PIs: Ramey, Lo, Award Number U01NS106655-01A1)

Examining within-person temporal stability and coupling of physical activity and cognition

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Recent decades have seen growing interest in the pro-cognitive effects of physical activity, driven largely by findings from controlled, laboratory-based assessments. However, traditional between-subject approaches may oversimplify the dynamic, within-person relationships linking physical activity and cognitive function in daily life. Multimodal intensive monitoring designs offer a promising framework for ecologically capturing these time-varying relationships in naturalistic settings. This pilot study examined day-to-day associations between objectively measured physical activity and ecological momentary cognitive function using an intensive longitudinal design. Fifteen healthy young adults completed repeated assessments of accelerometer-derived physical activity and momentary cognitive performance over one week. Cross-panel path models were used to examine the temporal stability of each construct and their within-person temporal coupling across days. Physical activity demonstrated relatively low stability, with only two autoregressive paths reaching significance, indicating substantial within-person variability ($\beta_s = .67-.84$, $p < .05$). In contrast, cognitive function exhibited consistent stability across all autoregressive paths ($\beta_s = .54-.74$, $p < .05$), indicating short-term reliability in momentary cognitive performance. Cross-panel associations between physical activity and cognitive function were limited, with a significant coupling observed on only one day ($\beta = .63$, $p < .05$). These findings highlight distinct temporal dynamics of

physical activity, cognitive function, and their coupling that emerge at the within-person level. Rather than reflecting stable, trait-like associations, physical activity–cognition relations may depend on contextual and individual-level factors that mean-level analyses are insensitive to. Together, these results demonstrate the utility of multimodal intensive monitoring for capturing meaningful variability in health behaviors and cognitive function and outline a framework for next steps to examine contextual processes through which physical activity may support cognitive function in real-world settings.

Movement-based curriculum: Effects on motor and cognitive performance

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Research has found motor abilities and play to be related to important learning processes, supporting its role in cognitive development. A good example of how learning and moving are connected relates to paying attention. Studies have found that right after active motor play, children are more attentive in the classroom than they are when they have been sitting for a long period of time. This may be because children have been able to expend energy and are now ready to learn. Another view is that during motor play, children learn to regulate their behavior. They must control their own body movements and inhibit impulses on the playground, and these skills transfer to the classroom to help them better focus on learning tasks. The present study examined a movement-based preschool curriculum and its effects on motor skills and cognitive performance. It was hypothesized preschoolers participating in a movement-based curriculum would show an increase in fundamental motor skill and cognitive performance. Preschoolers ($N = 46$, $M_{age} = 4.3 + 1.3$ years) at a rural school completed a movement-based curriculum for 12 weeks (3 days per week for 40 min). Participants were assessed using the Test of Gross Motor Development 3rd edition (motor) and the Early Years Toolbox (cognition). Paired sample t -tests revealed TGMD-3 results - Pre-Test M: 38.74 ($SD = 5.3$) and Post-Test M: 54.39 ($SD = 4.8$), $p = .001$. The Early Years Toolbox showed a Pre-Test M: .50 ($SD = .08$) and Post-Test M: .66 ($SD = .07$, $p = .001$). These results suggest a movement-based curriculum is highly effective in improving motor and cognitive abilities, particularly executive functioning, language and self-regulation. Future research is needed to track students over several years to confirm movement directly leads to sustained cognitive gains and to investigate whether “open” motor skills provide greater cognitive benefits than “closed” motor skills.

Environmental affordances to infant postural development across the first year of life

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From the time infants are born, caregivers help mold infants' brains into learning new skills and translating them into daily life. Whether it involves trying to get from one place to another or seeking the attention of a caregiver, infants consistently need to adapt to their environment. Despite past literature supporting the role of environmental affordances, observations on the influence of caregiver behavior and toy placement on infant postural development are limited. In this naturalistic, observational research investigation, we hoped to identify trends between; [1] caregiver distance (within arm's reach and beyond arm's reach) and posture (floor, kneel, standing) influence on infant development of the upright poster, [2] age and posture related changes to toy placement (amount of toys and location), [3] developmental changes in infant locomotion within the first year of life in relation to postural control. Infants ($n = 22$) were observed

monthly until the onset of walking within their home environment. Each data collection included a 30-60 min video recording of the infant and their environment. Post-visit, the video was analyzed and continuously coded via Datavyu to determine the placement of the infant, caregiver, and toys. It was hypothesized that the more infant engagement produced by caregivers and toy placements, the more locomotion within the upright posture will occur and develop during the first year of life. Pilot data suggest that once infants begin to reach their upright postures (cruising and walking), onsets, toy placements were observed in a scattered position, closer to the floor. It also suggests that caregivers provided more opportunities for independent growth by gradually increasing their distance as the infant aged, promoting more exploration of their environment. Infant locomotion is suggested to have increased with more practice in postures as the infant aged and new environmental spaces became available. Further analyses, such as an ordinal GEE, were used to examine developmental changes in infants' abilities, postures, and ages, which may influence the placement of toys and the positioning of caregivers. Results provide a better understanding of the role environmental affordances play in shaping infant postural development and locomotion during the first year of life.

Tools of growth: Investigating equipment use and motor development in children

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Parents shape infants' early motor experiences through everyday decisions about space, activities, and the use of infant equipment. Devices such as walkers, bouncers, jumpers, and rockers are commonly used in the home, yet empirical evidence regarding how these structured environments influence infants' spontaneous motor behavior remains limited. Understanding how equipment use compares to unrestricted, free-roaming conditions is critical for informing evidence-based guidance on early motor experiences. The primary aim of this investigation is to determine whether commonly used infant equipment facilitates or constrains spontaneous, self-initiated movement compared to free-roaming conditions in the home. A secondary aim is to examine whether equipment use is associated with differences in the characteristics of infant movement patterns across early development. This longitudinal, naturalistic observational pilot study follows approximately 10 infants, assessed monthly from early infancy until the onset of independent walking. Data collection occurs in the home environment. Infant movement is captured using synchronized video recordings and MBientLab MetaMotion inertial measurement units (IMUs) placed on the infant's limbs and trunk, as well as on the caregiver's arm. Caregiver-mounted sensors allow differentiation between infant-generated movements and movements resulting from external assistance. Video data are coded using Datavyu with non-continuous coding to identify the type of equipment used, the presence or absence of movement, and the source of movement generation. Spontaneous motor activity is the primary outcome measure, while environmental condition serves as the primary comparison factor. We hypothesize that supportive equipment will reduce postural demands, allowing infants to generate spontaneous limb movements with greater ease compared to unsupported sitting or free-roaming conditions. Specifically, infants in seated equipment are expected to demonstrate increased arm and leg movement relative to unsupported postures. Preliminary data indicate that, of the ten infants observed, only one used a walker in which movement was entirely self-generated, while the remaining infants were not recorded using any infant devices. Findings from this pilot study will provide preliminary evidence on how structured equipment environments shape infants' spontaneous motor activity and will inform future studies and recommendations regarding infant equipment use and the optimization of early motor opportunities in the home.

Robot says: Effects of human and robot feedback on shaping reaching behaviors

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The rapid progress in robotics offers a great potential for innovative use in healthcare. However, whether children and adults respond to a robot's feedback as effective as to a human agent's (e.g., coach) feedback while playing a virtual-reality game is unclear. Twenty adults and 21 children volunteered to pop bubbles in seven virtual-reality games, including two in baseline, three in acquisition, and two in extinction. Participants were randomly assigned to received feedback from a robot or a human during acquisition. The 80% of the average movement time at baseline was used as the target time to determine feedback content, which was the same except for the agent (robot or human). Outcome variables included number of reaches that had a movement time less than the target time ("#successful"), percentage of successful reaches ("%successful"), and reaching kinematics (movement time, average speed, number of movement units, and path length). A mixed analysis of variance or analysis of covariance was used to determine significance. Results showed children's responses differed between agents: Children followed a robot's feedback to move their arm faster ($p = .04$) as did adults ($p < .003$); however, children showed no carry-over once feedback stopped. Children in the Human group did not follow the expected trend with delayed response to show faster movement, whereas adults followed human's feedback to move arm faster ($p < .003$). In adults, acquisition and extinction had significantly higher "#successful" and "%successful" than baseline (all $p < .001$). In children, there was no statistical difference among baseline, acquisition, and extinction phases on "#successful" (all $p > .05$), whereas acquisition phase had higher "%successful" than extinction ($p = .03$). Additionally, only in the Human group did children have a longer trajectory and slower and jerkier reaching than adults (all $p < .02$). Children showed higher intrinsic motivation than adults ($p < .03$) with a trend that children in the robot group had higher motivation than children in the Human group (all large effect sizes). This study indicated the potential to use a robot as a feedback provider to train adults and children but children might even prefer a robot than a human. Funding source: This study was supported by the Field Initiated Program of the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR) under award number 90IFST0009.

Relationship or situationship? Product and process motor competence scores differentially relate to health outcomes

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Given the variety of assessment tools available, motor competence may show differential associations with health outcomes. The purpose of this study was to determine the differential correlations between motor competence scores, as assessed by four different tools on multiple health-related outcomes. A total of 127 children in two age cohorts (4-5 years, $n = 65$; 10-11 years, $n = 62$) participated in the study. Participants' motor competence was assessed on the same skills via three process-oriented tools (Test of Gross Motor Development-2, Get Skilled Get Active, Developmental Sequences) and a composite of product-oriented scores (hop, run, throw, kick, jump). Health-related outcomes included physical activity (accelerometer), muscular strength (handgrip), body composition (body fat percentage), cardiorespiratory fitness (PACER), and perceived

motor competence (Pictorial Scale of Perceived Competence and Social Acceptance or Self Perception Profile for Children). Initial Pearson correlations were calculated, followed by bootstrapping with 10,000 iterations with replacement to provide confidence in the final result (i.e., average difference in correlations with 95% confidence intervals). In the younger age cohort, 7 of 30 correlations were significantly different from one another. Product assessments showed stronger correlations in five cases. The outcomes associated with these differences were body fat percentage ($n=2$), grip strength ($n=2$), cardiorespiratory fitness ($n=1$), and physical activity ($n=2$). The range of differences in these statistically significant correlations were $\pm .18$ to $.29$. In the older age cohort, 7 of 30 correlations were significantly different from one another. In four instances, product assessments were the superior measure. These differences were related to grip strength ($n=2$), cardiorespiratory fitness ($n=1$), physical activity ($n=2$), and perceived motor competence ($n=2$). The range of differences in these statistically significant correlations were $\pm .13$ to $.28$. There is value in using both product and process assessments, as they can provide unique information and be used for different purposes. However, product assessments tended to lead to stronger associations with health outcomes in the current study. Product assessments may also offer greater practicality and feasibility, which may further support their continued and more widespread use. Funding source: This work was financially supported by the National Institutes of Health grant 1R21HD055621-01A2.

Longitudinal associations between gross motor competency and physical activity in early childhood: Child and Mother Physical Activity Study (CAMPAS)

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Gross motor competence is thought to play a critical role in the development of physical activity behaviors during early childhood; however, longitudinal evidence using sensor-based physical activity measurement in this age group remains limited. The study objective was to examine longitudinal associations between gross motor competence and sensor-measured physical activity across early childhood. Data were drawn from the Child and Mother Physical Activity Study (CAMPAS), a prospective cohort with up to five waves of assessment conducted approximately every six months from ages 1 to 3 years. Physical activity was assessed using ActiGraph accelerometers, from which moderate- and vigorous-intensity physical activity (MVPA) and total physical activity were derived. Gross motor competence was assessed using the Ages and Stages Questionnaire (ASQ) gross motor subscale and categorized into low versus high competence. Linear mixed-effects models were used to examine associations between gross motor competence and MVPA and total physical activity, adjusting for child age, sex, and neighborhood resources. A total of 139 children (74 females) participated in CAMPAS. Mean age was 13 ± 2 months at baseline and 39 ± 3 months at Wave 5. Across all waves, children with higher gross motor competence consistently demonstrated higher mean levels of MVPA and total physical activity, compared with those with lower competence. In adjusted mixed-effects models, low gross motor competence was associated with lower MVPA ($\beta = -5.7$ min/day; $p = .049$) and lower total physical activity ($\beta = -12.3$ min/day; $p = .028$). MVPA increased with age overall; however, this age-related increase was attenuated among children with lower gross motor competence (interaction $\beta = -.6$ min/day; $p = .021$). No significant interactions between gross motor competence and age were observed for total physical activity (interaction $\beta = -.1$ min/day; $p = .874$). Children with lower gross motor competence demonstrate persistently lower physical activity levels and slower age-related gains. This study provides high-quality evidence that gross motor

competence is an important factor for physical activity development in early childhood, particularly for MVPA. Early interventions that promote gross motor development may be important strategies for supporting healthy physical activity trajectories during early childhood. Funding source: NHLBI.

Implications of motor assessment choice: Comparing MABC-3 and BOT-3 outcomes across neurodevelopmental groups

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Motor challenges are common in autism spectrum disorder (ASD) and developmental coordination disorder (DCD); however, variability across standardized assessments may influence how motor challenges are identified and interpreted. This study compared the outcomes of two widely used motor assessments, the Movement Assessment Battery for Children-Third Edition (MABC-3) and the Bruininks-Oseretsky Test of Motor Proficiency-Third Edition (BOT-3), in autistic, those with DCD, and neurotypical (NT) children. Seventy-eight children aged 5–18 years completed both assessments. Results revealed significant main effects of group and assessment, with autistic children and those with DCD scoring significantly below NT peers on both measures. Overall scores were lower on the MABC-3 than the BOT-3 across groups. Classification agreement between instruments was poor ($\kappa = .20$), with particularly low agreement within the ASD and DCD groups, indicating frequent discrepancies in identification of motor difficulty. In contrast, neurotypical children showed consistent classification across assessments. These findings suggest that although both assessments differentiate neurodevelopmental groups at the group level, they are not interchangeable for individual classification. Differences in task demands and construct emphasis may lead to divergent identification of motor challenges. Careful assessment selection is therefore critical in both clinical and research contexts involving neurodevelopmental populations. Funding source: Pearson Assessments.

Motor Learning and Control Abstracts

Evaluating the transfer effects of skill acquisition principles on performance in academy soccer players

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Despite growing evidence for the benefits of skill acquisition principles (SAPs), their effects in real-world settings remain understudied. This study examined the transfer effects of SAPs on youth soccer players' performance in an elite academy environment. Fifty male youth players (16.54 ± 1.01 years) and eight professional coaches from an elite soccer academy participated in a within-subject design comprising a control phase followed by an experimental phase in which coaches applied seven evidence-based SAPs within their regular weekly microcycles. Performance was assessed at four time points (pre-test [T0], post-control [T1], post-experimental [T2], retention [T3]) using (i) match statistics quantifying 12 in-game performance indicators and (ii) a video-based test assessing declarative (tactical) knowledge. SAP implementation was verified via a manipulation check, and internal and external training load were monitored during all training sessions. Changes over time were analyzed with generalized linear models and t -tests. Following successful SAP implementation ($p < .01$, $\eta^2 = .11$), no significant improvements were found on the 12 match performance indicators between T1 and T2 or between T1 and T3, although several indicators displayed delayed positive trends at T2

and/or T3. In the declarative knowledge test, response time improved from T0 to T2 ($p < .05$), whereas response accuracy remained unchanged. Compared with the control phase, SAPs-based sessions were shorter in duration, imposed lower external load, and elicited higher mental demand ($p < .05$). Overall, embedding SAPs into coaches' daily practice produced clear and measurable changes in training dynamics and suggested near-transfer effects in declarative tactical knowledge. However, transfer to on-field match behavior remained limited within the intervention timeframe, potentially reflecting the longer consolidation required for complex performance adaptations and/or the limited sensitivity of the match performance indicators used. These findings highlight the feasibility of implementing a multi-principle SAP approach at scale in elite youth soccer, while emphasizing the need for longer interventions and/or more sensitive match assessment strategies to capture potential on-field transfer. Funding source: Mitacs International.

Cognitively demanding breaks hinder micro-offline gains during motor sequence learning

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Motor learning occurs through both micro-online gains (performance improvements during practice) and micro-offline gains (performance improvements after brief rest periods) within a practice session. However, it remains unclear whether the structure of these brief rest periods can optimize motor memory consolidation. Therefore, the purpose of this study was to examine whether activities performed during wakeful breaks within a practice session influenced micro-offline learning of a sequential motor task. Participants were randomly assigned to learn either a deterministic or a probabilistic sequence and then to one of three break conditions: unstructured, cognitive, or motor. We used the Serial Reaction Time Task (SRTT) to assess learning of the motor sequence and the NASA-TLX to assess perceived workload. Preliminary results from 34 participants (M_{age} : 21.6 ± 2.4 years; 24 females) who learned the deterministic sequence are presented here. We found intact learning in the unstructured break group and impaired learning in the cognitive break group. The motor break group exhibited motor learning, but also evidence of interference from randomized trials. While there were no significant differences between micro-online and micro-offline learning in the unstructured and motor break groups, the cognitive break group exhibited significantly lower micro-offline learning than micro-online learning, suggesting that cognitively demanding breaks hindered consolidation. There were no significant differences in perceived workload across groups. These preliminary findings suggest that reducing cognitive load during rest periods may be essential for optimizing performance gains. Understanding the processes involved during rest may inform the design of training protocols that enhance motor learning outcomes.

Age-related changes in gait variability and the relationships between common gait variability measures

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Measures of gait variability are frequently interpreted as indicators of aging-related decline and neuromuscular health. Different techniques of quantifying variability include linear and non-linear measures, with non-linear measures often being found to be more sensitive between certain populations. However, it remains unclear which measures are most sensitive to age-related changes in gait and whether different variability

metrics quantify a common underlying construct. This work aimed to (1) determine which variability measures are sensitive to aging, and (2) identify whether these measures quantify shared constructs of gait variability. Healthy adults (34) spanning the adult lifespan walked at their preferred speed on an instrumented treadmill. Linear variability measures of magnitude (standard deviation and coefficient of variation), regularity (sample entropy), persistence (detrended fluctuation analysis), complexity (correlation dimension), and local dynamic stability were computed for stride parameters, knee kinematics, and pelvis motion. Relationships between variability measures, age, and preferred walking speed were examined using Pearson correlation, and principal component analysis with varimax rotation was used to identify common constructs among measures of gait variability. Gait variability measures demonstrated limited sensitivity to chronological age but were strongly associated with preferred walking speed. Sample entropy ($R^2 = .17$) and coefficient of variation of stride parameters ($R^2 = .12$) were the two variability measures significantly correlated with age. More and stronger relationships were found between gait speed and measures of variability including sample entropy of stride time and length ($R^2 = .51$, $R^2 = .14$), knee and pelvis complexity ($R^2 = .34$, $R^2 = .26$) and pelvis local dynamic stability ($R^2 = .11$). Principal component analysis revealed four independent constructs explaining 72% of the variance linking (1) stride variability and vertical pelvis stability, (2) knee stability, dimensionality, and stride regularity, (3) medial-lateral stability of pelvis motion, and (4) statistical persistence of strides. Linear and nonlinear measures loaded together when quantifying the same construct, rather than separating by methodological class. Gait variability does not reflect a singular measure of locomotor health nor is it inherently sensitive to aging. These findings emphasize the importance of selecting variability measures based on the specific construct of gait control under investigation.

The effects of varied step lengths on thorax angles, front thigh angles, and trail thigh angles in walking lunge exercises

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The purpose of this project was to analyze the effects of utilizing varied step lengths during walking lunge exercises on thorax angles (relative to the vertical), and front and trail leg thigh angles (relative to the horizontal). It was hypothesized that step length would not have a significant effect on thorax angles or front leg thigh angles. However, it was hypothesized that trail thigh angles would be significantly smaller in the shorter step length condition. Thirty-six males and thirty-six females ($M_{age} = 19 \pm 2$ years; $M_{mass} = 75 \pm 20$ kg; $M_{height} = 1.75 \pm .10$ m) performed four sets of unloaded walking lunges at their normal depth and pace. Participants' movements were recorded with eight Basler cameras at 100 Hz. The video files were processed using The MotionMonitor software in order to compute kinematic data. The thorax and thigh angles at the lowest vertical position of the entire body center of mass were used in the analysis. The relationships between thorax and thigh angles and lunge step lengths were assessed using two-way factorial ANOVA tests. Statistical significance was set at $\alpha = .05$. Step length values ranged from $.745 \pm .070$ m (short) to $.962 \pm .075$ m (long). Thorax and thigh angles showed statistically significant ($p < .001$) differences between front step limbs, but limited (1 to 3 degrees) practical differences. The thorax angles were significantly larger ($p < .001$) using a short step (26.0 ± 8.5 degrees) than a long step (23.8 ± 7.3 degrees). The front thigh angles were significantly larger ($p < .001$) using a short step (9.6 ± 7.0 degrees) than a long step (7.5 ± 5.4 degrees). The trail thigh angles were significantly smaller ($p < .001$) using a short step (77.2 ± 13.0 degrees) than a long step (101.0 ± 12.6 degrees). Results indicate that there is a significant relationship between walking lunge step length and thorax, front thigh, and trail

thigh angles. These results suggest that walking lunge step length alters the mechanical characteristics of performers.

Women at the plate: Association between visual skills and batting performance indicators in highly trained female baseball players

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Visual skills (VS) are crucial for successfully executing interceptive motor skills such as baseball batting and are proposed to span a continuum from fundamental (e.g., visual clarity) to cognitive skills (e.g., decision making). Understanding their contribution to batting performance is therefore essential for talent selection and development. Evidence regarding the predictive values of VS for batting performance remains inconsistent in male athletes and has not yet been investigated in female athletes. This absence of empirical investigation in female athletes highlights a sampling bias in the current literature and emphasizes the need for sex-specific research to support broader generalization. Therefore, the aim of this study is to associate a continuum of visual performance indicators with a set of batting performance variables in female baseball players. Twenty-eight highly trained female baseball players from the same club (aged 15-23 years; $M_{\text{age}}=17.27$) underwent a standardized generic sensorimotor test battery (Senaptec Sensory Station). Six generic VS spanning from fundamental to cognitive were assessed: visual clarity, contrast sensitivity, near-far quickness, multiple object tracking, reaction time, and perceptual span. These measures were associated with eight performance indicators, including game statistics (at-bats, hits, batting average) across 34 matches in a single season (GameChanger), coach-designed players' ranking, age, and years of practice. A total of 48 Bayesian correlations between VS and performance indicators were computed using JASP. Bayesian analyses identified one association (2.1%, between perceptual span and players' ranking) with moderate level of evidence in favor of the positive association. Therefore, this study provides limited support to the hypothesis that performance on generic VS predicts batting performance on female athletes. When considered alongside previous evidence in male athletes, these findings suggest potential non-linear or inexistant relationship between generic VS and batting performance. Those results may be explained by the fact that generic VS, assessed under well-controlled laboratory conditions and relying on declarative or simple motor responses, may not reflect the perceptual and motor demands of baseball batting. This study represents a first step toward addressing the sampling bias and encourages future research to examine the predictive value of sport-specific perceptual-cognitive skills assessed in representative scenarios in female athletes. Funding source: This study was supported by the Ministère de l'Éducation du Québec, Baseball Québec and Mitacs.

Acquisition of a dyadic interpersonal coordination pattern through self-organizing processes

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Interpersonal coordination can occur spontaneously or emerge through training processes. The reported experiment employed a discovery training program by minimizing individualized movement instructions in favor of instructions that emphasized using dyadic visual information. The dyad goal was to produce a 90° relative phase pattern between partners' arm motions. The task required each partner in a dyad to trace a 12 cm line (using a hand-held dowel) in an anterior-posterior direction

along the body midline while standing. LED diodes on the dowels produced a 3D motion signal recorded by a camera that was displayed in the form of a cursor moving on a computer screen. The 90° dyad relative pattern was defined by a circle on the screen. Dyads were instructed to trace their individual lines together so that the cursor moved around the circle, thus in a 90° pattern. The diodes provided the data used for analysis and for generation of the cursor as feedback. The dyads had full visual of each other. Forty-two college-age adults (21 dyads: 9 male; 18 right-handed, 2 left-handed, 1 mixed-hand) volunteered for the study. Dyad training consisted of 40 trials with 10 retention trials after a 10 min break. Concurrent visual feedback of the cursor was available on all trials, and terminal feedback of the cursor was provided on 20 trials. An auditory metronome paced movement at 1 Hz. The frequency and amplitude data of the individual dowel trajectories and the relative phase data between the dowel trajectories were analyzed with linear mixed-effects models with trial and condition-related factors as fixed effects with dyad included as a random effect. Movement amplitude increased with training: trial 1, 9.9 cm to trial 40, 11.3 cm ($p < .001$). Movement frequency decreased with training: trial 1, .4Hz to trial 40, .33Hz ($p < .001$). Dyadic coordination improved toward the target of 90° increasing from 36% time on target (trial 1) to 77% time on target (trial 40) ($p < .001$). Visual coupling supports the spontaneous formation of a dyadic coordination pattern without the need for complex verbal instructions that often are a burden to attentional processes.

Motor conflict resolution in the basal ganglia

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Decision making in the face of conflict requires coordination between the cortices and the basal ganglia. Theta oscillations communicate conflict, signaling the need to accumulate more evidence, whereas beta oscillations reflect motor intent and execution. The present study seeks to understand the network patterns in the basal ganglia using recordings from the subthalamic nucleus (STN) and globus pallidus internus (GPi) to understand motor conflict resolution and provide insight regarding executive control strategy. The Simon conflict task was administered intraoperatively to twenty-one participants with Parkinson's disease receiving deep brain stimulation devices targeting either the STN ($n=10$) or the GPi ($n=11$). Response locked local field potentials in the theta and beta bands were analyzed as a function of subregion (STN/GPi) and condition (conflict/non-conflict) using permutation-based statistics. Across conditions, significant ($p < .05$) modulations of theta and beta power were observed across the cue to response window. STN theta power was relatively higher in non-conflict trials earlier in trial period [-475:-258 ms before the response], and lower as the response approached [-119:0 ms before the response]. The same pattern was seen in the conflict trials with greater theta power in the STN further from the response [-447:-115 ms] and significantly less power approaching the response [60:0 ms]. Beta power was consistently elevated in the STN across conditions. In short, closer to the cue presentation theta and power was higher in the STN, as the response approached the GPi presented with a sharp increase in theta power. The STN shares neural projections with the motor cortex and projects onto the GPi, which serves as the final output structure in the basal ganglia. The late increase in theta power in the GPi suggests event related synchronization implying (general) task resolution whereas the sustained desynchronization in the STN implies ongoing communication of conflict at the task as opposed to trial level. Persistent beta activity in the STN further supports its role in maintaining motor control during decision making under conflict.

Gait and turning deficits in people with Parkinson's disease with and without rapid eye movement sleep behavior disorder

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Gait deficits are seen in about 50% of people with Parkinson's disease (PD) and these deficits may relate to other behavioral outcomes such as isolated rapid eye movement sleep behavior disorder (RBD). However, little work has been done to characterize gait and turning impairments in PD with probable RBD (PD+pRBD) and without probable RBD (PD-pRBD). In this study, we used wearable inertial sensors to identify differences in gait and turning between PD+pRBD and PD-pRBD. Although data analysis is ongoing, current data are available from 13 participants with PD+pRBD and 42 PD-pRBD collected at Oregon Health and Science University ($n=46$) and Arizona State University ($n=10$). In this cohort, probable RBD was evaluated using the RBD Sleep Questionnaire, which has shown high diagnostic accuracy. Participants completed a 2-min walking assessment with and without dual tasking; in which participants walked in a 10-m hallway at a comfortable pace, with 180 degree turns at each end. Both parametric (linear model) and non-parametric (rank-based) regression approaches were applied to the data since several gait and turning outcomes were not normally distributed. Age, sex, and MDS-UPDRS part 3 scores were included as covariates in both regression models and corrected for multiple comparisons using false discovery rate. Preliminary analysis suggests that PD+pRBD have faster turning velocity during both the single-task and dual-task 2-min walking assessments compared with PD-pRBD (pFDRs < .01). Secondly, during the dual-task walking assessment, PD+pRBD had a faster turn duration and larger lateral step variability compared with PD-pRBD (pFDRs < .05). All other single-task and dual-task gait and turning metrics were not statistically significant (pFDRs > .05). These data are part of an ongoing project with continued data collection over the next 3 years. Expanded sample size will enable more robust analysis and improved reliability of results. These data will help determine if an added diagnosis of pRBD, on top of a PD diagnosis, negatively impact mobility. Funding source: NIH 5R01AG086533-02; NIH AG077380.

Are athletes and musicians better at planning ahead? A construct validation of planning span

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This study investigated whether Division I collegiate athletes and expert musicians exhibit advanced motor planning span and planning time compared to the control group of non-athletes/non-musicians. Motor planning, the ability to anticipate and sequence movements to achieve specific goals, is critical in sports and music, requiring rapid adaptation. Previous research suggests athletes and musicians may develop enhanced planning abilities and cognitive strategies through intensive training, potentially leading to transferable skills beyond specific contexts. Ninety participants were recruited, aged 18-25, without motor or neurological impairments. Participants were 30 Division I athletes from open-skill ball sports, 30 high-level musicians (e.g., university School of Music major), and 30 control participants. Athletes were required to be practicing at least 7 hours a week in sport-specific training, and musicians at least 10 hours a week. Participants completed a motor planning task involving grasping and rotating a cylindrical object to align illuminated pointers with

corresponding target locations, under randomized conditions. The task assessed planning span (how far in advance movements are planned) and planning time. Data was collected using motion capture. Preliminary analysis ($n=17$) showed no significant differences between groups for our span measure, Mean Distance from Comfortable (MDC; $p=.24$, partial $\eta^2=.19$). However, athletes were closer to comfortable at the end-state for all spans and exhibited more time planning before initiation of movement. Whereas the other groups tended to plan more during reach time. Planning before the initiation of movement allows the athlete to conserve energy and prepare for future movements. These findings demonstrated that open-skilled athletes plan movements more efficiently than controls, which supports the construct validity of our task and suggests that athletic experience enhances cognitive-motor integration. If athletes show more efficient movement planning, this could highlight the broader impact of athletic experience on brain adaptability and skill transfer, with implications for rehabilitation and coaching. Funding source: Utah State University.

Investigating motor cortex contributions to music-evoked emotion with continuous theta-burst stimulation

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Despite past neuroimaging work showing activation of the motor network is associated with emotional response to music, it remains unclear whether the primary motor cortex (M1) directly influences our emotional response to music. Here, we tested the influence of M1 activity on emotional response to music. Sixteen right-handed participants ($N=16$; aged 23.9 ± 7.9 , planned $N=22$) engaged in two sessions whereby they underwent active (Day A; delivered to contralateral M1) or sham (Day B; placebo) continuous theta burst stimulation (cTBS; a form of non-invasive brain stimulation used to transiently inhibit focal regions of the brain) with the order of Day A and B counterbalanced across participants. Following cTBS, participants engaged in an active listening task, whereby they listened to musical stimuli and rated their perceived valence and arousal. A total of 32 stimuli from a validated database were used for four affective stimuli conditions (8 per condition), high valence/high arousal (HVHA), high valence/low arousal (HVLA), low valence/high arousal (LVHA), and low valence/low arousal (LVLA). We hypothesized that M1 inhibition would result in reduced arousal and valence during the active listening task vs. sham regardless of condition. For each participant, valence and arousal ratings were z-scored using all individual trial-level data to enable group-level comparisons of standardized means. To quantify changes in valence and arousal following M1 inhibition, effect sizes (Cohen's d) were calculated (cTBS minus sham) separately for each stimulus condition. Results indicate decreased arousal during active listening following M1 inhibition, but only for HVLA stimuli ($d = -.25$; other $ds < 1.21$). No evidence of change in valence was found (shown via negligible effect sizes; all $ds < 1.21$). Findings suggest that M1 selectively contributes to arousal related aspects of music. Overall, this work contributes to understanding the role of M1 to emotional response to music.

Adaptive visual fixation subserving successful basketball shooting

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While proper technique is foundational for basketball shooting, performance is ultimately guided by the visual information used to calibrate shot

parameters like force and trajectory. The “quiet eye” (QE), the final pre-movement fixation, is a key marker of this visual-cognitive process and is linked to superior performance in domains from athletics to surgery (Gonzalez et al., 2017; Harvey et al., 2014). However, QE research has predominantly focused on fixations prior to movement, neglecting those during the action itself. This study investigated visual search patterns during the basketball shot across different distances (free-throw and three-point line) and outcomes (successful “swishes” vs. misses). Eleven recreational basketball players performed a shooting task while wearing Tobii Pro Glasses 3 eye-trackers. Fixation locations and durations were analyzed for seven predetermined areas of interest (AOIs) on the basketball board. Although mean trial duration (~1.4 s) did not differ by distance or outcome, the mean proportion of fixation duration (fixation duration/trial duration) showed significant main effects for AOI, $F(6, 60) = 9.92$, $p < .001$, and Distance, $F(1, 10) = 6.21$, $p < .05$, as well as significant AOI x Accuracy, $F(6, 60) = 4.20$, $p < .005$, and AOI x Distance, $F(6, 60) = 3.66$, $p < .005$, interactions. Further analysis revealed that participants fixated only on the inner backboard, net, and rim during the shot, with significantly longer fixation on the rim than on other AOIs ($p < .01$). Moreover, successful free-throw “swishes” were associated with the greatest proportion of time spent fixating on the rim. These findings suggest that successful shooting is not governed by a single visual strategy but involves an adaptive process tailored to specific constraints, such as shooting distance. Future research should integrate gaze data with motor-coordination metrics to define a comprehensive, optimal visuomotor control strategy for shooting accuracy.

“That’s a mouthful!” Increased energy output, but not input, impacts estimates of oral cavity volumes

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An individual’s perceived ability to act is mediated by factors related to the environment (e.g., object height and distance) and the body (e.g., handspan and leg length). Although there is some debate over the veracity of these particular effects, there is also reported evidence that the perceived ability to act is influenced by physiological factors that are internal to the body such as energy states and fatigue. These physiological factors might have an important role in shaping perceived ability to act for actions that involve body parts that are not readily visible, such as when estimating how much water can fit in one’s mouth. Previous work from our lab has shown that people tend to over-estimate the volume of their oral cavity, but are more accurate in these estimations after experience drinking. The present studies were designed to determine if exercise (Experiment 1) or the consumption of a food with dehydrating properties (Experiment 2) impacts estimations of oral cavity volume. To measure the estimates of oral cavity volume, participants poured the maximum amount of water into a glass that they believed they could hold in their mouth. Participants in Experiment 1 made estimations before and after exercising for 10 mins at either a Low ($n = 20$) or a High ($n = 20$) rate. A significant Group by Time interaction ($p < .05$) revealed that estimations significantly increased for the High group, but not for the Low group. In Experiment 2, participants made estimations before and after eating 10 saline crackers. Participants in the Hydrated group ($n = 15$) were able to rinse their mouth with water prior to the estimation task, whereas participants in the Dehydrated group ($n = 15$) were not allowed any water prior to the estimation task. Neither group showed any pre/post differences (Group by Time interaction, $p = .55$). These data indicate that physiological processes associated with increased energy output might impact estimates of oral cavity volume, whereas processes associated with energy input do not.

Motor memory interference occurs independently of temporal and kinematic execution dynamics

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Interference refers to a reduction in the retention of a previously acquired motor memory developed to a new sensorimotor mapping A following exposure to a competing sensorimotor mapping B. In visuomotor adaptation studies, interference is typically characterized using spatial error-based measures, such as hand angle deviations from the target. It remains unclear whether such interference is also accompanied by changes in the temporal and kinematic measures of movement execution. We examined how learning context influenced reaction time, movement time, and peak velocity during a visuomotor adaptation paradigm. Participants first adapted to a visuomotor rotation A, then adapted to an opposing rotation B, and were retested on A after a 24-hour delay. Learning context was manipulated such that competing motor memory developed under context B was acquired through predominantly implicit (clamped feedback), explicit (delayed feedback), or combined processes (abrupt). Across learning contexts, reaction time was reliably reduced during relearning of A relative to initial exposure, indicating retention through improved efficiency in movement preparation with task familiarity. Importantly, this reduction occurred regardless of whether prior A learning was preserved, overwritten, or partially retained. This finding demonstrates that changes in reaction time were dissociable from motor memory interference. Movement time showed limited modulation across A blocks and was primarily influenced by feedback availability during B. Peak velocity remained stable across all experiments and learning contexts, indicating preserved scaling of the feedforward motor command during movement execution despite differences in learning mechanisms and memory interference. Overall, these results reveal a dissociation between motor memory competition and movement execution dynamics. While learning context determines whether competing motor memories interfere, temporal and kinematic execution-related measures indicate that planning efficiency, feedback-dependent control, and motor command scaling are largely conserved. This dissociation suggests that motor interference arises from learning-related changes that are not expressed through temporal or kinematic execution characteristics, raising questions about which components of the motor representations are disrupted during competing motor learning or if the reaction time, movement time, and peak velocity data reveal different learning processes.

Does direction matter? An examination of transfer of learning between virtual and real world environments

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Previous research has indicated that practicing in immersive virtual reality (VR) facilitates positive transfer of learning to real-world (RW) environments, supporting VR as a viable training modality across multiple contexts (e.g., sport, medicine, military). However, the reciprocal relationship – whether RW practice effects transfer to VR – remains unexplored. Some studies suggest “one-way transfer” where changes in performance facilitated through VR training transfer to RW environments, but practicing in the RW does not transfer to VR. This study examined bidirectional transfer between VR and RW environments following multiple days of practice. It was hypothesized that both practice conditions (VR & RW) would demonstrate significant improvements in their respective practice environments (retention) and alternate environment (transfer). Participants ($n = 47$) were randomly assigned to golf putting practice in

immersive VR ($n=25$) or RW ($n=22$) environments across 3 days (180 total practice trials). All participants completed a pretest consisting of 10 trials in both VR and RW environments before each day of practice on Days 1-3, and in both conditions (VR & RW) on Day 4 as posttests. Separate 2 (Group) $\times$ 4 (Test) mixed ANOVAs revealed significant main effects of Test in both environments ($p < .001$). In the RW environment, no main effect of Group emerged, $F(1, 41) = .33$, $p = .569$, and both groups showed similar performance at posttest, suggesting VR-to-RW transfer. Conversely, in the VR environment, a significant main effect of Group was observed, $F(1, 38) = 6.86$, $p = .013$, with the VR practice group demonstrating superior performance ($p = .035$) following the cessation of acquisition. Both groups showed significant Test 1 to Test 4 improvements in both environments (all $ps < .01$). These findings suggest an asymmetric bidirectional transfer. Specifically, VR practice effects positively transferred to RW performance, whereas RW practice, while producing significant improvements in VR performance ($d = .87$), led to relatively inferior transfer to the VR environment. These findings challenge claims of “one-way transfer” by demonstrating that transfer occurs bidirectionally, albeit asymmetrically. VR practice appears sufficient for RW skill learning, while RW practice produces significant, but less pronounced transfer to virtual environments.

Human sensorimotor synchronization while walking on the treadmill with various metronome beats

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The human sensorimotor system entrains to external rhythmic cues, and this phenomenon is well-studied in finger-tapping paradigms [Repp and Su, 2013]. This sensorimotor synchronization (SMS), known as the coordination of timing between movements and external rhythms [Ross & Balasubramaniam, 2014], forms the basis of rhythmic auditory stimulation (RAS), a rehabilitation tool that uses structured sounds like metronome beats to entrain and stabilize movement patterns. SMS is fundamental to adaptive gait, enabling real-time changes in speed and direction on a variety of surfaces. However, the relationship between auditory cueing and motor output in children in terms of SMS remains unclear. Although children achieve adult-like walking by seven, they exhibit variability in meeting timing constraints. This study investigates how RAS affects gait dynamics in typically developing children by analyzing vertical ground reaction force (vGRF) parameters, including peak forces, and braking/propulsive impulses. Six children aged between seven and eleven years walked barefoot on an instrumented treadmill at their preferred speed under four conditions: without a metronome and with a metronome at 75, 100, and 125 beats per min (BPM). vGRF parameters were measured and normalized by body weight. A one-way repeated measures ANOVA (1 Group $\times$ 4 Condition) was conducted, and there was a significant main effect of auditory cueing condition ($\alpha = .05$). The normalized first peak vGRF was greatest without a metronome and significantly decreased as BPM increased ($p < .001$). A similar but no significant difference was observed for the normalized second peak vGRF ($p = .087$). Both normalized braking and propulsive impulses decreased significantly with increasing BPM tempo ($p < .001$). These results indicate that RAS directly modulates gait kinetics, leading to more constrained force production. Consistent with previous findings, children displayed significant differences in motor output that were modulated by auditory cueing, potentially reflecting ongoing integration of auditory perception and motor control. RAS may structure locomotor output even in typically developing children, suggesting rhythm-based treadmill interventions can be an effective tool for children with ongoing brain development [Colling et al, 2017]. Future research should investigate these effects in children with gait impairments or developmental disorders.

The influence of attentional focus on mid-thigh pull performance

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A holistic focus of attention has been shown to benefit performance and learning across a variety of tasks; however, its effects have been examined less frequently in isometric tasks such as the mid-thigh pull. The mid-thigh pull is commonly used within athletic populations to assess maximal isometric strength. The purpose of this study was to investigate the effects of different attentional foci (internal, external, and holistic) on mid-thigh pull performance. Twenty-three individuals performed a total of eight mid-thigh pulls under control (CON), internal focus (IF), external focus (EF), and holistic focus (HF) conditions. The mid-thigh pull was performed on a Contemphas portable force plate with a Synergee barbell placed on a Fringe Sport rack. The primary dependent variables were maximum force production (MFP) and rate of force development (RFD). Two separate one-way ANOVAs with repeated-measures were used to analyze the data. Bonferroni post-hoc corrections were applied to significant main effects. A significant main effect of condition was observed for MFP ($p = .006$), with post hoc analysis revealing that HF produced greater force than IF ($p = .016$). No significant differences were observed between conditions for RFD ($p > .05$). These findings suggest that a holistic focus of attention may enhance force production during high-explosive tasks such as the mid-thigh pull and may serve as an effective cueing strategy for athletes undergoing strength assessment or training.

The effect of external and internal focus of attention on the performance of VR punching task

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There is ample evidence showing the advantage of external focus of attention on performance and learning. According to common coding theory (Prinz, 1997), this depends on the availability of information that specifies the anticipated effect of the movement on the environment. In punching a boxing bag, a salient source of information for the movement effect is the felt impact. Using virtual reality (VR), we manipulated the haptic information specifying the impact, to test if indeed the advantage of external focus relates to attention toward the movement effect. Twenty-four participants (18 males, 6 females) performed a punching task under both internal and external focus of attention instructions across two separate sessions. Participants wore a VR headset and punched a bag in two conditions: a virtual bag (without haptic information on the movement effect) and physical bag (with haptic information). Peak hand velocity at impact functioned as performance measure. linear mixed-effects model analysis revealed a significant main effect of attentional focus, with higher punching velocity under external compared to internal focus of attention, $b = -.68$, $SE = .11$, $z = -6.18$, $p < .001$, 95% CI $[-.90, -.47]$. A significant main effect of bag condition was also observed, with higher velocity when striking the physical bag compared to the virtual bag, $b = .74$, $SE = .11$, $z = 6.79$, $p < .001$, 95% CI $[.53, .95]$. Critically, the interaction of attentional focus and bag condition was not significant, indicating that the advantage of external focus of attention was preserved regardless of the availability of haptic information related to movement effects, $b = -.09$, $SE = .16$, $z = -.58$, $p = .56$, 95% CI $[-.39, .21]$. This finding challenges the explanation from common coding theory that the advantage of an external focus is due to its attention for movement effect. Rather, it is consistent with an ecological perspective that external focus is superior because it fosters attunement to task relevant information specifying the dynamic relation between the fist and the target.

Examining the effects of attentional focus on mental workload within a clinical psychomotor skill

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Literature within motor learning has shown that external focus of attention (EFA; focusing on the effect of the movement) enhances automaticity of movement and requires less conscious control compared to an internal focus of attention (IFA; focusing on movement characteristics) for various types of motor skills. While attentional focus has been applied to improve clinical skill acquisition, application to athletic training clinical skills requires greater attention. The primary aim of this pilot study was to examine the effects of attentional focus instructional cues and feedback on athletic training students' (ATSS) ability to perform C7 triceps deep tendon reflex testing; this secondary analysis examined the effects of attentional focus on mental workload for the task. We hypothesized that ATSS receiving EFA cues would report lower mental workload compared to subjects receiving IFA cues. Thirteen ATSS (EFA = 6; IFA = 7; $M_{\text{age}} = 23.31$, $SD = 2.06$) learned to perform reflex testing through a prerecorded video followed by either EFA or IFA instructional cues. Participants then performed 10 trials on a standardized patient, received either EFA or IFA verbal augmented feedback paired with video footage, and performed another 10 trials. Following the task, they completed the NASA-TLX as a measure of mental workload. Mann-Whitney U tests indicated significantly lower Raw TLX scores ($U = 6.5$, $p = .038$, $r = -.58$) and Frustration subscale scores ($U = 5.5$, $p = .024$, $r = -.62$), but also lower Performance subscale scores ($U = 4.5$, $p = .017$, $r = -.66$) for the EFA group compared to the IFA group; all with large effect sizes (Cohen's r). There were no significant differences for mental demand ($U = 20.0$, $p = .886$, $r = -.40$), physical demand ($U = 16.5$, $p = .511$, $r = -.18$), temporal demand ($U = 8.5$, $r = -.51$), or effort subscales ($U = 11.5$, $p = .170$, $r = -.38$). Collectively, these findings provide limited support for the use of EFA instructional cues and feedback in athletic training education, suggesting potential reductions in perceived workload and frustration during early skill acquisition, while highlighting important considerations regarding learners' perceived performance.

The effects of directionally focused strategic self-talk cues on broad jump performance

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The self-talk matching hypothesis predicts that instructional self-talk is most effective for tasks involving precision and accuracy and that motivational self-talk is most effective for tasks involving endurance, strength, and power (Theodorakis et al., 2000). This predictive effect has not been consistently observed throughout the literature (Hardy et al., 2018). Recent research has suggested this lack of consistency may be due to the use of various attentional foci (e.g. internal and external) within instructional self-talk (Sampson et al., 2025). Motor learning research has consistently shown a benefit for an external focus of attention across a variety of tasks when compared to an internal focus over multiple decades (see Chua et al., 2021). To better investigate the matching hypothesis, the present study examined the effects of internally and externally framed instructional self-talk in comparison to motivational self-talk on broad jump performance. 15 participants ($M = 20.7$ years) performed a series of eight jumps, two under each condition, in a counter-balanced order: control (CON), internal-instructional self-talk (IIST), external instructional self-talk (EIST), and motivational self-talk (MST). Jump distance

was the primary dependent variable and was analyzed using a one-way ANOVA with repeated measures. If a significant main effect was observed, Bonferroni post-hoc tests were used to examine performance differences across conditions. Results revealed a main effect for jump distance ($p = .002$), with post hoc tests revealing significant differences between CON and EIST ($p < .001$), all other comparisons were not significant. These results suggest that EIST benefits broad jump performance more than IIST or MST. These findings are in line with the attentional focus literature and the constrained action hypothesis. Interestingly, these results oppose the predictions of the self-talk matching hypothesis as externally framed instructional self-talk was most effective for the performance of a discrete motor task involving power in comparison to motivational self-talk. It is suggested that more research examining the effects of directionally focused instructional self-talk and comparing it to motivational self-talk is needed to further re-examine the predictions made by the matching hypothesis.

Bridging theory and practice: How coaches apply motor behavior principles across sports

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Coaches play a critical role in shaping motor skill acquisition and long-term learning; however, the extent to which motor learning principles are intentionally integrated into coaching practice remains unclear. This study examined how coaches across sports and competitive levels implement evidence-based motor learning strategies related to attentional focus, feedback, practice structure, and motivation. Fifteen coaches ($M_{\text{age}} = 39.7 \pm 14.3$ years) from youth, elite, and collegiate programs participated in an anonymous survey created in Qualtrics and distributed via email to coaching organizations, schools, and professional networks. The survey included structured items and open-ended questions. Responses were compiled, percentages were calculated for close-ended questions, and thematic analyses were conducted to identify commonalities in qualitative responses. Quantitative findings indicated widespread use of practices consistent with contemporary motor learning research. Most coaches reported using both external (77%) and internal (69%) attentional focus cues, while the majority emphasized variable or random practice (92%), cognitively challenging training environments (92%), and athlete motivation (100%). Despite this alignment, only 40% reported intentionally applying formal motor learning frameworks such as OPTIMAL theory or the Constrained Action Hypothesis. Qualitative analysis revealed applied reasoning underlying coaches' instructional decisions. A dominant theme was the deliberate design of competitive, game-like practice environments intended to promote decision-making, adaptability, and learning. Coaches frequently described manipulating task constraints, increasing cognitive load, and progressively scaling task complexity to enhance retention and transfer. Individualized instruction was also prominent, with coaches tailoring feedback, cueing strategies, and demonstrations to athletes' developmental level and cognitive readiness. External focus cues, analogies, and visual modeling were commonly used to minimize conscious control and support self-organization. Coaches highlighted motivational and relational factors, including autonomy support and confidence building, while identifying barriers such as inadequate time and formal training. These findings reveal a disconnect between formal theoretical knowledge and applied motor learning practice, highlighting coaches' reliance on tacit motor learning knowledge. This work provides novel insight into coaches' tacit motor learning knowledge and identifies opportunities for translational, practice-relevant coach education. Future work should expand data collection and develop translational, relevant educational resources tailored to coaching contexts and developmental stages of athletes.

Associations between balance, vision, and psychological resilience in Special Operations Forces soldiers

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Anterior reach balance performance is associated with sport and military injury risk. During balance tasks, stroboscopic glasses can increase somatosensory and vestibular system input by challenging visual reliance. Psychological resilience represents flexible self-regulation and might influence magnitude-of-difference in balance performance under increasing levels of visual occlusion. We cross-sectionally studied a Special Operations Forces (SOF) cohort by 1) measuring balance on the Y-Balance Test (YBT) under non-stroboscopic (no occlusion) and stroboscopic (moderate visual occlusion) conditions, 2) asking participants to complete the Ego Resilience Scale (ER89), and 3) collecting self-reported physician diagnosed mild traumatic brain injury (mTBI) history. Participants ($n=207$; age: 35.5 ± 7.7 , range: 24-73 years) were categorized by SOF career stage: early ($n=118$), in-career ($n=76$), and retired ($n=13$). Ego resilience scale responses and YBT anterior reach outcomes in non-stroboscopic and stroboscopic conditions were analyzed. Participant ER89 scores (46.04 ± 5.19) ranged from 28-56 (scale range: 0-56; higher scores are better), and mTBI history were categorized as none ($n=123$), 1-2 ($n=63$), or 3+ ($n=21$). Specific YBT outcomes were absolute reach distance (YBT-ARD) and limb asymmetry (YBT-ASYM). Magnitude-of-difference by stroboscopic condition (non-stroboscopic vs. stroboscopic) in YBT-ARD and YBT-ASYM were computed to then examine ER89 influence. Demographics and key variables were analyzed via descriptive statistics; paired-samples t -tests compared outcome means by stroboscopic conditions. General linear models evaluated ER89 score influence on magnitude-of-difference in YBT-ARD and YBT-ASYM while controlling for mTBI history and career stage (a priori $\alpha \leq .05$). Reach distances were significantly different between non-stroboscopic and stroboscopic conditions in both dominant (non-stroboscopic: 62.95 ± 9.37 ; stroboscopic: 60.87 ± 9.14 , $p < .0001$) and nondominant (non-stroboscopic: 63.20 ± 8.93 ; stroboscopic: 60.93 ± 9.33 , $p < .0001$) limbs. The YBT-ASYM was low and similar ($p = .55$) in non-stroboscopic ($-.25 \pm 5.63$) and stroboscopic conditions ($-.02 \pm 4.47$). There was no influence of ER89 on YBT-ARD magnitude-of-difference in dominant ($R^2 = .008$; $p = .67$) or non-dominant ($R^2 = .011$; $p = .52$) limbs nor in YBT-ASYM ($R^2 < .007$; $p = .70$). Psychological resilience was not related to YBT magnitude-of-difference for reach distance or asymmetry by stroboscopic condition. Investigating other factors affecting reach distance and movement symmetry when visual reliance is challenged may inform future injury prevention and rehabilitation efforts. Balance testing with and without stroboscopic effect is not associated with trait psychological resilience. Funding source: This work was originally supported by the US Army Medical Research and Development Command under Contract No. W81XWH-20-C-0022 and continues under an effort sponsored by the Government under Other Transactions Number W81XWH-15-9-0001.

Does dosage matter? An examination of immersion level on motor performance in virtual reality

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Immersive and non-immersive virtual reality (VR) technologies have been accepted as training tools for a wide variety of fields ranging from education, clinical medicine, and military combat preparation. In a previous study, we demonstrated when participants completed a single bout of stationary cycling in both immersive (IVR) and non-immersive (NIVR) VR conditions, their motor behavioral and physiological outcomes remained consistent while their psychological outcomes significantly differed. Therefore, the primary aim of the present study was to expand on previous findings by investigating the effects of virtual reality immersion on motor behavioral, physiological, and psychological factors across multiple practice sessions. Participants ($N=20$) performed a total of four 8000-meter stationary cycling trials on four separate days within a two-week period. NIVR trials were completed utilizing a three-dimensional, Android tablet application, and IVR trials were completed using the Meta Quest 2 head-mounted display. Completion time (s), success rate, heart rate (bpm), speed (kph), cadence (RPM), power output (W), and rated perceived exertion (RPE) were collected at 1000-meter intervals (i.e., laps) during each trial which spanned a total distance of 8000-meters. Participants also completed the Intrinsic Motivation Inventory (IMI) and Engagement Scale Questionnaire (ESQ) following each trial. Analyses revealed there were no significant differences between the first and second day of participation within each condition, and there were no significant differences between conditions for completion time, success rate, and heart rate. There were however significant condition (i.e., NIVR & IVR) X lap interactions for speed, cadence, and power, specifically across laps three through six. No significant differences in RPE, intrinsic motivation, or engagement were found between conditions, but condition order did impact the Perceived Competence and Pressure/Tension subscales of the IMI. These findings suggest VR immersion may impact within-task dynamics more than overall task performance, providing both researchers and practitioners with valuable insights into VR practice behaviors. The findings of this study also suggest that overall VR immersion level does not have a meaningful global impact on motor, psychological, or physiological behaviors. These findings are important for those considering adopting VR as a training platform.

Is it worth waiting? How time between trials affects visuomotor adaptation

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Visuomotor adaptation has been shown to be impacted by the duration of time between consecutive trials (i.e., intertrial interval; ITI). The current research looked to determine if ITI duration influences the contributions of explicit (i.e., conscious strategy) and implicit (i.e., unconscious) processes underlying visuomotor adaptation. Participants were randomly assigned to a Short ITI ($n=20$, $M_{age}=20$; 13 females) or Long ITI ($n=20$, $M_{age}=19$; 11 females) group. Testing was completed over two sessions, separated by a 24-hour interval, to evaluate initial learning on Day 1 and retention on Day 2. On Day 1, all participants were trained to reach with cursor feedback that was rotated 40° relative to their hand trajectory. Explicit and implicit contributions were established using the Process Dissociation Procedure (PDP), such that participants were instructed to reach using any strategies learned during the rotated cursor training trials (explicit + implicit contributions) or reach without using any strategies (implicit contributions). Results indicated both groups adapted their reaches to a similar extent on Day 1 during initial training. This adaptation was preserved for a longer time on Day 1 in the Long ITI group, as these participants demonstrated greater reach adaptation during initial washout trials in which cursor feedback was aligned with their hand motion. Reaches did not vary between groups when the cursor rotation was re-introduced on Day 1 or again on Day 2, though participants in the long ITI showed

improved retention on Day 2 compared to Day 1. Changes in retention were not accompanied by changes in explicit and implicit contributions across groups. Specifically, explicit and implicit adaptation were comparable between both the Short ITI and Long ITI groups across initial training and retention trials. Together, our findings suggest that longer ITIs encourage more stable visuomotor adaptation compared to shorter ITIs without impacting the contributions of explicit and implicit adaptation.

The effect of postural threats simulated through virtual reality on muscle contraction

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Virtual Reality (VR) is a safe and affordable tool for simulating postural threats, such as standing at a certain height. Virtual height exposure has been shown to elicit postural and emotional responses comparable to those elicited by real-life height exposure. Previous data from our lab demonstrated that such height-exposure environment led to reduced center of pressure (COP) magnitude, a shift to higher frequencies, and increased irregularity. These findings were interpreted as evidence of ankle stiffening or potentially as an indication of increased automatic control in response to postural threat. Building on this, our study investigated whether VR-induced postural threat elicits ankle stiffening, measured through co-contraction of the gastrocnemius and tibialis anterior. We hypothesized that the height-exposure condition would increase co-contraction, indicating ankle stiffening. 15 healthy young adults stood with their feet together. They attempted to remain still across 3 conditions: baseline (No VR), inside an elevator VR simulation, and a VR plank extending from the 80th floor of a building using Richie's Plank Experience. Electromyography (EMG) was collected from the dominant leg's tibialis anterior and gastrocnemius. COP data were analyzed using both linear (area, mean velocity) and non-linear measures (Sample Entropy (SampEn), wavelet-derived frequency composition). Contrary to our hypothesis, EMG data revealed no increase in co-contraction during the postural threat condition, indicating that ankle stiffening did not occur. However, the height-exposure condition did produce a higher SampEn than both other conditions in the anteroposterior direction, and a higher SampEn compared to the baseline condition in the mediolateral direction. Thus, these findings indicate greater sway irregularity and suggest that participants adopted an exploratory and automatic balance strategy. In conclusion, this study contributes to our understanding of how VR-induced postural threat alters balance strategies, to inform future interventions aimed at improving stability in individuals with impaired postural control. These findings suggest that increases in sway irregularity and reductions in sway magnitude do not result from ankle stiffening but rather reflect a shift towards a more automatic control system, which is believed to promote proprioceptive control. Future studies should further investigate this mechanism using dual-task paradigms and additional VR-based height exposure conditions.

A peripheral perception in action test: Anchoring facilitates detection

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The ability to process information peripherally can sometimes facilitate faster responses in sports (Vater et al., 2020). Yet this ability is rarely assessed in field settings, and when tested, eye-tracking is seldom used to verify peripheral vision processing (Vater & Strasburger, 2021). This study addressed both limitations by measuring peripheral reaction times in controlled field conditions with targets at different eccentricities. Twenty-three healthy sport science students ($M_{\text{age}} = 20.5 \pm 1.6$ years; 10

females) completed a peripheral detection task using Fitlight targets at 30°, 60°, and 90° eccentricity to the left and right. Participants stood under a basketball hoop and fixated a central LED at free-throw distance. Six peripheral target LEDs were positioned on the 3-point line and one response LED beneath the participant. Across three blocks, participants fixated the central LED and named its changing colors. At random intervals, a peripheral target illuminated; participants indicated detection by deactivating the response LED with their hand, then ran to turn off the peripheral target and returned to step on the response LED. Each peripheral target appeared once per block (18 trials). Heart rate (Polar H10), eye-movements (Pupil Labs Neon), and response times (Fitlight) were recorded. The primary dependent variable was peripheral detection time (interval between peripheral onset and hand deactivation). Mixed-effects models with random intercepts for participants determined predictors of detection time. Movement time (from peripheral onset to foot deactivation) and gaze validity (percentage of trials with fixation maintained) controlled for individual differences in movement speed and peripheral vision use. Additional predictors were eccentricity, block, and heart rate. Gaze validity showed a trend toward faster detection ($\beta = -62.93$, $p = .09$). Higher heart rate ($\beta = -131.00$, $p < .001$) and shorter movement times ($\beta = 117.17$, $p < .001$) predict faster detection. Eccentricity ($\beta = 39.21$, $p = .17$) and block ($\beta = -13.96$, $p = .62$) were non-significant. Anchoring gaze appears advantageous for detecting peripheral targets, suggesting peripheral monitoring may outperform foveal search. The findings also highlight the need to account for movement capabilities and gaze strategy when interpreting reaction-time measures in sport-specific contexts. Funding source: Swiss National Science Foundation (SNSF).

Prior expectations shape actions: Bayesian integration in complex sensorimotor behavior

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As a fundamental challenge in motor control, humans perceive the world and act upon it under uncertainty arising from noise and delays in neural signals. A functional strategy to reduce uncertainty – formalized by Bayesian integration – is to combine current sensory information and prior expectations, weighting each source according to its reliability. However, empirical evidence for Bayesian principles is largely limited to simple laboratory tasks investigating constrained hand and arm movements. Here, we test Bayesian integration in a complex sensorimotor task: returning tennis serves. We developed an immersive extended reality (XR) task that enables the study of unconstrained, full-body movements under naturalistic spatiotemporal task demands, while maintaining precise experimental control. In two experiments, participants returned serves delivered by a virtual opponent. In the first experiment ($N = 24$), the opponent's serve locations followed a bimodal (two-peaked) distribution. To manipulate sensory uncertainty, we varied ball speed across three levels. In the second experiment ($N = 48$), the opponent's serve locations depended on a contextual cue (first vs. second serve), allowing us to test context-specific priors. As a behavioral measure of movement bias, we used motion capture to compute the horizontal deviation between the ball and the racket's sweet spot at ball-racket contact. Our results show that, after repeated exposure to the opponent's serves, participants' movements were systematically biased by the learned prior distribution. Consistent with Bayesian model predictions, the magnitude of the bias scaled with sensory uncertainty. Explicit knowledge tests further indicated that participants incorporated prior knowledge of the opponent's serve distribution into their behavior without explicit awareness of the serve pattern. These findings show that humans can implicitly learn and exploit complex priors in full-body, naturalistic motor behavior.

Trigger-finger behavior and visual attention during high-risk firearm decision-making

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Special interventions operators respond to high-risk incidents involving dangerous situations, and are often required to use their firearms. The use of firearms carries inherent risk, including Involuntary Firearm Discharges (IFDs). While IFDs present an operational risk, little is known about the perceptual-motor processes associated with them. 'Trigger-finger movements' represent the final motor action leading to discharge, making them valuable observable behavior for understanding the mechanisms underlying IFDs. A common operational procedure is 'trigger-finger discipline', requiring the trigger-finger to remain along the muzzle unless a conscious decision to shoot has been made. In this study, the focus is on movements that deviate from trigger-finger discipline, i.e., the trigger finger slides down along the muzzle or is placed inside the trigger guard or on the trigger while no shot is fired (yet). The aim of this study was to investigate whether trigger-finger deviations occur in potentially high-threat scenarios and to what degree they are associated with differences in gaze behavior. Twenty-eight operators performed twenty-four short, simulated shoot/don't-shoot scenarios under two levels of ambiguity (low vs. high) and firearm procedures (SUL, a safe-ready position vs. holster) while wearing an eye-tracker. In 43% of the 672 analyzed scenarios, trigger-finger movements that deviated from the trigger-finger discipline were observed, underscoring their frequent occurrence and relevance for IFDs. To compare trials with and without trigger-finger deviations, 78 "matched trials" were created: pairs of trials executed by the same participant with identical ambiguity, starting position, and decision outcome, but differing in trigger-finger behavior. Results indicate that, across most conditions, fixation duration is higher and search rate is lower in trials involving trigger-finger deviations compared to trials without such deviations. Analysis of areas of interest indicates a gaze pattern in which operators initially focus on weapon-related regions (i.e., hips and hands), before shifting attention to target locations (i.e., torso and legs) later in the trial when the decision to shoot or not is made. These analyses provide insights into the perceptual-motor and attentional processes underlying trigger-finger movements in high-risk situations. Understanding these processes may inform training interventions aimed at reducing the risk of IFDs and enhancing operational safety. Funding source: Special Interventions Division of the Dutch National Police.

Examination of functional connectivity in individuals during collaborative execution of sequential tasks under various demands with an AI-enabled humanoid robot

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With increasing human interactions with AI-based systems through virtual platforms and physical robotic agents, it becomes important to examine neurocognitive processes to further understand human behavior in social settings. While multiple previous human-robot studies have examined cognitive-motor processes, as far as we know, only one investigation has studied the associated cortical regional activity through electroencephalography (EEG) when individuals jointly execute sequential problem-solving tasks with an AI-powered robot. However, no prior efforts have

investigated the modulation of functional connectivity in such collaborative contexts. Thus, by assessing cerebral cortical networking, this study investigated the deployment of neurocognitive resources when solving task-puzzles with an AI-powered humanoid robot under two levels of task demand. Twenty-four right-handed individuals executed task-puzzles jointly with a humanoid robot in a customized 3D virtual environment as performance and electroencephalography were continuously recorded. The weighted phase lag index (WPLI) was computed to capture the functional connectivity between the planning region and other regions involved in sensory, motor and cognitive processes across the theta, alpha, beta, and gamma frequency bandwidths. The findings revealed that engagement versus disengagement of the robot resulted in an overall enhancement of performance and an attenuation in gamma functional connectivity under elevated demand. Also, enhanced task demand led to a degradation of performance and an elevation in cerebral-cortical networking across the theta, alpha, and beta bands regardless of whether the robot was engaged or not. These findings suggest that collaboration with the robotic teammate led to a reduction in the integration of various sensory, cognitive and motor mechanisms required for planning which were critical for maintaining performance when the robot was disengaged. However, engagement of the robotic teammate could not fully compensate for the elevated demands, thus resulting in greater engagement of inhibitory, attentional control, and memory mechanisms to facilitate high-level planning. This work can inform human cognitive-motor behavior in a social context as well as the design of brain computer interface-based robotic systems with the aim of maintaining satisfactory mental workload levels while simultaneously enhancing cognitive-motor performance.

Offline EEG-based mental workload classification during human-robot collaborative execution of sequential problem-solving tasks under varying demands

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With increasing human collaborations with AI/robotic systems, it is important for these systems to understand and respond to changes in individuals' neurocognitive states to maintain acceptable mental workload (MWL) levels. Prior efforts, mainly conducted in aviation settings, have employed electroencephalography (EEG)-based machine learning techniques to decode MWL levels under varying task demands, providing a threshold index which when exceeded, triggered an assistive automation system. However, these past efforts did not consider a collaborative robotic teammate with AI capabilities, nor the joint execution of problem-solving tasks. Thus, this work is a first step in designing an EEG-based monitoring system to classify offline MWL levels when individuals complete a problem-solving sequential task with an AI-enabled humanoid robot under two demand levels. Twenty-six individuals executed under low and high demand conditions the task alone and with a robot in a 3D virtual environment while EEG was continuously recorded. After denoising the EEG data, relevant spectral features (theta, alpha, beta, and gamma power for all scalp electrodes) were extracted. These features were used for classification to obtain a threshold index which separated MWL across the two levels of task demand when individuals performed alone. Then, this index was utilized to identify the percentage of epochs exceeding the MWL threshold (PEET) as individuals collaboratively executed the task when the robot was engaged and disengaged under varying demand. The findings revealed that robot engagement versus disengagement led to an attenuation of the PEET. Additionally, elevated task demands resulted in

an increase in PEET irrespective of the engagement/disengagement of the robot. Thus, this EEG-based monitoring system was able to decode offline modulations of MWL driven by the participation of the robot as well as the levels of task demand. This suggests that while the robot's involvement was assistive in reducing human MWL, it was not enough to fully compensate for the elevated task demands. This effort is a first step in developing a real-time MWL prediction system based on which a robotic teammate can dynamically modulate its level of involvement to maintain individuals' MWL levels.

Interactive agents improve the representativeness of embodied decision-making behavior

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Decision-making is a crucial skill in team sports, yet remains challenging to measure and train in controlled settings. For instance, video-based testing of decision making has long neglected the interpersonal interactions between athletes and their opponents, with opponents seen in the video footage not able to respond to the movements of the participant and therefore the test fails to represent the conditions observed in the natural environment. Virtual reality offers a vital opportunity to implement interactive agents that respond to the actions of participants (Janssen et al., 2023), but to-date implementation has proven elusive. The aim of this study was to examine the extent to which interactive agents improve the representativeness of behavior in a decision-making task. We designed an interactive avatar opponent (the 'agent') in a virtual-reality setting who responded – in real-time – to the movements of the participant, and conducted an empirical study to evaluate the impact of the agent. Twenty soccer players performed a 1v1 decision-making task, and we manipulated the opponent's level of interactivity to compare participant behavior to that observed when facing an actual opponent in situ. Decision-making behavior was compared across four conditions of increasing interactivity (virtual no-opponent, virtual non-interactive opponent, virtual interactive opponent, and in situ opponent). Results strongly support the idea that adding an interactive agent makes the task more representative of in situ behavior. Specifically, key measures of behavior (e.g., trial duration and number of touches) measured with the virtual-interactive opponent ($\beta_{\text{Touches}} = 4.3$, $\beta_{\text{Duration}} = 2.7$) were much more representative of the in situ behavior ($\beta_{\text{Touches}} = 5.4$, $\beta_{\text{Duration}} = 3.6$) than they were against the virtual non-interactive opponent that would usually be seen in video footage ($\beta_{\text{Touches}} = 2.9$, $\beta_{\text{Duration}} = 1.84$), and in the virtual no-opponent condition ($\beta_{\text{Touches}} = 2.5$, $\beta_{\text{Duration}} = 1.4$). The results validate the call to incorporate interactive agents as a key component of decision-making tasks. To support implementation, we provide the open-source code necessary for others to implement interactive agents in their tests of decision making and other tests of sport performance. Janssen, T., Müller, D., & Mann, D. L. (2023). From natural towards representative decision making in sports: A framework for decision making in virtual and augmented environments. *Sports Medicine*, 53 (10), 1851–1864.

Impact of an adapted pickleball program on balance and physical fitness among young adults with intellectual disabilities

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Individuals with intellectual disabilities (ID) have an increased risk of falls due to motor impairments and balance deficits, restricting physical functioning and independence. These challenges are often linked to limitations in physical fitness, highlighting the need for structured, developmentally appropriate interventions. Pickleball (PB) incorporates continuous perceptual tracking, movement planning, and multidirectional locomotion. The purpose of this study was to examine the effects of a 14-week PB intervention on balance and physical fitness in young adults with ID. Twelve young adults with ID (76.48 ± 19.37 kg; 163.48 ± 9.37 cm; 20 ± 1.61 years; 4 females, 8 males) participated in a 14-week PB program consisting of biweekly 50-min sessions. A repeated-measures design (pre-, mid-, and post-intervention) was used to assess changes in balance and physical fitness. Static balance was evaluated using bilateral stance (firm/foam, eyes open/closed) and unilateral stance on dominant (DL) and non-dominant legs (NDL). Center of pressure (COP) metrics included average displacement (AD), average velocity (AV), 95% ellipsoid area (EA), and path length (PL). Dynamic balance was assessed using Timed Up and Go (TUG) movement time (MT). Physical fitness measures included cardiovascular endurance (6-min walk), flexibility (modified sit-and-reach), muscular strength (hand-grip dynamometry), and muscular endurance (30-s sit-to-stand). One-way repeated-measures ANOVAs with Greenhouse-Geisser corrections and Bonferroni adjustments ($\alpha = .05$) were applied. Significant decreases were observed in unilateral balance, with reductions in COP for both DL (AD $p = .038$; AV $p < .01$; EA $p = .047$; PL $p < .01$) and NDL (AD $p = .017$; AV $p = .047$; EA $p = .008$; PL $p = .023$), demonstrating improvements. MT significantly decreased from pre- to post-intervention ($p = .044$), indicating improvements. No significant effects were identified for bilateral conditions or physical fitness. Results reflect the nature of PB and its effects on balance, with frequent shifts in COG across the base of support, quick paddle skills with asymmetrical upper-limb movements, and multidirectional lower-limb actions. The dynamic demands challenge and train the postural control system. Although physical fitness did not change significantly, PB remains a promising intervention for improving balance and reducing fall risk in ID.

Effect of attentional focus on isometric plank performance and stability

Juliette Jacobi, University of Tennessee/United States Army; Jared M. Porter, University of Tennessee

Since the early 1900s, the United States Army has used standardized physical fitness tests to assess physical readiness across its formations. In 2025, the Army replaced the Army Combat Fitness Test (ACFT) with the Army Fitness Test (AFT), designating the forearm plank as the primary assessment of static strength and core endurance. Because Soldiers frequently sustain prolonged isometric postures during operational tasks (e.g., maintaining a stable firing position during marksmanship), this study aimed to investigate the influence of attentional focus on isometric performance and postural control. Although decades of attentional focus research have consistently demonstrated performance benefits of adopting an external focus, its effects on isometric plank performance and postural stability remain underexplored. In this study, participants received internal (INT), external (EXT), and holistic (HOL) attentional focus instructions while completing three counterbalanced plank trials to failure on an AMTI AccuPower-Optimized (ACP-O) force plate. Center-of-pressure (COP) metrics were used for stability, including the 95% confidence ellipse, path length, and average velocity. Linear mixed models were used to analyze both full-length trial outcomes and time-segmented intervals (initial vs. final 30 s). Plank duration and overall COP stability did not differ across

attentional focus conditions. However, a significant interaction between condition and time segment was observed for COP path length ($p < .001$). During the initial 30 s, EXT produced greater path length than INT and HOL ($p < .001$), followed by a substantial decrease in the final 30 s ($p < .001$), indicating improved postural stability later in the trial. This temporal pattern suggests that an EXT focus may prompt an early exploration phase that transitions into a more efficient postural organization as fatigue sets in. This interpretation aligns with the constrained action hypothesis, which proposes that external instructions facilitate automatic control processes during effortful tasks. The results of this study are relevant to military training, in which Soldiers are required to maintain stable isometric positions for extended periods, especially under various states of fatigue. Future research should examine whether similar attentional focus effects emerge during marksmanship performance in military populations.

Effects of attentional focus on marksmanship performance and stability among ROTC cadets in a virtual reality environment

Juliette Jacobi, University of Tennessee/United States Army; Jared M. Porter, University of Tennessee

Marksmanship relies on the intricate combination of cognitive decision-making, postural regulation, and precise motor control. Attentional focus strategies have been shown to influence motor performance across a wide range of tasks; however, limited research has examined the distance effect of attentional focus on marksmanship performance and stability in military populations, particularly within non-immersive virtual reality (VR) training environments. Thus, this study investigated the focus of attention distance effect on marksmanship performance and postural control among ROTC cadets using a non-immersive VR marksmanship simulator. Participants performed four 15-shot marksmanship trials on the InVeris FATS 100MIL simulator using a standardized baseline condition (BAS) and three counterbalanced attentional focus conditions, internal (INT), External-Near (EXN), and External-Far (EXF) while standing on an AMTI AccuPower-Optimized force plate (ACP-O). Marksmanship outcomes included radial error (i.e., precision), shot-to-shot variability (i.e., variable error), and response time. Postural stability was assessed using center-of-pressure (COP) metrics (i.e., 95% confidence ellipse, path length, and average velocity). A Related-Samples Friedman's Two-Way Analysis of Variance by Ranks (nonparametric analysis) was used due to the non-normality of the data. Results indicated that all attentional focus conditions produced significantly greater precision and displayed lower shot-to-shot variability than baseline instruction ($p < .001$); however, no significant differences were observed among INT, EXN, and EXF conditions. A significant effect on response time indicated that EXF was slower than the other conditions ($p = .029$). No significant differences were found across attentional focus conditions in COP metrics. Self-reported measures indicated that participants preferred externally focused instructions for precision ($p = .013$) and stability ($p = .062$). The overall findings suggest that providing task-relevant attentional focus instructions enhances marksmanship precision and consistency relative to standardized baseline instructions. The EXF extended response time suggests a deliberate, quality-focused strategy rather than rushing the trigger, indicating potential for improved trigger discipline in real-world combat scenarios. The preference for externally focused instructions, despite similar objective performance, highlights a potential role of perceived effectiveness and motivation in training contexts. Future research should examine the effects of structured, externally focused marksmanship training in a non-immersive VR environment on skill acquisition and transfer to live-fire performance among ROTC cadets.

Influence of visual target cues on landing ground reaction forces and instructional compliance

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Attentional focus instructions influence movement effectiveness and efficiency; however, less is known about how adherence to internal versus externally directed visual target cues relates to biomechanical outcomes such as landing ground reaction forces (GRF). Although visual targets are frequently used in applied settings, limited work has examined whether directing attention toward an external target is associated with changes in landing force magnitude. This secondary analysis evaluated whether self-reported compliance with visual target instructions corresponded with differences in landing GRF during vertical jumping. Twelve young adults completed a dynamic warm-up followed by three jump conditions: baseline (no instruction), an internal focus cue (reach the fingers as high as possible), and an external visual target cue (jump toward a ceiling target). Jumps were performed on dual force plates, with left and right landing GRF summed and averaged across three trials per condition. Participants completed self-report compliance and explicit-knowledge ratings (1–7) following each condition. A repeated-measures ANOVA assessed differences in GRF across conditions, and Spearman correlations examined associations between GRF and compliance. Descriptively, the external visual target cue produced the highest landing GRF ($M = 2780.72 \pm 1182.55$ N), followed by baseline ($M = 2475.18 \pm 496.79$ N) and internal focus ($M = 2470.71 \pm 483.68$ N). Mauchly's test indicated a violation of sphericity ($p < .001$); therefore, Greenhouse–Geisser corrections were applied. The main effect of condition was not significant, $F(1.13, 12.46) = 1.01$, $p = .35$, with a small-to-moderate effect size ($\eta_p^2 = .084$). Pairwise comparisons were nonsignificant, although descriptively higher forces (~ 300 N). Although effects were not statistically significant, the descriptive pattern suggests that visual target cues may influence landing force magnitude independently of perceived instructional compliance. The absence of associations between compliance and GRF highlights challenges in isolating attentional mechanisms using self-report measures alone. Future research using larger samples and incorporating kinematic, loading-rate, and objective attentional measures is needed to clarify how instructional cues shape landing force characteristics.

Too short to notice: Two separate processes underlying targeted movements in response to motion-to-photon latency in virtual reality

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Motion-to-photon latency is the delay between the movement of the user in the physical world and resulting rendering of this movement in the virtual world. This latency is inherent in real-time virtual reality (VR) systems, yet its impact on targeted manual actions is not understood. The aim of this study was to quantify how latency alters movement accuracy and timing in VR and localizes where in the movement these costs arise. Participants executed visually guided pointing movements in an immersive virtual environment while an external motion-capture pipeline translated the movement of the participant's hand into the movement of a virtual hand. Fixed rendering delays (0–500 ms) were imposed on the hand animation relative to the movement of the real hand. Linear mixed-effects models were used to assess endpoint error (EE), movement time (MT), and timing for different kinematic markers (peak acceleration (PA), peak velocity (PV), and peak deceleration (PD)). Three robust patterns emerged. (1) Relative to the physical baseline, VR produced systematic overshoot; EE increased monotonically with delay in a front-loaded, saturating fashion. (2) MT was longer in VR than in the physical environment; delay-to-MT effects were nonlinear and threshold-like, emerging around 50–100 ms, peaking near ~ 200 ms, and diminishing at 500 ms delay. (3) Analyses on

the timing of kinematic markers localized costs to the late homing segment (PV to PD). Together, the data reveal two processes by which latency shapes targeted movements in VR: visual-proprioceptive misalignment inflating EE without slowing, and strategic slowing that trades time for accuracy yet only partially compensates. Design implications include developing a brief, effector-only latency acclimation procedure that leverages users' delay sensitivity to train appropriate homing control and improve VR performance. Funding source: NFRF.

Examining focus of attention in collegiate breaststrokers

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Swimming performance at the elite level is often decided by hundredths of a second, making even small performance enhancements highly consequential. One factor that may influence performance outcomes is attentional focus, or the direction of an athlete's concentration during skill execution. Previous research has demonstrated that an external focus of attention enhances freestyle swimming performance relative to an internal focus; however, its effects on breaststroke, a stroke with unique technical and coordinative demands, remain unexplored. The purpose of this study was to examine the effects of internal versus external attentional focus on 50-m breaststroke performance in collegiate swimmers. Sixteen highly trained swimmers ($M_{\text{age}} = 20.0$ years, $SD = 1.59$; 14 males) completed three randomized long-course trials under internal focus, external focus, and control (no specific attentional cues) conditions. Swim times were recorded and analyzed using a repeated-measures analysis of variance. Results indicated that both the external focus ($M = 37.28$ s, $SD = 2.77$, $p = .042$) and control ($M = 37.12$ s, $SD = 3.10$, $p = .006$) conditions produced significantly faster performance times than the internal focus condition ($M = 37.80$ s, $SD = 3.17$). No significant difference was observed between the external focus and control conditions ($p > .05$), and 62.5% of participants recorded or tied their fastest performance during the control condition. These findings support the constrained action hypothesis, suggesting that directing attention internally may disrupt automatic motor control processes in highly trained swimmers, whereas external focus cues or minimal attentional instructions help preserve performance efficiency. From an applied perspective, coaches and practitioners working with elite swimmers should prioritize external or outcome-oriented cues and, in some cases, consider providing no explicit attentional instructions when optimizing performance in highly automated skills.

Attentional focus strategies and tennis serve accuracy: Skill-level-dependent effects of internal, external, and holistic cues

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Verbal instructions influence motor skill performance by directing attentional focus during movement execution. Motor learning research has shown that an external focus of attention often enhances performance compared to an internal focus, while a holistic focus emphasizing overall movement sensation has emerged as a potential alternative. However, the effectiveness of holistic and external focus strategies for complex sport skills, such as the tennis serve, and their dependence on performer expertise remain unclear. Fourteen male university tennis players ($M_{\text{age}} = 20.57 \pm 1.34$ years; $M_{\text{playing experience}} = 11.19 \pm 3.99$ years) participated in the study and were classified as moderately skilled ($n = 5$) or highly skilled ($n = 9$). Participants performed tennis serves into the deuce

service box, with accuracy assessed using a modified Hewitt-style serve placement test. Following a baseline pretest under neutral instructions, participants completed four experimental blocks of serves under internal focus, proximal external focus, distal external focus, and holistic focus conditions. Serve accuracy was scored on a 0–6 point scale. Attentional focus was manipulated within subjects, with skill level included as a between-subjects factor. A mixed ANOVA revealed no significant main effect of attentional focus on serve accuracy when averaged across skill levels, $F(3, 39) = .72$, $p = .545$, $\eta_p^2 = .053$. However, a significant Attentional Focus $\times$ Skill Level interaction was observed, $F(3, 39) = 4.56$, $p = .008$, $\eta_p^2 = .275$, indicating that attentional focus strategies influenced serve accuracy differently based on expertise. A significant main effect of skill level was also present, $F(1, 12) = 15.15$, $p = .002$, $\eta_p^2 = .558$, with highly skilled players demonstrating greater accuracy overall. Follow-up analyses showed that highly skilled players performed more accurately under a proximal external focus than a holistic focus ($p = .039$), whereas although moderately skilled players demonstrated the highest mean accuracy under a holistic focus, no within-skill-level differences reached statistical significance after Bonferroni correction. These findings suggest that attentional focus effects during tennis serve performance are expertise-dependent, supporting individualized instructional strategies and motivating future research examining learning and retention effects.

Self-controlled feedback amplifies contextual interference benefits in sequence learning

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Self-controlled feedback (SCFB) and increased contextual interference (CI) have both been shown to facilitate motor learning. However, the combined effects have been underexplored, with research findings showing an additive benefit (sequence learning; Patterson et al., 2013) and no additive benefits (anticipation timing; Ali et al., 2012). The purpose of the present study was to better examine these effects within a sequence-learning task. Fifty-two participants ($n = 13/\text{group}$) were randomly assigned to a feedback (blocked or self-controlled) and CI schedule (blocked or random), creating four groups. Participants ranged in age from 18–30. They were instructed to learn three distinct 10-light sequences using a Dynavision reaction-time board and to complete the sequence as fast as possible with their dominant hand. Self-control participants were allowed to request feedback on overall timing after each trial, while yoked participants received feedback based on their self-control counterpart. Blocked practice consisted of completing 30-trials of each sequence before starting a new sequence, while random participants completed 30-trials of each sequence in an interleaved order. Participants completed a total of 90 acquisition trials, followed by 24-hr delayed retention and transfer tests with 9-trials. Transfer consisted of a new 10-light sequence. To analyze overall time, a 2 (feedback) $\times$ 2 (schedule) $\times$ 3 (blocks) factorial ANOVA with repeated measures on blocks was used to analyze acquisition, while separate one-way ANOVAs were applied to retention and transfer. The factorial ANOVA revealed a significant main effect for Block ($p < .001$), Schedule ($p < .001$), and Feedback ($p < .05$). A significant Schedule $\times$ Feedback interaction was observed ($p < .001$). The one-way ANOVA for retention was significant ($p < .001$) with post-hoc analyses revealing that random-SCFB performed significantly faster than all other groups ($ps < .001$). The same pattern of findings was observed for transfer ($ps < .001$). These findings suggest an additive benefit for combining SCFB and increased CI, which is contrary to previous findings (Ali et al., 2012). Research should continue to investigate these effects across a variety of tasks and manipulations to better understand the influence of SCFB and CI on skill acquisition.

Acute effects of repetitive subconcussive soccer heading on stepping in place gait and arm swing variability

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Repetitive subconcussive head impacts alter gait and postural control, including reduced stepping-in-place variability hours after exposure. Laboratory-based repetitive soccer heading also produces significant postural control changes. What remains unknown is the extent to which arm swing during gait may also exhibit changes in dynamics after head impacts. If so, wearables such as smartwatches could be used to monitor neuromotor changes. The purpose of this study was to examine the acute changes in stepping-in-place gait variability and arm swing variability following repetitive soccer heading. Ten college aged participants completed baseline stepping-in-place assessments under three conditions: eyes open, eyes closed, head shake. The participants then performed ten soccer headers from nine meters at approximately 25 mph, followed immediately by performing the same three stepping-in-place tasks. Variability was quantified as the coefficient of variation (CoV) of peak thigh and arm flexion, measured using a validated custom Android application called AccWalker. Hedges g was used due to the small sample size. For leg swing, moderate effects were observed across conditions: eyes open (pre: $.911 \pm 2.166$; post: $1.270 \pm .524$; $g = .561$), eyes closed (pre: 1.576 ± 1.343 ; post: $1.339 \pm .883$; $g = .670$) and head shake (pre: $1.293 \pm .658$; post: 1.417 ± 1.078 ; $g = .569$). For arm swing, effect sizes were moderate: eyes open (pre: $1.092 \pm .229$; post: $1.189 \pm .361$; $g = .605$), eyes closed (pre: $1.116 \pm .314$; post: $1.035 \pm .239$; $g = .641$), and head shake (pre: $1.065 \pm .232$; post: $1.028 \pm .249$; $g = .551$). Decreased variability following repetitive subconcussive impacts in the leg swing eyes closed condition aligns with prior research, whereas increases during the leg swing eyes open and head shake condition are not supported by previous findings. Differences between arm and leg swing variability suggest that arm swing using AccWalker requires further validation. These findings support prior evidence of acute postural control/gait changes following repetitive soccer heading and highlight the need for further evaluations of arm swing and eyes open CoV to determine if they are valid markers of postural control/gait variability after subconcussive head impacts.

Perceived pressure and self-confidence, but not anxiety, predict dart throwing performance

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Anxiety and psychological pressure are prevalent throughout the human experience with perceived and real effects on motor performance. Previous work in our lab showed that different types of psychological pressure conditions have unique impacts on perceived pressure, anxiety, and performance. Given the varied responses of individuals across conditions, the purpose of this study was to assess if differences in perceptions of pressure, anxiety, and self-confidence predict performance in dart throwing. It was hypothesized that when participants reported higher perceptions of pressure and anxiety and lower levels of self-confidence (relative to their own average values and to their peers), they would throw less accurately. 43 participants (Age: $M = 21.14$, $SD = 2.65$, Female: 21) threw darts in 7 different pressure conditions leading to a total of 301 observations. After throwing in each condition, participants reported perceived pressure, somatic anxiety, cognitive anxiety, and self-confidence. For each measure, a person's average across all conditions was calculated, which was then used to calculate a within-person centering score for each condition. This approach allowed for understanding if either between-participant and/or within-participant differences in self-report metrics predict dart throwing

performance. A multiple linear regression with backward elimination was used to examine if mean scores or within-person centered scores for each variable predicted radial error (RE) in dart throwing. The best model included self-confidence (between and within-person centered) and perceived pressure (between-participant) significantly predicted RE (adjusted $R^2 = .225$). Both self-confidence variables ($\beta = -.28, -.19, ps < .001$) exhibited a negative relationship with RE, while perceived pressure (between-subject) exhibited a positive relationship with RE ($\beta = .232, p < .001$). Somatic and cognitive anxiety scores were not significant predictors in any model. The results of this study suggest that lower perceived pressure and higher self-confidence predicted more accurate throws. Additionally, when self-confidence was higher than a person's own average it was also associated with more accurate throws. The current findings suggest some, but not all psychological metrics predict performance. In particular, perceptions of anxiety are important in understanding the human experience, but may not directly relate to performance.

End-state grasp selection is largely robust to task-irrelevant speech

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In a binary grasp-and-place task where participants were asked to hold the object prior to moving it and the end location was a large hole, around half of participants chose beginning vs. end-state comfort (Studenka, Gamble, Hernandez-Zuniga, 2023). In a follow up replication of this experiment, we attempted to see if we could measure predictors of beginning-state comfort, however, post-hoc analysis of the grasping videos revealed brief incidental speech in a subset of sessions. This raised the possibility that concurrent speech acted as a distraction that impaired motor planning, potentially biasing grasp selection away from end-state comfort. To test this possibility, we examined whether speech predicted grasp selection. Participants grasped a wooden dowel and placed a specified color into a hole on the right side of the workspace. Task constraints were manipulated via hole size: when the hole was small, participants could move the object immediately after grasping; when the hole was large, participants were required to hold the object for 5 s before moving it. Both the duration and intensity of speech (RMS amplitude) was quantified (1) immediately prior to the planning cue (300 ms), (2) during the planning cue, (3) during the reach phase, and (4) for 1 s after grasp initiation. End-state grasp choice was analyzed using generalized linear mixed-effects models with subject-level random intercepts (41 participants). Speech during the planning cue did not predict end-state comfort selection (likelihood ratio test, $p = .964$), nor did it interact with task constraints ($p = .820$). In contrast, small effects were observed for speech occurring during and after movement execution. Greater speech duration during the reach and post-grasp phases was associated with modest increases in the probability of an end-state comfort grasp (maximum $\Delta P \approx .03$). Speech intensity showed similarly small effects, with weak positive associations during the reach and weak negative associations following grasp completion ($|\Delta P| \leq .009$). These results indicate that end-state comfort planning is robust to concurrent speech. Variability in grasp strategy is driven primarily by task constraints rather than distraction from verbal activity, with speech-related measures showing only minor, predominantly post-movement associations.

Combined effects of precue and foreperiod variability on simple reaction time

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Reaction Time (RT) is an easily measurable variable that can provide insight into how musculoskeletal movement is prepared. Previous studies on foreperiod effect state that RT increases as the foreperiod is changed between trials. When interstimulus intervals increase, it has been shown that foreperiod variability and the range of the variability increase RT. In addition, precue has been shown to decrease RT, as individuals tend to prepare their response movement in advance. Therefore, the purpose of this study was to explore the combined effects of precue and foreperiod on RT among healthy college students. A total of 45 participants ($M_{\text{age}} = 20.4$, $SD = 1.4$; 27 females) from a midwestern University were recruited and randomly assigned to one of the four groups: constant foreperiod with precue (CFP), blind constant foreperiod (CFB), variable foreperiod with precue (VFP), or blind variable foreperiod (VFB). Participants conducted five baseline trials, then 15 condition trials according to their assigned group. RT was measured in ms, using the Multi-Operational Apparatus for Reaction Time (MoArt) board Model 35601 and PsymCon control panel Model 35500 (Lafayette Instrument, IN). One-way ANCOVA with baseline RT and prior experience as covariates was conducted to examine the difference between groups. Results indicated that there was a significant difference between groups on RT, $F(1, 4) = 5.71$, $p < .05$. Specifically, there was a significant difference between CFP and VFB groups, and between CFB and VFB groups. Although not statistically significant, when attenuated for covariates of prior experience and baseline reaction time, VFB group had a slower RT than VFP group. Results from this study reinforce the importance of a foreperiod effect especially in athletics. In races, foreperiod length varies between practice and the race, or between races which could result in false starts. It is necessary to train athletes' sense of critical moment - when they expect the imperative stimulus to be presented - to have them align with the imperative stimulus to decrease the likelihood of a false start. Funding source: UMN Undergraduate Research Program Opportunity.

Past, present, and future of motor learning: A citation analysis

Eric Griessbach, Champalimaud Foundation; Alfredo Hernandez, Champalimaud Foundation

Motor learning research is a diverse field spanning sports science, rehabilitation, psychology, neuroscience, robotics, and computational modeling. With almost 27,000 publications and counting, no single researcher can keep track of them all. But what does the full landscape look like? Which research communities are in conversation with each other, and which have developed in parallel? What are the emerging topics within these communities? To answer these questions, here, we present a comprehensive citation network analysis to reveal the field's underlying structure and temporal dynamics. We compiled publications on motor learning spanning from the field's inception through 2025. To identify distinct research communities and to reveal patterns of intellectual exchange and isolation, we used the Leiden community detection algorithm, which partitions the citation network into clusters of papers that frequently cite one another. To characterize each cluster, we conducted a TF-IDF (Term Frequency-Inverse Document Frequency) analysis of keywords, which identifies the most distinctive terms for each research community. To trace temporal dynamics, we analyzed the number of papers and the most frequent keywords in each cluster over time. Our analysis identified several major research clusters within motor learning. These communities range from neuroscience-focused approaches concentrating on specific brain regions, to diverse behavioral paradigms spanning basic science to applied research contexts. The citation patterns reveal both densely interconnected subfields, such as rehabilitation research, and less integrated communities. Temporal analysis reveals the evolution of foundational approaches, including early computational work, and shows multiple emerging topics gaining

momentum across different research traditions. With this characterization of the field, we provide a bird's-eye view of the academic ecosystem of motor learning. By visualizing where different research communities sit in relation to one another and what they are currently exploring, we aim to spark curiosity about neighboring areas of research and inspire unexpected connections across their boundaries.

Two hands tell the story: Robotic assessment of upper limb coordination in people with multiple sclerosis

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Upper limb impairments are common in people with multiple sclerosis (MS) and can substantially impact everyday activities that require coordination within and across limbs, such as reaching for a cup (unimanual coordination), or tying shoelaces (bimanual coordination). However, in individuals with mild disability, these impairments are often subtle and may go undetected by standard clinical assessments. With the advent of robotic technology, there has been increasing interest in developing high-resolution measures to precisely characterize motor impairments and underlying deficits. The purpose of this study was to characterize unimanual and bimanual coordination in people with MS using a robot-based dynamic object hitting task. Thirty-six participants were recruited: 12 people with MS who self-reported upper limb impairment (MS-R), 12 people with MS who did not report upper limb impairment (MS-NR), and 12 age- and sex-matched controls without MS (CTL). Participants completed the task unimanually with each hand and bimanually with both hands. In the unimanual condition, the MS-R group made fewer total hits than the CTL group ($p < .001$), whereas the MS-NR group did not differ significantly from either group ($p > .091$). During the bimanual condition, the MS-R group made fewer total hits than the CTL group and demonstrated clear coordination asymmetries ($p = .004$), with the dominant hand moving faster ($p < .001$) and contributing disproportionately more hits ($p < .013$) than observed in the non-dominant hand compared to both the MS-NR and CTL groups. Notably, in a secondary analysis of individual participant performance, 75% of MS-R and 58% of MS-NR participants were classified as impaired during bimanual performance based on task success (i.e., total hits). Total hits in the bimanual task were significantly correlated with scores on the ABILHAND questionnaire, performance on the 9-Hole Peg Test (dominant-hand), and the Symbol Digit Modalities Test. Together, these findings suggest that robot-based measures are sensitive to subtle coordination impairments, particularly in a bimanual coordination task, and relate meaningfully to established clinical measures. Robotic tasks can complement clinical measures by providing a more nuanced characterization of upper limb impairment, with implications for planning rehabilitation protocols and tracking responses to physical therapy. Funding source: This work was supported, in part, by an endMS Doctoral Studentship Award from the Multiple Sclerosis Society of Canada awarded to Mr. Darrin O. Wijeyaratnam, as well as by the Faculty of Health Sciences at the University of Ottawa awarded to Dr. Erin K. Cressman and Dr. Lara A. Pilutti.

Assessment of cognitive-motor processes underlying physical and virtual execution of action sequences under varying task demands

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While virtual systems are increasingly being used in various domains such as industry or rehabilitation, the neurocognitive processes engaged by individuals when performing in such settings still warrant further investigation. Specifically, it is unclear how cognitive-motor mechanisms differ between physical versus virtual execution of a task. Thus, this study investigates the cortical dynamics underlying cognitive-motor performance during physical and virtual execution of a sequential task under two levels of demand. Twenty-two participants had to generate action sequences to complete the Tower of Hanoi task which consists of moving disks of different diameters between three evenly-spaced pegs under a low (3 disks) and a high (4 disks) demand. For the physical and virtual execution modality the disks were moved manually or via basic point-and-click computer-mouse controls, respectively. Video and electroencephalography were collected to evaluate performance and cortical activity. Performance was also evaluated using the number of puzzles completed (NPC) and the Levenshtein distance (LD; lower values indicate higher performance). The findings revealed that, irrespective of the demand, physical execution led to reduced NPC and enhanced theta power over the anterior-frontal, temporal and occipital regions compared to virtual execution. Moreover, regardless of the execution modality, greater task demand resulted in an increase of the LD, a reduction of NPC and an elevation of anterior-frontal, frontal and occipital theta power. Our findings suggest that, during physical versus virtual execution, individuals may have prioritized the quality of the produced sequence structure (similar LD) over task speed (lower NPC). Moreover, the need for more complex digit coordination during physical execution likely engaged attention, working memory, action monitoring and visual feedback control processes (higher theta power). Finally, independently of the execution modality, higher task demand led to sequential performance decrement and a greater recruitment of cognitive-motor resources, especially those frontally-mediated which are critical for planning, ultimately resulting in an attenuation of neural efficiency. This work has the potential to inform the neurocognitive processes underlying physical and virtual execution of action sequences of various complexities as well as the relevant applications of virtual technology such as, but not limited to, rehabilitation and tele-medicine.

Postural threat and dual-tasking elicit distinct postural control strategies: Evidence from stock market indicators

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Postural-threatening environments, such as standing at heights, are known to influence postural regulation. However, the interpretation of these changes remains debated. Increased fear of falling has often been associated with greater postural constraint aimed at limiting movement. Others, however, suggest that threatening environments may redirect attention toward fear-related cues rather than posture itself, thereby reducing conscious control and promoting a more automatic control, similar to what is observed under dual-task conditions. One limitation in resolving these interpretations lies in the reliance on global centre of pressure (COP) variables that inadequately capture the temporal dynamics of postural control. The present study aimed to investigate how a height-induced threatening environment interacts with attentional demands by combining a dual-task paradigm with exposure to a virtual height and by using an analytical approach capable of characterizing such temporal dynamics (i.e., the Trend Change Index (TCI)). Originating from stock market analysis, TCI quantifies directional changes (i.e., trend changes) in COP trajectories in the time domain. Unlike traditional global COP measures, TCI provides insight into the timing and structure of postural adjustments.

Twenty-three young adults (20.70 ± 1.22 years) stood as still as possible for 60s, with feet together, on a force platform at ground level and at the top of a virtually simulated 80-story building using a virtual reality headset (Meta Quest 2), with and without a concurrent cognitive task. TCI analysis revealed distinct patterns across conditions: TCI amplitudes increased during height exposure but decreased under dual-task conditions ($p < .05$). Moreover, time-domain classification of the intervals between consecutive trend changes revealed a shift toward longer intervals (.2–.5 s) in the height condition, whereas shorter intervals (0–.1 s) were more prevalent during dual-tasking. Together, these findings suggest that postural threat and dual-tasking elicit distinct control strategies that are not fully captured by traditional global measures. Exposure to height does not appear to influence trend changes in the same way as dual-task conditions, challenging the view that fear-related postural changes reflect increased automaticity.

Precision of hand localization decreases during visuomotor adaptation regardless of explicit strategy engagement

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Changes in the motor system are accompanied by changes in the sensory system. Specifically reaching with altered visual feedback results in reach adaptation and proprioceptive recalibration, such that one's sense of felt hand position is shifted in the direction of visual feedback provided (Cressman & Henriques, 2009). We have recently shown that the precision with which localizes their hand decreases when reaching with distorted visual feedback, even after reaches have been adapted. In the current research we asked whether this change in hand precision is impacted by the engagement of explicit reach strategies. Participants trained to reach with cursor feedback that was rotated 40 degrees (explicit reach strategies engaged) or 20 degrees (no explicit reach strategies engaged) clockwise relative to their hand position. A localization task to establish participants' perceived hand position and the precision with which they localized their hand was completed over the course of the training trials and within a retention interval. During both the training and retention trials, all participants adapted their reaches to the cursor rotation and shifted their felt hand position in the direction of the distorted cursor feedback. Moreover, participants in both the 40 and 20 degree groups exhibited an immediate decrease in proprioceptive precision when the cursor rotation was introduced. This decreased precision persisted across training trials and was also observed in the retention interval. Together, our findings indicate that the precision with which one localizes their hand decreases, independent of the engagement of explicit reaching strategies. Thus, diminished proprioceptive precision appears to be a general characteristic of visuomotor adaptation, rather than a consequence of explicit strategy engagement.

Assessment of performance and mental workload in individuals practicing a complex problem-solving task in dyad with an AI-powered humanoid robot

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While dyadic practice has been traditionally examined in human teams formed of two novices, a more recent body of work has also investigated such a form of practice by combining a human novice and robotic expert. Although informative, these previous human-robot dyad practice investigations mainly involved sensorimotor tasks without considering the performance of action sequences when executing complex problem-

solving tasks having associated mental workload and a humanoid robot teammate with AI capabilities. Thus, this work is a very first step towards investigating performance and mental workload dynamics when individuals practice a complex sequential task alone or with an AI-powered humanoid robotic teammate skilled at completing the task. Twenty-four healthy participants completed a complex problem-solving puzzle-task alone before (pre-test), immediately after (post-test) and 24 hours after (retention test) a practice session. Participants were randomly assigned to either the human-robot dyadic practice group ($n = 12$) where they trained with the skilled AI-powered humanoid robot (participants alternated trials between execution/observation) or to the practice alone group ($n = 12$) where they performed all practice trials by themselves. During the pre-test, post-test and retention-test, all participants executed the task alone (with a 3-min time limit) while their cognitive-motor performance was evaluated. Performance was assessed by recording the time to complete the puzzle-task and the number of occurrences for which it was successfully solved. Additionally, once the task was executed, participants completed the NASA-TLX questionnaire to assess the perceived mental workload. Results revealed that relative to the pre-test, performance was improved along with an attenuation of mental workload during both the post-test and retention-test for both experimental groups, although this was more pronounced for the human-robot dyad. Although as a first step this work warrants further analyses (e.g., sequence structure, cortical activity analyses), these findings are consistent with the view that dyadic practice may be advantageous relative to practicing alone, while extending this notion within a context where individuals practice a complex cognitive-motor task in dyad with an AI-powered humanoid robot. Funding source: Artificial Intelligence Interdisciplinary Institute at Maryland (AIM) 2025 research seed award grant program.

Gaze behavior and passing performance during a soccer-specific passing task

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Research on visual attention in soccer has largely focused on laboratory-based tasks or full match play, with comparatively little attention given to structured drills that dominate daily training environments. This study examined how gaze behavior relates to passing performance during a structured, soccer-specific task and contextualized these patterns with prior laboratory- and match-based research. Specifically, we investigated associations between key gaze metrics and performance during the Loughborough Soccer Passing Test (LSPT) in collegiate women soccer players. Fourteen athletes completed three trials of LSPT while wearing mobile eye-tracking glasses. Gaze behavior was analyzed using the Vision-in-Action framework, through calculation of fixation characteristics, allocation of viewing time to task-relevant areas of interest (AOIs; ball and passing targets), and Quiet Eye (QE) behavior. Linear mixed-effects models were used to examine trial-level relationships between gaze behavior and performance outcomes (i.e., LSPT score, execution time, and passing accuracy), accounting for repeated measures within players. Gaze metrics were normalized within eye-tracking devices to account for hardware differences. Results indicated that a greater allocation of viewing time toward passing targets was associated with poorer overall LSPT performance, such that a 1 SD increase in passing target viewing time (~6% increase) corresponded to a 5.6-point increase in LSPT score. Longer QE duration was neither associated with LSPT Score nor execution time but showed an interesting trend toward reduced passing accuracy ($\beta = -.58$, $SE = .3$, $p = .06$), indicating that prolonged final fixations may be less functional in fast-paced, time-pressured tasks. Average fixation durations

were shorter ($M = 142.57 \pm 57.37$ ms) than those typically reported in laboratory paradigms (332 – 1002 ms) and full match play (242 ms), highlighting context-dependent adaptations in gaze behavior. Flexible visual exploration, rather than prolonged engagement with any single AOI, emerged as an efficient gaze strategy within the context of LSPT performance. These findings emphasize the importance of examining perception-action coupling within representative training tasks and suggest that drill-based tasks such as LSPT could provide an intermediate context for understanding visual attention in soccer and informing sport-specific vision and cognitive training design.

Novice $\neq$ beginner: Definitional inconsistency in motor learning research

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Terminological ambiguity can undermine research quality, theoretical interpretation, and translation to applied practice. In motor learning, the term ‘novice’ is widely used to describe early-stage learners, yet considerable variability exists in what constitutes an early-stage learner. For example, ‘novice’ may describe someone with no prior experience in a task or related activities, or an individual with limited experience, such as an athlete who has played for only one season. This scoping review examined how contemporary motor learning research operationally defines novice status. An electronic search of peer-reviewed articles published within the past five years was performed using SportDiscus, selected for its comprehensive coverage of sport and movement science. Search terms combined descriptors of motor learning (e.g. motor skill learning, motor skill acquisition, movement learning, skill development) with terms describing early-stage learners (e.g., novice, beginner, inexperienced, unskilled). Studies were included if they explicitly identified participants as novices or beginners in a motor learning context. Forty-seven articles met inclusion criteria. Data extraction focused on how novice status was defined and determined. Inductive coding identified five categories of novice definitions: (1) true task novices (no prior experience in the target skill); (2) limited-experience novices (experience allowed below a defined threshold); (3) task-novices with previous experience in other movement contexts that could influence learning; (4) novices classified based on program enrollment (e.g., beginner classes); and (5) undefined novices (labeled ‘novice’ with no operational definition). Most studies relied on self-report with only a few using performance-based verification, such as objective pre-tests or expert ratings. Across studies, operational definitions were inconsistent or absent, and in some cases, participants classified as novices had meaningful prior experience in the task or related activities. This variability in defining novice learners undermines confidence in interpreting results and applying findings, as participants labeled “novice” may represent fundamentally different learner populations.

The influence of task demands and skill level on perceived attentional focus effectiveness

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Previous research on attentional focus has widely examined the performance benefits when adopting an external or holistic focus compared to an internal focus. However, it remains unclear whether the task dynamics and learner skill level affect how an attentional focus cue is perceived. Therefore, we conducted two experiments to explore how the demands of a task and the learner’s experience influence the perceived effectiveness of attentional focus cues. Experiment 1 examined how attentional focus impacted duration and stability during a maximal effort isometric plank

test in a novice university population. Participants completed three trials of the plank test while receiving external (EXT), holistic (HOL), or internal (INT) attentional focus cues. Participants were then asked to select which condition they believed produced the longest duration plank. Experiment 2 examined how attentional focus impacted performance in a marksmanship task in skilled Reserve Officers' Training Corps (ROTC) cadets. Participants completed four trials of a marksmanship task using a non-immersive virtual reality marksmanship simulator. Participants completed a baseline trial followed by three trials incorporating external-far (EXF), external-near (EXN), or INT attentional focus cues. Participants identified which cues they believed led to the highest precision, fastest response time, and greatest stability. Results from Experiment 1 found that perceived effectiveness of each attentional focus condition did not differ significantly, although 44.2% of participants perceived EXT as producing the longest plank duration compared to INT (32.6%) and HOL (23.3%). In Experiment 2, there was a significant result for perceived marksmanship precision with the EXF instruction selected most frequently (48.4%), followed by EXN (33.9%), and INT (17.7%). For response time, the distribution of responses did not differ significantly. Differences in perceived stability were marginally significant ($p = .062$) with EXN the most frequently selected (41.9%), followed by EXF (38.7%), and INT (19.4%). These findings suggest that perceived effectiveness is not a stable construct but is context dependent. Perceived effectiveness may vary in the relationship between skill type and skill level. Future studies should explore whether perceived effectiveness of attentional focus cues aligns with or predicts future motor performance.

Red light, no bite: Measuring response times in K-9 handlers

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Law enforcement dogs (K-9s) have many roles, including searching for contraband and explosives, locating missing persons, and assisting with crowd control. However, K-9s are primarily tasked with the pursuit and apprehension of individuals suspected of violent crimes. Most commonly, K-9s are trained to bite and hold the suspect until their handler commands to release. K-9s can exert up to 10x more bite force compared to an untrained dog of a similar breed. The use of K-9s, paired with their tremendous bite force, has led to major injuries, complications, litigations, and settlements. Little is known about the interaction between handler response and suspect response that influences K-9 call-offs. The purpose of this study was to investigate the response time of handlers during a simulated suspect apprehension. Handlers and K-9s ($n = 27$) completed six trials of a go/no-go task. Out of the six trials, 2 or 3 were randomly assigned as "no-go" trials, with the remaining as "go" trials. K-9s were first stimulated by the suspect (decoy) located 25 m away from the start position of the K-9 and its handler. Next to the decoy, an LED screen would turn green, prompting the handler to release the K-9. On the "go" trials, the screen would remain green, and the K-9 would bite and hold the decoy. On the "no-go" trials, a red signal appeared at random intervals between 500 and 2000ms prompting the handler to call off the K-9 as quickly as possible. All trials were video recorded. Response times ranged between 210-1200ms with an average response time of (370 ± 26 ms). Pearson's correlations revealed response times increased with greater training frequency, but this relationship was not statistically significant ($r = .330$, $p = .093$). A one-sample proportion test indicated that the observed "go" response rate (62.5%) significantly differed from the expected 80% distribution ($p < .001$). These results provide insight into the fast-paced nature of handler and K-9 work and may help to guide training strategies to mitigate litigations and settlements associated with law enforcement dog bites.

Effects of AI-powered humanoid robot explanations during a collaborative teaming task on human trust, performance and neurocognitive processes

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Physical human-robot collaboration has been a critical task domain over the past decade. However, in recent years, the influence of a robot's explanations on the human teammate has become decisive in yielding safe, efficient, and transparent team environments. While a large body of work has technologically developed explainable AI systems, there is only limited understanding of human neurocognitive mechanisms when individuals process explanations provided by their synthetic teammate and how these mechanisms are affected by human skill level. Therefore, this work examines how the presence versus absence of explanations from an AI-enabled humanoid robot teammate affects performance, trust and neurocognitive processes in humans having a different skill level during the execution of a collaborative task. Before and after practicing a sequential task, twenty-nine right-handed, healthy individuals completed it with an AI-powered humanoid robot where the synthetic teammate provided or not an explanation about the actions it had just executed. Thus, humans jointly executed this sequential task with the humanoid robot before (i.e., novice individuals) and after (i.e., skilled individuals) practice (after robustly learning verification) while performance, trust ratings and EEG were collected. Once EEG was denoised, the theta power was computed and averaged over the frontal, central, temporal, parietal and occipital regions as well as both hemispheres. The results revealed that i) performance errors decreased with experience, but remained unaffected whether the explanation was provided or not; ii) regardless of experience, trust improved with explanations versus without explanations; iii) theta power increased in novice individuals suggesting a greater recruitment of neurocognitive resources (e.g., working memory, attention, action monitoring) in presence versus absence of explanations; and iv) theta power decreased with experience independently of whether explanations were provided or not, indicating an attenuation of the engagement of neurocognitive processes. This work suggests that explanations enhanced trust without negatively affecting learning, but resulted in an initial neurocognitive cost that disappeared when individuals became skilled at completing the task. This work can inform human neurocognitive processes during collaborative execution of sequential tasks as well as human-robot teaming applications such as, but not limited to, navigation, problem-solving or rehabilitation tasks. Funding source: Maryland Robotics Center and the Institute for Systems Research; University of Maryland's Department of Kinesiology Gebhardt-Phillips Graduate Student Fund.

The limits of aerobic fitness in supporting bilateral transfer

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Bilateral transfer refers to changes in motor performance in one limb that occur after practice with the opposite limb and is typically asymmetrical in younger adults (Pan & van Gemmert, 2013). Asymmetrical transfer, in which one limb shows greater change than the other after opposite-limb practice, may be a product of lateralized hemispheric control (Sainburg, 2002). With age, more symmetrical transfer caused by additional ipsilateral activation as a compensatory mechanism is observed (Cabeza, 2002). Higher aerobic fitness appears to help preserve lateralized hemispheric activation in older adults, suggesting asymmetric transfer could be retained

with age (McGregor et al., 2013). Similar effects have been observed in younger adults with high reported physical activity (Cochran & Aiken, 2026). 28 right-handed young adults aged 18-35 and 15 older adults aged 65-81 were grouped by aerobic fitness determined via the 2-min walk test (Rowley et al., 2022). Individuals performed a 30° visual rotation drawing task assessing movement parameters. Pre-test of two trials with each limb established baseline performance, followed by 40 practice trials on the left limb. Post-tests mirror pre-tests assessing the transfer of motor planning parameters including movement time, normalized jerk, trajectory length, and initial direction error. Results showed samelimb learning for all variables except trajectory length in both older adult fitness groups ($ps < .05$), though only the high fitness group improved on trajectory length ($p < .05$). Older adults also demonstrated bilateral transfer for movement time regardless of fitness level ($p < .05$). No other variables showed transfer across age or fitness groups ($ps > .05$). Findings suggest aerobic fitness may not facilitate bilateral transfer in a younger population. The limited transfer observed in older adults indicates that aerobic fitness alone may not be sufficient to preserve asymmetrical transfer associated with hemispheric specialization. Contrary to previous literature, current results imply that the influence of aerobic fitness on bilateral transfer may be more task specific. Findings present the need for further investigation into how fitness, aging, and bilateral transfer processes interact.

Not everything is as it appears: Oculomotor behavior following repetitive loading

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Repetitive loading can have detrimental effects on neurological functioning. United States Coast Guard (USCG) personnel undergo comprehensive training in various sea states, exposing individuals to a variety of loads with physiological effects that remain unclear. Oculomotor tasks assess neurocognitive function by evaluating visual tracking and control. Thus, the present study examined oculomotor behavior in conjunction with load exposure for USCG personnel following a standard underway operation. Twenty-one participants ($M_{age} = 28.29 \pm 6.41$ years) from three locations completed two trials each of circular and horizontal smooth pursuit (CSP/HSP) and horizontal and vertical saccade (HSA/VSA) tasks in an immersive virtual reality headset before and after a routine underway. Participant helmets were outfitted with unique configurations of inertial measurement units (IMUs) based on their helmet design to capture head loads. Angular velocity, angular error, and percent of the trial spent on target were calculated from the combination vector of gaze point transforms. A composite head loading score was developed combining peak head acceleration, cumulative energy absorption, characteristics of periods of high acceleration (> 2 g; intensity and duration), and overall exposure consistency from IMUs. Bivariate correlations were computed to determine relationships between oculomotor performance metrics and composite score from IMUs. Moderate positive correlations were observed between composite score and percent time on target for CSP, HSA, and VSA tasks, and between composite score and angular velocity for CSP ($r = .45-.54, p < .05$). A moderate negative correlation was found between composite score and angular error for VSA ($r = -.50, p < .05$). No significant correlations were found for HSP. The association between greater loading exposure, as represented by composite score, and increased time spent on target paired with angular velocity in some tasks may reflect increased oculomotor effort and anticipatory control deficiencies. Moreover, a negative association between composite score and angular error

may not indicate overall neurocognitive enhancement, as prior work suggests compensatory adaptations in response to sustained loading. Moderate correlations suggest loading exposure explains only part of oculomotor behavior, emphasizing the need to include metrics describing other underway variables and preexisting impairment to better explain neurocognitive strain. Funding source: Office of Naval Research.

Population-level motor pattern shifts and individual learning variability in sprint freestyle swimming

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Optimizing performance through refined movement strategies is central to motor learning. In competitive sprint freestyle swimming, modern coaching emphasizes transitioning rapidly into propulsive phases to minimize non-propulsive time in the stroke cycle. This study examined whether modern coaching emphases have altered stroke phase patterns in competitive swimmers relative to the benchmarks reported by Chollet et al. (2000), and the extent of individual variability in these patterns. Twenty-five collegiate swimmers performed three maximal 50-meter freestyle trials with palm-mounted sensors recording bilateral hand kinematics and pressure data ($n = 144$ trials analyzed, 72 per hand). Each stroke cycle was segmented into four phases: Entry & Catch (hand entry to propulsive initiation), Pull (early propulsion), Push (late propulsion), and Recovery (aerial arm repositioning). Phase durations were calculated as percentages of total cycle time. In this study, swimmers spent substantially more cycle time in propulsive actions compared to previously reported benchmarks ($59.5 \pm 4.6\%$ vs. 49.0%). This 10.5% shift was driven by an extended Pull phase ($33.2 \pm 4.3\%$ vs. $24.5 \pm 3.0\%$, $p < .001$) and compressed Entry & Catch ($17.8 \pm 10.6\%$ vs. $24.5 \pm 5.0\%$, $p < .001$), demonstrating adaptation in response to coaching emphasis on a quick transition over the past two decades. Additionally, movement variability patterns revealed how individual swimmers have adapted to this constraint differently. Propulsive phases showed tight consistency (Push $CV = .12$, Pull $CV = .25$), indicating well-learned motor patterns. In contrast, Entry & Catch showed substantial variability ($CV = .73$). Subsequent variance decomposition revealed this variability reflected primarily between-swimmer differences (45%), indicating some swimmers have adopted the quick transition strategy more than others. Additional within-swimmer variation (33%) likely reflects contextual factors such as breathing mechanics. While these results suggest all swimmers learned to prioritize propulsive phases, considerable individual differences currently exist in how aggressively they compress the first phase. These findings highlight how coaching constraints drive population-level motor pattern changes while permitting substantial individual variation within movements.

“Land softly”: The effects of verbal cues to decrease fatigue response

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Verbal cues guide movement development and enhance performance. However, few studies have examined verbal cues during fatigue, with findings varying by cue type and the motor skill used (Lohse & Sherwood, 2011). Understanding how fatigue influences movement and the impact of verbal cues on performance may support practitioners in developing effective cues critical for optimal performance. Therefore, this study investigated the effects of an attentional focus verbal cue on biomechanical and perceptual responses during fatigue-inducing exercise. It was hypothesized (H1) that fatigue would reduce motor control, supported by decreases in vertical stiffness and increases in perceived

exertion. It was further hypothesized (H2) that a holistic verbal cue would attenuate these fatigue-related changes compared to no verbal cue. Twenty-two participants (20 ± 2 years of age) completed two fatiguing trials of continuous vertical jumping at 60 bpm for one min under one of two conditions: no verbal cue (NVC) or verbal cue (VC). The verbal cue instructed participants to “land as softly as possible,” promoting a holistic focus of attention. Vertical stiffness (Kvert) was calculated from force plate data, and Ratings of Perceived Exertion (RPE) were recorded. Separate 2 (Condition) $\times$ 2 (Fatigue Trial) $\times$ 4 (Time Interval) mixed-factors ANOVAs examined differences in Kvert and RPE. Kvert decreased from the 0-15 s interval during both trials ($p < .001$), indicating initial movement adjustment. However, a three-way interaction showed differing late-stage responses: in the second trial (45–60 s), the NVC group reverted toward a stiffer, “hard-landing” pattern, whereas the VC group maintained the intended softer-landing response ($p < .05$). RPE increased across all intervals ($p < .001$) and was significantly higher in the second trial overall ($p < .001$). Notably, the VC group reported lower RPE compared to NVC during fatigue progression ($p < .05$). A simple holistic verbal cue (“land as softly as possible”) prevented late-stage stiffness rebound and reduced perceived exertion. Excessive stiffness under fatigue can compromise landing strategies and potentially increase injury risk, while softer landings support safer lower extremity mechanics. Practitioners can implement this cue during training to maintain movement quality and reduce injury risk.

Translational neuroscience: The impact of “smart” optic flow on gait and freezing in Parkinson’s disease

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The underlying mechanisms of impaired gait, and specifically the puzzling phenomenon of freezing of gait continue to be the most poorly understood issue facing people with Parkinson’s disease. Thus, the lack of effective strategies is likely associated with the variety of approaches that aren’t founded in translational neuroscience. While our previous research has demonstrated that optical flow is essential to elicit gait improvement, the aim of the present study was to evaluate the delivery of “smart” optic flow, that takes into consideration both sensory and attention deficits present in PD. The present study manipulated how optic flow was provided, and how this might influence freezing of gait (FOG) in PD. Therefore, three groups; 15 PD FOG, 16 PD non-FOG, and 16 healthy controls were tested in 3 walking conditions; baseline walk with turns, optic flow cues from ground lines (Ground), and “smart” laser optic cues that recognize when the patient is moving (Smart). A significant interaction was found between group and condition, $F(4, 88) = 2.52$, $p < .0465$ with post-hoc analysis revealed that PD FOG significantly decreased their velocity in the “smart laser” condition (from 78.91 ± 32.6 cm/s to 59.13 ± 25.98 cm/s) ($p < .008$) while velocity did not be affected by ground lines (97.43 ± 25.27 cm/s). Yet, both PD FOG and PD improved step length with ground lines to the level of healthy controls, while the laser increased gait variability and double support time. These results suggest that optic flow in itself is not enough to elicit gait improvement in PD. When PD patients use visual cues, gait becomes less automatically controlled and hence may place more demand on cognitive resources for conscious gait control, which may be an important factor contributing to gait improvement.

RDE-GAIT: A novel mirror-equilibrium approach to 3-D gait symmetry

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Digital wearables enable large-scale, ecological gait assessment, yet many commonly used gait metrics lack mechanistic interpretability and cannot identify stride-level symmetry dynamics. To this end, we developed the Reza Dimensional Equilibrium for gait (RDE-GAIT) method and evaluated it using a public mobile brain-body dataset of healthy young adults from Hanada, Kalabic, Ferris (2024). Using a left ankle IMU with a waist reference, we reconstructed the 3D leg vector at the forwardmost (anterior) and backward-most (posterior) locations during each gait cycle, and flipped/mirrored the posterior vector so that it could be directly compared to the anterior vector in 3D space. We defined the similarity between the vectors as ranging from 0 to 1, with values near 0 indicating less similarity and values near 1 indicating more similarity. Next, we normalized to a robust local baseline computed as a rolling median over a -cycle moving window, so each gait cycle is evaluated relative to nearby cycles (cycles before and after) within the session. Lastly, an equilibrium value at each gait cycle termed was calculated as, with 0 indicating a consistent gait cycle relative to the surrounding gait cycles and 1 indicating inconsistent gait cycles (either an inconsistent gait cycle embedded within consistent gait cycles or a consistent gait cycle embedded within inconsistent gait cycles). Outdoor walking sessions contained gait cycles (median 4,632). The median proportion of consistent gait cycles was (range), and the median proportion of inconsistent gait cycles was (range). Twenty-eight participants exhibited consistency in of gait cycles, with eight of those exhibiting a very high proportion of consistency in their gait cycles. On the contrary, three participants exhibited a high proportion of inconsistency in their gait cycles. What remains an empirical question is the extent to which the similarity metric (0 to 1) and/or the equilibrium metric relate to adaptive and/or functional gait patterns. Future work should explore these questions, as well as compare RDE-GAIT to traditional symmetry gait metrics across diverse locomotor tasks (e.g., running, irregular terrain, etc.).

Acute effects of walking on executive function and motor performance in young adults with autism spectrum disorder

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Adults with autism spectrum disorder (ASD) show higher rates of executive function deficits, exhibit a range of motor impairments relative to typically developing peers, and experience a 42% lifetime prevalence of anxiety disorders. Chronic exercise is an effective intervention to reduce anxiety and improve executive functioning and motor skills in individuals with ASD. However, limited research has examined whether acute exercise interventions produce similar benefits for these comorbidities in adults with ASD. Therefore, the purpose of this study was to investigate the effects of 20 min of walking on executive function, motor performance, and anxiety for young adults with ASD. Eight adults with ASD (24.0 ± 3.6 years; 6 women) completed two walking sessions: one at light intensity and one at moderate-to-vigorous intensity. Intensity was determined by heart rate (HR) targets: 35-60% of maximum HR for light-intensity and 60-95% of maximum HR for moderate-vigorous-intensity. Before and after each session, participants completed assessments of state anxiety, executive function, and motor performance, including the State Trait Anxiety Inventory, Trail Making Test (TMT-A & TMT-B), Stroop Color and Word Test, Digit Span, maximum grip strength, and the Purdue Pegboard test. Data were analyzed using a 2 (Intensity: light, moderate-to-vigorous) $\times$ 2 (Time: pre, post) repeated-measures analyses of variance. The results revealed an effect of time for the Purdue Pegboard Assembly task ($p = .024$), indicating improved performance following walking regardless of intensity. The results revealed an effect of time for TMT-A ($p < .001$), indicating faster performance following walking regardless of intensity. An intensity by time interaction for TMT-B ($p = .044$) demonstrated greater improvement following moderate-to-vigorous-intensity

compared to light-intensity walking. No other effects or interactions were observed. These findings suggest that acute walking improves processing speed and visual attention (TMT-A), and that moderate-to-vigorous intensity walking may be particularly important for enhancing higher-order executive functions such as cognitive flexibility and divided attention (TMT-B). Finally, improvements on the Purdue Pegboard Assembly task suggest that acute walking may enhance bimanual coordination and motor planning, skills that rely on integrated executive and motor control processes that are frequently affected in adults with ASD. Funding source: NASPSA Graduate Student Research Grant.

Factoring together sensori-motor measures of imagery ability in a large-sample of self-reported people with aphantasia

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There is growing interest in the identified phenomenon of aphantasia, characterised by an inability to visually imagine, affecting ~2% of the population. This inability is argued to be qualitatively different to hypophantasia, which is a poor ability to visually imagine and thus comprise unique characteristics. Whilst, by definition, aphantasia is the absence of visual imagery, it is unclear whether other domains of imagery (e.g., motor, auditory, emotions) are also impacted. We recruited people with self-reported aphantasia ($N=169$) through the Aphantasia Network to complete an online study. To assess imagery across different sensory domains, participants completed three validated measures; the Vividness of Visual Imagery Questionnaire (VVIQ), the Movement Imagery Questionnaire-3, and the Plymouth Sensory Imagery Questionnaire. We combined individual items from all these measures (63 items) and ran a factor analysis to determine how various items related to different sensorimotor imagery modalities. From an analysis of all items and across all self-identified people with aphantasia, three unique factors were identified: 1) visual-sensory and visual-motor imagery, 2) kinesthetic-motor, body sensations, and emotional imagery and 3) other non-motor sensory imagery. Despite the separation of these factors, scores on all items/subscales were in the low range. A specific analysis isolating individuals who had a complete absence of visual imagery (VVIQ score of 16; $n=109$), revealed an additional factor, where visual-sensory (e.g., imagining an object) was now distinct from visual-motor (i.e., imagining seeing yourself move). As such, there was evidence that visual imagery is a distinct impairment in people with aphantasia, with the ability to imagine seeing and feeling actions and other multisensory non-motor imagery being distinct. Nevertheless, the overall sample scored low on all subscales of imagery, with kinesthetic motor imagery ($M=3/7$) and imagery of sound ($M=2.7/7$) being the highest. These data further our understanding of the experiences of people with aphantasia and provide evidence for both the separability of visual from other sensory-motor imagery processes, in a manner similar to what we see in the non-aphantasic population, as well as a general imagery impairment. Funding source: NSERC.

Risk-dependent modulation of swing initiation timing in professional baseball games

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Successful baseball batting requires precise temporal coordination between pitch arrival and swing initiation. Laboratory studies suggest that batters prepare to swing based on fastball timing and subsequently modulate or inhibit this preparation when slower pitches are detected

(“sitting on a fastball”). However, whether such timing strategies manifest in real competitive games remains unclear. Here, we analyzed two seasons of in-game batting data from professional baseball players, focusing on the relationship between pitch ZoneTime (time from release to plate crossing), swing initiation timing, and batting outcomes. Swing initiation timing was estimated from high-speed game footage synchronized with pitch-tracking data, and batting outcomes were classified as hits (singles, doubles, triples, and home runs) or swing-and-miss. We examined outcome-dependent timing structures using a two-dimensional ZoneTime–swing onset space, separately for two-strike and non-two-strike situations. Under two-strike conditions, we observed a clear reversal in the relative swing initiation timing associated with hits and swing-and-miss around ZoneTime $\approx .42-.43$ s. For shorter ZoneTimes, hits were most likely when swings were initiated slightly earlier relative to the average timing, whereas delayed initiation was associated with swing-and-miss. In contrast, for longer ZoneTimes, hits were associated with delayed swing initiation, while early initiation increased swing-and-miss probability. This boundary corresponded closely to the transition between fastball-dominated and off-speed-dominated pitch distributions. Importantly, similar pitch-speed distributions were observed in non-two-strike situations, yet the pronounced reversal in success–failure timing structure emerged only under two-strike conditions. These findings suggest that while pitch-speed-dependent motor preparation may exist across contexts, its behavioral consequences are strongly shaped by task constraints. Specifically, under two-strike conditions where swing errors lead directly to strikeout, initiating the swing slightly earlier versus later relative to the typical timing had qualitatively different consequences depending on pitch ZoneTime, revealing context-dependent timing strategies consistent with fastball-oriented preparatory mechanisms and their modulation. Our results provide real-game evidence linking pitch-speed-dependent motor preparation to outcome-specific timing landscapes in elite baseball batting.

Exploring effects of soccer heading on oculomotor control

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Producing no clinically observable symptoms of concussion, sub-concussive loading increases the risk of neurological injury and degeneration. However, this lack of observable symptoms may not equate to a lack of injury. Measurable neurophysiological impairments have been identified in individuals who displayed no clinically observable signs or symptoms of mild traumatic brain injury (mTBI) following head impact. With exposure to repetitive sub-concussive loading (RSCL), as seen in contact-sport settings, these impairments may accumulate to mTBI-like impairment. In soccer, studies have identified associations between ball heading and changes to brain structure and function. The use of sensorimotor integration techniques, including oculomotor control and specifically circular smooth pursuit (CSP) assessments, may detect even mild impairments without clinically observable symptoms. The purpose of this study was to determine whether alterations in oculomotor control, specifically CSP, occur following a period of RSCL through soccer heading. It was hypothesized that significant increases in sample entropy and decreases in the percent of gaze points on target would be present following a soccer heading task. Thirteen adults (8 females, 5 males) with a mean age of 20.31 ± 2.40 years, who played organized soccer within the past 2 years, participated in this study. Oculomotor control was assessed prior to and following completion of a soccer heading task. Participants completed a series of eye tracking tasks, including CSP, in a custom-built virtual reality environment, completing two 15-s trials of each task. For the heading task, participants performed 10 directional headers into a goal following a ladder

drill, 5 delivered via kick and 5 delivered via throw. Oculomotor control data was processed using a custom MATLAB script and a paired samples *t*-test was performed to compare pre-heading and post-heading means. No significant differences were identified in mean sample entropy ($M_{PRE-X} = .055$, $M_{POS-X} = .043$, $p_X = .067$; $M_{PRE-Y} = .042$, $M_{POS-Y} = .047$, $p_Y = .342$) or mean percent of gaze points on target ($M_{PRE} = 75.37\%$, $M_{POS} = 83.39\%$, $p = .364$). These results suggest the performance of 10 headers with varied loads, including higher and lower force headers, does not cause significant changes in oculomotor control performance. Funding source: International Society of Biomechanics Matching Dissertation Grant.

Vertical plane presentation of a visual illusion can alter movement planning of but not displacement of horizontal manual length estimations

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Visual illusory effects on perceived length in the vertical–horizontal (V–H) illusion often reveal that the vertical segment of an inverted T (IT) is perceived longer than its horizontal segment. Deceptive visual cues can direct upper limb control when projection surface of the illusion and movement surface remain the same. Although they can guide step planning, deceptive cues do not guide final step estimations when projection surface of the illusion and movement differ. In this study, we examined whether the illusory influences affected manual estimations under gaze restrictions when visual presentation and movement surfaces differed. Twenty right-handed participants (18–34 years) moved their index finger to make curved point-to-point movements on a horizontal surface to estimate vertical segment lengths of IT configurations shown on a vertically oriented display. Configurations included a vertical segment length of 10% shorter (9 cm) or 10% longer (11 cm) than the 10 cm horizontal length of a vertically projected IT. After IT presentation and orally stating whether vertical and horizontal segments were equal or one was longer than the other, people made manual length estimations under three gaze conditions: gaze maintained on the IT intersection (target fixation, TF) or the remembered IT intersection after its removal (remembered target fixation, RTF) or gaze directed on the moving hand (movement fixation, MF). Illusory deceptions occurred 89% of the time. Movement displacement (D) and peak velocity (PV) for long IT exceeded short IT ($p < .05$). D did not vary by gaze condition ($p > .05$). PV for RTF exceeded TF and MF and remained similar to control trials ($p < .05$). Like step length estimations, deceptive visual cues can guide movement planning of reaching yet not final manual estimations when projection surface of the illusion and movement surface differed. Unlike step length estimations, asking people to gaze at remembered target intersection did not alter the planning of manual estimations. Vertical projections of IT illusory deceptions which can alter planning control over manual length estimations do not guide horizontal end positioning of the limb.

Heart rate variability in response to the Shoot/Don't Shoot Task under normal and high cognitive load

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Connectivity between the heart and brain is theorized to promote flexibility in a changing environment, such that higher heart rate variability (HRV) reflects greater adaptability (Thayer & Lane, 2000). There is evidence for the benefits of higher HRV on cognition (e.g., Keen et al., 2020) and those

with lower HRV show reduced inhibitory control under high working memory conditions (Yang et al., 2020). The response of HRV to the shoot/don't shoot task (SDST) is presently untested. We examined HRV in 56 civilians at baseline and in the SDST under normal cognitive load, and higher cognitive load induced by processing a complex dispatch message. Performance measures were errors (shoot; fail-to-shoot), response time, and accuracy. HRV measures included root mean square of successive differences (RMSSD), standard deviation of RR intervals (SDRR), high and low frequency (HF; LF), and heart rate (HR). The Wilcoxon test found that participants made more shoot and fail-to-shoot errors under higher cognitive load ($p < .01$). Repeated measures ANOVA with Bonferroni post-hoc tests revealed effects of test condition. RMSSD and SDRR were lower in both SDST conditions than at baseline ($p < .001$), suggesting the SDST reduced parasympathetic modulation and autonomic flexibility. Evidence for increased sympathetic modulation in the task was shown in lower HF, and higher LF, LF/HF, and HR in both SDST conditions than at baseline (all $p < .001$). The addition of high cognitive load exacerbated this effect, with lower HF ($p < .01$), and higher LF ($p < .03$), LF/HF ($p < .01$), and HR ($p < .001$) with the dispatch message than without. Participants responding faster in shoot conditions under normal cognitive load showed lower RMSSD than those responding more slowly ($p < .05$), implying that higher parasympathetic modulation slowed responses. Tests of differences in HRV between those making more or less shoot errors revealed few effects. Participants making more shoot errors under normal cognitive load had higher LF scores than those making fewer errors ($p < .03$), implying that higher sympathetic modulation negatively impacted shoot errors. This, however, contrasts non-significant trends across multiple variables suggesting that under higher cognitive load, higher sympathetic modulation was associated with less shoot errors.

Attentional focus distance and practice structure shape motor learning: Evidence from performance and kinematic analyses

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Although attentional focus and practice structure both influence motor learning, little research has examined how attentional focus at varying distances interacts with contextual interference. This study addressed this gap by investigating how attentional focus distance (i.e., distal, middle, proximal) and practice schedule (i.e., random vs. blocked) jointly affect accuracy and consistency during the acquisition of dart-throwing. A secondary aim was to determine whether attentional focus distance produced systematic differences in performance or movement kinematics, including peak accelerations of the upper arm, forearm, and hand segments, and peak shoulder and elbow flexion angles, measured using a Noraxon MR3 3D motion capture system. These analyses evaluated whether outcome or kinematic differences across attentional focus distances explain mechanisms underlying contextual interference. Participants ($N = 48$) were assigned to a Blocked or Random practice group and completed 150 acquisition trials under three focus of attention (FOA) conditions: distal (target's bullseye), middle (dart trajectory), and proximal (release angle). The Blocked group performed 50 consecutive trials per FOA, whereas the Random group completed all 150 trials with FOA cues presented in a randomized order. All participants returned 24 hours later for retention and transfer testing. During acquisition, both groups improved in accuracy (MRE) and consistency (BVE), demonstrating performance gains regardless of schedule. In retention, the Random group showed superior accuracy and consistency compared to the Blocked group, revealing a contextual interference effect. No meaningful group differences emerged in the transfer test. Analyses of attentional focus distance during practice indicated that the Blocked group performed more accurately under the distal focus than under the proximal and middle foci,

whereas the Random group showed no FOA-specific performance differences. Kinematic results revealed that distal focus produced greater peak segment accelerations than the middle and proximal foci in both schedules. For peak shoulder flexion angles, the Blocked group showed higher values under distal focus, whereas the Random group showed no differences. These findings indicate that attentional focus distance influenced throwing accuracy and movement kinematics during practice. Additionally, the findings revealed that practice schedule played a meaningful role in shaping learning, as reflected in enhanced retention following random practice.

Phase-specific stroke timing consistency across the freestyle stroke cycle in competitive swimmers

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Stroke-to-stroke consistency is commonly known as a trait of skilled swimmers. However, variability in stroke timing may change across phases of the stroke cycle. The purpose of this study was to compare coefficients of variation (CV) of stroke duration across freestyle stroke phases to determine which phases exhibit the greatest timing consistency and variation in competitive swimmers. Competitive swimmers ($n=25$) completed two to three freestyle or frontcrawl trials while wearing palm-mounted sensors that recorded pressure and kinematic data. Data was recorded separately on both the left and right hands. Linear acceleration, relative velocity, and relative displacement were measured separately with three-dimensional vectors along the x, y and z axes. Stroke phases were segmented separately into entry and catch, pull, push and recovery parallel to that of Chollet 4-phase model. Stroke-to-stroke consistency for each phase was found using the CV across valid strokes for each swimmer. Stroke temporal variability differed across phases of the freestyle stroke cycle. Mean CVs were greatest during the entry-catch phase, ($68.9 \pm 21.0\%$), followed by the pull phase ($31.1 \pm 16.9\%$). Substantially lower variability was observed during the push phase ($13.3 \pm 4.7\%$) and the recovery phase ($12.8 \pm 4.6\%$), indicating greater temporal consistency during the later portions of the stroke cycle. Temporal consistency varied greatly across the freestyle stroke cycle, with the most pronounced difference found during the entry-catch phase and progressively greater consistency found throughout the pull, push and recovery phases. These findings show that early stroke phases represent the least stable component of the freestyle stroke in swimmers and improving timing during the early stroke could be important for technique development.

Impact of disease progression on postural fractal dynamics in individuals with Parkinson's disease

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People with Parkinson's disease (PwPD) experience progressive impairments in mobility and postural stability given the disease's progressive neurodegenerative nature. As an alternative to traditional postural measures, fractal dynamics quantify the relationship between variability magnitudes at short (e.g., 200 ms) and long (e.g., 3 s) temporal scales and theoretically represent one's level of postural adaptive capacity. However, fractal dynamics in the early stages of Parkinson's disease (PD) are understudied. The purpose of this study was to compare postural fractal dynamics in individuals in early stages PD to their control peers. Twelve

PwPD were classified using the Hoehn & Yahr (HY) disease progression scale and identified as HY stage 1 (HY1; unilateral impairments; $n=7$; 3M/4F; age 67.2 ± 6.8 years) or stage 2 (HY2; bilateral impairments; $n=5$; 2M/3F; age 68.2 ± 7.6 years), with HY2 indicating greater mobility impairments. The control group consisted of 9 age- and sex-matched controls (5M/4F; age 68.1 ± 10.39 years). Participants performed 3 quiet standing postural tasks, each for 30 s; eyes open (EO), eyes closed (EC), and EO standing on a foam surface (EOF). Fractal dynamics were obtained using detrended fluctuation analysis of medial-lateral (ML) and anterior-posterior (AP) accelerations collected using an accelerometer at the 5th lumbar region. Differences in fractal dynamics between groups within postural tasks were assessed via Cohen's D effect size (ES) calculations. Across postural tasks, the HY1 and control groups displayed similar fractal dynamics (ES range .1-.35 for all comparisons except EO in the ML direction [ES=.81]), while HY2 displayed lower fractal dynamics compared to the control group (ES range .38-1.06 and .40-.67 for ML and AP directions, respectively) and HY1 group (ES range .78-1.19 and .64-1.19 for ML and AP directions, respectively). These findings provide preliminary evidence that PwPD classified as HY1 have similar postural adaptive capacity to those without PD. Moreover, fractal dynamics appear to distinguish HY1 from HY2 categorization across all three postural tasks and in both AP and ML directions. The discrepancies between HY1 and HY2 suggest that fractal dynamics may serve as an indicator of disease progression and mobility impairment in PwPD, and these findings substantiate the need for a larger-scale study.

Acute effect of balance board standing on postural control in adults and typically developing children

Yeon-Joo Kang, Georgia State University; Jianhua Wu, Georgia State University

Acute balance-board practice can alter postural control, but center-of-pressure (COP) responses across age remain unclear. This presentation reports COP outcomes during single-leg standing in children (7–12 y; $n=12$) and young adults (18–35 y; $n=15$) before and after practice on two boards: a multi-directional wobble board (WB) and a mediolateral (ML) rocker board (RB). In a randomized pre-post crossover design, participants completed both board conditions. Each condition included five 40-s standing bouts with 60-s rest. Single-leg standing COP was recorded on a force plate before and after each condition. Outcomes were sway range and sway velocity in the anteroposterior (AP) and ML directions, and 95% ellipse area with ellipse height and width. Children demonstrated larger COP values than adults across measures, reflecting greater sway magnitude and speed. No board type $\times$ time interactions were detected. AP sway range was greater in children and larger during RB than WB for both groups. A group $\times$ time effect emerged for AP sway range and AP sway velocity; however, pairwise tests did not confirm the interaction. Descriptively, children tended to decrease AP sway from pre to post, whereas adults tended to increase. ML sway range showed significant group $\times$ time and group $\times$ board-type effects. Children reduced ML sway range from pre to post for both boards, WB yielded smaller ML sway range than RB, and adults showed minimal change. ML sway velocity was higher in children, and both groups exhibited higher ML sway velocity during RB than WB. Ellipse measures were larger in children. Ellipse area and width were larger on RB than WB in children, while adults showed little board dependence. Ellipse height decreased from pre to post in children, whereas adults remained stable. Overall, brief balance-board standing yielded acute reductions in ML COP metrics in children, suggesting short-term improvements in ML postural control, whereas adults exhibited limited consistent change. Board selection influenced sway primarily through task demands (RB > WB), especially for AP range and ML velocity. These findings inform age-appropriate balance-board prescriptions and

interpretation of COP outcomes when comparing pediatric and adult balance responses. Funding source: NASPSA Graduate Student Research Grants 2024.

Effects of supramalleolar orthoses and joint laxity on obstacle-crossing behavior in young children with Down syndrome

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Individuals with Down syndrome (DS) demonstrate delays in motor, cognitive, language, and social development, with nervous system impairments contributing to joint instability, compensatory motor strategies, and inefficient gait mechanics. Supramalleolar orthoses (SMOs) are commonly prescribed to provide ankle and foot stability; however, limited evidence exists regarding their effects on obstacle-crossing behavior and strategy selection. This study examined obstacle-crossing strategies in young children with DS who either wore SMOs or did not wear orthoses during daily walking. Children aged 2-6 years attempted to cross obstacles set at 10% and 30% of leg length under barefoot and externally supported conditions (shoes for the non-SMO group; shoes with SMOs for the SMO group). Attempts were classified as successful or unsuccessful cross-over, bear crawl, walk-around, or refusal, with walk-around and refusal defined as avoidance behaviors. Joint laxity was quantified as a composite of bilateral elbow and knee hyperextension and ankle dorsiflexion. Thirteen children were allocated to the non-SMO group (age: 42.9 ± 16 months; walking experience: 23.3 ± 13.8 months) and 10 children to the SMO group (age: 39.6 ± 14.3 months; walking experience: 22.8 ± 14.1 months). The SMO group demonstrated greater joint laxity ($81.8^\circ \pm 19.9$) compared to the non-SMO group ($58.5^\circ \pm 14.2$). Across groups, increasing obstacle height was associated with fewer successful cross-over attempts and greater execution failure. During barefoot 10% trials, the non-SMO group demonstrated more successful crossings (4.8 ± 1.7) than the SMO group (3.1 ± 2.5), whereas the SMO group demonstrated more unsuccessful cross-over attempts (2.0 ± 2.1 vs $.6 \pm 1.2$). At the 30% obstacle height, both groups exhibited increased unsuccessful cross-over attempts, with higher frequencies in the SMO group during barefoot conditions (1.3 ± 1.3 vs $.7 \pm 1.0$) and similar frequencies during shod conditions (1.3 ± 1.9 vs 1.5 ± 1.9). External support increased cross-over attempts and fewer avoidance behaviors at both obstacle heights for each group. These findings suggest that joint laxity influences execution and strategy selection during obstacle crossing, and that SMOs may enhance perceived stability, enabling children with greater joint laxity to engage in more challenging crossing strategies. Funding source: APTA Georgia.

Creation and validation of a de novo motor learning assessment using a virtual reality bimanual task

Emily Roduin, Montana State University; Tyler Whittier, Montana State University

Motor learning reflects experience-dependent improvements in performance and the formation of internal control models; however, many laboratory paradigms rely on simplified tasks that limit ecological validity to practical learning situations. De novo motor learning tasks, which require the formation of entirely new control patterns, provide a rigorous framework for examining skill acquisition, particularly when combined with bimanual coordination demands. The purpose of this study was to design a de novo motor learning assessment that is both internally valid and ecologically relevant. To address this purpose, we developed a virtual reality motor learning assessment in which participants move a virtual cursor into targets at varying locations using a novel, non-intuitive

bimanual mapping. In this altered mapping, cursor right-left motion is controlled by the Y-axis movements of the left controller, while cursor up-down and forward-backward motion are controlled by the right controller's Z- and X-axes movements, respectively. These mappings require participants to develop entirely new control strategies to reach targets across ten training blocks. To assess learning in this experiment, movement time, normalized path length, bimanual control metrics, and trial-to-trial variability were calculated for each participant and each training block. It was hypothesized that healthy young adults would show progressive improvements in movement efficiency and bimanual coordination over the course of training, reflecting motor learning. Seven participants ($M_{\text{age}} = 22.658 \pm 1.578$) completed a single session of our VR-based bimanual coordination task using an HTC Vive Pro 2 headset and controllers. Results from a paired *t*-test indicated participants reduced movement time, $t(6) = 4.779$, $p = .003$, normalized path length, $t(6) = 3.444$, $p = .01$, trial-to-trial variability, $t(6) = 3.772$, $p = .009$, and increased in bimanual control, $t(6) = 5.449$, $p = .001$, across training blocks. These findings indicate successful acquisition of a novel motor control strategy within a single session. Compared to simplified laboratory tasks that rely on constrained movement or compensation for a visuomotor perturbation, this VR-based paradigm better captures the complexity of real-world skill acquisition by requiring the formation of new sensorimotor mappings, coordination of three-dimensional movement, and control across multiple degrees of freedom.

The Bayesian knee: Uncertainty of knee position and reliance on sensory priors increases as sensory feedback increases during stepping movements

Colten Smith, Montana State University; Trysten Morse, Montana State University; Lilith Mueller, Montana State University; Emily Roduin, Montana State University; Tyler Whittier, Montana State University

Despite efforts to understand risk factors for knee injuries, these injuries remain prevalent in athletic, military, and healthy adults of all ages. Additionally, the rate of re-injury following a primary Anterior Cruciate Ligament (ACL) rupture remains high at approximately 40%. While traditional rehabilitation has yielded minimal effects, a growing body of evidence suggests that ACL injuries result in significant neural damage. One impactful impairment is the loss of crucial sensory receptors that inform the central nervous system (CNS) of vital information regarding the state of the knee during movement. This damage leads to impaired motor control leaving the knee susceptible to additional injury. In this study, we applied a Bayesian model to estimate the degree of uncertainty in the state of knee in healthy individuals during a full body stepping movement. The purpose of this study was to apply the principles of Bayesian motor control to understand knee control in healthy individuals performing an ecologically relevant stepping task. To address this purpose, 15 healthy young adults (5 females, $M_{\text{age}} = 21.3 \pm 1.5$ years), performed a full body visuomotor adaptation task in virtual reality (VR). In this assessment, participants attempted to move a cursor into various targets with a one-legged step, compensating for cursor shifts under different levels of visual uncertainty. Motion of the cursor was controlled by a marker attached to the participant's knee. We hypothesized that as sensory feedback became more uncertain, participants would rely more on their sensory priors and have increased intrinsic uncertainty regarding the state of their knee. Results showed a significant increase in the bias toward sensory priors when estimating knee position as visual uncertainty increased ($p < .001$, $d = 3.95$). Additionally, the amount of intrinsic sensory uncertainty in estimates increased significantly as their visual feedback was limited ($p < .001$, $d = .60$). These findings support the use of Bayesian modeling to quantify sensory uncertainty during whole-body movement and provide a foundation for detecting clinically meaningful sensory

impairments. Future work will extend this approach to ACL-injured individuals to uncover sensorimotor consequences of injury.

The impact of feedback schedules on brain adaptations during motor imagery neurofeedback training in competitive soccer players

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Neurofeedback technologies, such as brain-computer interfaces (BCIs), are gaining popularity for training cognitive skills, such as motor imagery. While motor learning literature suggests that reduced feedback frequency can improve motor skill acquisition (i.e., the guidance hypothesis), it is unclear if this also applies to cognitive skills training. This study investigates how different amounts of concurrent neurofeedback influence neural adaptations during training with i-BrainTech, a BCI soccer-training platform that links motor imagery to an avatar's successful kicks. In this preliminary analysis, competitive soccer players ($N = 16$, $M = 21.63$, $SD = 2.42$) were randomly assigned to a ~3-week training intervention (22.6 ± 5.6 days) with either 100% or 50% concurrent neurofeedback. Participants completed pre- and post-training tests with 100% or 0% feedback for soccer (trained) and basketball (transfer) tests in i-BrainTech. EEG data were recorded from C3, C4, and Cz (motor execution), Fz (attentional control), and Pz (attentiveness) electrodes. Changes in alpha power and performance from pre- to post-training tests were analyzed using linear mixed-effect regressions. Analyses revealed a significant Group $\times$ Sport interaction for all electrodes ($ps < .027$). Post hoc analyses revealed no significant changes during post-training soccer tests ($ps > .056$), though trends toward desynchronization (increased attentional engagement) were observed at Fz and Pz for the 50% group. For the basketball transfer test, the 100% group exhibited significant alpha desynchronization relative to the 50% group at Fz ($p = .048$), and Cz ($p < .001$). At central electrodes, the 50% group showed trends for increased synchronization, and the 100% group showed trends for desynchronization. No significant effects emerged for performance ($ps > .123$). Findings reflect a potential advantage of reduced neurofeedback in generalizing neural efficiency for cognitive functions such as attentional control (Fz), attentiveness (Pz), and motor simulation (C3). This may suggest that reduced feedback frequency encourages athletes to adopt internal processing strategies that increase attentional control and attentiveness when facing the cognitive demands of trained BCI tasks. Further data are necessary to elucidate these findings. Future studies should investigate if changes in concurrent neurofeedback over time can optimize sport-specific neural engagement for long-term skill retention.

Increased neural activity in Coast Guard members during a balance perturbation following fast boat rides

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United States Coast Guard service members frequently operate fast boats, exposing them to repetitive loads, potentially increasing their risk of neurological injuries. Postural control deficits have been associated with cognitive damage after acute TBI. Although repetitive loading may not produce traditional concussive symptomatology, it does not preclude the development of long-term neurological deficits. To further examine these effects, sensorimotor perturbations are often used to challenge static

postural control, while neural activity provides an important index of the higher-order cognitive processes required to maintain balance. Thus, this study examined neural responses to sensorimotor perturbations using electroencephalography (EEG) before and after fast boat exposure in Coast Guard personnel. Twelve healthy Coast Guard members (66% male, 27.83 years ± 6.43 years) completed virtual reality-induced balance tests pre- and post-fast boat exposure while wearing a 27-channel EEG to record neural activity. EEG data was processed and analyzed in the time-frequency domain using MATLAB. Event-related spectral perturbation (ERSP) analysis revealed significant pre-post differences in neural activity in the theta (4-8Hz), alpha (8-12Hz), beta (12-30Hz), and gamma (30-50Hz) frequency bands (all $p < .05$) across multiple brain regions. Findings indicated that more neural activation and areas of the brain are required to maintain postural control during balance perturbations following fast boat exposure, as neural activity is significantly higher in Coast Guard members from pre- to post-fast boat rides. Significantly larger differences were found within the Beta frequency band as it is known to play a role in regulating postural control through a redistribution of executive function following concussion. Overall, the results support that neural activity increases as balance tasks become more challenging, and highlight the potential impact of repetitive loading on neurological function.

The influence of different types of fatigue on anticipation reaction timing

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Anticipatory reaction time (ART) is best described as the ability to respond to an environmental stimulus before it occurs by interpreting external information and initiating movement. Individuals experiencing mental fatigue, physical fatigue, or combined (i.e., mental and physical) fatigue have shown decreased performance of motor skills. Thus, the purpose of this study was to investigate the influence of different fatigue types on anticipation timing. It was hypothesized that mental fatigue, physical fatigue, and combined fatigue groups would have slower ART compared to a control group. It was also hypothesized that combined fatigue would have slower ART compared to isolated mental fatigue and physical fatigue conditions. In this study, motor performance and perceived exertion were compared while completing an ART simulated baseball bat swing task between four conditions. Participants were pseudo randomly placed into one group condition (i.e., control, mental fatigue, physical fatigue, or combined fatigue). Participants completed a 5-min walking warmup on a treadmill before performing their assigned fatiguing task. The control group watched a 30-min nature video (i.e., Seasonal Forests). The mental fatigue group performed 600 trials of the Stroop task on a computer. The physical fatigue group performed a treadmill walking task in which the speed of the treadmill was set to 4.86 km/hr, and every 5-min the treadmill grade was increased two degrees. The combined fatigue group performed the Stroop task and treadmill walking tasks simultaneously. Following the nature video or fatigue inducing treatment, participants performed the ART swing task and ART was recorded using the Bassin anticipation timer. Participants completed 30 ART swing trials and after every 5 trials, participant rating of perceived exertion (RPE) was recorded. A 4 (condition) $\times$ 6 (trial block) repeated measures ANOVA revealed no significant differences for ART or RPE between all group conditions. Although the results did not support our hypothesis, they suggest that the introduction of fatigue may influence ART over time. Specifically, the results of the current study demonstrated that the control group display faster ART over time, whereas the ART of the fatigued groups did not improve.

Effect of visual biofeedback scaling on center of pressure displacement during a dual-leg standing task: Assessing ankle co-contraction

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Maintaining an upright standing posture requires the integration of complex physiological processes that counteract internal and external forces by adjusting the center of pressure (COP) relative to the body's center of mass (Diener & Dichgans, 1988). One strategy for controlling posture is real-time visual biofeedback (vBF), which provides a visual representation of one's COP displayed in real time on a screen, commonly used in tasks that require individuals to regulate their COP within a target boundary. This approach has been shown to improve postural control by reducing COP sway and enhancing postural stability (Imaizumi et al., 2018). Additionally, scaling vBF (view of an amplified COP on screen) has been associated with further improvement. As increasing the scale also increases the difficulty of the task, the effect of scaling may be explained by use of ankle muscle co-contraction (Michaud et al., 2024). However, the role of ankle co-contraction combined with vBF has remained unexplored. Therefore, this current research will assess the role of ankle co-contractions (tibialis anterior (TA) and medial gastrocnemius (MG)) during feet-together vBF scaling tasks. 12 healthy young adults (23.33 ± 1.87 years) completed ten 1-min trials for each of four visual feedback conditions, for a total of 40 trials, presented in randomized order: (1) no visual feedback (static target), (2) visual feedback magnified by 1, (3) visual feedback magnified by 5 (X5), and (4) visual feedback magnified by 10 (X10). Participants were not informed of the vBF scaling. TA and MG muscle activity were recorded using surface electromyography (EMG), rectified, and smoothed via RMS envelope. Mean EMG and co-contraction index were calculated. Preliminary data suggest that vBF significantly reduced postural sway, particularly in the anterior-posterior direction. This significant reduction in COP sway was further reduced in the X5 condition, while the effect in the X10 condition was less prominent compared to the No Feedback condition. Differences in co-contractions were not significant, suggesting the reduction in sway could be due to sensory reweighting, particularly an increased modulation of the proprioceptive input to align with visual feedback. Future research could investigate vBF as a tool for neuromuscular injury rehabilitation.

Beyond internal vs. external focus: The role of imagined action effects in motor performance

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The benefit of an external focus for motor performance has typically been viewed as a stable finding in the motor behavior literature. However, some scholars have recently proposed this benefit may be overstated or more nuanced than previously thought (Herrebrøden, 2022; McKay et al., 2024). The constrained action hypothesis proposes that an external focus enhances self-organization by planning the movement at the level of the action effect instead of controlling specific actions. A consideration that may explain discrepancies in prior findings is that not all action effects are created equally. Action effects can occur within our outside of the body, and can be real or imagined. The present study was designed to test whether focus statements that direct attention to literal or imagined action effects differentially impact motor performance. Volunteers ($N=28$) performed maximum effort isometric contractions using a grip dynamometer. All participants first completed baseline trials, and then completed 2 trials in each of four

focus conditions presented in a counterbalanced order. Conditions included a literal-internal focus (squeeze your hand as hard as possible), literal-external focus (squeeze the handle as hard as possible), imagined-internal focus (imagine squeezing so hard your hand goes right through the handle), and imagined-external focus (imagine squeezing so hard the handle breaks). Peak force production was recorded for each trial, and averaged for each condition. A repeated-measures ANOVA on focus condition indicated a significant main effect of focus ($p=.002$). Sidak post-hoc tests indicated both an imagined-internal focus ($p=.008$) and imagined-external focus ($p=.013$) resulted in higher force production than a literal-internal focus. A literal-external focus did not differ from any other condition ($ps > .05$). The present findings suggest that all action effects are not equally effective. Focusing on a literal action effect provided no performance advantage over focusing on the literal body action. However, focusing on imagined outcomes referencing either the body or an external object provided similar advantages to motor performance in a simple grip task. Future work should expand this framework to determine if focusing on imagined action effects enhances performance for more complex motor tasks.

Eye-hand coordination in multisensory sequence learning: Goal-directed reaching improves independent of visual search

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Successful performance of many everyday tasks requires performing sequences of goal-directed actions that are guided by efficient eye-hand coordination. Initially, learners explore the environment visually to locate action-relevant targets. As the sequence order is acquired, target selection requires less time, facilitating more efficient sequence performance, including faster reaches. Current understanding of motor sequence learning has focused on measuring visually guided sequences, but relatively little is known about eye-hand coordination and the integration of sound to enhance eye-hand coordination. To gain a better understanding of the perceptual-motor processes supporting motor sequence acquisition, the present experiment measured eye-hand coordination during a 3-dimensional goal-directed reaching version of the Serial Reaction Time Task (SRTT) under visual compared to audio-visual conditions. 17 neurotypical adults (18-36 years) practiced a five-target twelve-item explicit SRTT on two consecutive days. The SRTT required participants to reach to a series of visual targets appearing on a touchscreen monitor in a radial array. Group 1 was guided by visual cues only, while group 2 received additional auditory feedback in the form of spatially congruent tones presented as action-effects. A two-way mixed ANOVA on day one revealed a main effect of practice where both groups performed the sequence faster in late compared to in early practice ($p \leq .05$). No effect for group or group-by-practice interactions were identified for manual sequence performance, indicating that faster sequence execution was not influenced by sensory condition. The omnibus analysis of visual search time did not reveal any significant effects for practice or group. In addition, an exploratory analysis using a paired t -test indicated a significant decrease in visual search time for the visual group. That said, target search time was more variable in the audio-visual group both in early (238 ± 71 ms) and late (252 ± 118 ms) practice. Overall, manual sequence execution improved independently of visual search times, with some preliminary evidence that those learning with visual cues spent less time locating targets later in practice. Given the variability of visual search in the audio-visual group, future work should investigate if sensory preferences contribute to inter-individual differences in multisensory sequence learning. Funding source: Natural Sciences and Engineering Research Council of Canada.

Smart trainers and virtual environment: Co-action effects in a dyad practice protocol

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The experimental approach of co-action effects examines the behavior of dyads who are engaged simultaneously in the same activity. Co-action gains performance advantages when individuals have the full view of each other and the task requires highly energetic/motor processes (cycling). A new field of research and co-action occurs in virtual environments (VR) on indoor smart trainers. The present study examined if co-action effects in a dyad practice protocol depends on the physical presence and the full view of each other or if a companion depicted as an avatar is sufficient to facilitate motor performance in an indoor VR bicycle all-out race. Athletes ($n=22$, males, students of sport-science) were instructed to perform a 3000 m all-out race with a racing bike that was fitted onto an indoor Tacx roller trainer. The race and the athletes' performance was controlled by an on-line platform and provided on a tablet placed in front of them. To control the maximal workload (maximal heart rate, maximal power) a step test was conducted prior to the experiment. In a within-subject design the athletes perform the (a) 3000 m race individually, (b) as dyads in VR (dyads were on-line connected and the performance of the partner was provided as an avatar), or (c) as dyads in physical presence of the partner (dyads had full view of each other). According to the findings on performance advantages in groups, dyads with moderate performance differences ranging from 20% to 35% were formed on the basis of the individual 3000 m time. The bikes were identical for the dyads and the order of the dyad conditions was counterbalanced. The 1000 m split times and the 3000 m final time were analyzed. As expected the split times and the final times were significantly shorter for the dyads in physical presence compared to the dyads in the VR and individual condition. However, no significant time differences were observed for the individual and the dyad VR condition. These findings suggest the co-action effect depends on the physical presence and full view of each other in a dyad practice indoor cycling protocol. Funding source: Saarland University program: 'Anschubfinanzierung 2025'.

The effect of imagery dominance on dart-throwing with various attentional focus strategies

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Both imagery and attentional focus have been shown to enhance motor performance and learning (Wulf, 2013; Ladda et al., 2021); however, limited research has examined their combined influence. Sakurada et al. (2016) proposed that imagery modality dominance may interact with attentional focus, such that visually dominant individuals benefit more from an external focus, whereas kinesthetically dominant individuals benefit more from an internal focus. The present study investigated the joint effects of imagery dominance and attentional focus on fine motor skill performance and learning using a standard dart-throwing task. Sixty-five participants were classified according to imagery dominance using the MIQ-R (Hall & Martin, 1997) and assigned to one of five experimental groups. Participants completed a five-trial pre-test, followed by 50 imagery training trials using an assigned attentional focus strategy. A five-trial post-test was then conducted, followed by 24-hour retention and transfer tests. Task performance was assessed using mean radial error to index accuracy and bivariate variable error to index consistency. Repeated-measures ANOVAs and Pearson correlations were used to analyze performance outcomes and adherence to attentional focus strategies.

Results showed that participants adopting an external focus demonstrated significantly greater improvement in dart-throwing accuracy ($p < .05$) and consistency ($p < .05$). A significant interaction between imagery dominance and attentional focus was observed for accuracy ($p < .05$) and consistency ($p < .05$), where kinesthetic-dominant individuals benefited significantly more from adopting an external focus of attention. Pearson correlations revealed a positive association between visual imagery scores and adherence to internal focus cues ($p < .05$), indicating that individuals with stronger visual imagery dominance showed greater adherence during mental imagery training. These findings support the established benefits of external focus in fine motor tasks (Wulf, 2013). The direction of the interaction did not support the expected pattern of results based on the optimal individual attentional strategy proposed by Sakurada et al. (2016). While imagery dominance did not consistently moderate attentional focus effects, the findings reinforce the effectiveness of external focus for fine motor performance and highlight the need for further research examining combined imagery and attentional focus interventions across different motor skills. Funding source: New Mexico State University, College of HEST.

Expertise shapes motor variability adaptations under pain constraints

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Road cycling is a repetitive endurance task that imposes substantial cumulative loading on the lower limbs. Knee pain is one of the most common overuse complaints in cyclists and is known to alter neuromuscular control. Motor variability (MV), the natural fluctuation in muscle activation and movement patterns, is a key feature of neuromuscular control and is typically lower in experts, who stabilize movement more effectively than novices. Experts are defined by structured, high-volume, goal-oriented training over years. While pain alters MV in athletes through exploratory or stiffening responses, little is known about how expertise modifies these pain-related adaptations. This study examined whether cycling expertise shapes MV responses to experimentally induced knee pain. We hypothesized that high-skilled (HS) cyclists would display greater MV than low-skilled (LS) participants under pain, reflecting more adaptable neuromuscular strategies. Thirty-two HS cyclists (14 females, 18 males, 42 ± 11.64 years) and 30 LS participants (15 females, 15 males, 27.64 ± 6.41 years) performed a five-min bout at self-selected submaximal intensity while alternating between pain-free and painful 30-s recordings. Pain was induced via 4-Hz electrical stimulation to the dominant knee. Surface electromyography (EMG) recorded activation amplitude and onset angle from the vastus lateralis, biceps femoris, tibialis anterior, and gastrocnemius lateralis. 3D motion capture quantified trunk and lower-limb angles, and cadence was extracted from the smart bike. MV was quantified using coefficients of variation. Linear mixed models assessed group $\times$ conditions main and interaction effects. Preliminary analyses revealed a significant effect ($p < .05$) of pain on MV in both muscle activity and kinematic variables. Pain increased variability in the tibialis anterior amplitude during contraction, trunk angles, and in cadence in both groups. Expertise also influenced MV significantly ($p < .05$): HS showed reduced variability in vastus lateralis activation and gastrocnemius onset angle, trunk angles and cadence compared to LS. Interaction effects emerged for vastus lateralis, gastrocnemius onset angles, and the trunk frontal angle. These results demonstrate that pain and expertise alter neuromuscular control during cycling. Contrary to our hypothesis, HS cyclists did not show greater MV than LS participants under pain; rather, both groups exhibited different adaptations.

Talent identification for technological savviness: Learning behaviors and experiences related to rapid learning of novel digital interfaces

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Advanced technologies are becoming more intelligent and ubiquitous across industry and sport domains. Consequently, rapid skill acquisition with new technologies ("tech-savviness") may put individuals at an advantage through enhancing competence with assistive technological tools (e.g., simulation platforms). Scant research exists on what makes an individual tech-savvy, especially with regards to black-box technology like AI. In this study, we investigate individual differences, developmental histories related to technology, and behaviors associated with learning a novel digital interface. Participants ($M_{\text{age}} = 24.0 \pm 4.0$; $N = 75$) completed our newly developed "Robot Maze Test" (RMT), where they navigate a robot through grid-based mazes with obscure, learnable obstacles. User interactions with the RMT (i.e., behaviors) were measured. Learning outcomes were derived from a comprehension questionnaire about learned features after the RMT. Participants completed questionnaires assessing technology-related dispositions/life experiences (e.g., video gaming, programming). We used linear regressions to examine relationships between behaviors, individual differences, and learning outcomes, adopting a conservative significance threshold ($p = .005$) given our exploratory design. Behaviors associated with learning outcomes included fewer attempts to success, more attempts after success, more time spent, and more seconds per action ($ps < .001$). Video gaming and computer use before adulthood were positively associated with learning outcomes ($ps < .001$). More time playing action and role-playing video games in the past year was related to more time spent per trial ($ps < .002$). Perceived ability to use AI and frequency creating phone content in the past year were associated with less seconds per action ($ps < .002$). More time playing action games before adulthood was related to more attempts after success ($ps < .004$). Findings highlight slower, more deliberate actions are superior to faster actions for best digital learning outcomes, which may be developed through early-life interactions with computers and video games. We also found evidence that phone and AI-use may be associated with more impulsive learning behaviors (i.e., faster actions). Findings provide potential markers for selection in technology-based jobs (e.g., military) as well as insights for interventions developing digital problem-solving skills that can transfer across platforms and domains (e.g., performance simulation environments). Funding source: U.S. Army Research Office.

Planning ahead: How cognitive function predicts motor planning in aging adults

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Aging is accompanied by declines in both cognitive and motor function, yet the relationship between these domains—particularly as it relates to motor planning—remains insufficiently understood. Motor planning, a cognitive-motor process essential for coordinating purposeful action, deteriorates with age and may contribute to increased fall risk, reduced mobility, and decreased independence in older adults. Although prior research has established age-related differences in anticipatory motor planning (Wunsch et al., 2015) and linked these differences to executive

functions (Stöckel et al., 2017), specific components of planning, such as how far in advance one can plan a movement, have not been systematically examined across adulthood. 60 participants, aged 50 years and older, with no known neurological impairments, were collected. Cognitive function was assessed using the Montreal Cognitive Assessment (MoCA) for general cognition, the Trail Making Test (TMT) for processing speed, and the Numerical Activities of Daily Living – Short version (NADL-Short) for numerical cognition. Motor planning was assessed using Vicon motion capture during a grasp-and-rotate task with varying planning demands (spans 1–3). Participants wore occlusion goggles to isolate planning time. Planning span was quantified by comparing end-state postures across spans and rotation conditions to individual baselines. Preliminary analysis ($n = 10$) using multiple linear regression models were conducted (nine in total), predicting motor planning span/rotation condition from MoCA score, TMT-B completion time, and NADL-Short score. Only one model reached significance (span 3, 45° rotation; $R^2 = .91$, $p = .02$). In this model, MoCA score ($\beta = -2.08$, $p = .01$), TMT-B completion time ($\beta = -.14$, $p = .03$), and NADL-Short score ($\beta = 1.84$, $p = .006$) significantly predicted performance. Findings show that general cognition, processing speed, and numerical cognition significantly predicted the performance of planning three steps ahead in a sequential movement. Indicating that under higher task demands, motor planning performance is jointly associated with these executive functions in older adults. This provides further understanding of how cognitive abilities shape motor planning, which offers critical insight for interventions aimed at preserving functional independence, mobility, and social engagement in aging populations. Funding source: NASPSA Graduate Student Research Grant.

Distal-only versus mixed-distance attentional focus under random practice: Performance, learning, and kinematic effects

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Prior research shows that a distal external focus of attention can enhance motor performance and learning. However, under high contextual interference, it is unclear whether keeping attention exclusively distal provides added benefits over alternating focus distances. This study compared distal-only versus mixed-distance attentional focus cueing under random practice in a dart-throwing task and tested whether the various cues altered movement outcome and production characteristics. Participants ($N = 57$) were randomly assigned to a Mixed-Distance or Distal-Only group. Both groups completed 150 acquisition trials on Day 1 under a random schedule, and a 30 trial retention and transfer test the following day. The Mixed-Distance group received three cues at different distances from the body (bullseye, dart trajectory, release angle), each presented 50 times in a randomized order. The Distal-Only group received three distal cues (bullseye, target ring, impact spot), each presented 50 times in a randomized order. Performance accuracy (mean radial error; MRE) and consistency (bivariate variable error; BVE) were recorded. During acquisition, peak accelerations of the upper arm, forearm, and hand segments and peak shoulder and elbow flexion angles were measured using a Noraxon IMU system. Both groups improved in MRE and BVE across acquisition. In Block 1, the Distal-Only group showed significantly better accuracy and consistency than the Mixed-Distance group. To test whether this early advantage reflected learning, average performance across Blocks 1–3 was compared with retention and transfer. The Mixed-Distance group showed a significant improvement in MRE from acquisition to retention, whereas the Distal-Only group did not; no meaningful group differences emerged in retention and transfer. Kinematic analyses indicated that, in the Mixed-Distance group, distal cues produced greater peak segment accelerations relative to middle and proximal cues. In the Distal-Only group, the impact-spot cue elicited greater peak segment accelerations compared to the other

distal cues. Overall, distal-only cueing provided immediate performance benefits early in practice but did not yield superior long-term learning relative to mixed-distance cueing. Moreover, the findings of the present study suggest that attentional distance effects are expressed more clearly in acquisition kinematics compared to delayed post-testing outcome measures.

Ocular-motor coupling reveals selective use of visual information in coordinated rhythmic movement

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Previous studies on coordinated rhythmic movement have demonstrated that when both visual and kinesthetic information are available, coordination performance of in-phase and anti-phase movements at high frequencies primarily relies on kinesthetic information. However, the specific functional role of vision across a frequency spectrum remains unclear. This study investigated the use of visual information to sustain visually defined in-phase and anti-phase patterns as movement frequency increased. Using eye tracking and behavioral measures, nine participants controlled two joysticks to produce the coordination patterns at six frequencies (.5-3 Hz in .5 Hz increments) under visual feedback (with two moving dots) and no-visual-feedback (no dots) conditions. Coordination performance was assessed using proportion of time-on-task (PTT). Ocular-motor coupling was quantified using Fisher Z-transformed peak cross-correlation between gaze trajectories and the corresponding joystick movement trajectories. PTT showed a significant interaction between coordination pattern and frequency, $F(5, 40)=20.66$, $p<.001$, with the in-phase coordination declining as movement frequency increased, while the anti-phase coordination remained stable. Coordination performance in both patterns was largely unaffected by removal of visual feedback. However, ocular-motor coupling showed a significant three-way interaction among coordination pattern, frequency, and feedback condition, $F(5, 33)=8.75$, $p<.001$. For in-phase coordination, coupling was strongest at lower frequencies with visual feedback but became weakest at high frequencies, regardless of feedback. For anti-phase coordination, coupling was consistently weak and invariant to both frequency and feedback. These findings support the previous research (Huang et al., 2020) showing that the visually defined in-phase coordination is less stable and more susceptible to movement frequency, than the visually defined anti-phase coordination during joystick movements in the frontal plane. Furthermore, we demonstrated that visual information, enhanced by visual feedback, is used selectively to sustain the visually defined in-phase coordination at lower frequencies, which suggests that kinesthetic information likely becomes the sole resource for maintaining both in-phase and anti-phase coordination patterns at higher frequencies.

Differences in gaze-stimulus coupling in perceiving intrinsic coordination patterns across movement frequencies

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Previous studies on coordinated rhythmic movement have shown that visual information contributes to the control and learning of coordination patterns. However, how visual information is utilized during the perception of these patterns across a frequency spectrum remains unclear. In the current study, we examined the use of visual information in perceiving intrinsic coordination patterns using eye tracking. Nine participants

completed visual discrimination tasks judging in-phase (with two dots moving horizontally on the screen in the same direction at the same speed) and anti-phase (with two dots moving in the opposite direction at the same speed) patterns at six frequencies (.5-3 Hz in .5 Hz increments). Gaze-stimulus coupling was quantified using Fisher Z-transformed peak cross-correlation between gaze trajectories and stimulus motion. Repeated-measures ANOVA revealed significant main effects of coordination pattern, $F(1, 8)=111.14$, $p<.001$, movement frequency, $F(5, 40)=129.85$, $p<.001$, and a significant interaction between coordination pattern and frequency, $F(5, 40)=76.01$, $p<.001$. During in-phase judgments at low frequencies, gaze behavior showed sustained coupling with stimulus motion, indicating consistent visual tracking behavior. In contrast, during anti-phase judgments, gaze-stimulus coupling remained low across all frequencies, reflecting a stable gaze strategy rather than continuous tracking. Across both coordination patterns, gaze-stimulus coupling decreased as movement frequency increased, suggesting a transition in visual processing strategies at higher speeds. These findings demonstrate that visual information is used during the perception of intrinsic coordination patterns at low frequencies, and that its use is expressed through pattern-specific gaze strategies. Specifically, perceiving in-phase coordination at low frequencies relies on gaze alignment with stimulus motion, whereas anti-phase coordination is perceived using a stable gaze pattern without continuous tracking. Funding source: Experiential Student Scholars Program, Texas Woman's University.

Does time-series kinematic features improve classification of Parkinson's disease during backward walking?

Ke'Vaughn Waldon, Texas Christian University; Morteza Farivar, Texas Christian University; Adam King, Texas Christian University

Parkinson's disease (PD) is associated with impairments in gait and motor control that increase fall risk and challenge early clinical detection. While previous work using three-dimensional motion capture has demonstrated altered gait characteristics during forward walking in individuals with PD, less is known about motor control organization during backward walking. Backward walking reduces visual guidance and alters interlimb coordination demands, requiring greater reliance on proprioceptive feedback and anticipatory motor control. The purpose of this study was to use a machine learning approach to examine whether incorporating time-series-based biomechanical features improves the classification of PD during backward walking compared to spatiotemporal measures alone. Twenty-three older adults consisting of those diagnosed with PD ($N=11$) and those without PD ($N=12$) performed backward walking trials on a treadmill at a self-selected pace while three-dimensional motion capture data were collected. Two model configurations (Model 1 - spatiotemporal parameters 1; Model 2 - spatiotemporal parameters with time-series kinematic features) were evaluated using multiple machine-learning classifiers. The first configuration included only spatiotemporal parameters, whereas the second was combined. Classification performance was assessed using logistic regression, support vector machines, random forest, and gradient-boosted models, with and without principal component analysis. Models using spatiotemporal parameters alone demonstrated limited classification performance, with accuracies ranging from 43.4% to 56.5%. In contrast, models that incorporated time-series features showed improved classification accuracy across classifiers, including logistic regression (73.9%), support vector machines (69.5%), and gradient-boosted models (69.5%). These findings suggest that temporal characteristics of movement provide additional information relevant to distinguishing PD-related motor control alterations during backward walking. Time-series features likely captured temporal structure and coordination patterns not reflected in discrete spatiotemporal measures. The inclusion of time-series biomechanical

features enhances classification performance and offers insight into movement organization under increased motor control demands.

Effects of visual feedback on rambling and trembling components of postural control

Chieh Yin Kao, Texas Christian University; Adam King, Texas Christian University

Visual feedback has been shown to improve postural control; however, few studies have examined its effects during single-leg stance, even though single-leg balance is more sensitive for detecting subtle differences in postural control. The purpose of this study was to examine how visual feedback affects balance control during single-leg stance. Twenty young adults performed single-leg stance tasks on a force plate with and without visual feedback using both the left and right legs. Visual feedback was provided only in the anterior-posterior (AP) direction of the center of pressure (COP). Each condition included three 30-s trials separated by 30-s rest intervals. This study analyzed linear – standard deviation (SD) – and nonlinear – detrended fluctuation analysis (DFA) – measures of the rambling and trembling COP components. Rambling reflects shifts in the equilibrium reference, whereas trembling represents oscillations around the reference point. Results from the linear measures showed that visual feedback had significant effects on the rambling component in both the AP and medial-lateral (ML) directions. In contrast, within the trembling component, a significant effect of visual feedback was observed only in the ML direction. The DFA results showed that, under the visual feedback condition, the DFA values of Rambling ML, Trembling ML, and Trembling AP exhibited an increasing trend and reached statistical significance. Our findings indicate that visual feedback influences both the rambling and trembling components of postural control. Specifically, visual feedback induced migration of the reference point during single-leg stance, thereby altering the body's instantaneous equilibrium point. In addition, visual feedback appears to play a more critical role in modulating rapid error corrections and fine-scale postural adjustments in the medial-lateral (ML) direction. Furthermore, DFA results suggest that visual feedback may alter the temporal structure of rapid corrective mechanisms, leading to deviations from the original fluctuation characteristics.

In my mind's (Quiet) Eye: Imagery practice influences motor planning in novice golf putting

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The ideomotor idea of perceptual-cognitive scaffolding posits that learning via imagery is driven by scaffolding of (quasi) action effects. According to this, repeated anticipation through imagery practice (IP) promotes perceptual-cognitive scaffolding, and this in turn guides the planning and monitoring of motor actions. One marker of motor planning is the Quiet Eye (QE; final fixation on target before movement onset), spanning QE_{pre} and QE_{post} phases (before and after movement onset). This study investigated whether IP influences motor planning (QE) and performance when learning a novel task. Participants ($N = 29$; 18 males; 26.0 ± 4.3 years) were assigned to intervention (IG) or control group (CG) and completed four measurement times (MT1-MT4) of a golf putting task. Between MT1 and MT2, IG performed IP, while CG did not practice. Between MT2 and MT3, both groups participated in physical practice. Linear mixed effect models analysed QE (final fixation before backswing) and performance (distance from ball to target), including group, MT with their interaction, age and sex as covariates, and random intercepts and slopes for MT by participant. The model revealed higher QEs at MT3 and MT4 compared to

the MT1 for QE_{total} ($p = .029$; $p = .027$) and QE_{post} ($p = .001$; $p = .006$). For QE_{post}, a group $\times$ MT interaction at MT2 ($p = .049$) indicated a greater increase in the IG relative to the CG. No significant effects were found for QE_{pre}. Performance improved at MT2 ($p = .002$), MT3 ($p < .001$) and MT4 ($p < .001$) compared to MT1. The increase in QE_{post} from MT1 to MT2 in the IG suggests that IP primarily enhanced movement-related monitoring aspects of planning rather than planning prior to movement onset. This interpretation is supported by the absence of significant effects for QE_{pre}.

Effect of combined fatigue on military marksmanship: Performance, skill retention, and transfer to simulated combat environments.

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During sustained military operations, Soldiers encounter prolonged physical, cognitive, operational, and environmental challenges. Despite these demands, the effects of simultaneous combined fatigue (i.e., physical & mental) on skill performance, acquisition, and transfer of learning in military settings remain underexplored within a military population. Investigating these issues is important to more thoroughly understand how to best train Soldiers in preparation for combat. Twenty-eight ROTC Cadets ($M_{age} = 20.43 + .92$ years, male = 23) were randomly assigned to a fatigued or non-fatigued practice condition. The fatigue practice group completed a novel, combined fatigue task (i.e., rucking on a treadmill & Stroop discrimination) prior to marksmanship practice. The non-fatigue practice group completed marksmanship practice following a seated rest period. Marksmanship performance was assessed using a non-immersive virtual reality marksmanship platform (i.e., InVeris 100MIL) at baseline, post-test, delayed non-fatigue and fatigue retention tests, and a fatigued transfer task which involved an active shooter room clearing exercise. Results revealed statistically significant differences between practice groups, with the non-fatigue practice group demonstrating superior marksmanship accuracy during non-fatigued retention testing ($p = .049$), and the fatigue practice group demonstrating improved marksmanship accuracy ($p = .006$), response time ($p = .008$), and qualification scores ($p = .009$) during fatigued retention testing. Most notably, the fatigue practice group demonstrated superior marksmanship accuracy ($p < .001$) and response time ($p = .010$) during the fatigued transfer room clearing task. These results suggest that the traditional training programs that emphasize skill acquisition in rested conditions may not adequately prepare the warfighter. By integrating combat-specific fatigue into training, military personnel will develop and execute the necessary skills for effective transfer to combat environments. Additionally, Soldiers will gain vital exposure to the increased mental and physical demands of combat, leading to an increased resistance to the perception of fatigue during extended operations. Funding source: NASPSA Graduate Student Research Grant.

Effects of experimentally induced hearing attenuation on gait variability during indoor and outdoor walking in young adults

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Hearing loss has been associated with an increased risk of falling, yet the mechanisms linking auditory input and locomotor control remain unclear, particularly during walking in real-world environments. Prior work has largely focused on laboratory walking tasks and clinical populations, often comparing aided and unaided hearing. The present study examined whether simulated hearing loss influences gait in young adults during indoor and outdoor walking using wearable inertial sensors. Young adults ($N = 13$) without self-reported hearing, vision, or mobility impairments

completed a 2×2 fully crossed, within-subjects protocol involving indoor and outdoor walking with and without experimentally induced hearing attenuation. Hearing attenuation was achieved using earplugs and noise-canceling construction-grade hearing protection. Linear acceleration was recorded in three dimensions using inertial measurement units affixed to each ankle. Standard spatiotemporal gait measures were computed, along with a spatiotemporal index (STI) derived from time- and amplitude-normalized ankle acceleration profiles to quantify gait variability. Preliminary analyses indicated that gait variability differed between indoor and outdoor walking contexts. Greater variability was observed during outdoor walking in both the hearing ($M_{STI}=4.55$) and hearing-attenuated ($M_{STI}=4.37$) conditions compared to indoor walking in the hearing ($M_{STI}=4.12$) and hearing-attenuated ($M_{STI}=4.21$) conditions. In contrast, experimentally induced hearing attenuation did not appear to substantially alter gait variability. Taken together, these findings suggest that, in young adults, basic locomotor control during relatively unchallenging walking conditions may be robust to acute reductions in auditory input, likely due to the availability of intact visual and somatosensory information and limited demands on navigation. In addition, experimentally induced hearing attenuation may not capture the long-term sensory changes, compensatory strategies, or comorbid factors that characterize populations with chronic hearing loss. Additionally, auditory contributions to gait may be expressed in kinematic, temporal, or coordination measures not captured by time- and amplitude-normalized variability. These results highlight the importance of participant characteristics, task demands, and outcome selection when evaluating auditory influences on locomotor control.

Gait differs during fast walking but not during dual-task walking between Black and White older adult women

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White older women fall more frequently and have worse falls-related sequelae than their black peers. Gait patterns (i.e. slower speeds, increased time in double support) can indicate increased fall risk, but racial differences in gait behavior across challenging gait tasks (e.g., dual-task and fast walking) remain underexplored. Clarifying these racial differences would aid practitioners to tailor fall-prevention interventions. In this study, we examined racial differences in fall-risk factors and gait in community-dwelling older adult women (> 55 years). We provided 155 free community-based health assessments to examine older adult fall-risk. We matched a sub-sample of black women ($N=16$, $M_{age} (SD)=73.2 (5.2)$) to a comparable sub-sample of white women ($N=16$, $M_{age} (SD)=71.4 (8.0)$) based on age and education. Participants walked along a 16-foot pressure-sensitive mat (Zeno, Protokinetics) in a fixed order: self-selected (SS), with a semantic verbal fluency dual-task (DT), and fast walking (FW). Participants also completed the Flanker Task, Timed Up and Go (TUG) test(s), and the 16-item Falls Efficacy Scale-International (FES-I). Hierarchical Linear Modelling analyzed gait parameter [Velocity (cm/s), Step Length(cm), Step Width(cm), Step time(s) and Double Support(%)] differences across tasks (SS (reference), DT, and FW) with dichotomized race, TUG, Flanker and FES-I scores as level 2 predictors. Faster TUG time predicted increased Velocity ($p=.003$) and Step Time ($p<.001$) and reduced Double Support ($p<.001$) but after controlling for TUG,

Flanker, and FES-I scores, no task differences existed across race in gait outcomes. However, an interaction for Double Support between gait task and race ($p=.04$) indicated that from SS to FW, white women exhibited more time in Double Support in FW than black women (SS: black = 32.7, white = 33.3, FW: black = 28.6, white = 30.5), but DT walk was similar (DT: black = 36.7, white = 36.1); thus the negative slope for Double Support was steeper for black women between DT and FW. Overall, when fast walking, white women adopted gait patterns reflecting more caution than black women, yet both groups performed dual-task walking similarly. Prospective research should examine how differences during FW may influence falls incidence for tailored fall-prevention efforts. Funding source: George Mason University Office of Research, Innovation, and Economic Impact.

Sport and Exercise Psychology Abstracts

Best practice recommendations for preventing, identifying, managing, treating, and supporting recovery from eating disorders among athletes: A comprehensive review

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Sport participation can support physical, mental, and social health; however, athletes are at great risk of developing eating disorders (EDs). To support athletes in experiencing the benefits of sport participation, guidance on how to prevent and address EDs in sport is needed. Currently, no Canadian national sport organization has an ED policy or protocol, and international guidelines are similarly limited. Therefore, to inform the development of such a policy and protocol, the purpose of this study was to review existing literature and best practice recommendations regarding ED prevention, identification, management, treatment, recovery, and the return-to-sport. We searched the electronic database PsychINFO using terms related to (1) EDs, disordered eating, or relative energy deficiency in sport (RED-S); (2) prevention, identification, management, or treatment; and (3) athlete or sport. Papers were included if they provided practical recommendations on addressing EDs in sport and were published in English, resulting in 55 papers for analysis. The first author and two research assistants independently charted the data by recording each paper's title; author(s); year; country; target population; target audience; and recommendations for ED prevention, identification, management, treatment/recovery, and the return-to-sport. Most papers originated from Westernized countries and focused on high-performance female/women athletes in weight-sensitive sports, with growing attention on male/men athletes. Recommendations for ED prevention involve increasing ED education among sporting agents (e.g., coaches) and implementing policies on healthy management of athlete eating behavior, weight, and body composition. Papers suggested integrating ED or RED-S screening tools into athletes' annual preparticipation physical evaluations, emphasizing the importance of early intervention. Effective management involves a multidisciplinary team, establishing a therapeutic alliance with athletes, and respecting confidentiality. To support treatment and recovery, papers described determining appropriate levels of care and sport participation for athletes on an individual basis, using an athlete contract to ensure accountability, and addressing sport-specific considerations (e.g., compulsive exercise, athletic identity). Papers also listed resources to assess athlete readiness to return-to-sport. This review provides tangible direction on addressing EDs in sport by highlighting specific prevention, screening, management, and return-to-sport practices that can guide sport organizations in developing an ED policy and protocol. Funding source: McGill Sport Science Grant.

Incorporating self-compassion into exercise interventions for depression: Preliminary evidence for optimizing motivational outcomes

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Exercise combined with behavioral counselling has shown effectiveness in reducing depressive symptoms in university students, yet affective dysregulation remains a key exercise motivation barrier not adequately addressed in existing interventions. Emotion-focused adjuncts, such as self-compassion, may improve exercise motivation by shaping how individuals interpret setbacks and emotional discomfort. For instance, this novel approach may encourage more intrinsic motives that support pleasure and symptom relief, versus extrinsic or outcome-based motivations that can exacerbate guilt, withdrawal, or maladaptive cognitions. The purpose of this investigation was to examine if a peer-led group exercise program with self-compassion psychoeducation was associated with larger improvements in the quality of exercise motivation, compared to exercise and behavioral counselling approaches. Undergraduate students diagnosed with major depressive disorder ($N=50$; $M_{\text{age}}=21.6 \pm 3.2$ years) were randomized to three groups: exercise only, exercise plus behavioral counselling, and exercise plus self-compassion. Across 10 weeks, participants engaged in three virtual group sessions per week, consisting of aerobic and strength training activities. Exercise motives were assessed at baseline, immediately post-intervention, and at three-month follow-up. Effect sizes were calculated to compare difference scores from pre-intervention to post-intervention and follow-up. At follow-up, participants in the self-compassion condition reported large, sustained increases in exercising for competition, health, nimbleness, strength and endurance, and enjoyment ($d_s = .68-.98$) compared to participants in the exercise-only condition. At end-of-intervention and follow-up, the self-compassion group reported small increases in stress management motives ($d_s = .21, .29$) compared to the exercise-only group, who instead reported moderate increases in appearance-based motivation ($d_s = -.44, -.43$). Relative to participants in the behavioral counselling condition, those receiving self-compassion training demonstrated moderate increases in revitalization, stress management, and enjoyment motives ($d_s = .39-.45$) at follow-up. Further, compared to the self-compassion group, behavioral counselling participants demonstrated small-large increases in weight- and appearance-based motives ($d_s = -.30-.85$) at end-of-intervention and follow-up. Self-compassion adjuncts were evidenced to promote motives of greater intrinsic value, health, and functional purpose, with lower endorsement of externally-driven motives (e.g., appearance, social recognition) known to undermine sustained physical activity and consequential psychosocial benefits. This pilot, randomized control trial indicates the promise of new, compassion-based approaches to improve exercise motivation outcomes and optimize university mental health supports. Funding source: Western Strategic Support for CIHR Success Program.

Discover pleasure in exercise: Efficacy of the FEEL intervention program

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Interventions for promoting PA often show rather small and unsustainable effects on exercise. This may reflect limited consideration of affective processes during and after exercise (e.g., shame). Furthermore, people are insufficiently enabled to independently regulate their affective response to exercise. Yet, the competence for physical activity-related affect regulation (PAAR) is central to fostering sustainable exercise behavior. The FEEL program aims to promote PAAR, positive affective exercise experiences, well-being, and, consequently, exercise. Topics such as affect-based intensity regulation, attention strategies, or the individually “suitable”

exercise activity are addressed in the program and reflected upon based on participants’ experience. The study evaluates the efficacy of the FEEL intervention program in comparison with a Functional Fitness Training control group. Both exercise programs consist of eight weekly 90-min group-based sessions, with approximately 12–15 participants per session. 230 physically inactive young adults (18–35 years) were cluster-randomized (1:1) across two study sites. Primary outcome: PAAR. Secondary outcomes: affective exercise experiences, exercise behavior, habitual affective well-being. It is hypothesized that the intervention group will show higher scores than the active control immediately post-program (T2) and at the 8-week follow-up (T3). A linear multilevel model will be estimated across the three time points, under a modified intention-to-treat approach, with multiple imputation used to handle missing data. Preliminary multilevel analyses of the primary outcome (PAAR) indicated a substantial increase from T1 to T3 in both groups. The FEEL intervention showed a significantly ($p < .05$) larger increase than the active control group over time. Despite a larger-than-expected increase in the Functional Fitness Training control group, the FEEL intervention showed an additional benefit. These preliminary findings suggest that participants in the FEEL program may be better able to adjust their exercise behavior to encourage more positive affective responses. The FEEL program focuses on affective processes and targets PAAR. This approach addresses a key gap in traditional (social-) cognitive interventions and may foster the competence for PA-related affect regulation. Funding source: Federal Office of Sport – Switzerland.

Considerations for the development of a gamified fitness app for older adults

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Physical activity has been demonstrated to reduce the symptoms or prevalence of non-communicable diseases (e.g., osteoporosis) that largely impact older adults. However, physical inactivity is widespread in the older adult population. Evidence supports the use of behavioral strategies such as gamification (i.e., the application of videogame components to a non-videogame scenario) that increase psychological determinants of physical activity (e.g., positive affective valence) as a method to increase physical activity. To date, no gamified fitness app has been developed explicitly for use in older adults. The purpose of this study was to determine what potential topics may be of interest to a sample of older adults for potential inclusion in a future gamified fitness app. Sixteen older adults ($n=8$ women, 70.3 ± 3.9 years, 100% non-Hispanic White) participated in a virtual interview assessing topics for a hypothetical app that would be created specifically for individuals over the age of 65. Researchers applied thematic analysis to transcribed data and produced three primary content-related themes: (1) Natural Scenery (“I love flowers,” – Jumble; “the waterfalls are just around the corner...” – Swim), (2) Digital Media (“music, but the ability to pick the music,” – Cage; “I like listening to books,” – Grammy), and (3) Interactive Gaming (“like a golfing app,” – Sonny; “some kind of sports theme,” – Broadway). Further, researchers produced two general themes related to app development: (1) Simplicity (“it definitely needs to be simple”), and (2) Personalization (“I typically choose them [podcasts] myself”). Based on these results, a simple-to-use, personalizable gamified fitness application focused on digital media or nature may be acceptable and enjoyable in a population of healthy older adults. Future research should explore the relevance of these topics to subpopulations of older adults and focus further on understanding aspects of fitness applications that may effectively promote physical activity in this population. Funding source: Healthy Communities Research Institute Distinguished Dissertation Award.

Single-session resistance training behavior among college-aged women

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Little is known about how women engage in resistance training (RT) behavior, and how RT behaviors may differ by RT identity. The purpose of this study was to observe and describe what women do for RT and examine differences between RT identity groups. Twenty women (20.25 ± 1.07 years) completed a one-time, self-directed, 45-min RT session. RT identity was categorized as Low ($n = 8$), Moderate ($n = 6$) or High ($n = 6$) using a modified Exercise Identity Scale. Video recording and direct observation of participants occurred. We recorded the exercise, number of sets, number of repetitions, amount of weight, equipment used, and total session duration. Descriptive statistics are provided for the measured variables. Differences in measured variables between exercise identity groups were determined through one-way ANOVAs. Across all groups participants completed an average of $6.4 (\pm 3.9)$ total exercises, $2.7 (\pm .9)$ sets per person per exercise. The most common muscle groups targeted were Legs (1.7 ± 1.7 exercises per session), Shoulders (1.6 ± 1.7), Arms (1.5 ± 1.8), Abdominal (1.5 ± 2.0), and Hips/Butt (1.3 ± 1.3). With the least common muscle groups targeted being Back ($.6 \pm .5$) and Chest ($.9 \pm 1.1$). Participants with a Low, Moderate, or High RT identity spent an average of $27.4 (\pm 7.8)$, $36.7 (\pm 9.04)$, and $43.3 (\pm 3.5)$ min per session, respectively. Individuals with low RT identity completed an average of $5.3 (\pm 4.5)$ exercises compared to those with a high RT identity completing an average of $8.2 (\pm 3.8)$. RT identity did not differ significantly between groups' total number of exercises, $F(2, 17) = .98, p = .396$, number of muscle groups targeted, $F(2, 17) = 1.52, p = .247$, or average sets, $F(2, 17) = .28, p = .76$. However, session duration differed by RT identity, $F(2, 17) = 8.54, p = .003$, with Low identity participants having shorter sessions than high identity participants (Tukey post-hoc $p = .002$). These findings provide more detailed context about what women do for RT. This advances beyond simply understanding how many days women participate in RT activity. Longitudinal or multi-session studies are needed to confirm the consistency of RT activity among college women. Better assessment of muscle strengthening activity is vital for developing and implementing more informed interventions. Funding source: LSU School of Kinesiology Directors Research Award.

Pilot study of college student athlete burnout knowledge

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Athlete burnout is a cognitive-affective syndrome characterized by emotional and physical exhaustion, reduced accomplishment, and sport devaluation with implications for athlete health and well-being. Although commonly understood by sport science professionals, there is a lack of understanding of the knowledge of core athlete burnout symptoms, antecedents, and outcomes by the general population. Accordingly, the purpose of the current study was to examine athlete burnout knowledge in a college student sample. We hypothesized that core burnout symptoms would be the knowledge area of which participants were most familiar and years of sport experience would be associated with more burnout knowledge. Participants were 208 college students ($M = 20.7 \pm 1.8$ years) who self-reported a history of at least one season of sport participation at the high school level or higher. Participants completed self-report items assessing demographics, sport participation history, and athlete burnout knowledge (including knowledge sources) via an online survey interface. Participants were majority female ($n = 167, 80\%$), White/Caucasian

($n = 174, 84\%$) and most reported their highest sport participation at the high school level ($n = 80, 39\%$). Following data screening, most participants identified emotional and physical exhaustion (91%) and sport devaluation (82%) as symptoms of athlete burnout; 66% endorsed all three core burnout symptoms. Participants most commonly reported competitive pressure (88%), excessive time and energy demands (91%), lack of enjoyment/meaning in sport (86%), and lack of motivation (84%) as athlete burnout antecedents. Contrary to extant sport science literature, most participants (96%) identified dropout/withdrawal/retirement from sport as a primary outcome of athlete burnout. Regarding knowledge sources, participants most commonly reported obtaining athlete burnout knowledge from college coursework (37%) or through personal experience with burnout (29%). The total number of burnout symptoms reported was associated with age ($r = .16, p < .05$) but not with years of sport experience ($r = .06, p > .05$). Overall, study participants mostly identified the core athlete burnout symptoms and many of its primary antecedents/outcomes. Yet, participants reported some athlete burnout knowledge incongruent with sport science literature and sport experience did not explain study results. Study findings have implications for continued athlete burnout knowledge survey development and inform burnout knowledge assessment in broader populations.

Athlete burnout among high school drill team members

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High school drill team is a physically demanding, year-round activity that blends athletic performance with artistic expression, yet little research has examined burnout within this population. The purpose of this study was to examine levels and patterns of athlete burnout among high school drill team members in the Rocky Mountain region. Using a mixed-methods design, 59 female drill team athletes who graduated in 2025 from 2A–6A schools completed the Athlete Burnout Questionnaire (ABQ), which measures emotional and physical exhaustion, reduced sense of accomplishment, and sport devaluation. Participants also responded to an open-ended question about factors contributing to burnout. Descriptive results indicated that emotional and physical exhaustion was the most prominent dimension of burnout ($M = 3.98, SD = .78$), followed by reduced sense of accomplishment ($M = 2.55, SD = .95$) and sport devaluation ($M = 2.46, SD = .94$). Overall, 90% of participants reported moderate to high burnout in at least one dimension. A one-way ANOVA revealed no significant differences in burnout levels across school classifications (2A–6A), suggesting that burnout is influenced more by training culture and interpersonal factors than by school size or competition level. Qualitative analysis revealed two dominant themes contributing to burnout: coaching issues (34%) and workload/time demands with physical exhaustion (28%). Athletes frequently described long practice hours, early morning sessions, limited recovery time, and strained coach-athlete relationships as major sources of stress. These findings suggest that many drill team athletes experience early-stage burnout, characterized primarily by exhaustion while still valuing their sport. Results highlight the need for structured recovery time, balanced training loads, and autonomy-supportive coaching practices to prevent burnout progression. This study provides foundational evidence to inform coaches, administrators, and policymakers seeking to improve athlete well-being in dance-based sports.

Bidirectional associations between sport participation and mental health difficulties among US children and adolescents: A longitudinal analysis

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There is considerable evidence linking organized youth sport participation with better mental health outcomes. Nonetheless, most studies are cross-sectional and/or do not explore bidirectional associations. Using longitudinal data, the purpose of this study was to investigate the bidirectional associations between organized sport participation and mental health difficulties among US children and adolescents. Data were from the Adolescent Brain Cognitive Development study—a longitudinal study examining health in a broadly representative sample of US youth. Analyses were based on data from four waves, collected between 2016 and 2021 (Wave 1: $n=10,753$, $M_{\text{age}}=9.9$ years; Wave 2: $n=10,587$, $M_{\text{age}}=10.9$ years; Wave 3: $n=10,574$, $M_{\text{age}}=12$ years; Wave 4: $n=9,855$, $M_{\text{age}}=12.9$ years). Eight youth mental health difficulties were measured using the Child Behavior Checklist (e.g., anxious/depressed, social problems). Data were analyzed using negative binomial (incidence rate ratios; *IRRs*) and logistic (odds ratios; *ORs*) mixed-effects regressions. Participation in exclusively team sports compared to non-sport participation predicted significantly lower scores across six syndrome scales (*IRRs* = .88 to .97). Participation in exclusively individual sports compared to non-sport participation predicted significantly lower scores across five syndrome scales (*IRRs* = .87 to .96). Participation in both team and individual sports compared to non-sport participation predicted significantly lower scores across five syndrome scales (*IRRs* = .80 to .95). These models accounted for 1-7% of the variance in syndrome scores. From the reverse perspective, higher scores on six syndrome scales, respectively, predicted significantly lower odds of participating in team sports compared to non-sport participation (*ORs* = .88 to .97). Higher scores on six syndrome scales, respectively, predicted significantly lower odds of participating in individual sports compared to non-sport participation (*ORs* = .88 to .96). Higher scores on seven syndrome scales, respectively, predicted significantly lower odds of participating in both team and individual sports compared to non-sport participation (*ORs* = .78 to .97). These models accounted for 3-41% of the variance in sport participation scores, with models predicting participation in both team and individual sports generally accounting for the greatest amount of variance (19-41%). The results suggest that youth mental health explains sport participation better than sport participation explains youth mental health.

The SHINE (Supporting Her In Navigating Exercise) program: Exploring virtual and in-person peer-support as an exercise promotion tool among undergraduate women initiates

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Social support is a well-established determinant of university students' exercise behavior. Undergraduate women (UW), in particular, have traditionally relied on social support to engage in exercise, a need heightened during the pandemic due to limited foundational skill development opportunities. Peer mentorship programs (PMP) that involve pairing knowledge-seeking participants with experienced mentors, may enhance engagement through in-person and virtual communication methods. Despite mentors' key role in these types of programs, their experiences remain underexplored, especially in the pandemic recovery era. This is important because without mentor perspectives, opportunities to improve program training, delivery, and sustainability are limited. Senior students in a health discipline with exercise expertise were recruited as mentors to support inactive UW involved in a 6-week randomized controlled PMP. Training involved a 2-hour workshop on motivational interviewing, exercise planning, and program logistics. Mentors exercised 1-1 with their participants weekly and offered interim support through text messages. Post program, mentors completed a demographic questionnaire and 30-45-min semi-structured interview to explore their experiences. Messages and interviews

were compared and analyzed thematically to provide a comprehensive understanding of program engagement and corroborate emerging themes. Six mentors were enrolled and paired with 2-3 UW each ($n=15$). On average, mentors were 24.5 years old ($SD=3.1$) and primarily kinesiology students (83%). Main themes involved perceived facilitators for participant engagement (e.g., on-campus location, tailored coaching). Personal benefits gleaned by mentors were also commonly expressed (e.g., enhanced communication, leadership, self-confidence). Participants sent fewer messages ($M=4.85$, $SD=3.33$) than mentors ($M=5.31$, $SD=2.91$), who used follow-up messages to promote conversation. While challenges such as fluctuating schedules and communication text delays ($M=18.67$ hours) impacted participant engagement, mentors saw participants' exercise habits improve over time and viewed the program as valuable. Overall, mentors' insights may be used to enhance program design and participant support in future PMPs. Given exercise rates still fall below national recommendations, these findings offer practical guidance for campus health promoters and researchers alike to enhance engagement and support UW in establishing consistent exercise habits, essential for their current and future well-being. Funding source: Canadian Institute of Health Research- Masters, Centre of Applied Health Research, and a Lakehead University Faculty Research Grant.

An international evaluation of values-based anti-doping education

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Values-based anti-doping education (VBE) prioritises the development of athletes' ethical values, moral reasoning, and personal responsibility, rather than focusing solely on rules and sanctions. By promoting principles such as fairness, integrity, and respect, VBE seeks to encourage athletes to internalise clean-sport behaviors as part of their identity. Since 2021, the World Anti-Doping Agency (WADA) has mandated VBE within anti-doping education programmes, based on the assumption that it can help prevent doping by addressing underlying motivations and social influences, and by fostering autonomous decision-making beyond reliance on drug testing and sanctions. However, to date, there has been no evaluation of VBE from the perspective of anti-doping organisation (ADO) staff responsible for its delivery. To address this gap, semi-structured qualitative interviews were conducted with 27 ADO staff members (10 male, 17 female) from six organisations across five continents. In total, 1,609 min of interview data ($M=60$ min, $SD=19$ min) were transcribed and analysed using reflexive thematic analysis. Anti-doping educators consistently identified VBE as the most challenging component of WADA's International Standard for Education to implement. A central challenge concerned the lack of concrete guidance and practical examples for delivering VBE, particularly in relation to adapting values-based approaches to the needs of different learner groups. Staff reported particular difficulty engaging adult and high-performance athletes, whose values were perceived as already well-established, and highlighted concerns that values-based activities may be perceived as patronising. In contrast, VBE was viewed as more feasible with youth athletes, especially those transitioning into international competition, where educators perceived greater scope to influence ethical development. In response to these challenges, ADOs described an educational shift towards scenario-based and challenge-driven learning as a more effective means of promoting ethical reflection and decision-making. Overall, the findings point to a broader transition in anti-doping education from didactic, information-based delivery towards more interactive and transformative approaches, while underscoring the need for clearer frameworks and targeted resources to support the effective implementation of VBE across athlete populations. Funding source: World Anti-Doping Agency Social Science Research Grant.

Sleep continuity and next-day physical activity: The role of daily executive functioning

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Sleep continuity, or the ability to sleep soundly through the night without frequent awakenings, is an important yet understudied determinant of physical activity (PA). Although prior research has shown that poorer sleep quality is associated with reduced next-day PA, limited evidence exists surrounding other aspects of sleep continuity (e.g., sleep efficiency, sleep latency). Further, executive functioning (e.g., inhibitory control, planning) could influence relations between sleep and PA despite negative effects from poor sleep. This study aimed to determine if daily fluctuations in sleep continuity predicted next-day moderate-to-vigorous PA (MVPA), and whether executive functioning moderated this association within individuals. Using a 10-day daily diary design, emerging adults ($n=95$, 76% female, 38% Black) wore a waist-worn accelerometer to provide device-based daily min of MVPA and indices of sleep continuity (i.e., sleep duration, sleep efficiency, and sleep latency) and self-reported sleep quality and executive functioning. Executive functioning was operationalized as a daily composite derived from 9 items assessing planning, attention, and inhibitory control. Multilevel modeling was used to analyze daily associations between sleep continuity, executive functioning, and log-transformed MVPA, while controlling for person-level differences in those constructs. Daily executive functioning did not significantly moderate daily sleep-MVPA relations, so the interaction term was removed from models. The more reduced models revealed that on nights when emerging adults experienced longer sleep duration ($b=-.05$, $p<.01$) or greater waking after sleep onset ($b=-.08$, $p=.02$) than was typical for them, they engaged in significantly less MVPA the following day. In contrast, nights characterized by better perceived sleep quality than usual; participants reported greater next-day MVPA ($b=.08$, $p=.02$). Sleep efficiency and sleep latency were not associated with next-day MVPA. Daily executive functioning exhibited a positive trend in its association with MVPA regardless of sleep indices included in the model ($ps=.08-.17$). These findings suggest that specific sleep continuity dimensions are differentially related to next-day PA, and that self-reported executive functioning may be independently associated with daily MVPA but does not condition sleep-MVPA associations.

Service provider perspectives on engagement in community physical activity for newcomer youth

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Engagement and access to physical activity (PA) are shaped by social, organizational, and environmental contexts. These contexts are salient for newcomer populations navigating unfamiliar recreation and sport systems, where access is influenced by partnerships, institutional practices, and local infrastructure. While community organizations play a central role in shaping participation, less is known about how service providers understand the conditions that shape engagement. This study aimed to better understand service providers' perspectives on how relational and structural factors shape engagement in community-based PA across multiple municipal settings. Data were drawn from facilitated hub discussions conducted at the launch of regional Hubs, as a part of the newly developed CO-PLAY Network. One of the structured exercises included a Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis to identify how hubs

can be adapted to create more opportunities for young newcomers to engage in PA to aid in their settlement in Canada. Participants included community and academic partners from across four Canadian hubs: Greater Hamilton ($n=20$; 9 partner organizations), Halifax East ($n=13$; 8 partner organizations), Greater Victoria ($n=28$; 17 partner organizations), and Calgary ($n=29$; 12 partner organizations). A synthesis of responses indicated a shared motivation to reduce barriers by building inclusive, community-embedded systems that promote belonging, physical literacy, and sustained PA engagement. Motivation was reinforced by observed relational impacts, including strengthened family connections and productive partnerships. Relational capacity emerged as a central strength, reflected in community trust, cultural responsiveness, multidisciplinary expertise, and established partnerships within each hub. These strengths were supported by commitments to collaboration, knowledge sharing, and facilitating collective action across hubs. However, organizations identified persistent structural challenges to sustained engagement, including limited funding, leadership turnover, siloed practices, and unclear coordination. Overall, findings suggest that engagement in community PA is shaped by the interaction between relational capacity, collaborative practices, and ongoing structural constraints. Overall, the results of this work reflect a snapshot of organizational perceptions and future work is needed to examine how these perceived facilitators and barriers influence real-world access and sustained engagement necessary to support greater PA participation among young newcomers to Canada.

The influence of coaching styles on collegiate athlete motivation

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Coaching leadership behaviors are important factors influencing athlete motivation, yet less is known about how specific leadership dimensions relate to athletes' introjected motivation. This study examined the relationships between coaching leadership behaviors and introjected motivation among collegiate athletes. Participants included 27 NCAA Division I student-athletes from the University of North Alabama. Student-athletes completed the Leadership Scale for Sports (LSS), assessing perceptions of coaching behaviors across five dimensions: training and instruction, democratic behavior, autocratic behavior, social support, and positive feedback. Introjected motivation (IM) was measured using the introjected subscale of the Sport Motivation Scale (SMS-28). Data was collected via online surveys, and Pearson correlation coefficients were calculated to examine associations. Results indicated that several coaching behaviors were significantly positively associated with introjected motivation. Training and instruction demonstrated a moderate positive correlation with IM, $r(25)=.515$, $p=.006$, and democratic behavior also showed a strong positive association with IM, $r(25)=.630$, $p<.001$. IM was also positively correlated with a coach's social support, $r(25)=.435$, $p=.023$, and with positive feedback, $r(25)=.431$, $p=.025$. In contrast, autocratic behavior was not significantly associated with introjected motivation, $r(25)=-.061$, $p=.76$. This suggests that controlling or authoritarian coaching behaviors are not meaningfully associated with athletes' experiences of guilt or obligation motivation in this sample. Overall, training and instruction, democratic behavior, social support, and positive feedback were all significantly positively associated with introjected motivation, with significant moderate-to-strong correlations. These findings indicate that coaching behaviors characterized by structure, involvement, support, and positive reinforcement are associated with higher introjected motivation, whereas autocratic leadership appears unrelated. The results emphasize the importance of evaluating distinct coaching behaviors when assessing motivational outcomes and may inform coach education programs aimed at promoting adaptive leadership in collegiate sport. Future research with

larger, more diverse samples is needed to confirm these associations and further clarify the role of coaching behavior in athlete motivation.

Exploring the concept of body functionality within athletes through an integration of multiple research areas

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Body functionality includes everything the body can do or is capable of doing. Understanding how individuals experience and evaluate their bodies in terms of functionality has become increasingly prominent within areas of body image, physical self-concept, and disability. However, research on body functionality remains siloed, lacking integration among fields. This gap is particularly evident in sport, whereby body functionality is central to participation, performance, and identity, yet has not been directly examined within athletes in ways that integrate research areas. Through an integration of body functionality frameworks derived from body image and physical self-concept research, and the International Classification of Functioning, Health, and Disability, the purpose of this study was to investigate (1) how athletes describe body functionality and (2) how these descriptions differ across identity factors. Using an online survey, open-ended questions captured athletes' definitions of body functionality and aspects they perceived most important. Participants ($N=440$; $M(SD)_{age}=31.83$ (9.98) years; 35.1% female; 30.3% women) reported an average of 13.13 ($SD=9.79$) years of sport participation across various sport-types and competitive levels. Data were analysed using qualitative content analysis, combining deductive and inductive coding. A codebook was developed by integrating research areas to deductively code responses and was piloted for consensus among two independent coders. Responses that did not align were coded inductively. Athletes most frequently described body functionality in terms of physical capabilities (82.3%), including strength (59.6%), flexibility (57.3%), coordination (40.7%), and endurance (26.6%). Recovery and healing (27.5%) were also identified as important aspects of body functionality. Inductive codes emphasizing movement quality (8.3%) and injury prevention (4.3%) were present for some participants. Endorsing coordination as a key indicator of body functionality differed across self-identified sex ($\chi^2(1)=4.01$, $p=.05$) and race ($\chi^2(2)=15.34$, $p<.001$) suggesting meaningful variation across individuals. Each research area independently contributed to understanding body functionality, highlighting the benefit of integrated approaches. Athletes reported aspects of body functionality beyond existing frameworks, suggesting that this concept may not be clearly defined within this population. Overall, findings provide a foundation for re-conceptualization, future operationalization, and application to advance body-related knowledge and theory in athletic populations. Funding source: SSHRC; Inlight Fellowship.

Do music and exercise ameliorate mental fatigue-induced inhibitory control deficits? A preliminary analysis

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A single bout of aerobic exercise provides a transient (~60 min) executive function (EF) benefit and some work has shown that listening to 'upbeat' music can induce an EF benefit comparable to exercise. In contrast, prolonged work or study induce mental fatigue (MF) and contribute to impaired EF. The present study seeks to determine whether exercise or music listening ameliorate MF-induced EF impairments and determine whether combined exercise and music listening elicit an additive benefit. At present, we have collected data on six (of 30 projected participants) healthy young adults in 14-min conditions involving: (1) a non-exercise

and non-music listening control, (2) moderate intensity aerobic exercise, (3) listening to 'upbeat' music, and (4) combined aerobic exercise and music listening. Each condition was completed on a separate day and involved a baseline (i.e., T0) EF assessment (via the antisaccade task) followed by a 20-min psychomotor vigilance task (PVT; i.e., to induce MF) and a subsequent EF assessment (i.e., T1) prior to participants completing the condition-specific intervention. Subsequently, participants completed a follow-up EF assessment (i.e., T2). As expected, findings demonstrate that the last 5-min of the PVT rendered a reliable increase in reaction time (RT) compared to the first 5-min – a result evincing an objective increase in MF ($p<.05$). As well, antisaccade RTs at T1 – across all conditions – were longer than their T0 counterparts and thus indicates a MF-induced EF impairment ($ps<.05$). Notably, T2 antisaccade RTs for exercise, music, and exercise+music conditions ($ps>.05$) – but not the control condition ($p<.05$) – showed a return to T0 values. In other words, results provide preliminary evidence that exercise, music listening, and the combination of both may circumvent EF deficits associated with a sustained vigilance and MF-inducing task. Funding source: NSERC.

Effects of intensity, order, and instructional framing on affective responses to yoga

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Past research indicates that exercise enjoyment is influenced by intensity, changes in intensity within the session, and instructional framing; lower intensity, decreasing intensity, and mindfulness- or task-involving instructions are generally associated with greater exercise enjoyment. Because yoga incorporates physical postures, breath control, and awareness and is rarely competitive, these patterns may not apply to yoga in the same way they do to cardio- and strength-based exercise. The purpose of this study was to examine how yoga intensity (easy v. hard), order of yoga intensity (increasing v. decreasing intensity), and instructional framing (mindful, task-involving, ego-involving) influence participants' affective feeling state during yoga. Using a within-subjects design, 46 participants followed instructions in six 3-min yoga videos that varied in intensity and instructional focus. Half of the participants' yoga videos increased in intensity (three easy then three hard), while the other half's yoga videos decreased in intensity (three hard then three easy). After each video, participants rated their current feeling state from very bad to very good. Results indicated that easy yoga was generally perceived as more pleasant than hard yoga, $F(1, 44)=7.12$, $p=.01$, $\eta_p^2=.14$; however, this effect depended on the order of intensity. In the decreasing-intensity condition, easy yoga was rated as more pleasant than hard yoga, whereas in the increasing-intensity condition, easy and hard yoga were rated as similarly pleasant, $F(1, 44)=18.09$, $p<.001$, $\eta_p^2=.29$. Additionally, mindfulness and task-involving instructions resulted in more pleasant feeling states than ego-involving instructions, $F(2, 88)=10.76$, $p=.002$, $\eta_p^2=.13$. Overall, these findings suggest that yoga intensity, sequencing, and instructional framing all shape affective responses to yoga practice. Structuring sessions to end with higher-intensity poses, incorporating easier poses, and using mindfulness- (emphasis on awareness and acceptance) or mastery-focused (emphasis on personal improvement and effort) language may enhance yoga enjoyment and future engagement. Future research should examine these effects in more diverse samples and across other forms of physical activity.

"The ideal sport environment would be an accepting environment": Exploring youths' visions for body inclusivity in Canadian youth sport

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Body image concerns remain a significant barrier to sport disengagement among Canadian youth, with disproportionate impacts among those identifying as girls or gender diverse. Cultivating body inclusive sport environments has been identified by Canadian youth as an actionable strategy to reduce these concerns and sustain sport engagement. However, little is known about the specific strategies youth view as effective in this regard. Using a qualitative descriptive approach, this study explored how Canadian girls and gender diverse youth envision cultivating more body inclusive environments through their experiences as past or current sport participants. Semi-structured interviews were conducted with twelve girls ($N = 11$) and gender diverse youth ($N = 1$), ages 12-18 years ($M_{\text{age}} = 15.67$, $SD = 2.01$), recruited via maximum variation sampling. Data were analyzed using inductive content analysis and deductively mapped onto levels of the socioecological model (i.e., interpersonal, organizational, socio-cultural) based on identified data patterns. Interpersonally, youth envisioned body inclusive sport environments as those fostering positive relationships through encouraging, non-judgmental interactions (e.g., no body shaming) and open-minded, inclusive support for all (e.g., acknowledging differing abilities). Organizationally, youth envisioned intentional and communicative sport cultures (e.g., proactive rather than reactive inclusion), participation opportunities that accommodate diverse interests and abilities (e.g., free, non-competitive play), expanding diverse representation in sport leadership, and implementing inclusive policies and practices (e.g., adapting uniforms and equipment to diverse bodies). Socio-culturally, youth envisioned sport environments that normalized open-minded and inclusive norms, including acceptance of diverse bodies, experiences, and contexts for participation (e.g., valuing all bodies as capable regardless of weight, size, ability, skill level, and race), as well as greater visibility and representation of diverse bodies and identities (e.g., visible role models with diverse bodies). Findings suggest that cultivating body inclusive sport environments requires attention to intersecting levels of the sport ecosystem, including interpersonal interactions, organizational practices, and broader socio-cultural norms. Youth perspectives suggest that meaningful inclusion may arise when participants feel respected, capable, and represented, regardless of body type, ability, or identity. These findings present important implications for the design of body inclusive sport programs, policies, and practices, highlighting the need for coordinated action across the individuals and groups shaping sport environments. Funding source: The Social Sciences and Humanities Research Council (SSHRC).

Exploring body functionality in Black athletes through a critical race-conscious lens

Chelsi Ricketts, University of Toronto; Isabella L. Tremonte, University of Toronto; Roxy H. O'Rourke, University of Toronto; Delaney E. Thibodeau, University of Toronto; Catherine M. Sabiston, University of Toronto

Body functionality, defined as everything the body can or is capable of doing, is central to athletics. However, existing conceptualizations of body functionality applied in athlete-focused research are largely derived from predominantly White samples and frameworks rooted in Western biomedical traditions. Contrastingly, little is known about how these conceptualizations apply to non-White athletes, especially those situated in non-Western contexts. Guided by a critical race-conscious lens that interrogates the prioritization of body functionality domains within dominant frameworks, this study examined how athletes self-identifying as Black describe aspects of the body considered essential to its overall functionality, and how these descriptions align with or diverge from existing conceptualizations. Employing a cross-sectional design, participants ($N = 237$; $M_{\text{age}} = 29.92$ years, $SD = 9.61$; males = 51.1%; competitive athletes = 65.4%; from African countries = 93.2%) provided open-ended descriptions of body functionality via an online survey. Responses were

deductively coded using three frameworks: (1) domains of body functionality, (2) components of physical self-concept, and (3) the International Classification of Functioning, Disability, and Health. Responses not aligning with these frameworks were coded inductively. Deductive content analysis indicated that participants' responses largely aligned with domains emphasizing physical capacity (79%; e.g., mobility), movement (79%; e.g., physical activity), and performance-related aspects of the physical self-concept (23-55% across subdomains; e.g., strength). In contrast, aesthetic (e.g., appearance), expressive (e.g., creativity), social (e.g., communication), cognitive (e.g., mental functions), and most physiological domains (e.g., digestive functions) were rarely or not endorsed ($\leq 18\%$). Additional inductively coded domains (i.e., not captured by existing frameworks) included resilience, adaptability, movement mastery, and injury prevention. Findings suggest that Black athletes, particularly from non-Western contexts, emphasize body functionality aspects supporting physicality, performance, and movement, consistent with racialized expectations that place value on the physical skill and durability of Black bodies in sport and society. The inductively identified aspects highlight Black athletes' emphasis on maintaining bodily integrity, movement efficiency, and the capacity to adapt to varied physical and situational demands. Together, these results indicate that dominant conceptualizations of body functionality do not fully capture the culturally and contextually grounded priorities of Black athletes, signaling the need for frameworks that account for bodily capabilities as experienced across diverse, racialized settings.

Life-history narratives of movement and identity among transgender and gender-diverse adults

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Physical activity (PA) is often framed as a universal pathway to health, yet this framing obscures how movement is lived through power, identity, and belonging – particularly for transgender and gender-diverse (TGD) people navigating cisnormative spaces. Research with TGD populations primarily documents discrete barriers and facilitators to PA participation at a single point in time. Less understood are the evolving, identity-based narrative processes through which TGD adults make meaning of movement across shifting identities and social contexts. Guided by the Social Identity Approach and Transgender Theory, this study explores TGD adults' experiences with PA across the life course. Twenty-three TGD adults ($M_{\text{age}} = 35.35 \pm 10.21$), representing a range of self-identified genders (e.g., nonbinary, transgender woman, transmasculine, genderqueer), were recruited through community organizations and online networks in the Greater Vancouver area. Participants completed two life-history interviews supported by participant-generated timelines mapping key movement-related events. Data were analyzed using person-centered narrative analysis (PCNA), a constructionist approach that maintains the individual as the unit of analysis while attending to participants' positioning in relation to dominant social norms and cultural narratives. Through analytic interpretation, participants' movement stories were read as deeply individualized while also showing cross-case convergences in how movement is narrated and felt. In subsequent cross-case interpretation, PA and sport emerged as recurring sites of gender regulation, affirmation, and contestation, contributing to participants' sense of bodily legitimacy from childhood into adulthood. While movement provided connection and affirmation for some, for others it was entangled with surveillance and exclusion – experiences that continue to shape (dis)engagement with PA. These stories reflect a wide emotional range, including grief and ambivalence, alongside a future-oriented hesitancy related to the current sociopolitical climate, characterized by uncertainty regarding safety and belonging.

Concurrently, participants resisted deficit-based narratives, emphasizing trans joy and reclamation through movement. Specifically, seeing their bodies move in affirming ways was central to participants' assertions of agency and identities not defined solely by oppression. Taken together, these narratives demonstrate how centering community voices is essential for (1) challenging dominant gender norms in PA and, (2) supporting researchers and practitioners in reimagining movement spaces for all bodies and genders. Funding source: Social Sciences and Humanities Research Council (SSHRC), and the University of British Columbia.

Cash, cleats, and carpools: A multi-method investigation of resource allocation in families with multiple youth athletes

Emily DeLisle, Utah State University; Shannon Pynn, Utah State University; Jordan Blazo, Louisiana Tech University; Travis Dorsch, Utah State University

Parents play an important and supportive role in young athletes' sport participation (Côté, 1999; Knight, 2019). Given the continued and increasing professionalization of youth sport (see Gould, 2019; McLellan et al., 2022), parents are encountering greater time and financial commitments in supporting their children's participation. This can burden families with multiple children engaged in sport, especially those at the lower end of the income distribution. Despite this, research has yet to fully explore parents' time and financial commitments associated with their children's sport participation (cf. Dorsch & Blazo, 2025). The present multi-method study explored (a) whether the number of athletes in a family impacts parents' allocation of time and financial resources to children's youth sport participation and (b) the ways parents engage in decision-making around the allocation of resources to their children's participation. 1591 parents ($M_{\text{age}}=41.2$) of youth athletes aged 6-18 completed survey measures related to the time and money their families allocated to youth sport. As a follow up, nine parents ($M_{\text{age}}=44.4$) of youth athletes aged 4-18 completed in-depth, semi-structured interviews about decision-making processes related to the allocation of family resources to multiple children in sport. Linear regressions were used to analyze survey data, and an interpretive description approach (Thorne, 2016) was used to analyze and interpret interview data. Counter to our resource dilution hypothesis (see Blake, 1981), findings suggest that an increase in the number of youth athletes in a family does not significantly decrease the time and financial commitments afforded to the oldest athlete's sport participation. Insights from qualitative interviews help explain this finding as a consequence of parents' beliefs in the value and benefit of sport participation. In our study, that appeared to motivate families to employ a variety of strategies aimed at increasing their capacity to meet the time and financial demands supporting multiple athletes in sport. This is important because families with multiple children participating in sport – especially on club or travel teams – make significant financial, career, educational, and lifestyle choices based on their children's participation and the expected outcomes of that participation.

Family sport commitment in a professionalized landscape: An examination of parental pressures and beliefs

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Youth sport in the United States has become increasingly professionalized, heightening pressure on parents and caregivers to invest substantial time and financial resources in their children's sport participation (see Gould, 2019; McLellan et al., 2022). Despite widespread concern that pressures to

specialize and optimistic beliefs about a child's athletic potential drive such investments, few empirical studies have examined these relationships. Guided by Eccles's expectancy-value model (Eccles et al., 1991; 2024), the present study explored whether perceived pressures to specialize and parental beliefs about a child's sport potential predict parental commitment to youth sport. Parents and caregivers ($N=1229$) completed an online survey assessing perceived pressures to specialize (from children, parents, club coaches, school coaches, and society), beliefs about their child's sport potential, and sport commitments (financial spending, days per week, and months per year). Three multiple regression models were fit to examine the independent contributions of pressures and perceived potential to each investment outcome. Findings suggest that perceived pressure from club coaches significantly predicted greater financial spending ($b=49.96$, 95 % CI [11.15 – 88.76], $SE=19.78$, $p<.05$), while perceived pressure from school coaches predicted more days per week spent in sport ($b=.49$, 95 % CI [.21 – .76], $SE=.14$, $p<.01$). Contrary to theoretical expectations, parental belief in child potential did not predict greater financial investment. Results suggest that families' youth sport investments may be more structurally determined by sport system demands than shaped by parental beliefs. In line with Dorsch and colleagues (2022) integrated model of the youth sport system, scholars should consider a range of family, team, and contextual factors in future models of sport commitment. Additionally, in line with recent calls in the sport psychology literature (e.g., Burke et al., 2024; Milan et al., 2024), coaches and youth sport organizations should offer education to help parents navigate decisions without undue pressure. Funding source: Aspen Institute's Project Play.

A preliminary investigation on the effects of differences in knee-related quality of life on quadriceps muscle strength and performance

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Knee-related quality of life (QoL) is often reduced in two populations; (1) individuals who have previous knee injuries and (2) physically active individuals with knee pain but without a diagnosed injury. However, the extent to which perceived knee-related QoL relates to objective measures of quadriceps strength and functional knee performance remains unclear. Understanding these relationships would offer novel strategies for identifying individuals at elevated risk for impaired physical function. The purpose of this study was to examine the extent to which individuals with high vs low knee-related QoL metrics differ in quadriceps strength, jump height, knee moment and knee power. Fifteen healthy and 17 ACL reconstructed young adults completed the Knee Injury and Osteoarthritis Outcome Score (KOOS) survey followed by isokinetic (concentric) quadriceps strength testing at 60°/s. Participants then performed five single leg vertical jumps on a force plate while motion analysis captured the subject motion. Knee moments, knee powers, and jump height were calculated. Following data collection, participants were stratified into high and low QoL groups based on the median split of the KOOS QoL subscale score (High > 81.25; Low < 81.25). Between-group differences using independent-samples *t*-tests and Cohen's *D* effect sizes with Hedges *g* correction were conducted. Results indicated that in comparison to the low QoL group, the high QoL group demonstrated no significant difference, but moderate effect sizes were observed for quadriceps strength (High: $2.28 \pm .63$ Nm/kg, Low: $1.93 \pm .47$ Nm/kg; $d=.63$, $p=.079$, observed power = .406) and jump height ($d=.41$, $p=.245$, observed power = .202). Small effect sizes, with no statistical differences, were present for knee moments ($d=.24$, $p=.493$, observed power = .101) and knee power ($d=.097$, $p=.782$, observed power = .058). These preliminary results suggest that individuals with higher knee-related QoL exhibit meaningfully greater quadriceps strength and jump performance compared to those

with lower QoL, and this is regardless of injury status. Poor performance on the KOOS QoL subscale appears to reflect meaningful strength and performance deficits and offer a clinically accessible way to assess functional recovery following ACL injury.

Physical activity and mental health in college women: A systematic review of longitudinal and experimental studies

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College women experience elevated levels of anxiety, depression, and stress relative to their peers, underscoring the university years as a critical context for understanding mental health risk and protective behaviors. Physical activity (PA) is frequently proposed as a modifiable behavior that may support mental health; however, evidence specific to college women has yet to be synthesized. This systematic review examined longitudinal and experimental studies investigating the associations between PA and anxiety, depression, and stress among college women. Following PRISMA guidelines, searches of PubMed, Web of Science, CINAHL, PsycINFO, and SportDiscus identified 12,890 records, with 51 additional records from citation searching and grey literature. After removal of 6,459 duplicates, 6,480 records were screened and 291 full texts were assessed, yielding 131 included studies. Data extraction included the classification of study design, PA operationalization, mental health outcomes, and key findings, and methodological quality was evaluated using the Joanna Briggs Institute (JBI) critical appraisal tools. Methodological appraisal showed that most studies used appropriate outcome measures and analytic approaches; however, reporting quality was variable, with several JBI appraisal criteria related to intervention description, participant flow, and analytic transparency rated as unclear or unmet. No studies were excluded based on JBI critical appraisal. Across studies, PA was most commonly operationalized as a structured, in-person activity condition (e.g., assigned exercise or supervised movement-based interventions), rather than as habitual, self-selected activity. Anxiety was the most frequently examined outcome (85 studies), followed by depression (56) and stress (54). Positive effects of PA were consistently observed for depression. Findings for anxiety and stress were more heterogeneous, with both anxiety and stress outcomes split between positive and mixed effects. Overall, findings support structured PA interventions as a promising approach for supporting mental health among college women, particularly for depressive symptoms. Future research should prioritize theory-driven PA interventions with clearly specified activity parameters (e.g., type, dose, supervision) and transparent reporting of intervention delivery and analytic procedures to strengthen translation into sport and exercise psychology practice and campus mental health programming.

Physical activity motivation, barrier self-efficacy, and enjoyment changes after psychological plus exercise intervention aiming to reduce social physique anxiety in women with obesity: A secondary analysis

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In women with obesity, concerns that others may negatively evaluate their physique (i.e., experiencing social physique anxiety (SPA)) can drive their physical activity (PA) engagement due to external motivations such as the desire to match societal appearance ideals and avoid negative judgment. This external motivation, however, is not always associated with sustained PA. Indeed, the fear of judgment can also lead to avoiding PA programs and settings perceived as threatening, lower PA self-efficacy, and reduce enjoyment of PA. To encourage participation in PA and target psychological factors associated with PA (motivation, self-efficacy, and enjoyment), interventions addressing SPA in women with obesity are needed. Our study aimed to: (i) explore the effects of exercise training plus (a) cognitive-behavioral therapy with virtual reality-based exposure combined (group 1) or (b) psychological supportive intervention (group 2) aiming to reduce SPA on PA motivation, barrier self-efficacy, and enjoyment, and (ii) describe associations between changes in SPA and body appreciation (BA), and changes in PA motivation, barrier self-efficacy, and enjoyment. Secondary analyses using data from a pilot randomized controlled trial that included 45 women with obesity ($M_{age} = 39.3 \pm 5.7$ years) randomized to group 1 ($N = 22$) or group 2 ($N = 23$). Data were collected using questionnaires administered before and after the 12-week of intervention. Based on linear mixed model results, PA motivation (relative autonomy index (RAI); $\eta_p^2_{time} = .564$), barrier self-efficacy ($\eta_p^2_{time} = .168$), and enjoyment ($\eta_p^2_{time} = .118$) improved significantly in both groups ($p < .05$), with no significant difference between groups. Based on Pearson correlations using residual change scores, change in SPA was negatively associated with change in barrier self-efficacy, $r(39) = -.447$, 95% CI $[-.694, -.269]$, $p = .004$, and change in BA was positively associated with changes in RAI, $r(36) = -.521$, 95% CI $[-.303, .754]$, $p \leq .001$, barrier self-efficacy, $r(38) = .657$, 95% CI $[-.450, .825]$, $p \leq .001$, and enjoyment, $r(38) = .501$, 95% CI $[-.173, .748]$, $p = .001$. In conclusion, both interventions increased PA motivation, barrier self-efficacy, and enjoyment. Participants with more pronounced decreases in SPA and greater increases in BA showed larger gains in PA motivation, barrier self-efficacy and enjoyment. Funding source: Institut du Savoir Montfort.

Sport talent development in Quebec: The role of support networks

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The developmental environment in which young athletes evolve is a key determinant of their sport engagement and psychological well-being throughout their athletic pathway. Grounded in a holistic and ecological perspective, this study was conducted in the context of the 2025 Quebec Games Summer Finals. The present study aimed to examine the talent development environment of young Quebec athletes, with a particular focus on support networks involving coaches and parents. Data were collected through an electronic questionnaire administered to athletes ($n = 2,062$; $F = 1,094$; $M = 968$; $x = 14.5$ years). The questionnaire was based on three scientifically validated instruments. Athletes' perceptions of their talent development environment were assessed using the five dimensions of the Talent Development Environment Questionnaire (TDEQ-5). Perceived coach efficacy was evaluated using the Coaching Efficacy Scale (CES), particularly with respect to communication, clarity of expectations, and orientation toward athlete development. Athletes' perceptions of parental support were examined using the Parental Involvement in Sport Questionnaire (PISQ), focusing on the dimensions of active involvement, praise and understanding, and directive behaviors. Descriptive analyses indicated that the perceived sport development environment related to coaches, as assessed using the TDEQ-5, was generally viewed favorably by athletes. Specifically, for

the long-term development dimension, 80% of athletes reported being in agreement or strong agreement with items assessing their coaches' ability to support long-term athlete development. In contrast, perceptions were more nuanced for dimensions related to communication and alignment of expectations, with approximately 60% of athletes indicating agreement with the corresponding items, while a substantial proportion reported more neutral or disagreeing responses. Regarding perceived parental involvement, as assessed using the PISQ, results revealed high levels of agreement with items reflecting active involvement, reported by approximately 75% of athletes, as well as praise and understanding, endorsed by nearly 80% of respondents. In contrast, items related to directive behaviors elicited more contrasted perceptions, with nearly half of athletes reporting neutral or disagreeing responses. Overall, this study aims to contribute to a better understanding of environments that foster development and sustained engagement among young Quebec athletes in sport.

Adolescents' plans for the future: The influence of academy baseball and softball

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Established by the San Francisco Giants Community Fund in 2022 to build upon its Junior Giants program, The Academy (<https://www.mlb.com/giants/community/fund/programs/gcf-academy>) is a free, competitive baseball and softball program for middle school and high school boys and girls in under-resourced communities in San Francisco. The 8-week program occurs in summer and fall, includes one practice and one game per week, and served more than 750 youth in 2024. Notably, physical skill development is paired with psychosocial development focused on planning for the future. The Saturday Speaker Series invites individuals with careers in professional sports to discuss their career path and pursuing their passion. Previous evaluation studies have focused on Junior Giants, a free, non-competitive sport-based youth development program designed for children ages 5-14. However, evaluation data on the Academy and this free yet competitive opportunity for adolescents is needed. This study explored perspectives from 2025 Fall Academy participants about the Saturday Speaker Series and their plans for the future. The sample included 33 adolescents ages 11-17 ($M = 14.03$), identified from a variety of racial ethnic backgrounds (10 Asian, 9 Multiple Races/Ethnicities, 6 Latino, 6 White, 1 Some Other Race, 1 Preferred Not to Answer), and was split in terms of binary gender (23 boys, 10 girls) and sport type (23 played baseball, 10 played softball). Most (81.8%) had participated in the summer Academy and over half (57.6%) had previously participated in Junior Giants. Results from the online survey showed that participants definitely or probably planned to attend a four-year college (93.9%), attend a junior college (63.6%), or attend trade school (54.5%). Forty-two percent said they were interested in a sports-related career, and the majority (69.7%) agreed or strongly agreed they learned more about sports-related careers. Findings are best understood in the context of sport-based youth development, as planning for the future is a purposeful aspect of the program. This study offers ideas for how adolescents' plans for their futures can be cultivated through sport-based experiences. Funding source: Giants Community Fund.

Social media use and body talk as predictors of female young adults' body self-conscious emotions

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Self-conscious emotions including pride, shame, guilt, envy, and embarrassment are evoked through self-reflection and evaluation and can be directed towards one's body (i.e., body-related self-conscious emotions; BSCE). These emotions could be influenced by the social context, including use of social media and participation in body talk. Yet, there is a lack of empirical evidence supporting these claims. The purpose of the current study was to examine if use of social media and body talk influenced BSCE. Participants included 126 female young adults ($M_{age} = 20.57 \pm 1.75$ years). Participants completed online measures of social media use, body talk, and BSCE twice, with four weeks between time points. Multiple linear regression analyses indicated that fat talk ($\beta = .29$ to $.50, p < .001$) and positive body talk ($\beta = -.26$ to $-.46, p < .001$) at time one (T1) predicted shame, guilt, envy, and embarrassment at time two (T2). Comparison of oneself to others on social media at T1 predicted envy ($\beta = .21, p = .017$) and embarrassment ($\beta = .14, p = .043$) at T2. Finally, positive body talk at T1 predicted authentic pride ($\beta = .63, p < .001$) and hubristic pride ($\beta = .50, p = .017$) at T2. Results suggest that discussion of needing to lose weight and feeling fat along with comparing one's life and body with others on social media heightens feelings of shame, guilt, envy and embarrassment. Discussion about the positive attributes of one's body heightens body-related pride and reduces negative BSCE. As BSCEs are linked to health behaviors (e.g., disordered eating), it is important to improve BSCE. Results provide potential pathways for intervention, including changing how female young adults talk about their bodies.

Psychosocial risk factors for doping: Does theory translate to practice?

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Substantial research has examined psychosocial factors linked with doping in sport. However, to date, no systematic attempt has been made to synthesise these factors and examine whether anti-doping organisations (ADOs) utilise this evidence when identifying athletes most at risk of doping. Accordingly, the purpose of this study was to examine the extent to which ADO staff draw on the existing empirical evidence on psychosocial doping risk factors when identifying athletes most at risk. First, a literature review was conducted to identify psychosocial factors associated with increased doping risk. Semi-structured interviews were then conducted with 19 staff from eight ADOs from three continents, to explore the psychosocial factors they consider when determining which athletes are at highest risk of doping. Interview data were analysed by mapping ADO practices against the psychosocial risk factors identified in the literature, in order to determine which factors were incorporated into ADO decision-making. The literature review identified eight categories of psychosocial risk factors for doping: (1) demographic factors, (2) nutritional supplement use, (3) social influence, (4) training demands, (5) financial incentives, (6) body dissatisfaction, (7) doping attitudes, and (8) injury concerns. Analysis of interview data revealed that training demands were considered most consistently across ADOs; however, none of the remaining factors were utilised by all eight organisations. Nutritional supplement use and financial incentives were used by several ADOs, not because they were viewed as more important, but because their relative objectivity and data availability facilitated their inclusion in largely numeric assessment systems. For example, the number of supplements disclosed on doping control forms and prize money values could be easily incorporated into risk scoring. In contrast, demographic factors and injury were considered by only a small number of ADOs, while social influence, body dissatisfaction and doping attitudes were not used by any of the ADOs represented. Difficulties in accessing reliable and valid data were identified as one of the primary barriers to incorporating several of these psychosocial factors. Future research should therefore focus on developing and validating practical assessment solutions for these factors. Funding source: CENTRE-UB.

Physical activity preferences, attitudes, and beliefs among autistic individuals: Findings from a large cross-sectional survey

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Autistic individuals engage in lower levels of physical activity (PA) than typically developing peers, contributing to poorer health outcomes. While substantial research has examined the effects of PA on health and functional outcomes in autism, comparatively little is known about autistic individuals' PA preferences, attitudes, and beliefs. Understanding how these factors affect their PA engagement is essential for designing effective, acceptable, and sustainable PA interventions. This study examined PA preferences, attitudes, and beliefs among autistic individuals to inform future PA intervention development and promotion. This exploratory, cross-sectional study examined PA preferences, attitudes, and beliefs using a large survey sample. A total of $N=539$ respondents participated, including $N=272$ caregivers of autistic children and $N=267$ autistic adults. Participants completed a custom survey instrument designed to assess PA self-efficacy, beliefs, perceived barriers, access to opportunities, and preferences for activity type, location, and partner characteristics. Survey items were informed by previously published PA questionnaires and qualitative reports from autistic individuals and caregivers. Data were analyzed using descriptive statistics to examine means, frequencies, and proportions, with chi-square analyses used to compare variables within and between groups. Across groups, respondents reported greater participation in and preference for general exercise activities, particularly those that could be completed at home. Home-based PA was consistently rated as more accessible and desirable than activities requiring community or recreational settings. Both caregivers and autistic adults reported low preference for large group activities. Differences emerged in partner preferences, with caregivers more likely to prefer their child engage in PA with a single partner of similar skill level, whereas autistic self-advocates more frequently preferred independent participation. Autistic adults also reported fewer accessible PA opportunities compared to children, suggesting age-related disparities in available supports and programming. Overall, findings highlight substantial heterogeneity in PA preferences and access among autistic individuals. While common trends such as preferences for home-based exercise, smaller group PA or solo PA were observed, variability across respondents underscores the importance of avoiding one-size-fits-all approaches to PA promotion. These results provide actionable guidance for PA providers, emphasizing flexible, inclusive programs that offer adaptable exercise options, promote moderate exertion, and respect individual preferences. Funding source: NIH R01-MH125823.

Clustering of 24-hour movement behaviors and associations with health-related quality of life in Brazilian adolescents

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The 24-Hour Movement Behaviors, including sleep, sedentary behavior, and physical activity, play a fundamental role in adolescents' health and well-being. However, person-centred approaches examining daily movement profiles and their associations with health-related quality of life remain scarce. The purpose of this study was to identify distinct clusters of 24-hour movement behaviors and examine their association with health-related quality of life in Brazilian adolescents. This analysis pooled data from two waves of a repeated cross-sectional study—the ELEVA Project

(2019 and 2022)—and included Brazilian adolescents. Twenty-four-hour movement behaviors (sleep, sedentary behavior, light physical activity [LIPA], and moderate-to-vigorous physical activity [MVPA]) were assessed using wrist-worn accelerometry over seven days, with intensities derived using Hildebrand cut-points. Health-related quality of life was assessed using the KIDSCREEN-10 questionnaire. Clusters of 24-hour movement behaviors were identified using k-means cluster analysis on isometric log-ratio-transformed compositional data, and their association with health-related quality of life was examined using linear regression models adjusted for sex, age, socioeconomic status, and survey. A total of 1,282 adolescents (53.28% girls; $M_{\text{age}} = 16.43 \pm 1.15$ years) were included. The Active cluster ($n=384$) accumulated on average 47.8 min/day of MVPA, 292.0 min/day of LIPA, 654.9 min/day of sedentary behavior, and 445.1 min/day of sleep. The Moderately active cluster ($n=602$) presented intermediate values (27.1 min/day of MVPA, 277.0 min/day of LIPA, 684.3 min/day of sedentary behavior, and 451.5 min/day of sleep), whereas the Sedentary cluster ($n=277$) accumulated only 15.9 min/day of MVPA, 254.0 min/day of LIPA, 706.4 min/day of sedentary behavior, and 463.5 min/day of sleep. Compared with the Sedentary cluster, adolescents in the Active cluster ($\beta: 1.25$; 95% CI: .26; 2.24) exhibited significantly higher health-related quality of life scores, whereas no significant differences were observed for the Moderately active cluster ($\beta: .28$; 95% CI: $-.61$; 1.16). In conclusion, Brazilian adolescents characterized by higher engagement in LIPA and MVPA, lower sedentary time, and a balanced distribution of sleep reported better health-related quality of life than adolescents in the Sedentary cluster.

University athletes' preferences for leadership behaviors exhibited by formal and informal athlete leaders

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Athlete leadership is the process of an athlete occupying a formal or informal role on a team who influences team members to achieve a common goal (Loughead et al., 2006). Formal leaders are individuals designated as leaders by members of their team (e.g., captain) whereas informal leaders develop into a leadership role organically (e.g., veteran). Leadership scholars have commonly explored athlete leaders by specifically examining formal leaders (e.g., Dupuis et al. 2006) or informal leaders (e.g., Imholte et al., 2019) or by exploring formal leaders and informal leaders as one collective source of leadership (e.g., Hirsch-Agnew et al., 2026). However, limited research has explored both formal leaders and informal leaders while distinguishing their separate role types. Hirsch-Agnew et al.'s (2026) findings demonstrated that athlete leaders are expected to provide many leadership behaviors at a high frequency. Research is warranted that examines how each leader type is expected to meet these expectations. Therefore, the purpose of this study was to compare athletes' preferences for athlete leadership behaviors from formal leaders and informal leaders. Participants included 238 intercollegiate athletes who completed the formal athlete-referent and informal athlete-referent preferred-versions of the Leadership Scale for Sports (Chelladurai & Saleh, 1980). A repeated measures MANOVA was conducted to investigate differences between leader type and between leadership behaviors for each leader type. Significant differences were found between leader type, $F(5, 233) = 10.03, p < .001, \eta_p^2 = .177$. Participants reported a preference for formal leaders to exhibit training and instruction ($p < .001$), social support ($p < .01$), and positive feedback ($p < .001$) more often than informal leaders. Significant differences were found between behaviors for formal leaders and informal leaders. Participants reported the following order of preference for leadership behaviors from both leader types: (1) positive feedback, (2) democratic behavior, (3) social support, (4) training

and instruction, and (5) autocratic behavior. The findings extend our understanding of how formal leaders and informal leaders should share leadership responsibilities. Further, behaviors are identified for both leader types to prioritize which can enhance role clarity and leader effectiveness. Funding source: Emerging Scholars (New Mexico State University).

Should I stay or should I go? Psychological need satisfaction, thwarting, and sport persistence in girls and young women

Jessica Malcolm, Brock University; Philip Sullivan, Brock University

Girls leave organized sport at disproportionately high rates during adolescence, with participation declines only stabilizing in young adulthood (Canadian Women & Sport, 2020; Eime et al., 2019). This pattern deprives girls and young women of the well-documented physical, psychological, and social benefits of sustained sport participation during a critical developmental period. Self-Determination Theory (SDT; Ryan & Deci, 2000) posits that environments supporting the psychological needs for autonomy, competence, and relatedness foster motivation and persistence, whereas the thwarting of these needs undermines continued motivation and engagement. The purpose of this study is to investigate how the satisfaction and thwarting of psychological needs in sport impacts persistence among girls and women. Current ($n = 82$) and former ($n = 32$) athletes aged 14–25 who had participated in a variety of team-based girls' or women's sports completed an online survey assessing psychological need satisfaction (Basic Need Satisfaction in Sport Scale; Ng et al., 2011) and psychological need thwarting (Psychological Need Thwarting in the Sport Context; Bartholomew et al., 2011). Binary logistic regression on preliminary data examined whether satisfaction and thwarting of autonomy, competence, and relatedness predicted athlete status. The model significantly distinguished between current and former athletes, $\chi^2(6) = 19.21$, $p = .004$, explaining 23.4% of the variance in athlete status (Nagelkerke $R^2 = .23$) and demonstrating acceptable discrimination ($AUC = .76$). Relatedness thwarting emerged as the only significant individual predictor of athlete status ($OR = .48$, 95% CI [.28, .85], $p = .012$). These findings highlight relatedness thwarting as a critical risk factor for sport dropout among girls and women, suggesting that negative social experiences may be more influential for persistence than the presence of positive relationships alone. Efforts to retain girls and young women in sport should prioritize preventing harmful interpersonal dynamics such as experiences of social disconnection, exclusion, or bullying. Funding source: Social Sciences and Humanities Research Council of Canada.

Selecting basketball talent the fast-and-frugal way

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An aspiration in elite sports is to identify future top players. A selection of those athletes with the highest likelihood of achieving top performance ensures efficient allocation of resources. Selection decisions are accompanied by a high degree of uncertainty because decisions are made with limited individual information and uncertainty about players' future development. A decision process is often lacking transparency and is made by designated individuals such as coaches. Given the theory of bounded rationality, coaches' decisions are formed by heuristics. To understand the heuristics that coaches process during selection, we conceptualized this decision using fast-and-frugal trees (FFTrees). Current evidence indicates that selection decisions are primarily based on current and immediate multidimensional performance of players. Therefore, this study not only uses motor diagnostics but also objective game statistics to illustrate key predictors of selection decisions. Two independent raters

reviewed game video footage from the 3x3 basketball U17 German national selection tournament 2025. The analyzed sample included 74 basketball players ($M = 15.73 \pm .48$ years, 70% female) of whom 20 athletes were nominated. During the tournament, measures on anthropometry, motor speed-related performance diagnostics, subjective coaches' assessments of performance potential and current performance as well as in-game statistics were captured. All these measures were then used to conceptualize coaches' selection decisions via an FFTree. The FFTree identified three cues (coaches' assessment of performance potential, in-game statistics, coaches' estimates of current performance) which successfully classifying 83 % of selections. All three cues assigned athletes with higher cue values into the selected cohort. Anthropometry and motor speed-related performance were not incorporated in the final FFTree due to lacking statistical added value to classification accuracy. This study suggests that the selection is primarily based on subjective coaches' assessments as well as objective evaluations of in-game performance. The cue order identified coaches' estimates of performance potential as the cue with the highest priority for selection, while diagnostic data (e. g., anthropometry) played no determining role. In conclusion, this indicates that coaches' decision is predominantly influenced by in-game performance coaches observe during the tournament.

Intrinsic goals and autonomous motivation as antecedents to Theory of Planned Behavior constructs in predicting exercise intention among university students

Joonyeup Kim, Iowa State University; Sungho Kwon, Seoul National University; Doheung Lee, Seoul National University; Seungmin Lee, Iowa State University

University students are vulnerable to low levels of physical activity (PA) due to academic demands and developmental transitions. Although this life stage is characterized by increasing autonomy in behavioral decision-making, it is important to identify antecedent psychological factors that predict how students regulate and sustain motivation for PA. While the Theory of Planned Behavior (TPB) has been widely used to explain exercise intention through attitudes, subjective norms, and perceived behavioral control, it has been recognized as an opportunity to further enhance the model by incorporating important antecedent factors that contribute to the formation of these determinants. The present study examined the role of exercise goal content as an antecedent, first to autonomous motivation and subsequently to TPB constructs, in predicting exercise intention among university students. Data from 243 university students enrolled in PA courses in South Korea were analyzed. Participants self-reported (a) exercise goal content; (b) autonomous motivation; and (c) TPB variables (i.e., attitudes, subjective norms, and perceived behavioral control). Relative intrinsic goal index (RIGI) and relative autonomy index (RAI) were computed from the first two measures. Structural equation modeling was used to test a single sequential mediation model in which RIGI predicted exercise intention first through RAI and subsequently through TPB constructs. The Type I error rate was set at .05. Results showed that the model provided evidence for adequate fit to the observed data. RIGI exerted significant positive sequential indirect effects on exercise intention, first through RAI and subsequently through TPB constructs: attitude ($B = .134$, $p < .001$), subjective norm ($B = .025$, $p < .01$), and perceived behavioral control ($B = .025$, $p < .05$). These findings indicated that relative intrinsic goals indirectly predicted exercise intention, first through RAI and subsequently through each of the TPB constructs among university students. This observational study underscores the importance of intrinsic exercise goals in predicting autonomous motivation and their relation to attitudes, subjective norms, and perceived behavioral control, which are key determinants of exercise intention. From

a practical perspective, PA courses that emphasize intrinsic exercise goals may be particularly effective in promoting PA among university students.

Changes in the perceived empowering and disempowering motivational climate as predictors of young soccer players' basic psychological needs and indicators of well- and ill-being across a sport season

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Throughout the sport season, young sport participants are influenced by significant others, such as coaches and teammates. The Integrated Model of Empowering and Disempowering Motivational Climate (Duda, 2013; Duda et al., 2024) assumes the motivational climate created by such significant others can be more or less empowering and disempowering, which influences the extent to which sport participants feel that their basic psychological needs for autonomy, competence, and relatedness are satisfied (BPNS) or thwarted (BPNT). Degrees of BPNS and BPNT are expected to hold implications for athletes' experiences of well- or ill-being. Previous studies based on this integrated model, and foundation theories such as Achievement Goal Theory (Ames, 1984; Nicholls, 1992) and Self Determination Theory (Ryan & Deci, 2017), have mainly focused on examining the role of coaches in cross-sectional studies. The purpose of this study was to longitudinally test if changes in perceived empowering and disempowering climates created by one's coach and teammates predicted changes in young athletes' basic psychological needs satisfaction and thwarting, and if these changes predicted changes in their feelings of subjective vitality and physical and emotional exhaustion. Participants (515 Spanish male and female soccer players; $M_{\text{age}} = 14.31$; $SD = 1.19$ years) completed questionnaires capturing the variables of interest at the beginning and end of a soccer season. Path analysis revealed changes over the season in perceptions of an empowering climate from the coach and peers positively predicted changes in BPNS and negatively predict changes in BPNF. Changes in perceived disempowering coach and peer climates positively predicted changes in BPNF. Changes in BPNS positively predicted changes in vitality; BPNF negatively predicted changes in vitality and positively related to changes in physical and emotional exhaustion. These findings are in concordance with theoretical tenets and provide evidence for the promotion of more empowering and less disempowering motivational climates with the aim of facilitating young athletes' welfare. Funding source: CIAICO/2021/248 "Entorno social y funcionamiento óptimo: Un estudio longitudinal en fútbol base". Conselleria de Educación, Universidades y Empleo. Generalitat Valenciana.

The divergent role of functional and moral attitudes in moderating the risk of doping associated with performance supplements and medicine use

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An enriched body of research has shown that dietary supplement use is known to pose a risk for doping, and athletes who develop habitual use of performance supplements (e.g., stimulants, muscle-building) tend to report not only greater intention to dope but also actual doping behavior. Besides performance supplements, athletes may also use medication (e.g., painkillers) to enhance performance or recovery. The potential risk for doping associated with such maladaptive use of medication is rarely examined and compared with that of performance supplement use. In this

context, while greater supplementation or medication use has been conceptualised as contributing to more favourable doping attitudes, research has yet to devote much attention to distinguishing between functional and moral doping attitudes, especially how these distinctive attitudes modulate the risk of doping associated with performance supplements and medication use in athletes. In a sample of 527 Chinese competitive youth athletes ($M_{\text{age}} = 14.84$, $SD = 2.20$; 73% male, 40% with a national grading in their sport) from 14 sports, we examined the potential divergent role of functional and moral attitude in modulating the risk of performance supplements and medication use in relation to doping willingness. The clustered path model, adjusting for sport-level differences when performing hypothesis testing at the athlete-level, revealed that performance supplements use contributed to greater doping willingness as functional attitude increased, whilst medication use contributed to reduced doping willingness as moral attitude increased. In contrast, a functional attitude did not amplify the medication-related risk for doping willingness to the same extent as for performance supplements, and a moral attitude did not mitigate the risk for doping willingness to the same extent for performance supplements as for medication use. Representing the first reported research examining the divergent role of functional and moral attitude in performance supplements and medication use in relation to doping, our findings suggest that high functional attitude is more at risk for those who develop habitual use of performance supplements, whilst low moral attitude is more at risk for those who develop maladaptive use of medication. Different doping attitudes should be tackled for individualised considerations in doping prevention practices.

Co-creating inclusive sport systems for newcomer youth: Community perspectives from the CO-PLAY Hamilton Hub

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The Co-creating Opportunities through Physical Literacy for All newcomer children and Youth (CO-PLAY) Network is a pan-Canadian initiative designed to reduce inequities in sport and recreation access for newcomer children and youth by developing community-embedded hubs that mobilize cross-sector partnerships. Grounded in equity, co-design, and collective action, CO-PLAY seeks to strengthen local capacity to support newcomer children's participation in physical literacy programs as part of their broader settlement and well-being. This study examined the Greater Hamilton hub as a case example of early-stage partnership development within the CO-PLAY Network. Using a qualitative case study design, we conducted a thematic analysis of facilitated group discussions with 20 hub members representing nine community organizations across settlement, sport, public health, and social services. Discussions were centered on partnership strengths, barriers, and priorities for collective action. Data were analyzed inductively to identify key themes relevant to program implementation and sustainability. Findings highlighted several partnership assets, including strong community trust, culturally responsive practices, diverse organizational representation, and shared commitment to meeting families within their lived contexts. Participants emphasized reducing structural barriers to participation, expanding access to facilities and programming, and fostering long-term engagement in physical movement among newcomer youth. The group demonstrated clear orientation toward collaboration, knowledge sharing, and non-competitive resource use. Concurrently, participants identified sustainability limitations, including staffing and volunteer capacity, language and communication barriers, funding uncertainty, leadership turnover, evaluation capacity, and difficulties addressing complex intersecting community needs. The City of Hamilton has several priority areas, including

newcomers, girls, and Indigenous youth, noting the need for dedicated and knowledgeable staff. Similarly, sport programmers like Sportball named barriers like quality staff retention. Coordination and role clarity across sectors could be improved using infrastructure such as digital resource mapping to support collective action and awareness. For example, McMaster University-based newcomer programming could benefit from partners with access to newcomer families to aid recruitment. Overall, findings suggested that hub-based, cross-sector collaboration may advance equitable access to sport and physical activity for newcomer children and youth. However, sustained impact requires intentional investment in partnership infrastructure, coordination mechanisms, and long-term resourcing to address persistent structural barriers.

An in situ analysis of scanning and gaze behavior in junior elite ice hockey

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In invasion sports like ice hockey, effective performance relies on perceiving relevant information within complex and constantly evolving performance environment. Scanning refers to the head and eye movements athletes perform away from the ball or puck to gather contextual information about teammates, opponents, and space. It is an active, multimodal process by which an organism's perceptual systems (i.e., eyes, head and body) seek out information from the environment. While head movement (macro scanning) indicates when information is sought, the eye movement (micro scanning) reveals what might be prioritized. To date, no research to date has examined scanning and gaze behavior in situ during game-representative ice hockey play (5v5) using mobile eye-tracking technology. This absence is particularly limiting, as constraints in ice hockey challenge the direct application of findings from other sports like soccer or basketball. Thus, the purpose of this study was to examine gaze characteristics during scanning behavior in situ during scrimmage play. We analyzed 563 scans across 72 trials from 16 players competing in the United States Hockey League (USHL). Specifically, we sought to determine how visual search strategies adapt to the distinct demands of offensive versus defensive phases and differ between positional roles. Results revealed similar mean scan durations for defensemen (.73s) and forwards (.72s), as well as comparable scan excursion values. However, divergent trends were observed regarding game phase. While total scanning rates appeared consistent between groups (10.2 scans/min), forwards tended to engage in higher scanning rates during offensive phase (5.91 scans/min) compared to defensive phases (4.29 scans/min), whereas defensemen displayed relatively balanced rates. Metrics such as fixation counts, unique AOIs per scan, and viewing time highlighted nuanced understanding of gaze behavior driven by player position and game phase. We discuss how this eye-tracking study in ice hockey provides unique theoretical and practical insights into the perceptual processes underlying complex movement. Furthermore, we outline the directions for future research and provide guidelines for the practice of mental performance consultants and coaches.

The SCORE! Youth-Sustainability Healthy Active Living (SCORE! YOUTH SHAL)

Ava Hsieh, Brock University; Alexandra Leslie, Brock University; Jayden Miller, Brock University; Taylor Rowe, Brock University; Sujane Kandasamy, McMaster University; Matthew Kwan, Brock University

Cardiovascular disease is the global leading cause of death, although largely preventable. In childhood, maintaining a healthy lifestyle is critical,

as poor diets and low levels of physical activity increase susceptibility to these diseases. The SCORE! Youth SHAL project seeks to utilize youth as key agents of change towards the development of interventions targeting sustainable healthy active living within at-risk communities in Canada and Jamaica. The focus of the current study was to describe the findings from co-design workshops with youth participants from our Calgary site. A total of 29 youth ($M_{age} = 16.52 \pm 2.06$; 51.7% male) participated in one of two 3-hour co-design workshops in Fall 2025. Using the MoSCoW method for prioritization, youth identified the "must haves", "should haves", "could haves", and "won't haves" as it relates to healthy eating, physical activity, and sustainable actions. The collaborative work was collected on paper, transcribed digitally, interpreted, and synthesized to uncover emerging themes of youth-obtainable actions. Four overarching themes emerged across the three categories. A prominent theme was the need for community-based activities in which participants acknowledged a perceived benefit from social, group-structured, and community-based physical activity. The "3 big Rs" (reduce, reuse, and recycle) identified efforts to reduce single-use plastics and increase eco-friendly behaviors. Intentional and mindful eating and habit-building based-responses were the most prevalent, involving individual-level initiatives to learn about proper nutrition, cooking, and engaging in movement-based behaviors. Outdoor physical activity (e.g., Biking, walking, hiking) was identified as a meaningful form of exercise. Interestingly, active transportation was not included in the must-haves; however, repeatedly reported within the two other categories. Findings from these workshops will directly inform strategies for the next phase of the project, working with community partners to adapt current programs and practices to better support youth sustainable healthy active living. Future research will aim to understand the broader reach and effectiveness of this youth-driven approach to behavior change across these multiple health domains.

Is the step-defined sedentary lifestyle index linked with flourishing reported by women living with osteoporosis?

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Steps counts have emerged as public health targets designed to counteract physical inactivity. Researchers have advanced the step-defined sedentary lifestyle index (SD-SLI; Tudor-Locke et al., 2012) yet few studies have explored links with psychological well-being to this concept in clinical groups. The objective of this study was to explore links between the SD-SLI (threshold < 5000 steps/day) and psychological well-being reported by women living with osteoporosis. Data were collected from women living with osteoporosis during a single outpatient clinic visit ($N=48$; $M=68.52 \pm 8.80$ years; Range 53.00 to 89.00 years). Participants were mostly common-law/married (64.60%) and self-identified as White (83.30%). Body Mass Index (BMI) values derived from height (cm) and weight (kgs) measured by the clinic nurse ranged from 16.09 to 31.05 kg/m² ($M=22.42 \pm 3.48$ kg/m²; 83.30% < 25.00 kg/m²). Each participant wore a Yamax CW-700/701 pedometer (Yamax Corp., Tokyo, Japan) positioned over the midline of the right thigh during a 7-day (consecutive) free-living monitoring period. Each participant also completed the 8-item Flourishing Scale to assess psychological well-being. Daily step counts ranged from 535 to 9332 ($M=6811.59 \pm 3729.21$ steps/day). More than one third of this sample (37.50%) reported less than 5000 steps/day on average during the 7-day monitoring period. On average, women living with osteoporosis reported less flourishing ($t=-1.63$, $p=.06$, Cohen's $d=.64$) if they walked less than 5000 steps/day ($M=5.91 \pm .74$) versus greater than or equal to 5000 steps/day ($M=6.21 \pm .57$). This study provides initial evidence favoring the use of a SD-SLI that may identify subgroups of women living with osteoporosis reporting lower psychological well-being. Given the broad appeal of step counts in public health campaigns, further research using device-based assessments combined

with a broader range of psychological well-being markers to evaluate the utility (or futility) of the SD-SLI seems justified.

Negative and abusive coaching: A scoping review

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Substantial research has shown that coaches have significant impacts on athletes' development. Although coaches can positively influence athletes' experiences, coach behaviors can also be abusive, which have highly detrimental impacts on athletes such as their sport discontinuation and/or mental ill-health. Abusive coaching is defined as behaviors that result in, or have the potential to harm, an athlete's health, survival, development, or dignity, and can include neglect (e.g., ignoring athletes' needs), sexual abuse (e.g., inappropriate touching), emotional/psychological abuse (e.g., belittling athletes), and physical abuse (e.g., shoving athletes). Due to the wide range of research on these behaviors, a review of the literature is needed to understand how abusive coaching has been conceptualized and what methods have been used to study these behaviors to identify gaps and direct future research. Therefore, the purpose of this scoping review was to examine the current body of literature on abusive coaching. Relevant studies were identified using five electronic databases (e.g., SPORTDiscus), then through manual searches of various journals. Studies were included if they examined abusive coaching behaviors that take place in formal sport settings (e.g., formal practices working towards competition). Studies examining behaviors of other sport personnel (e.g., athletic directors, athletic trainers) were excluded. Of the 1,280 articles found in the initial search, 174 articles were retained for full-text review. Findings suggest that most prior research focused on sexual and/or emotional abuse and the long-term impacts of these behaviors. Almost exclusively, study participants have been athletes, most of whom were retired. Most research used qualitative methods, specifically semi-structured interviews, though some recent research has focused on developing measurements to quantify the experience of abusive coaching. Many studies discussed the prevention of abuse via policy evaluation or intervention (e.g., education programs). In contrast, there has been limited research that examined coaches' perspectives of abusive behaviors and the antecedents of coach abuse. These findings suggest that it may be useful for future research to examine how these behaviors have been perceived and justified in the sport context to better understand why coaches engage in these behaviors and identify methods to protect athletes.

De-gendering the push-up test: Relationship between test position and self-efficacy across the gender spectrum

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The push-up test uses binary-gender positioning for fitness grading. This protocol is at odds with exercise trends and gender diversity. Engagement in resistance training, once thought of as masculine, has increased rapidly in female-identifying exercisers (Kraemer et al, 2025). This format excludes transgender individuals. An inclined push-up could accommodate all genders and provide similar fitness ratings (Adams et al, 2025). The new test could improve self-efficacy (SE), especially for less trained populations. The purpose of this study was to examine SE ratings by test position (knees, incline, toes) and training status. A secondary aim was to investigate the validity of the inclined test. Healthy college students ($n=58$) were recruited for testing in 3 push-up positions (knees, toes, 18" incline). 14% identified as

transgender. Test position was randomized and sessions were separated by 2-5 days. Participants performed maximal repetitions with good form under supervision. Participants rated their SE (0-10) for doing that push-up in the future. 39 participants (15% transgender) completed all three sessions. SE was significantly associated with repetitions at each position. Incline push-up relationship ($r=.44$) was stronger than the knee push-up ($r=.37$) and less than the toe push-up ($r=.51$). SE differed significantly by position ($F=7.5, p=.001$) with a trend towards interaction with gender ($p=.08$). Cis-males reported higher SE for all positions. SE for knee and inclined push-ups were similar for cis-females and transgender participants. Participants with ≥ 2 training sessions a week (67%) reported significantly higher SE ($p=.05$). Interactions with position and gender were non-significant. Interestingly, cis-males had lower SE for the incline push-up compared to other positions regardless of training. Normative fitness categories (ACSM 12th edition) for cisgender participants ($n=33$) were compared to proposed incline push-up rankings (Adams et al., 2025). Fitness categories matched exactly 30% of the time. Intraclass correlation, ICC(3, 1) = .48, indicates acceptable agreement between the tests. The incline push-up test shows promise as a replacement for the current gender-binary protocol. Cis-females and transgender participants viewed it with greater SE than toe push-ups, while cis-males were slightly less confident. Lack of exposure to the incline position may account for lower than expected SE.

Grades as markers of affective experiences in school physical education: An integrative model

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This contribution suggests an integrative model conceptualizing grades in physical education (PE) as indicators of affective-motivational experiences that unfold across the school years. Drawing on Control-Value Theory (CVT) and two affect-oriented theories from exercise psychology (Affective-Reflective Theory, ART; Affective Experiences from Exercise, AFFEXX) the proposed model systematically links perceived competence, core affective experiences, and motivational tendencies. Central to this approach is the assumption that PE grades can be understood as evaluative reflections of repeated affective-motivational experiences that develop and consolidate over the course of students' school careers. To empirically explore key assumptions of the model, questionnaire data from 569 secondary school students ($M=14.6$ years, $SD=1.68$) were analyzed. Measures included PE-related cognitive antecedent appraisals (emotionally charged experiences, e.g. feelings of perceived competence), core affective PE-experiences, and approach versus avoidance tendencies (hedonic wanting). Results indicated that perceived competence was the strongest predictor of affective valence in school PE (pleasant vs. unpleasant experiences; $\beta=.69$, 95% CI [.60, .79]). As expected, PE grades were significantly associated with approach-avoidance tendencies ($\beta=.53$, 95% CI [.32, .73]), supporting their conceptualization as markers of affective-motivational classroom experiences. Across variables, female students showed less favorable profiles in terms of perceived competence, core affective experiences, and hedonic wanting. Overall, these model-explorative findings point to self-reinforcing cycles in which evaluative feedback through grades, affective experiences, and perceived competence are closely intertwined. Within the proposed framework, competence is not primarily conceptualized as a cognitive belief but as a somatically and affectively experienced state closely linked to pleasure in movement. From this perspective, PE grades may serve as diagnostic signals identifying students at risk for unfavorable affective-motivational developmental trajectories. Accordingly, fostering students' feelings of perceived competence emerges as a central leverage point for preventing long-term motivational disengagement in school PE.

Bridging eating disorder and muscle dysmorphia symptoms: The role of fear

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Eating disorders (EDs) and muscle dysmorphia (MD) are impairing mental health conditions that are highly relevant to sport and physical activity contexts, where muscularity and leanness are often emphasized. Although EDs and MD are typically studied separately, they share many features, such as body image and exercise disturbances. However, little is known about which specific symptoms are shared versus distinct across the two conditions, or which transdiagnostic mechanisms may help maintain symptoms across both. Anxiety-related processes, such as body image-related fears, represent a promising but understudied pathway linking EDs and MD. This pre-registered study used network analysis to explore associations among ED symptoms, MD symptoms, and related fears. A sample of 1,432 university students (68.72% women; 47.91% White; $M_{\text{age}} = 20.36$) completed self-report measures. Following rigorous item selection, two networks were estimated at the item level using regularized partial correlations: one with ED and MD symptoms, and another including fears. Central and bridge symptoms were identified using strength and expected influence indices. Networks were re-estimated separately for women and men and compared using the Network Comparison Test. Across both networks, body image concerns, exercise-related guilt, and weight gain-related fears were most central. Bridge symptoms linking ED and MD symptoms were body fat dissatisfaction, counting calories, overvaluation of food, and fear of social eating. Gender-stratified analyses revealed differences in network structure, but not global strength, between women and men. Overall, findings highlight the centrality of body image and exercise-related cognitions and emotions in EDs and MD, and identify fear-based processes that may facilitate symptom overlap or progression between conditions. Results have implications for assessment, prevention, and intervention in sport and physical activity settings, suggesting the importance of targeting maladaptive exercise-related cognitions and emotions across both thinness- and muscularity-oriented psychopathology. Funding source: Data collection was partially funded by internal research funding provided to Lindsay Bodell from Western University.

More than muscle: Women bodybuilders' body image across a competitive season

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Disturbances in body image are commonly cited in athletes, particularly those participating in aesthetic sports where physical appearance contributes to performance judgement, such as bodybuilding. These disturbances are often more pronounced among women due to gender-specific societal pressures related to their appearance. During the 16–24 week competition preparation phase, bodybuilders adopt rigid dietary practices and intensified training, prioritizing physical appearance to present highly muscular and lean physiques on stage; a process that can intensify body image concerns. While bodybuilders attempt to portray confidence and comfort with their bodies, research shows they simultaneously experience many symptoms of a negative body image, such as body dissatisfaction, muscle dysmorphia, and an obsessive desire for leanness; however, this research has only explored body image at acute moments in time. Therefore, the purpose of this longitudinal phenomenological study was to explore women bodybuilder's lived experiences of body image across a competitive season. Four women bodybuilders preparing for competition during the 2025 season took part in two one-on-one semi-structured interviews, one at the start of their competition preparation, and the second following their competition or at the end of their preparation. Between

interviews, participants completed electronic diary entries on a bi-weekly basis throughout their preparation. Interview and diary prompts explored participants' daily perceptions, emotions, and experiences in relation to their body and body image. Using interpretive phenomenological analysis we generated four themes: (1) Living under surveillance: The tyranny of the scale, mirror, and gaze; (2) Never enough: Internalized pressure, perfectionism, and conditional worth; (3) Sacrificing self for the sport: Health, function, and life; (4) Strength beyond aesthetics: Prep as proof of capability. These themes illustrate the complex and dynamic ways women bodybuilders navigate body image, identity, and self-worth throughout the competitive season. Findings from this study extend body image literature by providing a nuanced longitudinal understanding of body image fluctuations in women bodybuilders. The results may also inform coaching staff on how to better support athlete well-being throughout competition preparation.

Perceptions of psychological mattering in adaptive sports

Mathew Gibbs, Auburn University; Rachel Williams, Auburn University

Defined as a sense of feeling significant to those around you, psychological mattering can have a positive impact on one's well-being and health-related quality of life (HRQOL; Hamby et al., 2020). Psychological mattering draws upon well-established literature highlighting the importance of social connectedness for increased mental health (Tonini et al., 2025). Sport represents a salient context for the development of psychological mattering due to the close interpersonal relationships, shared goals, and challenges inherent in athletic participation. Adaptive sport constitutes a distinct sport environment in which additional barriers may shape the nature and meaning of these relationships, as well as their influence on athletes' sense of mattering, health-related quality of life (HRQOL), and well-being (Lape et al., 2017). Despite this, adaptive sports remain underrepresented in literature, limiting understanding of the adaptive sport experience. The purpose of this exploratory qualitative study was to examine how adaptive sport athletes and coaches understand and experience psychological mattering. Adaptive sport athletes ($n = 3$) and a coach ($n = 1$) participated in semi-structured interviews. A thematic analysis (Braun & Clarke, 2006) was used, and three themes were identified. Blazing A Path for Others reflected perceptions of mattering through purpose and influence, as participants described feeling valued when contributing to awareness, representation, and opportunities within adaptive sport. Social Inclusion and Belonging highlighted relational and environmental factors central to mattering, including social support, equitable participation, and a growth-oriented culture. Participants emphasized the importance of "variety without variance," wherein necessary accommodations enable equal involvement without differentiation that undermines belonging. This sense of inclusion fostered self-efficacy, perseverance, and positive well-being. Threats to Mattering captured barriers that diminished perceived value, including inaccessible equipment, misperceptions about ability, and experiences of being underestimated or excluded, all of which were described as negatively impacting motivation, well-being, and quality of life. Findings extend understanding of psychological mattering in underrepresented sport populations, indicating that feeling valued and included in adaptive sport is linked to well-being and HRQOL.

Reallocating time between sedentary behavior and physical activity predicts cognitive growth across childhood: A longitudinal isotemporal substitution approach

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The balance between sedentary behavior (SB) and physical activity (PA) is critical for childhood brain health and cognitive development; however, a paucity of work has modeled whether time allocation between these behaviors predicts long-term cognitive outcomes. The current study examined whether isotemporal substitutions between SB and PA predict concurrent and later cognitive performance. Data were drawn from 3,989 participants (49.5% boys) in the UK Millennium Cohort Study. SB and PA were determined from accelerometry data at age 7. Domain-general (i.e., spatial working memory, decision-making and spatial problem-solving ability) and domain-specific cognitive performance (i.e., word recognition, verbal reasoning, math and vocabulary) were assessed across childhood and adolescence, with different cognitive measures implemented at 7, 11, and 14 years of age. Isotemporal substitution models using isometric log-ratio (ilr) transformations adjusted for demographic, socioeconomic and baseline cognitive performance at age 5 were used to estimate standardized cognitive score differences when reallocating 5–50 min between behaviors, with additional sex-stratified analyses. Replacing 30 min/day of light intensity PA with moderate-to-vigorous PA was associated with improved verbal ability at age 7 ($\beta = .12$, 95% CI [.06, .18]) and stratified analyses showed this reallocation benefited math performance in boys ($\beta = .10$, 95% CI [.02, .18]) – but not girls – at age 7. In turn, replacing SB with LPA was associated with poorer verbal ($\beta = -.09$, 95% CI [-.12, -.07]) and math ability ($\beta = -.03$, 95% CI [-.06, -.00]) at age 7, as well as vocabulary ability ($\beta = -.07$, 95% CI [-.09, -.04]) at age 14. Reallocating daily time from low- to moderate-to-vigorous intensity PA predicts measurable domain-specific cognitive gains during childhood. Accordingly, the present findings emphasize the developmental significance of movement intensity and time-use balance for cognitive growth. These findings also suggest that future investigations into the associations between SB and cognition should move beyond total sitting time and carefully examine the types of SB and PA activities children become involved with.

Does physical activity foster mental health among adolescents, or vice versa? Bidirectional associations disentangling within- and between-person effects using Swiss and Canadian longitudinal data between 2016 and 2024

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Physical activity (PA) has been increasingly recognized as an effective strategy for fostering mental health (MH), but it is also possible that MH is a determinant of PA. However, information on the bidirectional or reciprocal relationships between PA and MH is scarce. In addition, a mix of between- and within-person effects is usually reported, neglecting different processes. This study therefore examines within-person bidirectional associations between PA and MH among Swiss and Canadian adolescents, including the Covid-19 pandemic. We used longitudinal data from two cohort studies: the Swiss Household Panel (SHP; $n = 349$) and the Canadian MATCH study ($n = 538$). Random-intercept cross-lagged panel models (RI-CLPM) were applied to estimate reciprocal within-person effects between PA and MH while accounting for between-person differences. Measurement waves covered periods before, during, and after the Covid-19 pandemic. Separate RI-CLPMs were estimated for each cohort after establishing longitudinal measurement invariance. Half of the variance in PA and MH was explained by between-person effects among Swiss adolescents, with intra-class correlation (ICC) of .50 for both variables. Although considerably lower among Canadian adolescents, this proportion was still high (ICCPA = .21, ICCMH = .22). The latent cross-sectional PA-MH correlation was low in Swiss data ($r = .03$), but moderate in Canadian data ($r = .35$). For within-person effects,

autoregressive paths display significant positive effects among Canadian adolescents except for PA from the pre-pandemic into the pandemic phase. In contrast, data among Swiss adolescents showed positive autoregressive paths from the pre- to post-pandemic phases, but not during the pre-pandemic phase. Most cross-lagged paths indicated small and non-significant associations between PA and MH, except for a large effect size of PA on MH ($\beta = .12$, $p = .07$) during pre-pandemic measures in the Canadian data. By disentangling between- and within-effects in longitudinal bidirectional PA-MH associations, results indicate that PA and MH were mainly stable at the within-person level over time. However, there was no evidence of MH predicting PA, and evidence suggesting that PA affects future MH was weak. Potential explanations, including cultural and measurement-related differences between the Swiss and Canadian cohorts, will be discussed. Funding source: Lindenhof foundation Bern (SLB).

Don't hold your breath: Autonomic, affective, and psychomotor performance responses to cold-water snorkel-assisted facial immersion

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Psychomotor performance during cold-water facial immersion (CWI) presents unique challenges, leading to shifts in autonomic and affective states. While most existing literature on CWI focuses on responses during apneic situations and the mammalian dive response, how psychophysiological responses are altered during CWI while respiration is preserved remains unclear. The present study implemented a protocol in which continuous breathing was permitted via snorkel-assisted facial immersion (SAFI) in cold (12 ± 1 °C) and thermoneutral (33 ± 1 °C) water. Physiological components of CWI were indexed by phasic heart rate variability (HRV), while affective components were indexed by self-reported valence and arousal. Both were used to elucidate the impacts of CWI on psychomotor performance. Cold-water SAFI was expected to increase HRV, decrease self-reported valence, and increase self-reported arousal. Additionally, greater HRV, greater increases in arousal, and greater decreases in valence were hypothesized to predict better motor performance, as assessed via mean reaction time (RT) on an auditory Go/No-Go task performed during immersion. In a randomized within-subjects design, healthy young adults ($N = 23$, $M_{\text{age}} = 20.0 \pm 1.36$ years) completed three conditions: cold-water SAFI, thermoneutral-water SAFI, and a posture/snorkel-matched control without water. HRV (RMSSD) and motor performance (mean RT) were measured during these conditions, while self-reported valence and arousal were assessed before and after immersion. Separate ANCOVAs were used to assess differences in dependent variables between conditions. Multiple regression was performed to assess relationships between autonomic and affective states and motor performance within the cold-water SAFI condition. Cold-water SAFI produced significantly greater HRV ($p < .001$) and significantly greater increases in arousal ($p < .001$) compared to thermoneutral-water SAFI. Within the cold-water SAFI condition, greater phasic HRV predicted faster reaction times during Go trials ($\beta = -.46$, $p < .05$). These results suggest cold-water SAFI evokes robust parasympathetic activation alongside elevated arousal, revealing a dissociation between autonomic regulation and self-reported affect. Overall, these findings position SAFI as a viable model for probing autonomic-affective-psychomotor coupling in aquatic environments and substantiate the complexity of the relationship between autonomic, affective, and psychomotor performance responses.

Investigating Australian women footballers' lived experiences in a semi-professional league

Oisin Foley, La Trobe University; Luke Wilkins, La Trobe University; Brett Taylor, Professional Footballers Australia; Clare MacMahon, La Trobe University

The A-League Women (ALW) is Australia's top-tier women's football (soccer) competition and currently operates on a semi-professional basis. This structure presents significant challenges for player well-being and talent development, with most players required to work outside football, low satisfaction with football-life balance, and high levels of sport-related psychological distress (Professional Footballers Australia, 2025). Research suggests that a move to professionalisation of a women's football league can have significant benefits for players but does not automatically lead to appropriate resourcing and environments for talent development and performance (Culvin et al., 2021). With the ALW expected to move toward full-time professionalism following an upcoming collective bargaining agreement, understanding players' experiences is essential for informing policy and practice. This study reports findings from a qualitative project exploring Australian women football players lived experiences during the ALWs 2025-2026 season. The aim was to examine players' current lived experiences in the semi-professional environment, their expectations and aspirations regarding a move towards professionalism in future seasons, and their perceptions of how existing league, club, and employment structures influence their development and well-being as football players. ALW players across multiple ALW clubs participated in online focus groups, each comprising 5 to 8 players. Data were collected using a semi-structured interview guide designed to capture players lived experiences, perceived challenges and supports, expectations of professionalism, and views on developmental environments within the ALW. Reflexive Thematic Analysis focused on identifying patterns in players' experiences of the semi-professional league structure, their aspirations and concerns regarding professionalism, and perceived impacts on their development, well-being and football-life balance. Themes were developed to reflect shared experiences across clubs while accounting for contextual differences across organisational environments. Findings provide a detailed snapshot of players lived realities in a semi-professional league and of how they perceive these realities affect their ability to develop and perform to their full potential. The results offer valuable evidence to inform clubs, player associations, and governing bodies as they plan for player-centred approaches to professionalisation in the ALW.

The talent development environment in A-League women's football in Australia: A longitudinal analysis of attitudes and perceptions

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In Australian women's football (soccer), the A-League Women (ALW) has experienced significant talent drain over recent years, with most of the top international-level Australian players leaving the league for overseas opportunities in leagues that have become more professional. This talent drain raises questions about how players' experiences of domestic football environments have changed or have failed to keep pace with the professionalisation of other leagues over this period. This study examines longitudinal and repeated cross-sectional changes in ALW players' attitudes and perceptions of their football ecosystem using secondary analysis of 1,214 responses from ALW players survey data collected by Professional Footballers Australia between 2018 and 2025. The sample represents a hybrid longitudinal design, with repeated cross-sectional samples across seasons and a subset of players contributing data at multiple time

points. Surveys captured players' perceptions of club environments, facilities, workload, remuneration, contractual security, sport-work-life balance, and the perceived viability of football as a full-time career. Descriptive statistics were used to summarise league-wide perceptions by season, while repeated cross-sectional comparisons and exploratory correlational and regression analyses examined trends and associations over time. Where repeated observations were available, within-player change was explored. League-wide analyses demonstrated a decline in satisfaction with the league, specific club development environments, and facilities over the past eight seasons. While player salaries have improved and the average number of hours worked outside football has decreased, this has not led to better perceived sport-work-life balance or financial stability. Regarding football environments, higher satisfaction with club culture and operations was positively associated with perceived club and league recommendation scores. Age, and ALW experience were also associated with significant differences in perceptions of players' football environments and sport-work-life conditions with younger, less experienced players more likely to perceive their club environment positively but less likely to perceive the overall league positively when compared with older cohorts with more ALW experience. Younger players were also more likely to report satisfaction with their work-sport-life balance than older cohorts. These results inform discussions around the conditions and environments needed for sustainable professionalisation, talent development, and talent retention in an evolving landscape in women's football.

Quality of life and wellbeing among people living with Parkinson's disease who participate in a boxing program

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Parkinson's disease (PD) is a neurodegenerative disorder with motor and non-motor symptoms that have previously been observed to impair quality of life (QoL) and wellbeing. Holistic support that complements traditional medicative treatments by focusing on overall QoL and wellbeing is critically important to help people living with PD maintain a more meaningful life. Specialized, adaptive group-based boxing programs (e.g., Boxing4Health) have been implemented in communities to support people living with PD, with evidence showing positive effects on PD symptoms; yet, fewer studies test the effects on physical, mental, emotional, and social dimensions of QoL and wellbeing. The primary aims of this study were to describe QoL and wellbeing levels among adults living with PD who participated in a Boxing4Health program (Ottawa, ON), and changes in QoL and wellbeing among participants who provided follow-up data. Additionally, given the importance of retaining participants in research to study changes over time, a secondary aim was to explore sociodemographic, medical, and program factors associated with study dropout to gain insight into who is most likely to dropout. Survey data collected from 83 participants at study entry and approximately 6 months after were analyzed via descriptive statistics, paired sample *t*-tests, and correlations. The Boxing4Health program has a tiered structure based on mobility and cognitive abilities; program tiers are: Fighter, Warrior, and Champion. At study entry, participants reported low negative impact of PD on QoL ($M = 23.67 \pm 14.58$, range = 0–100), low negative affect ($M = 16.40 \pm 5.31$, range = 10–50), and moderate positive affect ($M = 30.63 \pm 7.24$, range = 10–50). No significant ($p < .05$) changes in QoL or wellbeing were observed between timepoints ($n = 39$; $r(38) = -.56$ – $-.56$). No sociodemographic, medical, and program factors were significantly associated with dropout, except program tier, with the Champion program (designed for individuals with significant impairments) exhibiting a higher dropout rate than the Fighter (designed for high-functioning individuals) or Warrior (designed for individuals with

intermediate impairments) programs. Results offer further evidence that specialized, adaptive group-based boxing programs may foster QoL and wellbeing in people living with PD, and suggest strategies to enhance retention in the Champion program are warranted to retain adults with more advanced PD symptoms in studies. Funding source: Social Sciences and Humanities Research Council.

Impact of community programming targeting newcomer youth settlement and mental well-being: A scoping review

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Newcomer youth represent a growing proportion of recent immigrants to Canada and often face significant challenges as they navigate unfamiliar social, educational, and cultural environments. Community-based programs, particularly those centred on sport, physical activity, and creative engagement, are increasingly recognized as important mechanisms for fostering resilience, social belonging, and mental well-being. These programs provide embodied, movement-rich environments that support physical, social, and emotional regulation processes central to development. This review was initially registered as a systematic review; however, substantial heterogeneity in program types, study designs, and outcome measures necessitated a shift to a scoping review approach following PRISMA-ScR guidelines. Searches of PsycINFO, PubMed/MEDLINE, and Web of Science (2000–2025) yielded 1,680 records, of which 17 studies met inclusion criteria. Included programs ranged from creative arts and movement-based therapies to multi-tiered therapeutic and community support models. Overall, 13 of the 17 studies reported improvements in at least one domain of mental health and well-being, social connection, or settlement integration outcomes, including enhanced emotional expression, resilience, self-esteem, school belonging, and engagement in structured activities. A smaller number of studies identified potential risks, including re-traumatization and discomfort with disclosure, underscoring the importance of culturally safe and trauma-informed delivery. Taken together, this review demonstrates promising but limited evidence that community-based and activity-oriented programming can support newcomer youths' psychosocial and participatory development, while highlighting the need for more rigorous, longitudinal, and community-engaged research to inform policy and practice.

Coaching for character in youth sport: Coaches' beliefs and intentions across organizational contexts

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Sport is often seen as a vehicle for youth character development; however, these outcomes are not automatic and depend on intentional approaches across the environment (Bruner et al., 2021). Given that coaches play a crucial role in shaping these experiences, it is important to better understand coaching behavior in isolation. Organizational contexts shape how coaches understand and prioritize character development, influencing whether character-focused practices are embedded in coaching. Differences across settings affect how coaches interpret character and act on related initiatives. Guided by the Theory of Planned Behavior (Ajzen, 1991) and the PRIMED model (Berkowitz, 2017), this study examines coaches' understandings of character, the value they place on it, and how these beliefs shape intentions to coach for character. Using qualitative methods, semi-structured interviews were conducted with boys' soccer coaches across school and competitive settings. Preliminary findings ($n=14$) revealed several shared themes. Coaches in both contexts

described character development as a central part of their role, with responsibility to teach values such as respect and accountability, emphasizing athlete development beyond the field. As one coach explained, "We want these kids to be good human beings, not just good soccer players." While character seemed valued across contexts, differences emerged in how coaches approached it. School-based coaches often framed sport as a means to support athletes' broader development, emphasizing growth beyond wins and losses and the need to balance sport with education. As Coach Ellis explained, "We want to win, but ultimately it's about their mental development and preparing them for college and the workforce . . . they are student-athletes, and the student comes first." In contrast, coaches in more competitive settings emphasized character alongside performance, reflecting organizational expectations and pay-to-play pressures. While character mattered in both contexts, its priority shifted in response to performance demands. Overall, these findings suggest that coaching for character is shaped by context, organizational priorities, and coaches' beliefs about their influence. Addressing these differences is important for coaching-for-character efforts across contexts. Future phases of this study will include coaches from the recreational and middle school contexts, with analyses comparing childhood and adolescence to better understand developmental differences.

Effects of mental fatigue on physical activity decision making in older adults: An effort discounting study

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Older adults in Canada are not meeting current physical activity (PA) guidelines recommending 150 min of moderate-to-vigorous PA each week. Mental fatigue and PA characteristics such as increased intensity and duration, have been shown to decrease motivation to engage in PA in younger adults; however, research examining whether these barriers similarly affect PA motivation in older adults is limited. Furthermore, individual tendencies to approach or avoid physical effort may impact PA motivation in older adults. Using an effort discounting paradigm, this study investigated the effects of mental fatigue, intensity, and duration on PA motivation in older adults. A secondary purpose was to examine how individual differences in approach and avoidance tendencies towards physical effort affect PA motivation. Participants ($N=77$) responded to the Physical Effort Scale prior to completing an online survey. Next, participants made a series of hypothetical choices between engaging in PA of different intensities (light, moderate, vigorous) and durations (10, 20, 30, 40, 50, 60 min) for a fixed reward (\$20) or engaging in a sedentary task for varying rewards (\$2-\$20). Mental fatigue was measured before responding to questions for each intensity. A point of indifference score was calculated for each intensity-duration combination to measure motivation. Results from marginal models showed motivation was lower for higher intensity and longer duration PA ($p < .001$). Additionally, higher mental fatigue resulted in lower motivation for light and moderate intensity PA ($ps < .01$), but not vigorous intensity PA ($p = .08$). The secondary analysis revealed individuals who are approach-dominant showed decreased PA motivation as mental fatigue increased for all intensities ($ps < .03$), whereas individuals who are avoidance-dominant showed significant decreases in PA motivation as mental fatigue increased only for light intensity PA ($p < .001$). Findings replicate prior work in younger adults demonstrating lower motivation for engaging in PA of higher intensities and longer durations, and negative effects of mental fatigue on engaging in vigorous intensity PA. Results further highlight differences in people's willingness to engage in PA based on dispositional characteristics such as tendencies to approach effort and warrants further investigation. Funding source: Hallman Undergraduate Research Fellowship and UW/SSHRC Explore Grant.

Self and comparative efficacy across self- and other-referenced beliefs in a team performance rope-pulling task

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When performing in a team, athletes process various types of efficacy beliefs. Previous research indicates that an athlete can simultaneously consider judgements of (1) personal capability (i.e., self-efficacy; "I can pull as hard as I can"), (2) personal capability compared to others (i.e., comparative self-efficacy; "I can pull harder than anyone"), (3) a partner's capability (i.e., other-efficacy; "They can pull as hard as they can"), and (4) a partner's capability relative to others (i.e., comparative other-efficacy; "They can pull harder than anyone"). The purpose of this study was to examine the extent to which agent (self versus other) and ability reference (self-referenced versus comparative) differ across 2 trials of a team task. 60 tactical athletes consisting of university level Reserve Officers' Training Corps (ROTC) completed a 5-s tug-of-war style rope-pull as a 4-person team for maximal effort. Before each pull (trial), participants reported on self-efficacy, comparative self-efficacy, other-efficacy, and comparative other efficacy. A 2 (self, other) by 2 (self-referenced, comparative) by 2 (Trial 1, 2) RM-ANOVA indicated significant main effects for trial, $p < .001$, $\eta_p^2 = .25$, and ability reference (self-referenced versus comparative), $p < .001$, $\eta_p^2 = .57$. Further, the three-way interaction was significant, $p = .006$. Most notable was that reports of self-efficacy and comparative self-efficacy were significantly different from one another in the first trial, $M_{diff} = 2.36$, compared to the second trial, $M_{diff} = 1.65$. Whereas the difference between other-efficacy and comparative other-efficacy was consistent in both trials, $M_{diff} = 2.01$. The narrowing effect may suggest an increased alignment between self-efficacy and comparative self-efficacy with increased experience, implying a utilization of similar information sources such as mastery or vicarious experience after a performance trial. However, these results ultimately suggest a distinction among efficacy judgments, particularly when appraisals involve comparative evaluations of performance. Funding source: Funder: U.S. Army Research Institute for the Behavioral and Social Sciences. Funding statement: This work was sponsored by the U.S. Army Research Institute for the Behavioral and Social Sciences (ARI) and was accomplished under Grant # W911NF-24-1-0012. The views, opinions, and/or findings contained in this report (paper) are those of the authors and shall not be construed as an official Department of the Army position, policy, or decision, unless so designated by other documents.

Effects of moderate-intensity aerobic exercise duration and recovery period on affective responses in college students

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The benefits of aerobic exercise on general affective health were well documented. The present study was to address the research gap by focusing on immediate and short-term affective responses to exercise. 114 college students in good health voluntarily participated in this study. Participants signed an informal consent and were assigned to one of the three different physical activity levels based on IPAQ survey prior to an exercise protocol. Each participant was required to do a standard 30-min treadmill exercise protocol. Affective response evaluation (ARE) with six different facial expressions was measured six times during the exercise, where the Positive Affect and Negative Affect Schedule (PANAS) was tested for pre-exercise and 2-times post exercise. Results revealed significant temporal effects for ARE. Specifically, enjoyment peaked at 26–30 min, vigor at 6–15 min, and flow progressively declined after 15 min,

anxiety increased sharply at 0–5 and 21–25 min, despair peaked at 6–10 min, and fatigue was strongest at 16–25 min. For PANAS, the results showed that longer recovery periods are associated with improved positive affection and reduced negative affection. In addition, physical activity levels affected participants' affective responses. Higher physical activity level showed more prominent positive affective reactions. In summary, this study demonstrated that affective responses changes with exercise duration and recovery time.

Physical education students' perceived achievement goal orientations and competence profiles change from the middle to end of the semester: A latent transition analysis

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Individuals' psychosocial responses depend on the interplay of their task (self-referenced success) and ego (other-referenced success) goal orientations with their perceived competence. Goal orientations have been examined extensively in physical activity contexts. However, few studies have examined the interplay of goal orientations in perceived competence despite its centrality in achievement goal theory (Biddle et al., 2003). Consequently, the purpose was to use latent transition analysis to: 1) identify profiles based on students' perceptions of competence, task and ego goal orientations in physical education (PE) at the semester midpoint and again at the end of the semester, 2) explore profile stability and transitions over time, and 3) examine differences in perceptions of effort, enjoyment, and pressure by profile membership. PE students ($N = 453$, grades 7-12) were surveyed 6 weeks apart on motivational climates, goal orientations, perceived competence, effort, enjoyment, and pressure. Separate latent profile analyses identified 6 profiles at each time point. At mid-semester, the two most common profiles were a) moderately high ego, task, and competence ($n = 134$) and b) moderate ego, task, and competence ($n = 112$). In examining profile transitions, students in these two profiles were most likely to end the semester in one of two profiles defined by lower ego scores than their mid-semester profiles with minimal change in task and competence scores. Notably, an end of semester profile emerged that was characterized by high ego, task, and competence ($n = 51$). Students most likely to transition into this profile started in the lowest ego profile at mid-semester. A similar number of students from the lowest ego profile at mid-semester only slightly increased their ego orientation while their task and competence decreased at semester end. Lastly, differences in students' reported enjoyment, effort, and pressure experienced in PE were seen by profile membership. These results illustrate why latent profile analyses more realistically represent the interplay of goal orientations and perceived competence than variable-centered approaches. Theoretically, profile transitions over the semester would be influenced by the PE teachers' motivational climate.

Being in the moment: Experiences of adult leisure dance participants

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Immersion and escape, similar to the positive psychological state of flow, are considered creative embodied experiences in dance that contribute to lifespan health and well-being. In dance, research is primarily concentrated on the experiences of flow in professional performance contexts. There is little research about the experience of flow for adult recreational dancers who participate in dance as an autotelic activity without the goal of professional performance. Experiences of recreational dancers can provide insight into the contribution of flow experiences to lifespan physical

activity participation. The purpose of this study was to describe the experiences of flow for adult dancers during the everyday practice of participating in leisure dance. This study was grounded in a hermeneutic phenomenological approach focusing on descriptions of lived experience drawing from interviews with 20 adult leisure dance participants from a variety of dance disciplines including flamenco, contemporary, improvisation, and partnered dances such as salsa and bachata who described experiencing a flow-like state while dancing. Analysis was conducted through an existential thematic approach that explored the corporeality, spatiality, temporality, relationality, and materiality of the experience of being immersed in the moment during dance practice. Participant descriptions were organized into three themes that reflected the lived experience of being in the moment: (1) tuning in; (2) tuning out; and (3) meaningful movement. Adult dancers described tuning in as a whole-body involvement requiring temporal and corporeal immersion. The body and mind were felt as fully present in space and time separate from the world outside the dance studio allowing dancers to tune out from their daily lives. The interconnected elements of dance including music, intent, choreography, or technique provided meaning to otherwise abstract movement. Only once movements were learned and meaningful could a state of freedom, expression, and interpretation be achieved. These findings suggest that adult recreational dancers experience flow in their everyday practice in ways that are consistent with the literature. This study highlights specific characteristics of dance such as complex coordination and connection to music that appear to be important for achieving flow. Further research should explore these themes in relation to subjective well-being. Funding source: This research is supported in part by funding from the Social Sciences and Humanities Research Council.

Integrating social-emotional learning into youth sport: A pilot study of a recreational after-school program

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Whether sport promotes positive youth development (PYD) depends on how intentionally programs are structured to support specific developmental skills (Bruner et al., 2021). Research indicates that elementary school is a developmentally appropriate period for social-emotional learning (SEL), given rapid growth in self-regulation and social competence (Taylor et al., 2017); however, there is limited empirical work examining the intentional integration of structured SEL curricula within recreational youth sport settings. This pilot study addresses this gap by examining whether implementing the SEL curriculum in a six-week recreational after-school sport program resulted in measurable differences in youth social-emotional outcomes. Using a quasi-experimental design, youth who participated in an SEL-enhanced sport program ($n=26$) were compared to youth in the original sport programming delivered business-as-usual (BAU) at a different site within the same geographic region ($n=44$). Participants within both contexts completed a self-report questionnaire before and after the program, assessing PYD-related outcomes such as impulsivity and self-concept of ability. At the end of the program, youth reported on perceived program quality and motivational climate, while coaches completed post-test questionnaires assessing youth conflict management, coping with adversity, and coachability. Given the small sample size and potential violations of statistical assumptions, non-parametric analyses (Wilcoxon signed-rank tests) were used. Results indicated that youth in the SEL-enhanced program exhibited significant decreases in impulsivity and increases in self-concept of ability (effect sizes were small), whereas the BAU program showed no significant changes. At the end of the program, youth in the SEL-enhanced group demonstrated greater coping with adversity and coachability than youth in the comparison group. Since there were no differences in program quality or motivational climate, these

improvements were likely driven by the structured SEL curriculum embedded in the sport program. This pilot study offers early evidence that embedding an SEL curriculum into a recreational youth sport program can support youth social-emotional development. In the discussion, we explore the challenges and opportunities of integrating the curriculum, including the researcher-practitioner partnership, considerations for different youth populations, and finding the right balance between adapting the sport program and staying true to the sport itself.

Exploring latent profiles of masculinity in relation to men's drive for muscularity and body image

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Body image theorists suggest that men's pursuit of sociocultural body ideals (e.g., visible musculature/muscle mass) can signal traditional masculine traits like strength, dominance, and power. In support of this, research demonstrates that conforming to traditional masculine norms is associated with negative body image, drive for muscularity, and dysfunctional exercise in men. Nevertheless, research to date has used variable-centered approaches to explore these links. Gender theories suggest that men can embody multiple masculinities and conform to specific masculine norms to varying degrees. As such, person-centered approaches may unveil unique patterns of men's conformity to traditional masculine norms and how these patterns may be distinctly associated with men's body image and drive for muscularity. In this study, 333 men ($M_{age}=20.48$, $SD=3.46$) completed a measure of traditional masculine norms that assessed their conformity to 11 domains (e.g., dominance, violence, heterosexual self-presentation, power over women, risk-taking, winning, self-reliance, emotional control, status, primacy of work, playboy) and measures of drive for muscularity, body satisfaction, body appreciation, and functionality appreciation. Results from latent profile analysis supported a 4-profile solution with conceptually distinct masculinity profiles: "High Toxic Masculinity/Low Power Over Women" ($n=148$, 44.4%), "High Status-Seeking" ($n=95$, 28.5%), "High Toxic Masculinity/Low Emotional Control" ($n=46$, 13.8%), and "Low Toxic Masculinity/High Self-Reliance" ($n=44$, 13.2%). Examination of how profiles were associated with distal outcomes revealed that men in the Low Toxic Masculinity/High Self-Reliance profile endorsed lower ($p<.001$) drive for muscularity ($M=3.30$) compared to men in the High Toxic Masculinity/Low Power Over Women ($M=3.83$) and High Toxic Masculinity/Low Emotional Control ($M=4.09$) profiles. Additionally, men in the Low Toxic Masculinity/High Self-Reliance endorsed lower body appreciation, functionality appreciation, and body satisfaction compared to other masculinity profiles. Findings support that men may exhibit different degrees of conformity to masculine norms that are uniquely linked to drive for muscularity and body image. Further, lowering conformity to more "toxic" aspects of masculinity may be helpful to reduce drive for muscularity, a key predictor of dysfunctional exercise in men. Funding source: SSHRC.

Implementation of the International Olympic Committee Sport Mental Health Assessment Tool 1 for screening mental health symptoms in Taiwanese athletes

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Athlete mental health has received increasing attention in recent years. Understanding the prevalence of mental health symptoms and disorders

among athletes is essential for improving mental health resources, and developing supportive training environments and psychological interventions. In 2021, the International Olympic Committee (IOC) introduced the Sport Mental Health Assessment Tool-1 (SMHAT-1), which provides sport practitioners with a structured and systematic framework for mental health screening across multiple symptom domains. However, empirical data regarding the prevalence of mental health symptoms and disorders among Taiwanese athletes remain limited. This study aimed to investigate the prevalence of mental health symptoms and disorders among Taiwanese collegiate athletes using the IOC SMHAT-1 framework. A cross-sectional survey was conducted among 859 Taiwanese collegiate athletes using the IOC SMHAT-1, which assesses athlete psychological strain, anxiety, depression, sleep disturbance, alcohol misuse, substance misuse, and eating disorder. All questionnaires were completed anonymously. Descriptive prevalence rates were calculated for the total sample and stratified by sex (male vs. female) and injury status (injured vs. non-injured). Overall, a high proportion of athletes screened positive for psychological strain (86.73%) and eating disorder risk (58.32%). Female athletes demonstrated higher prevalence rates than male athletes across most mental health symptoms and disorders as well as injured athletes when compared with non-injured athletes. The findings indicate a high prevalence of mental health symptoms among Taiwanese collegiate athletes, with psychological strain and eating disorder being prominent. Female and injured athletes appear to be at greater risk across multiple mental health symptoms and disorders, suggesting that female and injured athletes represent populations with elevated mental health vulnerability. These results providing the whole picture of mental health symptoms and disorders using standardized tools such as SMHAT-1 and underscore the need for targeted prevention and intervention strategies within collegiate sport systems in Taiwan.

The effect of self-presentation on athletic identity and maladaptive health behaviors in female young adult runners

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Orthorexia (ON) is a form of disordered eating characterized by the preoccupation with the subjective quality of one's food. Orthorexic tendencies are prevalent in endurance athletes and may co-occur with compulsive exercise (CE). One's athletic identity and need to uphold that identity via self-presentation may be contributing factors to these maladaptive health behaviors. The purpose of this study was to (a) assess if athletic identity and self-presentation motives are linked with ON and CE and (b) examine if self-presentation motives moderated these relationships. Participants included 203 female competitive runners ($M_{\text{age}} = 28.56 \pm 3.89$ years). Participants completed established measures of study constructs. Hierarchical multiple linear regressions were used to test study hypotheses. Self-worth contingency, a property of athletic identity, was significantly associated with problems related to healthy eating ($b = .25, p = .005$) and knowledge related to healthy eating ($b = .23, p = .012$). No moderating effects were observed. Female runners whose self-worth was based on being viewed as a runner were more likely to report symptoms of ON. A protective-agentic motive significantly predicted exercise rigidity ($b = .29, p = .022$), a component of CE. Additionally, the interaction term (protective-agentic motive x athletic identity) was significant ($b = .06, p = .017$). The simple slope for those with higher endorsement of the protective-agentic motive was significant ($b = .19, p = .002$). Female runners who more strongly identified with the role of "runner" engaged in higher levels of disordered exercise behavior when they were motivated to avoid appearing as an inept runner. Presenting oneself as a runner was significantly associated with exercise rigidity ($b = .11, p = .016$). The interaction term (self-presentation x protective-

agentic motive) was significant ($b = .06, p = .026$). The simple slopes for those with higher ($b = .20, p = .001$) and average ($b = .11, p = .016$) endorsement of a protective-agentic motive were significant. Female runners who wished to appear aligned with the role of runner engaged in significantly higher levels of disordered exercise behavior when they were motivated to avoid being viewed as an incompetent runner. Funding source: Miami University Graduate Research Award.

The relationship between physical fitness and inhibitory control among middle-aged adults: The moderating role of APOE genotype

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This study examined whether different components of physical fitness (i.e., muscular strength and cardiorespiratory fitness) predict inhibitory control, and whether these relationships are moderated by apolipoprotein E $\epsilon 4$ (APOE $\epsilon 4$) genotype in middle-aged adults. Ninety cognitively healthy middle-aged adults (45–65 years; 43 APOE $\epsilon 4$ carriers, 47 non-carriers) completed a color-word Stroop task to assess reaction time and accuracy under congruent and incongruent conditions, while event-related potentials (ERPs) were recorded to derive mean P3 amplitude as an index of neurocognitive functioning. Physical fitness was assessed using one-repetition maximum (1-RM) tests for chest and leg press, and the YMCA submaximal cycle-ergometer protocol, reflecting upper-body strength, lower-body strength, and cardiorespiratory fitness, respectively. All continuous variables were z-standardized, and multiple linear regression models were used to estimate the independent predictive effects of the three fitness indicators on Stroop performance and P3 amplitude, adjusting for age, sex, years of education, body mass index, and APOE $\epsilon 4$ genotype, and to test interactions between physical fitness and APOE $\epsilon 4$ genotype as potential moderators. Greater lower-body strength and higher cardiorespiratory fitness significantly predicted shorter reaction times in the incongruent condition ($\beta = -.31, p = .047$; $\beta = -.24, p = .036$), whereas none of the fitness indicators significantly predicted accuracy in either condition. Cardiorespiratory fitness also significantly predicted larger P3 amplitudes in both congruent and incongruent conditions ($\beta = .43, p = .001$; $\beta = .49, p < .001$), and lower-body strength additionally predicted larger incongruent P3 amplitudes ($\beta = .36, p = .020$); upper-body strength was not a significant predictor of behavioral or ERP outcomes. APOE $\epsilon 4$ genotype did not significantly moderate any of the associations between physical fitness and behavioral or ERP outcomes. Better cardiorespiratory fitness and lower-body strength significantly predicted better inhibitory control, particularly as reflected in incongruent reaction time and P3 amplitude, and these associations were not moderated by APOE $\epsilon 4$ genotype in cognitively normal middle-aged adults.

Talking to a bot about your putt: Testing AI-guided self-reflection in golf putting

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Self-reflection is widely recognised as a core psychological process underpinning learning, confidence development, and sport performance. However, commonly used reflection tools are often cognitively demanding and typically completed after performance, limiting their potential to support learning in the moment. Advances in digital technologies offer new opportunities to facilitate structured, real-time reflection during skill execution. Golf provides a useful context to explore these possibilities, given the importance of psychological processes for golf performance.

This pilot study examined whether a novel chatbot delivering real-time, guided self-reflection could improve task-specific self-efficacy and putting performance in golf, as well as exploring usability perceptions. Fourteen recreational golfers (12 male, $M_{\text{age}} = 37$ years) completed five rounds of putting in a controlled indoor environment, engaging with the chatbot between rounds. Measures of self-efficacy and putting performance were assessed pre- and post-intervention, alongside usability assessments using the System Usability Scale and open-ended feedback. Results indicated significant pre-post improvements in self-efficacy magnitude ($p = .020$), self-efficacy strength ($p = .007$), and golf putting performance ($p \leq .001$). Mean usability was rated as 88.8 ($SD = 7.39$) out of 100, which was significantly above the benchmark for “good” usability ($M = 71$). Qualitative responses highlighted the value of structured prompts, ease of use, and support for in-task reflection, while identifying opportunities for refinement, including greater sport-specific tailoring and voice-based input. Together, these findings suggest that chatbot-support reflection is a feasible and promising approach for supporting psychological processes and performance in golf. Future research incorporating control conditions and larger samples is required to determine whether observed performance gains extend beyond practice effects and to establish the efficacy of chatbot-facilitated reflection as a performance-enhancement tool in real golf setting.

Using walk-along interviews to understand psychological and site-based influences on outdoor fitness equipment use

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Outdoor fitness equipment (OFE) has been increasingly installed by local councils as a population-level strategy to promote physical activity. Despite its widespread availability, engagement with OFE varies considerably. While previous research has identified individual-level barriers and facilitators to OFE use, less is known about how site-based characteristics interact with psychological processes to shape engagement with OFE. Research that captures these factors as they are experienced in context remains limited. Therefore, this study used walk-along interviews to examine how environmental features and psychological experiences interact to influence OFE use. Seventeen adults ($M_{\text{age}} = 66$ years, range = 35–84 years; gender identification: 10 females, 6 males, 1 non-binary; two were persons living with disability) participated in interviews conducted at OFE sites across metropolitan South Australia. Walk-along interviews followed a semi-structured guide prompting participants to reflect on their immediate impressions of the site, prior experiences (or non-use) of the equipment, and factors influencing motivation, confidence and comfort to engage. Prompts explored perceived barriers and facilitators related to equipment design, layout, accessibility, safety, social norms, and environmental conditions (i.e., time of day and atmosphere). Participants were also encouraged to consider how social presence, support, and inclusivity shaped their use of the equipment. Interviews were audio-recorded, transcribed verbatim, and inductively analysed using reflexive thematic analysis. The analysis identified a set of interrelated themes describing how environmental factors influence one’s psychological processes to shape engagement with OFE. Broadly, participants described how site and equipment design appeared to influence people’s desire to use the equipment; how signage and the presence of others shaped confidence, enjoyment and motivation around using the equipment; how visibility and location affected feelings of personal safety; and how co-located amenities created a sense of community that supported use. In addition, participants described preferring sites with a variety of equipment, targeting different exercises and muscle groups. By integrating environmental and

psychological perspectives, this study contributes to a more nuanced understanding of how people experience and engage with community-based physical activity infrastructure. Results also provide practical implications for councils when making decisions around installation and activation of OFE. Funding source: Local Government Association of South Australia.

The role of cohesion and tactical communication in doubles tennis performance: A structural equation modeling study

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Athletes are expected to deliver peak performance during competition. However, this depends not only on physical factors but also on various environmental and psychological factors. In team sports, communication and cohesion within the team are widely recognized as key determinants of performance. Yet, the interrelationships and underlying mechanisms of these factors remain poorly understood. This study examined the structural relationships among dyad cohesion, communication content, and performance indicators in competitive doubles tennis. Twenty-one pairs of male college tennis players participated, all newly formed dyads. Verbal exchanges during matches were recorded via lapel microphones, and match footage was captured by two cameras. Pair cohesion was assessed pre-match using a dyad-adapted Group Environment Questionnaire. Performance metrics included the percentage of points won (excluding first-shot errors) and the first-shot error rate. Results showed that rallies involving tactical communication had significantly higher point-winning rates (excluding double faults and return errors) and lower first-shot error rates. Winning pairs also exhibited smaller intra-pair differences in cohesion scores, reflecting stronger cognitive agreement compared to losing pairs. To explore these comprehensive relationships, structural modeling examined cognitive agreement regarding task cohesion, frequency of tactical communication, and performance metrics. Cognitive agreement and communication frequency did not directly predict performance; however, metrics such as point-winning percentage and error rate were predictive of match outcomes. Interestingly, performance showed a positive correlation with serve success rate while exhibiting a negative correlation with return success rate. This suggests that return games increase the difficulty of scoring in doubles. Over-allocation of cognitive resources may induce mental fatigue, potentially impairing subsequent serves and scoring opportunities. Furthermore, the positive association between point conversion rate and match outcome indicates that accumulating points through longer rallies positively influences match results in doubles, where rallies are typically shorter. Overall, these findings suggest that optimizing performance in collegiate doubles tennis requires both shared perceptions of communication quality and team cohesion, as well as strategic game management.

Examining within-day bi-directional associations between affect and physical activity: An ecological momentary assessment study

Ryan Chen, University of Waterloo; Steven R. Bray, McMaster University; Sheereen Harris, University of Waterloo

Extensive research has investigated the role of affect on physical activity (PA) and conversely the effects of PA on affect revealing significant and positive associations. This is consistent with dual process theories that suggest momentary affective states inform PA behavior. As an extension of research conducted in laboratory settings, research leveraging novel data capture methods such as ecological momentary assessment (EMA) has begun to examine these associations as they occur during people’s daily lived experiences. The present study examined within-day, bi-directional

associations between affect and PA using EMA. Undergraduates ($N=100$, 22 males) responded to EMA surveys assessing momentary affect up to four times daily for seven consecutive days. Participants simultaneously wore a wrist worn accelerometer assessing their light physical activity (LPA) and moderate-to-vigorous physical activity (MVPA) for 60 min following each survey prompt. Due to a large number of observations detecting zero MVPA min, a multilevel logistic regression was computed to determine if affect could predict the probability of engaging in some versus no MVPA. Models were computed to determine the effects of affect on subsequent LPA and (non-zero) MVPA as well as the delayed effects of LPA and MVPA on affect reported at the following EMA prompt. In each model, between- and within-person effects were disaggregated. Results revealed higher between-person ($OR=1.19$, $p=.01$) and within-person ($OR=1.14$, $p<.001$) affect were associated with a higher probability of engaging in some MVPA. Among observations reporting non-zero MVPA, higher within-person affect was associated with greater MVPA min (coef. = .28, $p=.01$). Higher within-person affect was also associated with greater LPA min (coef. = 1.13, $p<.001$). Greater MVPA min were associated with higher next timepoint affect (coef. = .02, $p=.007$) while LPA was not a significant predictor of next timepoint affect ($p>.14$). Results show that momentary affective states are related to acute LPA and MVPA levels. Furthermore, MVPA, but not LPA is associated with improved short term affective states. This expands the relationship between affect and MVPA beyond what dual-process theories initially suggest as findings suggest a positive bi-directional association on a within-day level implying a positive feedback loop. Funding source: Social Sciences and Humanities Research Council (SSHRC).

Selective contribution of health-related physical fitness and hot executive function in middle-aged adults

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Hot executive function (EF) refers to higher-order cognition that operates under emotionally salient conditions and is critical for real-life adaptation, whereas cool EF operates under emotionally neutral conditions. Health-related physical fitness (HPF), which encompasses capabilities essential for health and daily functioning, is positively associated with both behavioral performance and electrophysiological activation of cool EF. However, whether this association extends to hot EF remains unclear. This study aimed to identify which HPF components are associated with behavioral performance and electrophysiological activation of hot EF in middle-aged adults. Ninety middle-aged adults participated in this study. HPF components included cardiorespiratory endurance (CE), body composition (body mass index and body fat percentage [BFP]), as well as muscular strength (MS), muscular endurance (ME), and flexibility for both the upper and lower body. Hot EF was assessed using an emotional Stroop task with concurrent electrophysiological recording. The Balanced Integration Score (BIS) and P3 amplitude were calculated as indices of behavioral performance and attentional related electrophysiological activation of hot EF, respectively. Stepwise regression analyses were conducted to identify HPF components associated with BIS and P3 amplitude. CE, BFP, and lower body MS were significantly associated with both BIS and P3 amplitude. These associations were specific to the lower body HPF components, with no significant association observed from the upper body. These findings demonstrate the selective contributions of CE, BFP, and lower body MS to hot EF in middle-aged adults. Future exercise prescriptions for this population could be designed to reduce BFP, enhance CE, and increase lower body MS to optimise hot EF.

Leader development in youth sport: Exploring the role of agency

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The purpose of the study was to examine how youth athletes enact agency in leadership, how this enactment is shaped by personal and environmental factors, and how it contributes to their growth as leaders within sport contexts. Drawing on Cannon et al. (2023), leadership agency is conceptualized as individuals' capacity to act intentionally and purposefully to choose how they engage in leadership experiences, exercise leadership within a role, and interpret these experiences in shaping their leader identity. Guided by a constructivist grounded theory approach, semi-structured interviews were conducted with 12 participants: four youth athletes (ages 13–15) identified as having developed as leaders through participation in sport, four coaches, and four parents. Participants were from four sport teams that had competed at regional or national levels. Interview data were analyzed using Charmaz's (2023) constructivist grounded theory procedures, including initial coding with gerunds, focused coding, constant comparison, memo-writing, and theoretical integration. The results indicated that leadership agency of youth athletes was expressed through autonomous thinking, deliberate choice-making, and self-initiated leadership action in demanding situations. Athletes enacted agency by deciding when and how to speak, support teammates, and respond to mistakes. Their leadership agency was cultivated through interactions with the environment, including coaches' approaches, team climate, role experiences, and family support. The enactment of their leadership agency served as a catalyst that initiated changes in their behavior, the formation of trust and expectations from others, and transformations in self-perception. These changes, in turn, strengthened their leadership agency, creating a cyclical developmental process of their leader development. In sum, taking leadership agency as the central focus, the study yielded grounded theory of youth athlete leadership development. Our findings suggest that youth athlete leader development process reflects a co-evolutionary spiral in which agency and leadership experiences mutually reinforce one another. The results further implicate that youth leader development is fostered not simply by assigning roles but by creating environments that invite choice and legitimize initiative. Future research should examine assessment of leadership agency, the applicability of this model across different sports and age groups, and coaching practices that foster leadership agency.

The relationship between sport-confidence and sources of sport-confidence: An examination of gender differences

Risa Hashioka, *Osaka University of Health and Sport Sciences*; Moe Machida-Kosuga, *Osaka University of Health and Sport Sciences*

The present study examined gender differences in sport-confidence, the importance and availability of its sources, and the relationships among these variables. Participants were 477 Japanese collegiate student-athletes (206 females, 271 males). Sport-confidence was measured using the Japanese version of the Sport-Confidence Inventory (SCI; Vealey & Knight, 2002; Machida et al., 2015), and the importance and availability of sport-confidence sources were assessed using the Japanese version of the Sources of Sport-Confidence Questionnaire (SSCQ; Vealey et al., 1998). Factor structures were examined using confirmatory factor analysis (CFA) and exploratory structural equation modeling (ESEM). CFA of the SCI showed acceptable fit to the data, whereas ESEM was favored over CFA for the SSCQs assessing the importance and availability of sport-confidence sources. The final ESEM models demonstrated good fit, supporting a seven-factor, 32-item structure. Measurement invariance across gender was supported for both the SCI and the SSCQs. No latent

mean differences were observed in the importance of sport-confidence sources. However, the female group showed significantly higher latent mean for the availability of social support sources than the male group, whereas the male group demonstrated significantly higher latent means for all types of sport-confidence (i.e., physical skills & training, cognitive efficiency, and resiliency) as well as for the availability of mastery, demonstration of ability, and physical self-presentation sources than the female group. The results from multiple group path analyses examining gender differences in the relationships among sport-confidence and the importance and availability of its sources showed that the importance of sources generally did not predict sport-confidence, except for mastery in both groups and vicarious experience in the female group. In contrast, the availability of all seven sources was significantly and positively related to sport-confidence, except for coaches' leadership in the female group. The study provided preliminary construct validity evidence to the Japanese versions of the SCI and the SSCQs assessing the importance and availability of sources, as well as evidence of measurement invariance across gender. The findings also highlight gender differences in athletes' sport-confidence, its sources, and their relationships, and provide practical implications for coaching practice and athlete development.

The sport of Kī-o-Rahi as a pathway to wellbeing: Insights from two studies in Aotearoa New Zealand

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Physical activity is well established as a determinant of wellbeing, yet Indigenous communities often face inequities in access to culturally meaningful opportunities for participation. In Aotearoa New Zealand, the traditional Māori sport, Kī-o-Rahi, has re-emerged as both a recreational activity and a potential avenue for enhancing community wellbeing. Drawing on two complementary studies, we examined motivation for participating in Kī-o-Rahi. For Study One, we employed a cross-sectional design to explore motivation among 34 community players (24 Māori; 12 women, 22 men) during a regional tournament. For Study Two, which was guided by kaupapa Māori (Māori principles and values, we conducted three focus groups ($n = 13$; 7 women, 6 men) that explored motivation for beginning and sustaining participation in Kī-o-Rahi with the collaboration of Indigenous community organisers. Based on Study One, we found that participants reported very high intrinsic motivation ($M = 6.7$, $SD = .5$ on a 7-point scale). In Study Two, four interconnected themes were developed: (1) whakawhanaungatanga (building social connections), (2) pride in te ao Māori (Māori worldview), (3) wairua (spirituality and deeper meaning), and (4) enjoyment balanced with healthy competition. Participants described Kī-o-Rahi not merely as a sport, but as a culturally grounded practice fostering connection. Taken together, these studies illustrate that Kī-o-Rahi provides a unique avenue for health promotion that integrates physical, social, and cultural dimensions of wellbeing. Quantitative findings demonstrate that the sport elicits strong intrinsic motivation, which is a key factor linked to sustained participation. Qualitative insights revealed that the intrinsic motivation of participants was deeply rooted in cultural meaning, collective identity, spirituality, and relational connection. Within the New Zealand context, Kī-o-Rahi represents more than a form of exercise; it embodies an Indigenous framework for wellbeing that aligns with Māori values, such as kotahitanga (unity). These findings underscore the value of Indigenous sport as a culturally responsive strategy to reduce health disparities and revitalise traditional knowledge systems. Integrating Kī-o-Rahi into community and public health initiatives could thus support both individual and collective wellbeing across Aotearoa New Zealand, and may inspire similar initiatives for Indigenous Peoples of North America.

Viewing angles but not crowding affects decision making and reaction times in a soccer overrun-detection task

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Peripheral vision is crucial for monitoring multiple players in team sports (Vater et al., 2020; Vater, 2024). In soccer, skilled players anchor gaze on the ball carrier while using peripheral vision to track opponents (Vater et al., 2019). This study tested how peripheral vision is challenged in a counter-attack scenario by varying the viewing angle and crowding between a defender's direct opponent and a wing striker. We predicted that larger viewing angles and higher crowding would impair overrun detection. Twenty-two participants ($M_{age} = 20.81$, $SD = 1.81$; $n_{female} = 10$) with ≥ 5 years of experience as central defenders or midfielders viewed 3 vs. 3 virtual-reality counterattacks from a central-defender perspective. Their task was to detect when a wing striker overran his defender and indicate this via hand response. Detection performance indexed peripheral-vision use. Participants simultaneously defended their direct opponent, who could shoot or pass to either striker. A within-subjects 3 (Eccentricity: 20°, 40°, 60°) $\times$ 3 (Crowding: .25, .50, .75 Bouma) design was used with accuracy, response time, and peripheral detection as dependent variables. Body kinematics and eye-movements were recorded. Non-parametric H-tests were used to test significant effects of eccentricity and crowding. Eccentricity significantly affected accuracy, $p < .001$, $\eta^2 = .257$. Accuracy was higher at 20° and 40° than at 60°. Response times showed the same pattern, $p < .001$, $\eta^2 = .446$, being fastest at 20°, followed by 40° and 60°. Peripheral detection differed by eccentricity, $p < .001$, $\eta^2 = .372$, with greater reliance on peripheral vision at 40° and 60° than 20°. Crowding had no effect on any measure (all $p > .30$). Viewing angle strongly influenced performance, with larger eccentricities reducing accuracy and slowing responses. Although participants relied on peripheral vision at 40° and 60°, reduced accuracy at 60° suggests that initiating saccades to peripheral locations may be beneficial (Vater et al., 2020). Contrary to expectations, crowding did not affect performance, possibly due to sport-specific dynamic stimuli or the allowance of foveal search. Funding source: Swiss National Science Foundation (SNSF).

When the shortest path is not the best: Environmental structure and physical load guide joint-action decisions to optimize coordination efficiency in the object-transfer task

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Török et al. (2019, Psychological Science) demonstrated that in a virtual sequential object-transfer task, individuals selected among alternative paths so as to minimize collective costs. Here, we investigate coordination efficiency in a real-world social setting involving two partners who operated under varying physical loads. Forty participants (18 females, 22 males; $M_{age} = 22.5$ years) were instructed to transport a ball from a start location at one corner of a 10 $\times$ 10 m square to a goal location at the diagonally opposite corner. The square was segmented by a horizontal barrier containing two gaps to the left and right of the midline, as well as a vertical barrier of equal or unequal length on either side of the midline. Participants completed the task in dyads, while either none, one, or both partners wore a weighted vest to manipulate physical load. On each trial, Partner A selected one of the two gaps in the horizontal barrier, where she/he handed the ball over to Partner B, thereby determining the initial segment of the path. Due to asymmetries introduced by the vertical barriers, this choice led to paths that differed in overall length. In congruent conditions, the shorter initial subpath also yielded the shorter overall path for the dyad; in incongruent conditions, the shorter initial subpath resulted in a longer overall path; and in neutral conditions, both overall paths were

equal in length. This design allowed us to assess how environmental structure, social context, and physical effort jointly influence Partner A's initial movement decisions. The results revealed a strong bias towards selecting the initial subpath that minimized the aggregate path costs, with this preference becoming more pronounced as path-length asymmetry increased. Moreover, additional physical load systematically shaped motor decisions: participants adjusted their initial path choices to alleviate the burden on the more physically constrained partner. Taken together, these findings indicate flexible, cost-sensitive motor planning that integrates environmental constraints, effort-related costs, and shared task demands. They provide compelling evidence that coordination efficiency in joint action emerges from embodied, interactive processes that support the minimization of collective costs in real-world social behavior.

Social-cognitive determinants of physical activity in patients with depression: Do they work or are new approaches needed?

Markus Gerber, University of Basel

Social-cognitive theories have proven useful in predicting physical activity behavior in healthy populations. However, the determinants identified in these theories often have only limited predictive power. Research also indicated that interventions based on social-cognitive theories do not always lead to the desired behavioral change. Moreover, evidence suggests that patients with depressive disorders have poorer scores in social-cognitive determinants, particularly those with severe depression, and that traditional intervention approaches (e.g., goal setting, planning) can be associated with considerable pressure in this target population. This study therefore examined the extent to which determinants known from social-cognitive theories are associated with self-reported and device-based moderate-to-vigorous physical activity (MVPA) in patients with depressive disorders. Patients were recruited from four psychiatric clinics and examined three times: approximately two weeks after admission to the clinic for inpatient treatment (T1), six weeks (T2), and 12 months after discharge from the clinic (T3). The sample size was $n = 225$ at T1 (117 women, 108 men, $M_{\text{age}} = 41.3 \pm 12.8$), $n = 170$ at T2 (86 women, 84 men), and $n = 144$ at T3 (68 women, 74 men). The following variables were assessed via validated questionnaires: self-efficacy, (positive and negative) outcome expectancies, self-determined motivation (intrinsic, identified, introjected, extrinsic), goal intention, action planning, perceived barriers, coping planning and social support. MVPA was assessed with the Simple Physical Activity Questionnaire (SIMPAQ) and Actigraph devices. The most consistent bivariate correlations between self-reported and device-based MVPA were found for self-efficacy, intrinsic motivation and number of perceived barriers. Goal intentions were only correlated with MVPA at T1 and T2, whereas a negative correlation with MVPA was only found at T3 for negative outcome expectancies. When using linear regression analyses to predict MVPA at T3, only identified motivation (negative prediction of self-reported MVPA) and intrinsic motivation (positive prediction of device-based MVPA) turned out to be statistically significant predictors of MVPA. In summary, determinants from social-cognitive theories correlate only to a modest extent with physical activity behavior in patients with depressive disorders. This raises the question whether traditional intervention approaches that focus primarily on cognitive factors will lead to the desired outcomes in this specific population. Funding source: Swiss National Science Foundation (SNSF).

Targeting time perception under pressure: Examining perceptual and physiological responses to time pressure in biathlon

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The ability to perform under pressure is central to success in competitive sport. Time pressure represents one of the most pervasive and psychologically testing aspects of competition, yet limited research has examined how psychophysiological and affective factors influence time estimation in competition. Previous research examining time perception during physical activity in lab-based studies has consistently demonstrated that high intensity exercise and perceived exertion distort the subjective estimation of the passage of time. Further, negative affective valence (i.e., unpleasant feeling state), typically reported at high levels of exertion or fatigue, has been shown to slow perceived time; whereas positive affective valence (i.e., pleasant feeling state) is associated with the perception that time passes more quickly. This study investigated the influence of psychophysiological exertion, fatigue, and affective state on time perception during a simulated biathlon competition. Thirty-two biathletes (14 men and 18 women; $M_{\text{age}} = 21.3$, $SD = 3.9$ y) completed five laps of roller-skiing separated by four rounds of shooting (prone-standing-prone-standing). Participants completed a Prospective Time Production (PTP) task pre- and post- competition, as well as Prospective Verbal Estimation (PVE) tasks following each lap of skiing and round of shooting during competition. Heart rate, perceived exertion, fatigue, and affective valence were measured in combination with the measures of time perception. On the Prospective Time Production task participants significantly underestimated the target duration of 29 s post competition but not before, indicating the athletes perceived time to pass more quickly under cumulative psychophysiological load. Prospective Verbal Estimation differed by shooting position, with prone shooting perceived to be shorter in duration than standing. Within-person analyses revealed that higher heart rate, greater perceived exertion and fatigue, as well as negative affect predicted greater temporal distortions during roller-skiing and shooting. These findings suggest that time perception in biathlon is dynamically shaped by the interaction of physiological strain, affective state, and task-specific attentional demands. The applied implications of these findings highlight the value of integrating time perception calibration and affect regulation into pressure-training protocols to improve pacing, shooting rhythm, and performance consistency in sport competition. Funding source: Swedish Research Council for Sport Science; Umeå School of Sport Sciences.

Is better cardiorespiratory fitness associated with lower a physiological and psychological stress response to a mathematics exam in secondary high school students?

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School is widely recognized as one of the primary sources of stress among adolescents. While adolescents with better cardiorespiratory fitness (CRF) might be better equipped to deal with stress, research based on real-life stressors is currently lacking. Therefore, we examined whether CRF moderates the physiological and psychological reactivity in response to an academic exam in mathematics, while controlling for potential confounding factors. We employed a quasi-experimental, within-between interaction design to compare students with poorer vs. better CRF during a real-life stressor (mathematics exam) and a non-stress condition (baseline, normal mathematics lesson). Students were recruited from Swiss public schools (9th grade). The final sample included 67 students (58% female, $M_{\text{age}} = 15.09$ years). Heart rate (HR), heart rate variability (HRV), mood states, and state anxiety were used as indicators of physiological and psychological stress reactivity. CRF was assessed using the 20m shuttle-run test. Statistical analyses were based on repeated-measures analyses of covariance. Exposure to the mathematics exam resulted in decreased HRV and mood, alongside increased anxiety. While students with better CRF exhibited lower HR and higher HRV across both stress and baseline

conditions, no significant stress-buffering effect of CRF was observed. Additionally, better CRF was associated with lower anxiety, although this difference was present only at baseline. Further research employing other ecologically valid stressors is needed to better understand the impact of CRF on real-life stress reactivity. From a school health perspective, it is essential to support students in developing the capacity to effectively cope with academic stressors.

The relationship between athletic identity, exercise stress and leisure time physical activity of South Korean student-athletes

Jaegu Hong, Yonsei University; Minjae Kim, Yonsei University; Do Hui Chung, Yonsei University

The purpose of this study was to investigate the relationship among athletic identity, exercise stress, and leisure time physical activity for student-athletes in South Korea. In detail, it aimed to highlight the necessity of education that helps student-athletes shift from a competition-centric approach to a health-oriented one. This includes teaching them proper goal setting, exercise planning, and sport diversification. The findings are intended to provide useful insights for organizations related to collegiate elite sports. To conduct this study, a questionnaire was utilized as the measurement instrument. The participants were student-athletes currently enrolled at universities located in Seoul and Gyeonggi Province, and data were collected from a total of 283 valid responses using a convenience sampling method. The collected data were analyzed using SPSS and AMOS through confirmatory factor analysis, frequency analysis, reliability analysis, multiple regression analysis, correlation analysis, and structural equation modeling. The results were as follows. First, the athletic identity of collegiate student-athletes had a significant effect on exercise stress. Second, exercise stress did not have a significant effect on physical activity during leisure time. Third, athletic identity had a significant effect on physical activity during leisure time. Fourth, exercise stress did not show a mediating effect on the relationship between athlete identity and physical activity during leisure time. A key finding is that these South Korean student athletes suffer from significant psychological stress because they fail to differentiate between 'occupational' physical activity and leisure-time physical activity amidst such a hyper-competitive societal atmosphere. The intense pressure for results in Korean society affects the physical and mental well-being of student-athletes, impacting them as students first. This context allowed for a discussion regarding the significance of their general physical activity.

Structural effects of team identification on negative emotional responses and aggressive behaviors among professional sport fans: The moderating role of emotional intelligence

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The purpose of this study was to examine the structural relationships among team identification, negative emotional responses, and aggressive behaviors in professional sport fans, and to investigate the moderating role of emotional intelligence. With the rapid growth of professional sport fandom in Korea, fans' emotional reactions to game outcomes have intensified, and concerns regarding aggressive behaviors in both online and offline contexts have increased. From a sport psychology perspective, understanding the psychological mechanisms underlying these behaviors has become increasingly important. Grounded in social identity theory and emotion regulation frameworks, this study focused on how strong psychological attachment to a sport team influences fans' emotional and behavioral responses following game outcomes. In particular, the study

examined whether negative emotional responses function as a mediating mechanism linking team identification to aggressive behavior, and whether emotional intelligence serves as a moderating factor that buffers these relationships. Participants were 225 professional sport fans with experience attending games across multiple sport types. Data were collected via an online survey and analyzed using structural equation modeling. Prior to testing the hypothesized relationships, measurement validity and reliability were confirmed. The structural model was then examined to assess direct effects among variables, indirect effects through negative emotional responses, and moderation effects of emotional intelligence. The results indicated that team identification had significant positive effects on both negative emotional responses and aggressive behaviors. Negative emotional responses also had a significant positive effect on aggressive behavior and partially mediated the relationship between team identification and aggressive behavior. In addition, emotional intelligence moderated the relationships between team identification and negative emotional responses, as well as between team identification and aggressive behavior. Specifically, higher levels of emotional intelligence weakened the positive associations among these variables. These findings suggest that while strong team identification can increase fans' vulnerability to negative emotions and aggression, emotional intelligence functions as a psychological protective factor. This study contributes to the literature by clarifying the emotional mechanisms underlying fan aggression and highlights the importance of emotion regulation-based interventions aimed at promoting healthy fan culture and reducing aggressive behaviors in professional sport settings.

High school student-athlete perspectives on coach behaviors: Insights beyond the Xs and Os

Keith McShan, Missouri State University; Samantha Bates, The Ohio State University

The COVID-19 pandemic created uncertainty in the American youth sports ecosystem due to the cancellation of sports and social distancing guidelines. Athletes and coaches were thrust into new routines and were unable to engage in sport as traditionally offered during this period, potentially creating new needs and stressors within the system. As restrictions were lifted, opportunities existed to elevate the voices of student-athletes to better understand their needs. This study explored adolescent athletes' needs regarding coaching behaviors following the COVID-19 pandemic. Participants were 50 adolescent high school student-athletes ($n = 25$ males and $n = 25$ females) from a midwestern state in the United States. A majority of athletes played multiple sports ($n = 35$). Researchers conducted seven, 1-hour focus groups, using a semi-structured interview guide. Data was analyzed using Dedoose via an inductive approach and thematic analysis to create themes. Results indicated that athletes preferred coaching behaviors that focused on creating positive psycho-social climates, specifically caring, task-involving, and autonomy-supportive climates. Additionally, positive results indicated that student-athletes wanted coaches who focused on athletes' well-being, emotional support, and personal growth. Athletes highlighted negative behaviors shaping a post-COVID-19 world, including coaches who fostered an ego-involving climate, punished mistakes, devalued learning, engaged in authoritarian leadership, failed to prioritize rest and recovery for athletes, lacked appropriate boundaries, and demonstrated poor emotional regulation. Practitioners and scholars can glean from this research how post-COVID-19 adolescent athletes, now more than ever, want coaches who value the athlete's mental health and can manage their own coaching emotions. Athletes value environments that promote learning, where coaches value athletes and their time on and off the playing surface, and dislike coaches who do not know how to walk the line between caring and not overstepping their boundaries. To help coaches find training resources

to develop the strategies athletes are requesting, we will walk through nationally reputable training programs focused on youth development and mental health to continue to advocate for what athletes want in the post-COVID landscape.

NEKOMA Big Air Study: Autonomic and endocrine associations with competitive performance in elite snowboarders

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Elite athletes are thought to approach competition in a finely tuned psychophysiological state, yet evidence from truly consequential competitive settings remains limited. We quantified cardiac autonomic activity using electrocardiography (ECG) and salivary steroid hormones during an elite snowboard Big Air competition that offered prize money and followed international competition rules, with an on-site measurement setup established through collaboration with professional snowboarders and the host ski resort. Twenty-two male, top-level Big Air athletes completed three jumps during an official practice session in the morning and three jumps during the final session in the afternoon. Final placement was determined by the sum of the best two scores from the three final jumps, and each jump was scored independently by three professional judges. For every jump in both sessions, we recorded a three-min ECG and collected saliva 10 min before takeoff, followed by an additional saliva sample immediately after landing. Saliva was assayed for testosterone (T) and cortisol (C), and within-jump reactivity was calculated as the after–before change. ECG measures included mean heart rate (HR) and the root mean square of successive differences (RMSSD), a time-domain metric of heart rate variability. During pre-jump recordings, compared with the official practice session, the final showed a shift toward sympathetic predominance, with significantly higher mean HR and lower RMSSD. Among the three final jumps, only the second showed a tendency for a larger increase in $\log(T/C)$ to be associated with a better overall rank, whereas no such tendency was observed for the first and third jumps. In this format, the second final jump is pivotal. If the first jump is successful, landing the second can substantially increase the chances of winning; if the first jump is missed, the second leaves no margin for error and performance pressure intensifies. Taken together, these findings suggest that elite performance under extreme pressure may reflect a state in which athletes remain physiologically primed for action while shifting toward a more approach-oriented endocrine profile, as shown in this ecologically valid study of elite competition. Funding source: JST [Moonshot R&D][Grant Number JPMJMS2295-01, JPMJMS2295-02, JPMJMS2295-05].

Policy mapping of physical activity and education in hospital schools: Addressing the psychosocial gap in pediatric patients—an MSCA HOPE study

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Hospitalization during childhood and adolescence is associated with disruptions in psychosocial development, emotional well-being, and access to physical activity (PA). While exercise psychology highlights PA's benefits for psychosocial outcomes such as mental health, motivation, and quality of life, it remains unclear whether policy frameworks explicitly support PA and physical education (PE) during hospitalization. Within the Horizon Europe MSCA project “Hospital schools’ guidelines On Physical Education” (HOPE), this study presents a cross-national policy mapping examining how PA/PE in hospital school contexts are regulated and whether psychosocial dimensions are addressed. Policy documents were systematically identified through structured searches of international organization repositories (WHO, UNESCO, EU) and national governmental websites across 92 countries (Africa, $n=20$; Americas, $n=20$; Asia, $n=20$; Europe, $n=30$; Oceania, $n=2$). Eligible documents included laws and official guidelines addressing education during hospitalization or prolonged medical treatment. Records were analyzed using a theory-informed policy content analysis framework, integrating document analysis principles and governance- and scope-related dimensions. Deductive coding captured governance models, target populations, PA/PE provisions, and explicit psychosocial constructs (e.g., well-being, mental health, motivation, social interaction, quality of life). Results (1979-2026) revealed substantial heterogeneity across countries categorized in four health systems: (i) Bismarck (social-health insurance, 37%), (ii) Beveridge (tax-funded national-health service, 25%), (iii) Out-of-pocket (market-based, 20%) and (iv) NHI (national-health insurance, 10%) models. While 57% of frameworks guarantee educational continuity, fewer reference PA (43%) or PE (53%). Crucially, only 9% provide a mandate or recommendation for PA/PE delivery in hospital schools. Psychosocial constructs were inconsistently articulated: well-being (50%), social interaction (34%), mental health (26%), and motivation (23%). Quality of life was explicitly addressed in only 12% of documents. These dimensions appeared more frequently in general inclusive education policies than in hospital-specific regulations. Persistent policy-level gaps exist between the psychological evidence supporting PA during illness and its formal recognition within hospital education. The limited articulation of psychosocial dimensions may constrain the developmental and mental health potential of PA for hospitalized youth. Integrating psychological principles into policy frameworks represents a critical leverage point for supporting engagement and well-being among pediatric patients. Funding source: The “Hospital schools’ guidelines On Physical Education” project (ID: 101206417 — HOPE) is funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Research Executive Agency (REA). Neither the European Union nor the granting authority can be held responsible for them.

The Sport Injury-Impact of Event Scale: Measurement of trauma- and stressor-related disorder symptoms in response to sport injury in a sample of equestrian athletes

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Athletes experience trauma- and stressor-related disorder (TSD) symptoms post-injury. The field, however, lacks a tool supported by appropriate validity evidence to measure TSD symptoms to inform diagnosis and treatment. This study developed and explored validity evidence for a Sport Injury-Impact of Event Scale (SI-IES). Thirty-three items derived from the literature were developed and included in a symptom list. A convenience sample of equestrian athletes completed self-report measures of demographic variables, injury-related measures, and the 33-item symptom list. Most participants ($N=328$; $M_{\text{age}}=37.1$ years)

were female (88%). Participants rated their most distressing sport injury as “minor” (e.g., sprains; 11%), “moderate” (e.g., broken bones; 68%), or “serious” (e.g., spinal injury; 21%). The injury severity (IS) variable was binary coded where 0 = “minor and moderate” and 1 = “serious”. First, exploratory factor analysis (prioritizing parsimony) determined that a three-dimensional model may be best fitting for responses to the 33-item symptom list. The three dimensions were labeled as “disruption in cognition and/or mood” (F1), “intrusions” (F2), and “avoidance” (F3). Second, examination of item loadings resulted in the removal of 18 items because of low loadings (two), conceptual misfit (11), redundancy (two), and low clarity/relevance (three). The resulting SI-IES consisted of 15 items (i.e., five items in each dimension). Third, the three-dimensional SI-IES was tested (M1) using exploratory structural equation modeling with target rotation (i.e., 0 targets specified for items identified as not conceptually best-fitting to each dimension). M1 indicated evidence of close fit ($RMSEA = .039$). Construct reliability (H) for the dimensions ranged from .87-.90. Fourth, three-dimensional SI-IES was regressed on IS (M2) to examine external validity. M2 indicated evidence of close fit ($RMSEA = .036$). The direct effects of IS on F1 ($\beta_1 = .27, p < .001$), F2 ($\beta_2 = .24, p < .001$), and F3 ($\beta_3 = .29, p < .001$) provided evidence for external validity. The model explained 9-16% ($p < .01$) of the variance in the SI-IES dimensions. This exploratory study provided initial validity evidence for a 15-item, three-dimensional, SI-IES. Injury severity was positively associated with each SI-IES dimension.

Reconstructing the perception of talent through the view of scouts: An automated analysis of scouting reports with large language models

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‘Talent’ is a ubiquitous term in the field of competitive sport, as scouts are regularly required to identify and recruit promising athletes for further development. However, the definition of talent is often elusive and subjective varying over different contexts and stakeholders. Can we better understand the meaning of talent in an objective and perhaps quantifiable way? In this study, we present a novel paradigm that utilizes a state-of-the-art large language model (LLM) from artificial intelligence (AI) to investigate meanings of talent from scouting reports. As a starting point, we developed a computational pipeline that automatically extracts and interprets talent-related phrases and traits based on amateur scouting reports ($n = 839$) from the National Hockey League. The pipeline extracts structured information for reconstructing the scout’s perception of that player following three steps: 1) extraction of key phrases that describe the traits of a player (e.g., “hard working”, “big-bodied”); 2) generation of sentiment tags that identify a phrase to be strength or weakness; 3) categorization of each of the strength-based traits into one of the four talent criteria commonly discussed in the literature: physical, technical, tactical, and psychological. To our knowledge, this AI-informed high-resolution extraction method has not been explored previously in the field. With this approach, we discover clear positional patterns supported by statistical evidence from the dataset: defensemen were described more frequently with Physical and Tactical strengths, while forwards were associated with a stronger emphasis on Technical traits. We also observe a temporal trend in the data: in earlier draft years, the differences between defensemen and forwards were more pronounced, whereas in later years these positional profiles became more similar. Our study offers an efficient and scalable means to quantifying the potential meanings of talent from scouting reports, and it helps to strengthen the connection between AI and sports science in the automated analysis of athletic talent.

Comparing the influence of excellencism, perfectionism, and motivation on student-athletes’ psychological functioning

Janie Locas, Université du Québec à Trois-Rivières; Frédéric Langlois, Université du Québec à Trois-Rivières; Paule Miquelon, Université du Québec à Trois-Rivières

Student-athletes enrolled in sport-study programs face high academic and athletic demands, along with strong internal and external pressures to meet elevated performance standards. These conditions highlight the relevance of examining both the pursuit of high standards (excellencism and perfectionism) and the quality of motivation to better understand psychological functioning and inform targeted interventions. Grounded in the Model of Excellencism–Perfectionism (Gaudreau, 2019) and Self-Determination Theory (Deci et Ryan, 1985; 2000; 2017), this longitudinal study examined which predictors (motivations, excellencism, perfectionism) measured at Time 1 predict psychological adjustment (performance anxiety, life satisfaction, and symptoms of anxiety, depression, and stress) three months later (Time 2). A longitudinal correlational design was used with secondary school student-athletes enrolled in sport-study programs in Quebec. A total of 294 participants (14–17 years) completed both measurement waves. Structural equation modeling conducted separately for sport and academic contexts showed that, beyond orientations toward high standards such as excellencism and perfectionism, motivational processes play a central role in student-athletes’ psychological adjustment. These findings highlight the importance of targeting motivation when it comes to supporting student-athletes’ well-being.

The role of perfectionism and self-efficacy in athlete performance following negative feedback

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Athletes often respond differently to negative feedback, with some improving performance while others decline. Self-efficacy (SE) has been shown to predict performance following negative feedback, likely reflecting how feedback is perceived and interpreted. However, SE alone may not fully explain these differences, as changes in SE after feedback may translate into performance differently depending on an athlete’s perfectionistic profile, including the interaction between perfectionistic strivings (PS) and concerns (PC). The present study examined whether changes in SE predict performance following negative feedback and whether the interaction of PS and PC moderated this relationship. It was hypothesized that (1) higher SE would lead to better performance, (2) this effect would be stronger as PS increased and weaker as PC increased, and (3) the positive influence of PS on the relationship between SE and performance would be reduced when PC was high. 65 competitive athletes ($M_{\text{age}} = 20.1 \pm 1.77$ years) completed a beanbag-throwing accuracy task. Perfectionism dimensions (PS and PC) were assessed at baseline. Participants first completed an initial block of throws, then received experimentally manipulated negative feedback indicating poorer performance relative to competitors. SE was reassessed following feedback, and participants completed a second block of throws. Moderated linear regression analyses tested the three-way interaction between changes in SE, PS, and PC in predicting performance change following negative feedback. The moderated regression model was not significant, $F(7, 53) = .33, p = .94$, and explained a small proportion of variance in performance change ($R^2 = .04$). The three-way interaction between SE change, PS, and PC was not significant ($p = .76$), nor were any lower-order interaction effects. These findings suggest that, in this task and sample, other factors beyond SE and perfectionistic traits may drive variability in athletes’ responses to negative feedback. Future research should explore additional

psychological and situational influences to better understand why some athletes improve while others decline after receiving negative performance feedback. Funding source: Canada Graduate Scholarship – Masters.

Development of a novel on-ice speed performance paradigm to assess motor and cognitive dual-task performance in hockey players

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Sport-specificity of performance diagnostics and respective differentiation of expertise levels have recently been discussed. This highlights the need for tasks reflecting combined motor-cognitive demands of game play and allowing for individual-level assessment. Evidence indicates that concurrent task execution leads to performance decrements due to interfering processing resources. However, existing performance diagnostics in hockey insufficiently reflect on-ice dual-task (DT) demands. Aim of this study is to develop a paradigm to adequately capture performance in motor as well as cognitive measures. German male U17 hockey players ($n=16$, age: $M=15.9 \pm .7$) were assessed in cognitive single-task (ST) performance using an arrow flanker task with 24 stimuli, conducted on a computer as well as with stick-mounted buttons while standing. Motor ST performance involved speedy skating an eight-shaped course within 35 s. Lap times were measured via light barriers. In the DT condition, players responded to flanker stimuli using the stick-mounted buttons while skating. Motor performance in ST and DT was analyzed via mean lap times and cognitive performance in ST (computer and buttons) and DT conditions via stimulus congruency and correct reaction times. Repeated-measures ANOVA reveals an effect of condition on motor performance, $F(1, 15)=147.44$, $p < .001$, $\eta_p^2 = .908$, with faster lap times in ST than DT. For cognitive performance, 2×3 ANOVA shows an effect of congruency, $F(1, 15)=52.28$, $p < .001$, $\eta_p^2 = .777$, with lower reaction times for congruent than incongruent stimuli, and of condition (G-G corrected; $F(1.40, 20.98)=61.87$, $p < .001$, $\eta_p^2 = .805$) with highest reaction times in DT compared to both ST conditions. An interaction effect, $F(2, 30)=6.93$, $p = .008$, $\eta_p^2 = .316$) indicates a larger discrepancy between congruent and incongruent trials (flanker effect) in DT than in both ST conditions. Both motor and cognitive speed-related performance measures declined under DT conditions on-ice. While contributing to the development of sport-specific diagnostic and intervention tools, this paradigm will provide individualized training recommendations, reflecting both the detected type (skating, reaction time, accuracy, combination) and extent of DT-related performance decrements.

Mental toughness, sport anxiety, and principles for deep knowledge among athletes in Taiwan

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Numerous studies have demonstrated that Mental Toughness (MT) and Sport Anxiety (SA) are critical psychological factors influencing athletes' performance. The Principles for Deep Knowledge (PDK) questionnaire has shown potential as a tool for helping athletes reduce their SA. This study examined the differences by gender and competitive level, and relationships among MT, SA, and PDK in athletes from Taiwan. The study included 280 athletes ($M_{age}=19.69$ years, $SD=2.69$; 159 boys), categorized into two competitive levels (CL; Regional/University versus National /International). All participants completed the 48 item Mental Toughness Questionnaire 48 (MTQ-48, 4 subscales), the 21 item Sport Anxiety Scale (SAS, 3 subscales), and the 45 item Principles for Deep Knowledge (PDK, 9 subscales)

Questionnaires. Pearson correlation analysis revealed that the 9 subscales of the PDK were significantly positively correlated with the 4 subscales of the MTQ-48. Additionally, the 7 subscales of the PDK showed significant negative correlations with 2-3 subscales of the SAS. Four linear regression analyses used the 9 subscales of the PDK to predict each of the 4 subscales of the MTQ-48 (challenge, commitment, control, confidence). Specifically, the subscales of the PDK, Revolving & Cycling, Changing, and Cracking & Rippling principles explained 22% ($p < .001$) of the variance in Challenge. Deep & Down, Alternating, Revolving & Cycling, and Inner Connection ($p < .001$) accounted for 50% of the variance in Commitment and Revolving & Cycling and Changing predicted 27% ($p < .001$) of the variance in Control. Finally, Revolving & Cycling, Changing, and Cracking & Rippling explained 30% ($p < .001$) of the variance in Confidence. Three (SAS, MTQ-48, PDK) 2 Competitive Level X 2 Gender MANOVAs investigating CL and gender differences, revealed significant main effects for the SAS and MTQ-48. For the SAS, there was a significant ($p < .01$) main effect for CL but not Gender or the interaction. For the PDK, neither CL/Gender were significant. For the MTQ-48 there were significant main effects for CL ($p < .05$) and Gender ($p < .001$). These data have implications to working with athletes at all levels.

Mental health and psychological safety in ice hockey: A comparative study with team sports

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Prevailing norms in ice hockey that valorize toughness may undermine psychological safety and, in turn, athletes' psychological well-being. Accordingly, there appears to be a need for studies that examine the mental health of ice hockey players and compare them with other team-sport contexts. The objectives of this study were to: (1) compare overall mental health scores between adolescent ice hockey players and adolescents competing in other team-sports; (2) compare self-reported exposure to physical violence; and ; (3) compare fear of stigmatization as a barrier to seeking mental health support, including a male-only subgroup analysis given sex imbalances between groups. Participants were 69 ice hockey players ($n=46$ males $M_{age}=17.91$ [$SD=1.78$]) and 57 non-hockey team-sport athletes ($n=20$ males $M_{age}=17.6$ [$SD=1.92$]). Athletes completed self-report measures of overall mental health, hedonic well-being, exposure to physical violence, and fear of stigmatization related to seeking mental health support. Group differences were tested using Welch independent-samples t -tests. Ice hockey players reported higher overall mental health ($M=53.08$, $SD=10.36$) and hedonic well-being ($M=12.23$, $SD=2.31$) than other team-sport athletes ($M=45.40$, $SD=14.27$; $M=10.08$, $SD=3.35$), $t(92.7)=3.27$, $p = .002$, $d = .63$, and $t(89.66)=3.97$, $p < .001$, $d = .76$, respectively. Ice hockey players also lived more physical violence ($M=.40$, $SD=.73$) than the other team-sport athletes ($M=.32$, $SD=.33$), $t(90.41)=1.68$, $p = .096$, $d = .32$. When restricting analysis to male athletes, ice hockey players exhibited greater fear of stigmatization than athletes in other team-sports ($M=1.31$, $SD=1.59$ vs. $M=.29$, $SD=.47$), $t(44.31)=3.50$, $p = .001$, $d = .76$. Despite higher stigma concerns and higher exposure to physical violence, ice hockey players reported more favorable mental health indicators. These findings add evidence to an underrepresented population and highlight the importance of considering sport-specific norms when interpreting mental health and help-seeking among youth athletes.

Team Cohesion Inventory for Sport (TCIS): Content and substantive validity evidence

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Cohesion is a dynamic emergent state “reflected in the tendency for a group to stick together and remain united in the pursuit of its goals and objectives” (Carron, 1982, *p.* 124). Research on cohesion has a rich history in sport psychology, largely informed by the use of the Group Environment Questionnaire (Carron et al., 1985). However, in light of construct validity concerns, Eys and Beauchamp (2025) argued for a theoretical and operational realignment of the cohesion construct. The goal of this presentation is to introduce a preliminary version of the Team Cohesion Inventory for Sport (TCIS) by communicating the first three stages of instrument development and corresponding content and substantive validity evidence. In project 1, sequential item generation procedures were followed that drew directly from the revised conceptual model, yielding five progressively modified versions of the TCIS. In project 2, feedback was obtained from group dynamics experts. In total, 15 participants (10 faculty professionals, 3 doctoral students, and 2 masters students) provided structured qualitative (e.g., wording suggestions for items and instructions) and quantitative (e.g., match ratings to targeted dimensions) feedback across three modified versions of the new inventory. Finally, in project 3, cognitive interviewing (via a think-aloud protocol) with 20 athletes (12 female; basketball, rugby, cross-country running, track and field; youth and university level athletes) provided user feedback on the relevance, representativeness, framing/wording, and comprehension of all aspects of the TCIS (e.g., instructions, opening questions, items). Following these initial stages of instrument development, the TCIS consists of two parts. In part 1, participants identify, via open-ended questions, their team’s (a) task goals and (b) activities related to developing and maintaining relationships. In part 2, in addition to two response acquiescence detection questions, participants respond to 8 items designed to estimate their perceptions of (a) task integration (4 items) and (b) social integration (4 items). Overall, strong content and substantive validity evidence is offered regarding the initial development of the TCIS. Future research will examine structural, external, generalizability, and consequential validity evidence related to scores derived from this instrument.

A self-compassion physical education unit promotes high school students’ psychosocial development

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Research shows that self-compassion is an effective psychological resource for enhancing well-being and overcoming setbacks (Neff, 2023). Despite the burgeoning literature on self-compassion in youth, there is a lack of studies, especially interventions, on self-compassion in physical activity settings. This mixed-method study aimed to implement and test an empirically based self-compassion unit in high school physical education (PE). Participants were 42 ninth-grade students (10 boys, 31 girls, 1 preferred not to answer) enrolled in high school PE in the southeastern U.S. Participants completed an evidence-based six-session self-compassion unit (Bluth et al., 2016), two sessions a week, taught by their PE teacher. The sessions consisted of meditations, written exercises, and active mindfulness activities, modified to a PE context. Before and after the intervention, participants completed a survey with validated quantitative measures and qualitative questions about self-compassion (mindfulness, common humanity, and self-kindness), self-criticism, physical activity enjoyment, and their likes and dislikes of the unit. Pre-post mean comparisons indicated positive trends in common humanity, self-kindness, and physical activity enjoyment ($M_{diff} = .22$ to $.62$), as well as negative trends in self-criticism, overidentification, isolation, and self-judgment ($M_{diff} = -.06$ to $-.31$). Qualitative thematic analysis (Braun & Clark, 2019) revealed that most participants enjoyed mindfulness activities, especially the active ones, such as mindful walking and yoga. Participants also

enjoyed having the time to focus on themselves and learn to be kinder to themselves and others. Meanwhile, participants reported disliking the sitting time or meditations that lasted too long. Their perspectives on sports activities and group work were mixed. Some reported these activities as their favorite parts of the unit due to their competitiveness; others mentioned disliking them, given the negative feedback (i.e., a lack of compassion) from their classmates. Consistent with Bluth et al.’s (2016) work with youth in educational settings, the study findings suggest that a self-compassion PE unit has the potential to improve high school students’ physical and psychosocial well-being. Moreover, applying compassion to others beyond oneself needs to be a stronger emphasis in self-compassion interventions, especially when competitions and group work are involved (Backman et al., 2025).

Emotional responses to motivational climate feedback during a ROTC shooting task

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Reserve Officers’ Training Corps (ROTC) is a military-issued, university-based program that prepares students (i.e., cadets) to become commissioned officers while meeting demanding academic, physical, and psychological training standards. Cadets operate in high-performance environments in which motivational climate feedback plays a central role in shaping motivation, emotional responses, and performance. Sport-related research has demonstrated a caring and task-involving climate (CTI: valuing individuals, emphasizing effort, personal improvement, and skill mastery) fosters a more adaptive emotional and motivational climate than ego-involving climate (emphasizing comparison and outperforming others/norms). Research in military contexts is limited. Sixty-four cadets completed a simulated shooting task at baseline with no feedback and then returned for a second simulation while receiving either task-involving feedback or ego-involving feedback. Cadets reported their levels of enjoyment and shame after the baseline and feedback condition. It was hypothesized that cadets in the task-involving climate condition would report greater increases in enjoyment and smaller increase in shame compared to cadets in the ego-involving condition. Overall, correlations were in the anticipated directions. CTI climates were correlated positively ($r = .91$) and with the ego-involving climate negatively ($r_{care} = -.27$; $r_{task} = -.21$). Cadets’ caring and task-involving climate perceptions were significantly, positively correlated with enjoyment ($r_{care} = .49$; $r_{task} = .45$) and negatively with shame ($r_{care} = -.24$; $r_{task} = -.24$). Ego-involving perceptions were not significantly associated with enjoyment or shame. Lastly, changes in shame and enjoyment were significantly, negatively correlated ($r = -.24$). The hypotheses were tested with independent-samples, one-tailed *t*-tests of cadets’ changes in enjoyment and shame between feedback conditions. Cadets who received CTI feedback showed significantly greater increases in enjoyment ($M_{diff} = .86$) and smaller increases in shame ($M_{diff} = .91$) compared to cadets who received ego-involving feedback: minimal increase in enjoyment ($M_{diff} = .21$) and greater increases in shame ($M_{diff} = 1.15$). These differences were associated with moderate effect sizes for both enjoyment ($d = .63$) and shame ($d = -.48$). These findings demonstrate motivational climates meaningfully influenced cadets’ emotional experiences during a military task. This, as with sport athletes, tactical athletes respond more adaptively in a CTI than ego-involving climate. Funding source: AASP Research Grant.

Motivation in numbers: A test of social facilitation and loafing theories

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Various theories on teams behavior have emerged to explain why and under what circumstances individual effort will fluctuate. These theories largely fall under categories of social facilitation effects (i.e., effort gains) and social loafing effects (i.e., effort losses). The Köhler effect, for example, explains motivation gains within less capable/experienced members of a team that are indispensable. As another example, the Ringelmann effect explains motivation losses; the effort of a group member can decrease as the group size increases. Although these effects are often examined in isolation, they are likely present in practical settings. In the military, team size and tactical athletes' ability/rank may vary depending on the mission. Yet it is unclear what might happen to individual effort under these various circumstances. Therefore, the purpose of this study was to examine differences in individual effort across (a) 2- and 4-person teams and (b) individual rank to understand personal effort in a team task. Members of a university-level Army Reserve Officers' Training Corps (ROTC) program, 80 cadets (55 M, 25 F) completed a 5-s max effort tug-of-war style task in 2- and 4-person teams. These teams included ranks based on ROTC hierarchy that were essentially parallel to one senior (who was also designated as the leader), junior, sophomore, and freshman. Effort was measured using 3 electromyography sensors affixed on the erector spinae, forearm, and quadriceps of each person. Based on each participant's maximum value, we calculated the percent effort of their max for each pull to compare values across participants of varying body size. A 2 (team size) by 4 (rank) RM-ANOVA indicated effort was significantly higher in the 4-person groups, $M = 70.6\%$ of personal max, compared to the 2-person groups, $M = 64.8\%$ of personal max; $F(19, 1) = 9.99$, $p = .005$, $\eta_p^2 = .35$. No significant differences were observed for rank or the interaction between rank and team size. Results support social facilitation effects as group size increases in this ROTC setting. The findings of this study can guide best practices for the quick assembly of teams in the military. Funding source: U.S. Army Research Institute for the Behavioral and Social Sciences.

Mental health and help-seeking in international student-athletes: A scoping review

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International student-athletes (ISAs) in U.S. collegiate sport face unique stressors related to migration, cultural adjustment, and competitive demands, placing them at elevated risk for mental health challenges. Despite their growing presence, they remain underrepresented in sport and mental health research. The purpose of this scoping review was to map and synthesize research examining mental health and help-seeking experiences ISAs in U.S. collegiate sport contexts, and to identify key gaps and directions for future research. Guided by PRISMA-ScR, a database search identified 19 studies for thematic analysis. Although few studies directly assessed mental health outcomes, many reported stressors such as loneliness, discrimination, and academic pressure. Social support, particularly from teammates and other fellow international students, was the most frequently reported coping strategy. Some relied on self-endurance or emotional suppression, reflecting stigma and cultural norms that may discourage help-seeking. Limited attention to formal mental health services underscores gaps in research, as well as the need for culturally responsive mental health programming.

Evaluating associations between physical activity-related health competence and indices of physical fitness

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Physical fitness (PF) is an established predictor of health and well-being, which can be improved through participation in sufficient amounts of physical activity (PA). Contemporary scholars developed the physical activity-related health competence (PAHCO) model, outlining three sub-competencies: (i) control competence for physical load and for affect-regulation; (ii) self-regulation competence; and (iii) movement competence that support individuals in integrating health-enhancing PA in everyday life. Researchers have demonstrated associations between PAHCO scores and amount of leisure-time PA and self-reported physical health, however the degree to which PAHCO relates to PF has not been addressed. The purpose of this study was to assess how PAHCO relates to PF assessment outcomes. This secondary analysis involved data from a parent study wherein university students ($N = 214$, $M_{age} = 21 \pm 2$ years, 58% female, 92% non-Hispanic White) completed the PAHCO questionnaire as part of their baseline assessment followed by a battery of motor tasks and assessments related to muscular strength (handgrip dynamometer), flexibility (sit-and-reach test), and body composition (body mass index). Using a subset of PAHCO questions (PAHCO-12), an overall composite PAHCO score, and individual competency scores were calculated. Analyses included basic descriptives, Pearson's correlation, and linear regression. Effect sizes were interpreted following guidelines by Funder and Ozer: .05 very small; .10 small; .20 medium; .30 large, and .40 very large. Significance was set at $p < .05$. Scores were relatively high regarding control (14 ± 3 , range: 0-18), self-regulation (11 ± 2 , range: 0-14), and movement (13 ± 2 , range: 0-16) competencies. Compared to the relationship to sit-and-reach scores ($r_s = -.02$ to $.04$) and body mass index values ($r_s = .04$ to $.10$), stronger relation was observed for handgrip strength with the composite PAHCO score ($r = .14$, $p = .04$), control competence ($r = .11$, $p = .10$), self-regulation competence ($r = .12$, $p = .08$), and movement competence ($r = .14$, $p = .04$). The composite PAHCO score explained 1.5% of the variance in handgrip strength ($R^2 = .015$, $F(1, 212) = 4.23$, $p = .04$). The relatively weak relationship may be attributed to generally high competency scores in the sample population. Additional research is needed to evaluate how PAHCO relates to all domains regarding health-related PF (e.g., aerobic endurance, muscular endurance, lean body mass) and skill-related PF (e.g., power, speed, agility) in more diverse populations.

The relative importance of intolerance of uncertainty, perfectionism, and excellencism as predictors of competitive anxiety in elite hockey players

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Performance anxiety in sport, commonly referred to as competitive anxiety, is an aversive emotional response to perceived threat occurring before, during, or after athletic competition. Competitive anxiety is known to impair performance, reduce psychological well-being, and diminish enjoyment in sport, particularly among elite athletes. Despite its importance, limited research has examined the psychological mechanisms that contribute to competitive anxiety. The present study investigated how three psychological processes associated with anxiety, including intolerance of uncertainty, perfectionism, and excellencism, predict competitive anxiety, and examined whether self-determined motivation moderates these associations. Elite ice hockey players ($N = 75$) from the M18AAA Development League and the Quebec Major Junior Hockey League (QMJHL) participated in a cross-sectional protocol and completed validated self-

report measures assessing intolerance of uncertainty, perfectionism, excellence, and competitive anxiety. Multivariate regression analyses indicated that intolerance of uncertainty was the strongest predictor of both general anxiety ($\beta = .55, p < .001$) and competitive anxiety ($\beta = .50, p < .001$). When motivation was included as a moderator, the association between intolerance of uncertainty and competitive anxiety remained robust among athletes with low self-determination ($\beta = .62, p < .001$), but was non-significant among athletes with high self-determination ($\beta = .03, p = .874$). These findings underscore the central role of intolerance of uncertainty in competitive anxiety and highlight the protective influence of self-determined motivation. Implications for intervention are discussed, including the potential value of cognitive-behavioral strategies aimed at increasing tolerance of uncertainty and fostering self-determination among elite athletes.

Life skills in sport and their transfer across contexts: A cross-cultural comparison of adolescents in Botswana and the United States

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Sport participation is widely associated with positive youth development (PYD), particularly through the development of life skills such as leadership, goal setting, and self-regulation. PYD and life skills transfer frameworks emphasize that these outcomes are most meaningful when skills learned in sport are intentionally transferred to other life domains, including school, home, and community contexts. Extant research suggests that such transfer is shaped by social, structural, and cultural features of youth sport systems; however, limited empirical work has examined how national sport ecologies influence life skill development and transfer across cultural contexts. The purpose of this study was to compare life skills developed through sport and their transfer to non-sport contexts among adolescent sport participants in Michigan, United States ($n = 160$), and Botswana ($n = 46$). Multivariate analyses of variance, adjusted for unequal sample sizes, indicated that older adolescents reported greater life skill development across the overall sample, consistent with the PYD framework. Location emerged as the strongest predictor of life skills development, with adolescents in Botswana reporting significantly higher life skills across all domains than those in Michigan. These differences may be partially attributable to Botswana's national policies that emphasize broad-based, publicly supported youth sport participation and development. Despite these differences, the transfer of life skills to non-sport contexts did not differ significantly by location. This suggests that skill acquisition alone may be insufficient for transfer without intentional support. In the United States, youth sport environments often involve higher levels of competition, specialization, and privatization, which may shape the ways developmental experiences are structured and emphasized. In contrast, Botswana's more inclusive sport context may facilitate life skill development, while still requiring deliberate strategies to promote transfer beyond sport. These findings highlight the need for theory-led, culturally and contextually responsive approaches that intentionally support life skills transfer across youth sport systems and inform program design and policy.

Are children using the Feeling Scale differently than adults during playful exercise?

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Affective valence experienced during exercise can be theorized as a central building block of more global or summary affective exercise experiences

and, in turn, as a key determinant of future exercise participation. The Feeling Scale (FS) is the most widely used instrument to assess in-task affective valence; however, its empirical validation is largely based on adult samples. A formal scoping review conducted during the initial phase of the project revealed that the FS has only rarely been used with children and that little is known about its validity in youth populations. Against this backdrop, the Move with Joy (MoJo) intervention examined children's in-task affective responses during playful exercise as delivered through the Joy of Moving™ approach. Across three consecutively implemented weekly after-school game afternoons in 2025, more than 140 children participated in structured 60-min playful exercise sessions conducted in small groups of 5 to 25 children. In-task affective valence was repeatedly assessed using the FS, while exercise intensity was continuously recorded throughout the sessions via wearable Firstbeat Bodyguard 3 sensors, providing heart rate- and accelerometry-based intensity profiles across the entire game afternoons. More trait-like affective exercise experiences were assessed using the AFFEXX-kids questionnaire. Preliminary analyses are based on $N = 146$ children (8.67 ± 1.11 years; 53% female), yielding 2,118 in-task affect ratings. Linear mixed-effects models showed no association between objectively assessed exercise intensity and in-task affective valence ($b = 2.13, 95\% \text{ CI } [-.52, 4.78]$). In contrast, children's pre-activity affective state emerged as a robust predictor of how they reported feeling during playful exercise ($b = .24, 95\% \text{ CI } [.15, .34]$). The absence of the typically observed intensity-affect relationship challenges a core auxiliary assumption underlying FS use and suggests that, in playful exercise contexts, children may anchor FS ratings primarily in immediate emotional meaning (e.g., fun, enjoyment, social experience) rather than in intensity-related pleasure-displeasure. These findings highlight the need for a developmentally sensitive interpretation of in-task affect measures and call for a reconsideration of how affective valence during exercise is theoretically framed and methodologically assessed in childhood.

Exploring the role of cumulative childhood environmental stress in adolescent sport participation

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Low youth sport participation rates and high drop-out rates are public health problems in the U.S. Modifying sport systems themselves would be worthwhile, yet it is also important to understand how factors beyond sport affect children's likelihood of participation. Environmental stressors (e.g., household poverty; low neighborhood opportunity; exposure to negative life events) could impact sport participation. Specifically, high cumulative environmental stress over the course of childhood could set youth on a trajectory for not participating in adolescence and beyond. This secondary data analysis examined the relationship between cumulative childhood environmental stress and the odds of playing a sport in adolescence. We used data from the NET-Works 2 cohort study ($N = 534$; 64% Hispanic; household income $< \$65,000$ at baseline). Environmental stressors including family and neighborhood factors representing disadvantage were measured over 5 timepoints, beginning when children were ages 2–4 years and continuing through ages 7–11 years. The number of stressors present at each study visit were summed across the 5 timepoints to obtain a measure of cumulative environmental stress. The primary outcome was participation in an individual or team sport when children were ages 11–14 years based on parent report, which was available for 243 children. We fit logistic regression models to test whether cumulative environmental stress was associated with sport participation, stratified by assigned sex. Higher cumulative environmental stress predicted lower odds of sport participation for female adolescents ($OR = .94$ per additional stressor present at a timepoint, $95\% \text{ CI } = .90, .99, p = .02$), but not male adolescents ($OR = .99, 95\% \text{ CI } = .94, 1.04, p = .70$). Scholars studying youth sport

should consider measurement approaches capturing family stress holistically to better understand participation barriers for female children and adolescents. Policymakers may want to recognize youth sport does not exist in a vacuum; effective policies limiting family stress early and across development, such as those that address household stress and disadvantage (e.g., poverty; food insecurity), may support youths' engagement in sport and health behaviors.

An intersectional approach to quantifying sport and physical activity participation among Hmong American adolescents

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Sports have been important to Hmong culture since before Hmong communities arrived in the U.S. Yet, in limited studies describing sport or physical activity (PA) in Hmong American samples, engagement is low. Rates among Hmong American youth are rarely quantified. Though understudied, sport and PA behaviors may vary by social identities within this population. Prior research also suggests structural and interpersonal racism impact Hmong Americans' health behaviors; these may be important to consider alongside social identities. We quantitatively tested whether social identities and indicators of racism were associated with Hmong American adolescents' sport and PA behaviors. Hmong American adolescents ($N=1,897$) in middle and high school self-reported social identities of ethnicity; race; body mass index; assigned sex; and grade in school via a student survey in 2022. They also reported past-month experiences of race-based bullying (RBB) and socioeconomic status (SES; i.e., indicators of interpersonal and structural racism). Dependent variables included any sport participation in a typical week and days of at least 60 min of PA in the past week. We used Exhaustive Chi-Square Automatic Interaction Detections to identify groups of Hmong American adolescents with highest and lowest sport participation prevalence and highest and lowest average PA by social identities and experiences of racism. Overall prevalence of sport participation of Hmong American adolescents was 25%. Average PA was 3 days. Hmong American adolescents who were Multiracial had the highest sport participation prevalence (41%). Other high prevalence groups included Asian-only Hmong adolescents with high SES (31%) or with low-or-medium SES and who reported RBB (29%). All low sport participation prevalence groups included Asian-only Hmong adolescents who reported no RBB ($< 25\%$). Mean PA was highest in Multiracial male Hmong adolescents (4.5 days); Asian-only male Hmong adolescents; and female Hmong adolescents with high SES. PA was lowest for female Hmong adolescents with lower SES in all grades. Variability in sport and PA behaviors exists within the Hmong American community by social identities and experiences of racism. Sport and PA systems may need to adjust to meet Hmong American adolescents' needs and capitalize on sport and PA's potential to support their health. Funding source: Funding provided by the National Institute on Minority Health and Health Disparities of the National Institutes of Health under Award Number P50MD017342.

The impact of an exercise intervention and APOE-ε4 on subjective cognitive decline in midlife adults at risk for Alzheimer's disease: The Physical Activity and Alzheimer's Disease-2 (PAAD-2) Trial

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A person's perception of declines in cognitive abilities, subjective cognitive decline (SCD), is a risk factor for Alzheimer's disease (AD) and has potential as a pre-clinical indicant of responsiveness to behavioral interventions. In the PAAD-2 clinical trial (NCT03876314; NIH R01AG058919), we examined effects of a one-year exercise intervention on SCD in 180 cognitively-unimpaired low physically-active middle-aged adults with a family history of AD (FH+). Participants were randomized to exercise (strength and aerobic training 3x/week) or usual care control groups after pre-testing, and tested again at mid-test (6-months) and post-test (12-months). SCD data from the Everyday Cognition-12 scale (Ecog, lower score = better) were analyzed for 123 participants ($M_{age} = 56.31 \pm .56$, 87% females; 91% white). APOE-ε4 alleles categorized participants as non-carriers (0 ε4 copies, lowest risk), heterozygotes (1 ε4 copy, moderate risk), or homozygotes (2 ε4 copies, highest risk) to test for moderation by this genetic risk factor for AD. Mixed analyses of covariance assessed the effects of time, group, APOE-ε4, and their interactions on Ecog after controlling for pre-test Ecog, age, and education. Pairwise comparisons for time with Bonferroni adjustments were used to decompose significant interactions. The Time X Group X APOE-ε4 interaction was significant, $F(2, 112) = 3.97$, $p < .03$, and examination of the means showed that Ecog in the control group worsened for all carrier groups (non-carriers: $\Delta = .02$; heterozygotes: $\Delta = .03$) and worsened significantly ($p < .001$) from mid-test ($M = .94$, $SE = .14$) to post-test ($M = 1.60$, $SE = .17$) for homozygotes ($\Delta = .65$). Conversely, Ecog for exercisers improved for all carrier groups (heterozygotes: $\Delta = -.02$; homozygotes: $\Delta = -.06$) and trended toward a significant improvement ($p < .10$) for non-carriers ($\Delta = -.08$) from mid-test ($M = 1.35$, $SE = .04$) to post-test ($M = 1.27$, $SE = .05$). These findings demonstrate worsening of SCD for cognitively unimpaired middle-aged ε4 homozygotes in the control group while showing potential benefits of exercise irrespective of genetic risk for AD. The magnitude of change may be clinically significant given that an Ecog score of 1.36 is suggested as the optimal cut-point that differentiates cognitively impaired from cognitively unimpaired individuals. Given that participants are middle-aged with a FH+, these potential benefits from exercise may be particularly important. Funding source: NIH R01AG058919.

Group norms and social identity in recreational running

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The performance hypothesis within social identity theory describes the way in which sport performance may be shaped by way of the norms, values, resources, and goals associated with salient social identities (Haslam et al., 2020). Yet, no studies to date have explicitly measured or tested group norms in the social identity and performance relationship (Stephen et al., 2023). The purpose of this study was to identify the group norms perceived by runners and assess the extent to which social identification as a runner predicted strength of adherence to group norms. A total of 675 participants ($M_{age} = 39.02$) were asked to list "the norms that are associated with running and which runners adhere to" across three time-points, 4-weeks apart. Participants were also rated on a scale of 1 (strongly disagree) to 7 (strongly agree) if they made "every effort to conform to the norms mentioned above" and their social identification as a runner using the social identity questionnaire for sport (Benson & Bruner, 2018). A total

of 4322 norms were provided. The most frequently identified norms were the use of performance-tracking applications such as Strava and Garmin Connect (56%); owning or wearing the latest high-quality gear (40%); socializing or engaging in conversations with other runners (40%); displaying race shirts or medals (33%); engaging with running-related social media content (28%); adhering to “good” nutritional practices (10%); and setting performance goals (4%). With respect to the relationship between social identification and adherence to group norms, social identity as a runner at Time 1 significantly and positively predicted adherence to group norms at Time 2 ($b = .33, p < .001$). Similarly, social identity at Time 2 positively predicted adherence to group norms at Time 3 ($b = .37, p < .001$). Findings underscore the importance of recognizing the social norms that arise within running groups. As social identity is a key predictor of the norms that guide group behavior, clarifying these identity-driven processes is crucial for practitioners and researchers aiming to enhance group experiences and performance. Future work should investigate the relationship between social identity, group norms, and performance.

Interpersonal violence in organized sport and psychosocial adjustment problems: Results from a population-based sample of Quebec adolescents

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Interpersonal violence (IV) in sport has been associated with psychosocial adjustment problems (PAP) including anxiety, depression, eating disorders, and substance use (Tuakli-Wosornu et al., 2024). However, few studies have controlled for sociodemographic confounders or distinguished IV types and perpetrators, limiting understanding of these associations and the generalizability of the results, especially among adolescents. This study aimed to 1) examine whether PAP (anxiety, depression, binge drinking, drug use, eating disorders, and peer social problems) are associated with lifetime IV exposure in sport; and (2) explore if types of IV or perpetrators are associated with these problems. Data were derived from the ÉVAQ sub-study embedded within the 2022–2023 Québec Health Survey of High School Students. The analytic sample comprised 9120 (8369 weighted) adolescents aged 14–17 years engaged in organized sport who completed a short form of the Violence Toward Athletes Questionnaire (Parent et al., 2019). Participants (54.5% identifying as girls, 58.9% team sport) also completed sociodemographic questions, GAD-2, PHQ-2, EDE-Q7, and questions about peer social problems, alcohol use and lifetime drug use. Bivariate followed by multivariable analyses were performed, controlling for age, sex at birth, sexual orientation, place of birth, income, competitive vs. non-competitive sport, relocation for sport, and type of sport (individual/team). Overall, 51.1% of participants reported at least one instance of IV in sports by peers and 36.7% by authority figures. Objective 1: Peer and authority IV were significantly associated with an increased odds of reporting symptoms of anxiety, depression, binge drinking, and drug use as well as higher scores for eating disorder symptoms and peer social problems. Associations were strongest among adolescents who experienced both peer and authority IV. Objective 2: Among the types of IV considered, sexual abuse and harassment, psychological, and physical IV by peers and authority figures were generally significantly associated with increased odds and scores of these PAP. Experiencing IV in sport is associated with a broad range of PAP in adolescents. Future studies should clarify temporal associations between

IV and these problems, as bidirectional influences may contribute to their persistence and inform targeted prevention strategies.

The mental block paradox: High self-reported incidence, low conceptual fit

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Mental blocks are a phenomenon in which an athlete’s mind prevents them from performing a skill that was previously well-learned and automatic (Lawrence, 2016; Marsollier & Hauw, 2024). Paradoxically, in a recent study 82% of gymnasts report experiencing something like a mental block, but only 6 percent fit the established criteria of a true mental block. Related experiences like choking, slumping, and a fear of injury were far more prevalent (Maaranen et al., 2017). It is unclear if the classification of symptoms of a mental block is too rigid, or if athletes misuse the term in describing their experience. Therefore, the purpose of this study was to assess the incidence of mental blocks in cheer and stunt athletes using previously established methods and an exploratory cluster analysis. 105 participants (92 Female) indicated prior experience consistent with a mental block and completed an online survey adapted from Maaranen et al. (2017). The survey asked for details of their symptoms and experience with mental blocks that were previously used to distinguish a mental block from choking, slumping, and fear of injury. In our sample only one participant experienced a true mental block. Another 19 were considered choking, 13 fit slumping, and 84 reported fear of injury. To supplement these findings, we completed a 2-step cluster analysis to identify unique groupings of participant experiences in our sample. Results supported a model with four clusters with 20%, 16.2%, 43.8%, and 20% of participants fitting in each cluster. Questions that best distinguished clusters from one another were whether skill loss was sudden or gradual, whether they experienced fear, and their ability to perform the skill with and/or without pressure. These four clusters represent unique groupings of symptoms relative to the criteria proposed by Maaranen and colleagues (2017). Our findings suggest that what athletes perceive as a “mental block” may represent a more diverse set of experiences than what has been identified in prior literature. Future work with both quantitative and qualitative methods should clarify the nuanced experiences of athletes with mental blocks to identify best strategies to overcome this debilitating experience.

Interpreting performance setbacks as stable: Implications for emotion regulation in competitive athletes

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Athletes commonly encounter performance setbacks in competitive sport as they are required to navigate various demands such as performance pressures, uncertainty, and uncontrollable environmental conditions. The attributions, or causes, athletes ascribe to their setbacks may impact how they regulate their emotions after setbacks. Furthermore, how athletes engage in emotion regulation (ER, i.e., the ways they increase, decrease, or maintain their expressions, physiology, and subjective experiences of emotion) can play a role in setback management. However, attribution theory and emotion regulation are rarely examined together in sport research, particularly with respect to how attributional dimensions predict subsequent ER. Thus, we examined whether attributional dimensions (locus of causality, stability, and controllability) for explaining a performance setback predicted athletes’ reported ER strategies 5-weeks later. Since controllability is linked to effort and strategy use, and stability to

expectancy of success, we selected five ER strategies a priori as primary outcomes: positive reappraisal, refocus on planning, perspective taking, rumination, and catastrophizing. In addition, we tested four strategies (acceptance, self-blame, other-blame, positive refocusing) in exploratory analyses. Our sample consisted of 105 competitive athletes ($M_{\text{age}} = 27.1$; 48 women, 55 men, 2 did not report) dealing with a recent setback. We measured athletes' attributions for their sport setback at Time 1 and their ER at Time 2 (5-weeks later; Garnefski et al., 2001). Sixteen athletes reported no longer dealing with the setback at T2 and were excluded from analysis. We conducted multiple regression tests with attributional dimensions as predictors of ER and covarying for age. Setback attributions perceived as stable were positively associated with two ER strategies: catastrophizing ($\beta = .38, p < .001, 95\% \text{ CI} [.13, .46]$) and other-blame ($\beta = .31, p < .001, 95\% \text{ CI} [.08, .40]$). The locus of causality and controllability of attributions did not predict any ER strategies at follow-up. Our findings suggest that interpreting sport setbacks as enduring (i.e., stable) is linked to greater use of ineffective ER strategies, such as catastrophizing and blaming others. Targeting stable setback interpretations represents a promising avenue for attribution-based interventions to address unhelpful ER responses following performance setbacks in sport.

Making sense of mentorship across the athletic careers of high-performance women athletes

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Mentorship is showing promise for facilitating various positive outcomes for girls and women in sport (e.g., increased sport participation). Multiple figures, such as coaches and teammates, serve in mentoring roles at different points throughout an athlete's career. However, limited research explores how women athletes make sense of mentorship in sport settings and how their experiences with mentorship change over time. The purpose of this study was to understand high-performance women athletes' experiences with mentorship across their athletic careers and major sport milestones. Using an integrated qualitative approach, timeline and semi-structured interviews were used to elicit perspectives from 13 current or former Canadian high-performance women athletes ($M_{\text{age}} = 29.7$ years). Participants first engaged in a timeline interview ($M_{\text{length}} = 35.9$ min) where they co-constructed a visual timeline with the lead researcher by detailing their athletic journeys and identifying various mentor and non-mentor figures present within major sport milestones (e.g., entry into sport, injury). In the follow-up semi-structured interview ($M_{\text{length}} = 45.8$ min), participants reflected on their timelines to further unpack what mentorship means to them, their experiences with it, and recommendations to improve mentorship experiences for girls and women in sport. Interviews were analyzed using interpretative phenomenological analysis, which resulted in four themes: (a) the who and why of mentorship for women athletes; (b) mentorship as a continual, and sometimes retroactive, evolution of influence; (c) mentors as transitional anchors in sport and life; and (d) mentorship as a process of opportunity shaped by access rather than availability. Theoretically, this research advances a phenomenological understanding of mentorship for girls and women in sport, an area currently lacking within the sport literature. Practically, findings offer applied, gender-responsive insight to guide informal and formal mentorship structures across athlete development pathways for girls and women in sport. Funding source: Social Sciences and Humanities Research Council; Sport Participation Research Initiative.

A preliminary examination of how physical activity-related competencies and dichotomous thinking relate to self-reported physical activity

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Regular performance of moderate-to-vigorous physical activity (PA) confers numerous health benefits, warranting the identification of determinants that enable or thwart behavioral engagement. Within the last decade, scholars have developed the physical activity-related health competence (PAHCO) model, wherein integration of PA in everyday life is supported by strengthening three sub-competencies: (i) movement competence; (ii) self-regulation competence; and (iii) workload control competence for physical load and affect-regulation. The ability to volitionally engage in suitable forms of PA over time may be inhibited by dichotomous thinking (DT) – a cognitive distortion in which individuals evaluate behavior in extremes (e.g., good or bad, success or failure). To date, however, empirical examinations of such relationships have not been conducted. The purpose of this study was to evaluate associations between PAHCO, DT, and PA. A secondary analysis was conducted using survey data from adults ($N = 132$; $M_{\text{age}} = 32.8 \pm 13.6$ years; 75.8% female; 83.2% White, non-Hispanic). Validated instruments were used to measure self-reported PA (International Physical Activity Questionnaire), PA-related health sub-competencies, and an overall composite score (PAHCO-12 Questionnaire). Specific items from an existing DT instrument (Dichotomous Thinking in Eating Disorder Scale) were modified using exercise-related language to compute domain scores for general DT and exercise DT. Statistical analyses were conducted using basic descriptives and Spearman's rank correlation procedures. Effect sizes were interpreted using guidelines by Funder and Ozer: .05 very small; .10 small; .20 medium; .30 large, and .40 very large. Significance was set at $p < .05$. Time spent in vigorous PA was associated with competencies for movement ($\rho = .27, p = .002$), control ($\rho = .18, p = .044$), and self-regulation ($\rho = .30, p < .001$), as well as the PAHCO composite score ($\rho = .27, p = .002$). The composite PAHCO score was inversely related to general DT ($\rho = -.20, p = .024$) as well as exercise DT ($\rho = -.19, p = .02$). These findings warrant follow-up research to assess reciprocal effects (i.e., impact of performing vigorous PA on PAHCO versus the impact of strengthening PAHCO on the volitional performance of vigorous PA). Additionally, it would be useful to develop a specific instrument for assessing dichotomous thinking related to PA and exercise as a cognition of interest for behavioral interventions.

Youth interest in sport and other life domains: A scoping review

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Interest is a motivational state in which one is attracted and/or dedicated to an activity or object. Interest research originated in education domains and revealed that interest can be both situationally manipulated and cultivated over time to improve attention, learning, and engagement. However, interest research has very minimally been translated to sport literature which may be important for improving the rates of dropout and burnout in youth sport. The purpose of this scoping review was to examine the existing interest literature in common youth learning domains to identify gaps in the knowledge base. This will work as a starting point for the translation of interest to research for increasing youth sport participation. A scoping review was conducted in October 2025, adhering to the PRISMA extension for Scoping Reviews guidelines and reporting protocol. Five databases were searched for peer-reviewed articles published in English that discussed interest in common youth learning domains (i.e., academics,

art, music, psychology, physical education, physical activity, sport, leisure, and hobbies). Domains and topics of inquiry, research frameworks, and instruments used to measure interest were extracted from each study. The primary search returned 778 studies. After screening, 228 studies remained for extraction. Interest research has primarily been conducted in educational domains (87.3%), while other domains such as physical activity (8.3%) and sport (1.8%) have received less attention. Quantitative methods, taking up 68.9% of research, have been the dominant approach with qualitative only accounting for 6.6%. Interest has been defined inconsistently, is not always grounded in theoretical frameworks, and is studied using a variety of instruments. Interest has been widely researched in education and reading domains, which may be classified as involuntary and/or inherently less enjoyable. While voluntary/enjoyable activities such as sport and recreation have been researched less, interest in these domains may be important for improving participation and retention rates among youth. Qualitative methods may offer insights into the development of interest, as they illuminate the personal interpretation of lived experiences. The consistent conceptualization of interest and purposeful qualitative research represent meaningful next steps for the positive growth of interest research in sport. Funding source: Social Sciences and Humanities Research Council of Canada (SSHRC) Insight Grant (Grant 435-2020-0094); SSHRC Vanier Scholarship (Funding Reference Number 202410CGV-535665-424925).

Exploring perceptions of active esports participation for people with physical disabilities

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Electronic sports, also called esports, are the world's fastest growing sport with rapidly increasing cultural impact and growing acceptance from traditional gatekeepers to the sporting world, including the International Olympic Committee. Esports, in the context of the current research, consists of two types of participation: (a) traditional competitive gaming; and (b) active esports (i.e. when a participant physically moves to compete in sport as part of a virtual league). As both types of esports participation increase in prominence, it is important that inclusion of all community members, including people with disabilities, not be overlooked. The purpose of the current study was to explore perceptions of active esports participation for people with physical disabilities (PWPd), with the goal of identifying contextual factors that may contribute to quality participation. Semi-structured interviews were conducted with 24 participants, including PWPd who engage in active esports ($n=10$), PWPd who have not engaged in active esports ($n=4$), disability esports consultants ($n=3$), individuals that work in esports companies ($n=3$), healthcare providers ($n=2$), and gym staff ($n=2$). Data were analyzed using thematic analysis. Three themes were identified: (1) whether PWPd can participate in active esports, with sub-themes focusing on the types of activities and stigma within the social environment; (2) traditional esports (i.e. gaming) may be characterized as active esports for PWPd; and (3) integrity and fairness of competition within active esports. Factors influencing perspectives included disability status, type of disability, and knowledge or exposure towards types of esports participation. The results highlight two fundamental components of societal participation of PWPd previously identified in traditional sports and requiring further consideration from researchers, as well as esports game and league developers. First, there is a lack of knowledge of different esports participation options and how to adapt

esport participation, resulting in esports and active esports being viewed as a community in which PWPd are unable to participate. Second, as mapped to the Quality Participation Framework, foundational characteristics of esports and active esports (e.g. activity characteristics and the social environment) must be addressed to minimize barriers to entry and promote quality participation. Funding source: NICHD (R03HD111712).

The lasting effects of athletic participation: Positive and maladaptive coping pathways across gender

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Coping is a multidimensional repertoire shaped by individuals' stress appraisals and the contexts in which stressors occur. Research suggests people often employ both positive and maladaptive coping strategies, and participation in organized sports may foster a broader coping profile. The present study distinguishes between positive and maladaptive pathways to provide a deeper understanding of stress adaptation in emerging adulthood. Athletic participation was examined as a lasting developmental context, differentiating former sports participants from non-participants to capture how past involvement influences coping beyond active engagement. Guided by Ecological Systems Theory and the Transactional Model of Stress and Coping, this study examined whether athletic participation is associated with positive and maladaptive coping, and whether gender moderates these associations. Coping strategies were assessed using two measures adapted from Causey and Dubow's (1992) Self-Report Coping Measure. Positive coping was reflected in items including "I find a way to solve the problem", and maladaptive coping was reflected in items including "I get so upset that I can't talk to anyone". Athletic participation was measured by a single item asking participants to identify as having previously or never participated in organized sports. Participants were 262 undergraduate students ($M_{age} = 19.33$; 67.4% female) enrolled at a public university in the United States. Former athletes reported significantly higher positive coping, $t(260) = -3.32$, $p = .001$, $d = .44$, and significantly higher maladaptive coping, $t(260) = -3.02$, $p = .003$, $d = .40$, compared to non-participants. Regression analyses indicated that previous athletic participation was significantly associated with greater use of positive coping strategies; this association was moderated by gender, as this association was statistically significant only for female athletes. Additionally, former athletes reported greater use of maladaptive coping strategies compared to individuals who never participated in sports. Neither gender nor the interaction of gender and athletic participation was statistically significant. Results support developmental models suggesting that former athletes maintain elevated levels of both positive and maladaptive coping in emerging adulthood. Furthermore, findings indicate that gender differences emerge specifically in athletic participation, such that participation in organized sports differentially shapes positive coping pathways for men and women.

"Queerness is erased or our representation is depoliticized": Exploring queer women's interpretations of media representations and discourse on elite women athletes

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Media representations play a crucial role in shaping how queer identities in sport are constructed, perceived, and accepted. Research has documented how media discourses promote exclusion and discrimination for queer women and trans and gender diverse (TGD) individuals in sport. However, less is known about how queer women and TGD individuals themselves interpret and are impacted by dominant media representations and discourses surrounding queer identities in sport contexts. As such, the purpose

of this study was to explore how queer women and TGD individuals describe the power dynamics reproduced through media representation and discourse in elite and professional sport, their impact on individual psychosocial factors (e.g., body image, identity, social acceptance, sport participation), and desires for changes in future representation. We conducted semi-structured interviews with 16 participants (15 cisgender women, 1 trans femme nonbinary individual) and analyzed them using critical discourse analysis. Four discursive themes were constructed: (1) The Exception to the Rule: Athletic Excellence for Acceptable Queerness explored how sport discourse constructs the “acceptable queer athlete” as one who compensates for sexual identity and/or gender non-conformity through performance; (2) Playing it Straight: Assimilation and the Reproduction of Heteronormativity illustrated how sport is constructed as a space where queerness is tolerated only when it is not disruptive, political, or visible; (3) The Visibility Paradox described how media representations and sport discourses and policies regulate which bodies are scrutinized, vilified, or erased to uphold the gender binary; and (4) Imagining Embodied and Collective Queer Futures in Sport captures participant desires for representation that enable authentic self-expression, politicized visibility, and collective belonging in sport. These findings illustrate how media representations of elite women athletes transmit dominant discourses that shape queer women’s lived experiences and maintain systems of power, pointing to the need for representational change that supports diversity, safety, and belonging in sport. Funding source: Social Sciences and Humanities Research Council.

Applying social support training in group physical activity programs: Perspectives from instructors, administrators, and participants

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Social support contributes to positive experiences and sustained participation in group physical activity (PA) programs, yet instructors often receive limited training on how to intentionally foster social connection. This study examined how social support strategies were learned, implemented and experienced by instructors after completing an instructor education module and how these strategies were experienced by program administrators and adult PA participants. Using qualitative description grounded in a constructivist epistemology and relativist ontology, data were collected for various group PA programs across four sites in Canada (Calgary, Montreal, Toronto and Vancouver). Semi-structured interviews were conducted with instructors/program administrators ($n=17$) and PA participants ($n=41$), alongside observations of group PA sessions. Instructor and administrator interviews explored training experiences, feasibility and usefulness of social support strategies, implementation challenges and support needs, while participant interviews explored helpful and unhelpful support experiences and suggestions for improvement. Observations captured how instructors applied support strategies and how participants responded. Findings highlight the implementation of strategies, with PA participant accounts and observations illustrating and supporting how these strategies were experienced across contexts. We identified four themes. Actively listening to participants to understand needs and experiences involved instructors using check-ins and careful listening (including noticing silence and low engagement) to help learn what participants

needed. Getting to know participants to individualize social support over time included learning names, recognizing regular participants and respecting social comfort levels, which helped tailor support, but drop-in formats and inconsistent attendance were limitations. Responding and adapting in the moment based on participant feedback and cues involved instructors adjusting pacing, instructions, and exercise options, though large classes, time constraints and fewer visual cues online made adapting harder. Balancing care, encouragement and professional boundaries discussed how instructors aimed to be supportive without pressure while staying within their scope of practice. These findings show how implementing strategies from the training was impacted and influenced by the delivery mode and program structure. Strategies experienced by participants helped them feel heard, recognized, and safely supported. Future work should identify parts of training and program conditions that support consistent, participant-perceived social support across diverse PA settings. Funding source: Social Sciences and Humanities Research Council of Canada.

Examining Big Five personality profiles and training history among provincial level athletes across 22 sports

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Personality, via the big five framework, has been commonly used to investigate differences in trait expression and sport performance and development. Researchers have appealed for more integrated and interactional research to capture the complex relationships in personality psychology. Individuals do not express behavior due to one trait in an isolated manner, but rather they behave as a combination of our traits interacting with their environment. In this study, a Latent Profile Analysis (LPA) was conducted using big five personality traits within a cohort of provincial level athletes from 22 different summer and winter Olympic sports. The aim was to investigate the nature of personality profiles in a sample of athletes at a similar level of development from a variety of sports. After profiles were investigated, a secondary aim was to look at differences in training history, sport type, and sport exposure between these profiles. In total, 225 provincial level athletes ($M_{\text{age}} = 16.5$ years (± 4.87), 135 female and 90 male) were analyzed. The LPA found two personality profiles within this sample, characterized by lower overall trait expression in one, and higher overall trait expression in the other. Age, gender, and number of other sports played did not vary significantly between profiles. Additionally, sport type (individual vs team, contact vs non-contact, and high vs low risk) did not vary significantly between profiles. However, there was a significant difference for the age at which athletes started both organized, $F(1, 223) = 3.93, p = .049$, and unorganized, $F(1, 223) = 4.51, p = .035$, practice in their main sport. Athletes with higher personality trait expression tended to start both organized and unorganized practice in their main sport at a younger age ($M_{\text{org}} = 9.32 \pm 3.56$ years, $M_{\text{unorg}} = 6.51 \pm 3.50$ years) than those with lower personality trait expression ($M_{\text{org}} = 10.4 \pm 3.59$ years, $M_{\text{unorg}} = 7.64 \pm 3.69$ years). These results highlight that athletes from a variety of sports can have a similar personality type, and that individuals with higher levels of trait expression may be exposed to their main sport a couple of years younger than those who have lower levels of trait expression.

Impact of heat stress on athletic performance during hot weather

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In Hong Kong’s subtropical climate where summer temperatures can often exceed 35°C, heat stress can impair sports performance, increase fatigue and reduce enjoyment in outdoor activities. This study examined the

detrimental effects of heat stress on sports performance and psychological factors in basketball. Research outcomes were compared across three conditions: fully exposed (high heat stress), shaded (tree-shadow, moderate heat stress), and indoor (minimal heat stress). Amateur basketball players ($N = 30$; $M_{\text{age}} = 21.4$ years; $SD = 2.1$) were recruited and randomly assigned to perform standardized tests on three similar courts: one open to direct sunlight, one with substantial tree coverage providing intermittent shade, and one indoor facility with controlled temperature (approximately 22°C). Participants completed sessions in all conditions in a counter-balanced order, with at least 3 days between trials. Performance metrics included free throw accuracy (successful shots out of 10 attempts) for skill performance and the Yo-Yo Intermittent Recovery Test Level 1 for endurance. Subjective assessments comprised the Borg Rating of Perceived Exertion (RPE) scale for effort, a visual analog scale (VAS) for fatigue (0-10), and an enjoyment index (adapted from the Intrinsic Motivation Inventory's interest/enjoyment subscale) on a 10-point Likert scale. Environmental data (temperature and humidity) were monitored to confirm heat stress differences across sites. Results indicated that fatigue ratings and RPE were significantly lower in the tree-shadow and indoor conditions compared to the exposed condition ($ps < .001$). Enjoyment indices were significantly higher in the tree-shadow condition compared to both indoor and exposed conditions ($ps < .001$). Endurance distances were greater and free-throw accuracy were higher in the tree-shadow and indoor conditions than in the exposed condition ($ps < .05$). These findings reinforced the adverse effects of heat stress on athletic performance, fatigue and enjoyment during hot weather. More importantly, it shows that tree-shadow may be a sustainable and low-cost strategy that is alternative to indoor environments for alleviating such impacts in urban sports settings. Future research should include more participants and incorporate more physiological measures (e.g., sweat rate) in different sports populations. Funding source: Departmental Supportive Fund, SSPE, CUHK.

Intrateam conflict, commitment, and turnover intentions among collegiate student athletes

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Team dynamics play a central role in shaping athletes' sport experiences. Teams that function well tend to experience positive outcomes, while negative team functioning may prompt athletes to question their team membership and ultimately leave the group. In organizational psychology research, intrateam conflict has been positively associated with employee turnover; however, this relationship has received limited empirical attention in sport team contexts. Athlete turnover has become increasingly relevant in college athletics given the recent changes to transfer policies. The purpose of this study was to investigate the relationship between intrateam conflict and turnover intentions (e.g., intent to transfer and intent to quit) among collegiate athletes. The current research also explored if individual commitment would moderate this relationship. Participants in this study included 430 collegiate athletes, representing 11 different sports. All participants were competing at National Collegiate Athletic Association Division I, II, and III institutions ($n_{\text{DI}} = 72$, $n_{\text{DII}} = 160$, $n_{\text{DIII}} = 198$) in the Midwestern region of the United States. Athletes completed survey measures of intrateam conflict, affective commitment, intent to transfer, and intent to quit, and exploratory structural equation modeling was conducted in Mplus to analyze direct and moderated relationships among variables. Results indicated that only relationship conflict was a statistically significant, positive predictor of intent to transfer ($\beta = .20$, $p = .009$) and intent to quit ($\beta = .33$, $p < .001$). Task conflict was not a significant predictor in either model (transfer: $\beta = .161$, $p = .061$; quit: $\beta = -.11$, $p = .180$). Moderation analyses revealed affective commitment was a statistically significant moderator only in the pathway between relationship

conflict and intent to quit ($\beta = .704$, $p = .006$), such that relationship conflict was more strongly associated with intent to quit among athletes with lower commitment. These findings extend research on turnover from organizational psychology to the sport context and suggest affective commitment may be a potential protective factor against the detrimental effects of intrateam relationship conflict. This study advances the understanding of intrateam conflict in sport and offers implications for researchers and practitioners concerned with athlete retention and team functioning.

Remote physical activity counselling and its translation to adherence: Lessons learned from a pragmatic intervention in individuals with spinal cord injury

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Physical activity (PA) counselling is a key strategy to support adherence to PA interventions among individuals with spinal cord injury (SCI). However, little is known about whether, and how, specific counselling session characteristics relate to adherence during pragmatic PA trials. Our objectives were to descriptively analyze PA adherence and counselling session characteristics—frequency, mode (e.g., phone or video call), duration, talk-time ratio between participant and counsellor—and explore participants' perceptions of counselling and trial participation during the 6-month, home- and community-based Exercise guideline Promotion and Implementation in Chronic SCI (EPIC-SCI) trial. Nineteen participants with SCI (26% women/female; 52% tetraplegic; age 49 ± 15 years), who completed the intervention arm of the EPIC-SCI trial, were included. Weekly adherence (% of weeks) to the starting aerobic and strength SCI PA guidelines (2×20 min/week of moderate-to-vigorous aerobic PA + $2 \times$ week of strength training) was calculated for each participant. Counselling sessions were analyzed for frequency, mode, duration, and talk-time ratio across the adherence spectrum. Five participants completed interviews exploring counselling experiences and perceived adherence influences. We found that participants attended counselling sessions in $87 \pm 21\%$ of weeks (≈ 21 of 24), yet adherence was lower for aerobic activity ($76 \pm 22\%$; ≈ 18 of 24 of weeks), strength training ($61 \pm 29\%$, ≈ 15 of 24), and combined aerobic and strength training ($47 \pm 28\%$, ≈ 11 of 24). Counselling was delivered primarily via video (76%), followed by telephone (23%) and inperson meetings (1%). No clear patterns emerged between adherence and session frequency, mode, duration, or talk-time ratio. Interviews yielded three themes influencing adherence: (1) accountability to counsellors, (2) flexibility and person-centered PA prescriptions, and (3) access to inclusive, community-oriented facilities. In conclusion, counselling attendance was more consistent than PA adherence, yet counselling session characteristics did not explain adherence variability. Qualitative themes indicate that accountability and individualized support may help translate counselling attendance into actual PA engagement, while structural barriers—such as limited access to inclusive facilities—continue to hinder adherence. Future interventions should consider strategies beyond counselling session characteristics to support sustained PA adherence within real-world conditions similar to those of this trial.

Age-related performance trajectories in National Hockey League players

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The National Hockey League (NHL) represents the highest caliber of professional ice hockey worldwide and provides a unique context for examining performance trajectories across age. Entry into the NHL

requires extensive long-term investment from athletes and their families, while career longevity is influenced not only by individual ability and health but also by managerial decisions aimed at maximizing organizational success, such as winning the Stanley Cup. From a research perspective, this raises important questions about how player performance evolves over time and how aging affects career outcomes. Brander et al. (2014) analyzed NHL player data up to the 2011–2012 season to identify peak performance ages for forwards, defensemen, and goaltenders using metrics such as points per game, plus/minus, and save percentage. A self-identified limitation of this work was the inclusion of active players, which may have skewed results due to incomplete career trajectories. Moreover, the study was conducted over a decade ago and therefore does not account for subsequent changes in league practices. Building on this prior work, the present study examines age-related changes in NHL performance and their relationship to games played using a dataset sourced from eliteprospects.com. Linear mixed-effects models with quadratic age terms were applied to track performance metrics (e.g., goals, assists, total points, games played, save percentage, and season wins) across the three positions. Importantly, the dataset included only retired players, thereby avoiding issues related to incomplete careers, and incorporated athletes from the past decade. Compared to Brander et al. (2014), the findings indicate that scoring peaks now occur earlier. Forwards reached maximum goal scoring and total points at ages 25.29 and 25.60, respectively, while defensemen peaked at ages 25.32 and 25.73. In contrast, goaltenders peaked later, with save percentage maximized closer to their thirties. Despite declining offensive output, games played remained stable until approximately ages 27.63–27.80 for skaters, suggesting a potential “sunk cost fallacy” in managerial decision-making. Overall, these results demonstrate distinct age-related performance patterns across NHL positions.

Mental health stigma across competitive contexts in high-performance sport

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High-performance sport has traditionally addressed mental health through individual-level deficits such as mental illnesses, however, contemporary research advocates for cultivating assets in addition to reducing detriments (Whiting et al., 2012). Reflecting this shift, Canadian knowledge users endorsed an asset-based approach through the Mental Health Strategy for High Performance Sport (Durand-Bush & Van Slingerland, 2021). Asset-based approaches that equip individuals with the knowledge, skills, and attitudes needed to prevent mental health problems and promote well-being are particularly important for elite athletes, given that stigma has been identified as a significant barrier to help-seeking (Rao & Hong, 2020). Within this framework, stigma elimination represents a valuable asset. While Canadian policy emphasizes asset-based approaches to mental health, gaps remain in evidence-informed implementation. This study measured self-stigma using the 10-item Self-Stigma of Seeking Help (SSOSH; Vogel et al., 2006). A factorial ANOVA comparing stigma levels between sex and sport level (Olympic: males $n = 19$, females $n = 45$; Paralympic: males $n = 4$, females $n = 9$; Collegiate: males $n = 11$, females $n = 37$) was conducted. There was no significant interaction, or significant main effect for sex, however, there was a significant main effect for level, $F(2, 116) = 3.48$, $p = .034$, $\eta^2 = .06$. Stigma was significantly higher among USPORTS than Olympic athletes. These findings suggest that mental health stigma among athletes may be shaped by the competitive environment in which athletes are embedded. Relatively lower self-stigma among Olympic athletes may reflect greater exposure to mental health literacy or normalized access to psychological services within high-performance systems. In contrast, higher stigma among collegiate athletes

may highlight a critical gap in the implementation of asset-based mental health strategies earlier in the athlete development pathway. Although Canadian policy emphasizes stigma reduction as a protective mental health asset, the present findings indicate that such approaches may be unevenly distributed across sport levels and suggests the need for targeted action in one specific high performance sport context. Extending asset-based mental health initiatives into collegiate sport may therefore represent an important opportunity to reduce stigma, promote help-seeking, and support athlete well-being.

Leader humility predicts youth athletes' role commitment and intentions to return

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Athletes possess several roles that contribute to their team's functioning. When assigned a role by their coach, athletes respond in several ways, one of which is through committing to their responsibilities. Role commitment is “a dynamic and volitional psychological bond reflected in the dedication to and responsibility for one's role” (Eys et al., 2020, p. 91). When athletes are committed to their role, it is associated with positive outcomes such as greater intentions to return (Coleman et al., 2021). This is particularly relevant given ongoing calls to keep people engaged in sport across the lifespan (Sport Participation Research Initiative, 2025). As such, fostering athletes' role commitment in sport may be one path to continued sport engagement. One potential precursor is leader humility, which has been noted to contribute to group members' role commitment and autonomy to make decisions in organizational contexts (Ou et al., 2014). Leader humility is an “interpersonal characteristic that emerges in social contexts and connotes a willingness to view oneself accurately, an appreciation of others' strengths and contributions, and teachability” (Owens & Hekman, 2016, p. 1088). Considering the importance of coaching in sport (Hague et al., 2021) and leader humility in organizational settings (Ou et al., 2014), the purpose of this study was to examine whether perceptions of coach humility were associated with role commitment and intentions to return in a sport context. We hypothesized that athletes' perceptions of their coach's humility would positively predict their role commitment and intentions to return. The sample consisted of 158 competitive youth athletes (51 females; $M_{\text{age}} = 16.82$, $SD = .93$) from 17 task interdependent teams who participated across two time points. Accounting for interdependence in teammates' responses (i.e., teammates nested within teams), preliminary analyses suggest that coach humility positively predicts both role commitment ($\beta = .45$, $SE = .10$, $p < .001$) and intentions to return ($\beta = .85$, $SE = .24$, $p < .001$). These findings are discussed in light of theoretical (e.g., possible mechanisms explaining the coach humility-role commitment link), measurement (e.g., assessing dimensionality of leader humility inventories), and applied implications.

Determinants of motivational regulations for physical activity in Middle Eastern female college students: A follow-up study

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Understanding how motivation for physical activity develops and is maintained is a public health priority, particularly among young adults. Guided by Self-Determination Theory (SDT), this longitudinal study examined the stability and determinants of motivational regulations for physical activity over eight weeks among female college students in the Middle East. Participants were 254 Emirati women aged 18–24

($M = 20.41 \pm 1.97$ years) who completed measures of motivational regulation, MVPA, body composition, and handgrip strength at baseline, with motivation reassessed after eight weeks. Motivation was assessed with the Sport Motivation Scale II. MVPA was measured with accelerometers. Body composition was estimated using bioelectrical impedance, and muscular strength was assessed using handgrip dynamometry. Path analysis examined predictors of six motivational regulations (intrinsic, integrated, identified, introjected, external, and amotivation) and their stability over time. Results showed stability in all motivational regulations over eight weeks, except for amotivation, with intrinsic motivation being most stable. Baseline handgrip strength predicted more autonomous motivation types (intrinsic, $\beta = .18, p = .017$; integrated, $\beta = .16, p = .035$; identified, $\beta = .18, p = .017$). MVPA predicted identified regulation ($\beta = .18, p = .006$) and reduced amotivation ($\beta = -.15, p = .016$). Greater fat mass was linked to higher amotivation ($\beta = .27, p = .006$), while more muscle mass was related to lower amotivation ($\beta = -.21, p = .035$). Baseline motivation predicted follow-up outcomes, showing moderate stability. Handgrip strength remained a positive predictor of integrated regulation over time ($\beta = .28, p = .036$). MVPA and body composition had limited future effects beyond baseline motivation. These findings indicate stable motivational regulation among young Emirati women and highlight physical fitness, particularly muscular strength, as supporting self-determined motivation. Interventions to increase physical activity among Middle Eastern female college students could include strength-training exercises and autonomous motivation strategies. This study offers new longitudinal evidence from an underrepresented group and insights for culturally relevant physical activity promotion. Funding source: United Arab Emirates University.

The relation between purpose in life and physical activity for people aged 18 years and over: A scoping review

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Although purpose in life is generally not emphasized in mainstream health behavior change theories of behavior initiation and maintenance, some theoretical perspectives suggest that it may influence health behaviors, including physical activity. Yet, it remains unclear how empirical research on physical activity aligns with these theory-based propositions. Therefore, we conducted a scoping review, following established methodological guidance, to map empirical research examining purpose in life in the context of physical activity and assess its alignment with pathways proposed in existing theoretical perspectives among adults aged 18 years and older. The electronic database search included nine databases (i.e., PubMed, PsycInfo, SPORTDiscus, ProQuest Dissertations and Theses, AgeLine). Evidence sources comprised peer-reviewed journal articles and theses/dissertations, regardless of publication year or language. Two reviewers independently screened titles and abstracts and assessed eligibility by reading the full text. Two reviewers independently extracted data on study and sample characteristics, evidence types, study design, purpose in life, and physical activity. Data synthesis was guided by three specific theory-based pathways linking purpose in life to physical activity: a direct motivational pathway, an indirect (mediational) pathway through autonomous motivation, and an adversity-buffering pathway (i.e., purpose in life functioning as a moderator). We identified 53 studies for review, including 10 theses/dissertations and 8 qualitative studies. Most studies were cross-sectional ($n = 42$; 79%) and examined the direct link between purpose in life and physical activity ($n = 47$; 89%). Effect sizes were reported using various metrics, including Pearson correlation coefficients (ranging from .05 to .30) and odds ratios (ranging from 1.04 to 1.30). Notably, one study reported a negative association between purpose in life and sport participation. Few studies investigated an indirect ($n = 11$; 21%) or adversity-

buffering pathway ($n = 3$; 6%), and few ($n = 7$; 13%) explicitly examined physical activity maintenance. Overall, the findings of this review indicate a small positive link between purpose in life and physical activity. Future research is needed to clarify whether this association is mediated by other factors, such as autonomous motivation, to examine potential moderating effects, and to investigate its influence on physical activity maintenance.

“It changes the way I participate”: Women wrestlers’ menstrual cycle experiences

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Consistent with general trends in research from around the world, Western ideologies and methodological approaches dominate sport research focused on women athletes’ experiences with their menstrual cycles. The purpose of this community-based participatory research (CBPR) study was to employ an Etuaptmuk/two-eyed seeing (E/TES) approach to explore women wrestlers’ experiences with their menstrual cycles in training and competition. Purposeful sampling was used to recruit seven young women wrestlers between 16 and 20 years old. Centered in Indigenous methodology, data were generated via talking circles facilitated by two Indigenous Elders, along with follow-up one-on-one semi-structured interviews. Consistent with a CBPR approach, data were analyzed using a three-phase process of content analysis that included the preparation, organization, and reporting of data. Young women wrestlers’ experiences with their menstrual cycles are represented by five main themes: (a) “It changes the way I participate”, (b) “Is this sweat, or am I bleeding through?”, (c) “It’s not because we’re weak”, (d) “I feel like he takes it as a joke”, and (e) “It really feels like a family”. Findings highlight how young women wrestlers’ complex menstrual cycle experiences are shaped by feelings of inconvenience and fears of unexpectedly bleeding through clothing. Such experiences are compounded by participating in a sport environment that promotes a culture of no excuses and where coaches, particularly men coaches, may lack the necessary knowledge to support athletes’ as they navigate their menstrual cycle in sport. Finally, participants highlighted how teammate support from other woman wrestlers, as well as a change in narrative that focuses on the power of women’s menstrual cycles, are essential for supporting young women wrestlers navigate menstrual cycle experiences. This study serves as a practical example of how Indigenous perspectives and methodologies in sport research foster in-depth understandings of athletes’ menstrual cycle experiences. Funding source: Social Sciences and Humanities Research Council.

Development and initial validation of a mental demand questionnaire for physical activity

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Dual-process models of health behavior suggest that physical activity is regulated by both reflective processes (e.g., intentions, attitudes, self-control) and automatic processes (e.g., habits, identity). Contemporary theories of physical activity behavior increasingly emphasize automatic processes as critical for long-term behavioral maintenance because they reduce the mental effort required for physical activity. Accordingly, mental demand, the perceived mental effort needed for physical activity, may represent a key, yet understudied, mechanism underlying physical activity maintenance. Progress in this area has been limited by the absence of a validated measure of mental demand specific to physical activity. The purpose of this study was to adapt and validate a four-item mental demand

questionnaire originally developed in the self-regulated learning literature for application to physical activity. A sample of 150 adults (84 females; $M_{\text{age}} = 40.47 \pm 12.74$ years) completed an online survey assessing demographics, intentions, instrumental attitudes, trait self-control, habits, identity, moderate-to-vigorous physical activity (MVPA), and mental demand. Internal consistency was assessed using Cronbach's α , and structural validity was examined via exploratory factor analysis (EFA). Convergent and divergent validity were assessed with Spearman correlations. The mental demand questionnaire demonstrated good internal consistency (Cronbach's $\alpha = .90$). Sampling adequacy for EFA was supported ($KMO = .83$), and both parallel analysis and visual inspection of the scree plot suggested a single-factor solution. Mental demand was strongly negatively associated with habit ($rs = -.70$), and moderately negatively associated with identity ($rs = -.40$), trait self-control ($rs = -.36$), and MVPA ($rs = -.30$). Mental demand was not significantly associated with intentions, instrumental attitudes, or education. These findings provide evidence of reliability and initial structural validity for a brief mental demand questionnaire for physical activity. Convergent validity was supported through theoretically consistent associations with habit, identity, and trait self-control, while divergent validity was supported by null associations with intentions, instrumental attitudes, and education, which are not theoretically related to the mental demand for physical activity. This measure may offer a useful tool for advancing theory-driven research on physical activity maintenance. However, future research is needed to confirm the structural validity of the questionnaire.

“It was a journey, but it was an enjoyable journey”: Exploring the experiences of activity coaches in a 12-week randomized pilot trial comparing physical activity prescription and supportive coaching

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Chronic disease remains the leading cause of mortality and places substantial strain on healthcare systems. Although regular physical activity (PA) is a well-established strategy for chronic disease prevention and management, PA levels among Canadians have declined markedly since the onset of the COVID-19 pandemic. Prescription-based PA programs, such as Prescription to Get Active (RxTGA), aim to address this gap by linking patients with community resources, including facility access, structured exercise options, and virtual supports. However, prescription provision and redemption declined throughout the pandemic, underscoring the need for adjunct strategies that enhance accountability and support. Motivational interviewing via Co-Active Life Coaching (MI-via-CALC) is an evidence-based, client-centered, autonomy-supportive approach that may compliment PA prescription programs via its focus on behavior change. The purpose of this study was to explore the training and program delivery experiences of activity coaches involved in a 12-week randomized PA prescription program enhanced with MI-via-CALC. Individuals with backgrounds in health and fitness were recruited as volunteer activity coaches and completed a 2-day online MI-via-CALC training workshop. Coaches met virtually with participants for 30-45 min weekly during the first 6-weeks, and bi-weekly for the remaining 6-weeks. Sessions focused on relationship building, identifying PA barriers and facilitators, goal setting, and routine development. Upon program completion, coaches participated in individual semi-structured interviews (30-60 min), which were inductively analyzed to explore training experiences, perceived impact, and program logistics. Nine coaches were paired with 2-3 participants each. Coaches were predominantly female (78%), personal trainers (89%), and shared a passion for supporting

others' PA journeys. Coaches reported that study participation contributed to their personal and professional development, with many intending to integrate MI-via-CALC skills into their existing practice. Perceived gains included enhanced coaching competence, increased self-confidence, and improved perceived ability to support behavior change. Challenges included scheduling difficulties and variability in participant commitment. Overall, findings suggest that MI-via-CALC is a feasible and valuable adjunct to PA prescription programs, including RxTGA, offering a mechanism to enhance accountability and social support. These findings provide practical guidance for optimizing coaching-enhanced PA prescription models to support sustained engagement and reduce chronic disease risk.

Motivational profiles and young archers' affective responses and engagement intentions: A person-centered approach considering participants' motivation regulations and achievement goal orientations

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Sport research, based on achievement goal theory (AGT; Nicholls, 1989) and self determination theory (SDT; Ryan & Deci, 2017), has found achievement goal orientations and motivation regulations to predict participants' affective and behavioral patterns; autonomous motivation and task orientation is associated with positive outcomes and controlled motivation, amotivation and ego orientation tend to link to maladaptive outcomes. This research predominantly pulls from AGT or SDT and has taken a variable-centered analytic approach, focusing on average relationships between variables. Employing a person-centered analysis, the present study: (1) aimed to generate sub-groups of archers based on their motivation regulations and achievement goal orientations, (2) examined differences between emerging motivational profiles and affective responses (enjoyment, anxiety) and intentions to continue engagement, and (3) tested whether profiles could be distinguished in regard to age, gender and competitive status (participation, performance focus) of the archers. Participants ($n = 434$, 72% male; $M_{\text{age}} = 50.6 \pm 16$ years) were UK archers (57% participation; 43% performance); all anonymously completed an online survey assessing targeted variables with validated scales. Latent profile analysis was conducted with autonomous motivation, controlled motivation, amotivation, task orientation, and ego orientation as distal outcomes. Models specifying 2-6 profiles were estimated with model selection guided by AIC/BIC/SABIC, entropy, class sizes, and interpretability. A 4-profile solution was deemed optimal (entropy = .80): Moderate Autonomous-Task (53%, $n = 231$), High Autonomous-Task/Low Ego (30.9%, $n = 134$), High Amotivated-Controlled-Ego (6.0%, $n = 26$), and High Ego/Low Autonomous-Task (9.9%, $n = 43$). Targeted outcomes significantly differed across profiles ($F_s = 6.67-20.31$, all $p < .001$, $\eta^2 = .05-.13$). The High Autonomous-Task/Low Ego profile group reported significantly higher enjoyment and intentions to continue when compared to the other 3 sub-groups. The High Amotivated-Controlled-Ego profile group revealed significantly higher levels of anxiety, when contrasted with the other profiles. The four sub-groups did not significantly differ by age, gender, or participation/performance status. Findings are discussed in terms of conceptual and practical insights of a theoretically integrated and person-centered approach to ascertaining affective and behavioral patterns associated with motivation-related characteristics. Results also point to theoretically-predicted ubiquitous benefits of high autonomous motivation and task orientation. Funding source: Archery GB.

Social vs. solo physical activity: Does context matter for anxiety and depression symptom reduction?

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Physical activity (PA) is increasingly recommended as an evidence-based, accessible, and low barrier intervention for anxiety and depression. However, diversity in study methodologies (modality, duration, intensity, and frequency of PA) and methodological limitations create uncertainty about the characteristics needed for effective PA interventions. Integrating a social component into PA may be one such characteristic given that social connection lies at the base of many psychological theories of mental health and well-being. PA with a social component (i.e., done with others) may provide added benefits beyond PA alone. In this pilot study, we tested the feasibility and preliminary efficacy of a study design evaluating social vs. solo exercise in reducing anxiety and depression. Participants were adults with elevated anxiety, stress, and/or depression symptoms who were randomized into a social or solo PA intervention, both led by trained exercise professionals. Both interventions involved two weekly 60-min PA sessions at a gym for eight weeks. Participants completed measures of anxiety, stress, and depression at pre- and post-intervention. Participants also provided qualitative feedback post-intervention. While 14 participants consented to participate, only 8 ($M_{\text{age}} = 37.25$) successfully completed pre- and post-intervention questionnaires. Of these, five completed all sessions, one completed 13 sessions, one completed 9 sessions, and one completed 8 sessions. Post-intervention qualitative data suggested that participants found the intervention satisfying and helped them become more physically active. Results suggest the intervention is acceptable and appropriate, but only feasible for specific individuals. Participants that found the intervention convenient lived near the gyms and/or had flexible schedules. Those deeming it inconvenient indicated gym location (town outskirts), session times (early afternoon), and financial concerns as barriers. To increase feasibility and recruitment success, adjusting gym partnerships and session times is required. Pre- and post-intervention scores revealed that participants in the solo condition experienced reductions in stress. Participants in the social condition experienced reductions in anxiety, stress, and depression. This suggests that there may be different benefits to social and solo PA, and that their comparison is worth further exploration. However, feasibility data suggest that changes to study design are needed to effectively compare these interventions. Funding source: UNB University Research Fund (RF-EXP-2024-02).

Momentary associations of physical activity, affective states, and interoceptive awareness in adolescents' daily lives

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Insufficient physical activity is prevalent during adolescence, a developmental stage marked by increased risk for mental health issues. Beyond its physical health benefits, physical activity may influence affective states and interoceptive awareness, yet evidence from adolescents' daily lives remains limited. This study examined momentary associations between physical activity, affective states, and interoceptive awareness, and compared different types of activities. Ninety-seven adolescents (aged 12–18 years) completed a seven-day ambulatory assessment that combined accelerometry and smartphone-based ecological momentary assessment (EMA). Physical activity and inactivity were assessed using device-based methods, while affective states, interoceptive awareness, and activity type were assessed through questionnaires (3,699 EMA reports). Linear mixed-

effects models examined associations between physical activity, affective states, and interoceptive awareness. Physical activity during the five min preceding each EMA prompt was modeled as a predictor of subsequent affective and interoceptive states, and outcomes were compared across activity types (inactive screen-time, inactive non-screen-time, physical activity, and sleep). Inactive screen time was used as the reference category in these analyses. Preliminary analyses showed that physical activity predicted higher arousal ($\beta = 5.08, p = .002$) and interoceptive awareness ($\beta = 1.38, p < .001$) but was not related to calmness or valence ($ps > .05$). Compared with inactive screen-time, positive affect was higher after inactive non-screen-time ($\beta = 4.57, p < .001$), sleep ($\beta = 9.48, p = .006$), and physical activity ($\beta = 2.58, p = .044$). Arousal was higher after physical activity ($\beta = 3.61, p = .032$) and lower after inactive non-screen-time ($\beta = -2.37, p = .002$) than inactive screen-time. Calmness was higher after inactive non-screen-time ($\beta = 7.76, p < .001$), sleep ($\beta = 10.56, p = .005$), and physical activity ($\beta = 2.80, p = .046$) than inactive screen-time. Overall, physical activity primarily enhanced energetic arousal and interoceptive awareness, whereas screen-free and restorative activities were more strongly associated with calmness and positive affect. In contrast, screen-based inactivity was associated with less favorable affective states, suggesting that physical activity and screen-free periods in daily life may have mental health benefits. Funding source: University of Bern; internal funding.

Achievement goal congruence and youth athlete well-being: A latent profile analysis

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Youth mental health has reached crisis levels (CDC, 2024), prompting interest in non-clinical contexts that may support youth well-being (American Academy of Pediatrics, 2021). Sport has been identified as one such context (Office of the Surgeon General, 2021), yet evidence suggests sport may both protect and undermine mental health depending on individual and contextual factors (Jewett et al., 2014). More broadly, psychological research highlights the importance of person-environment fit for development and mental health (Lerner, 1989). Although achievement goal theory (AGT; Nicholls, 1984) has been widely used to examine how motivation and psychological outcomes are shaped by individual's goals or the motivational climate, little attention has been given to how their interaction relates to well-being. The present study examined whether configurations of athletes' achievement goal orientations, perceptions of the motivational climate, and perceived competence were associated with athlete burnout, life satisfaction, and well-being. Participants were 498 youth athletes ($M_{\text{age}} = 15.36$, 55.4% male, 55.2% White) who completed self-report measures of task orientation, ego orientation, perceived task climate, perceived ego climate, perceived competence, and mental health outcomes. Guided by AGT, latent profile analyses were conducted using task and ego orientations, task and ego climate perceptions, and perceived competence as independent indicator variables, resulting in six distinct latent profiles. Contrary to theories of person-environment fit, congruence between achievement orientation and motivational climate did not predict burnout, life satisfaction, or well-being. Rather, profiles characterized by stronger task orientations and climates, and particularly both, were associated with more adaptive outcomes. Ego-oriented profiles exhibited trends toward poorer outcomes only when accompanied by relatively low-task involvement. These preliminary findings reaffirm the enduring importance of task involvement for athlete mental health while indicating that ego involvement alone does not confer heightened mental health risk. Moreover, these results raise questions about the applicability of person-environment fit theory within AGT frameworks, suggesting that the presence of task-oriented structures may be more consequential for athlete

well-being than alignment between individual and environmental goals. Accordingly, youth sport coaches and organizations should prioritize task-focused structures (e.g., emphasizing effort, self-oriented goal setting) to support athlete well-being across levels of ability and competition.

Associations between resilience, coping, and mental health in young Brazilian athletes: A cross-sectional analysis

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The sports environment for youth athletes imposes demands that can foster mental health in salutogenic and pathogenic dimensions. Understanding how psychological skills are associated with complete mental health—operationalized as a dual-factor model comprising both well-being (salutogenic) and developmental problems (pathogenic)—is fundamental for effective interventions. Thus, the objective of this study was to investigate the association between psychological resilience and coping skills with the complete dimensions of mental health in young Brazilian athletes. A quantitative, cross-sectional study was conducted, with data collected between July and December 2023 using online forms. The sample comprised 248 athletes (161 boys; 65%; $M_{age}: 14.5 \pm 1.8$ years) from a large Brazilian club (team sports: $n = 149$: futsal, $n = 68$; basketball, $n = 20$; volleyball, $n = 61$; individual sports: $n = 99$: judo, $n = 20$; tennis, $n = 27$; swimming, $n = 36$; gymnastics, $n = 16$). The instruments used were the Athletic Coping Skills Inventory-28 (ACSI-28BR), Resilience Scale in Sport (RS-Sp), Sport Mental Health Continuum (SMHC) to assess well-being, and Strengths and Difficulties Questionnaire (SDQ) for development difficulties. The Structural Equation Modeling (SEM) analysis adopted coping and resilience as independent variables, well-being and developmental difficulties as dependent variables (joint dependent variables), and adjustments for age, sex (male; female), modality (team; individual), and practice time (in complete years). The results are expressed as unstandardized (B), standardized (β) and adjusted R^2 . A p -value $< .05$ was considered significant. Resilience showed a positive and significant association with well-being ($B = .291$; $\beta = .342$; $p < .001$), as did coping skills ($B = .388$; $\beta = .326$; $p < .001$). Regarding development difficulties, resilience ($B = -.065$; $\beta = -.203$; $p = .026$) and coping ($B = -.116$; $\beta = -.257$; $p = .001$) demonstrated a significant negative association. Furthermore, the independent variables explained 40.3% ($R^2_{adjusted} = .403$) of the variance in well-being and 19.2% ($R^2_{adjusted} = .192$) of development difficulties. The results indicate that both psychological resilience and coping skills are important for full mental health in the sports context. The expressive explanatory power over well-being reinforces the premise that strengthening internal resources may be more advantageous for promoting flourishing than for mitigating deficits. The sports environment should, therefore, act proactively in building positive experiences, prioritizing the development of these skills to foster the athletes' mental health.

Physical activity counseling in young adults following cancer treatment: Investigating patterns in mechanisms of action

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Physical activity (PA) is a key component of supportive care for young adults (18–39 years) following cancer treatment, yet many remain insufficiently active. To address this, we developed the *physicAl aCtivity Counselling for young adult canceR SurvivorS* (ACCESS) intervention,

grounded in self-determination theory (SDT), to target specific mechanisms of action (MoAs), namely PA-related needs satisfaction, motivational regulations, behavioral processes of change, and decisional balance, with the aim of increasing PA following cancer treatment. The aim of this study was to describe trends in presumed MoAs. A pilot randomized controlled trial was conducted with 42 young adults ($M_{age} = 33$ years; 81% female) allocated to a 12-week virtual person-centered behavior change intervention ($n = 20$) or usual care ($n = 22$). The intervention comprised six 60-min biweekly sessions that incorporated behavior change techniques intended to activate theory-driven MoAs. Data were collected at baseline, post-intervention, and follow-up (24 weeks post-baseline) via online questionnaires. Visual analysis was used to plot MoA trajectories (individual, group) and depict the magnitude and direction of change; no inferential statistics were conducted. The intervention group showed small, favorable shifts in relative autonomy index (indicating greater autonomous motivation), global needs satisfaction, behavioral processes of change, and decisional balance. Plots showed marked inter-individual variability not apparent in aggregated data, with distinct patterns of increasing, stable, and decreasing trajectories across all MoAs. These findings will be used to refine the ACCESS intervention to increase its likelihood of producing change in the targeted MoAs. A future large-scale efficacy randomized controlled trial is planned. The trial will embed a process evaluation to formally test MoAs through which the ACCESS intervention produces its effects on PA behavior and identify factors (e.g., participant characteristics, contextual issues) that help explain variability in MoA change between individuals.

Parental awareness and perceptions of 24-hour movement guidelines for preschool-aged children

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The 24-Hour Movement Guidelines include recommendations for the optimal proportion of each day individuals across the lifespan should engage in physical activity, sleep, and sedentary behaviors. Adherence to these guidelines is associated with better physical, cognitive, and socioemotional health – outcomes that are strongly shaped by parental knowledge and perceptions in early childhood. Original guideline versions contained recommendations for children aged five and above, but more recent updates now include information for infants, toddlers, and preschoolers. Because these guidelines were only recently extended to children under five and have not been formally adopted in the United States, parental familiarity remains largely unknown. Thus, the purpose of this study was to measure parental awareness and perceptions of the 24-Hour Movement Guidelines for preschool-aged children in the United States. Parents of 2–5-year-olds completed an online survey comprising measures of demographics, children's movement behavior engagement and socioemotional health, and parental guideline familiarity and recommendation guesses. Preliminary results include responses from $N = 49$ parents ($M_{child\ age} = 3.3 \pm 1.0$ years; 46.9% females) from seven U.S. states. Parents were largely unfamiliar with the 24-Hour Movement Guidelines: 76.7% reported they had never heard of them, 16% reported they had heard of the guidelines but did not know what they were, and 7% reported they had heard of the guidelines and knew what they were. Parental guesses varied widely and revealed systematic underestimation of physical activity recommendations, whereas sleep recommendations were more accurately perceived: physical activity (recommendation = 180 min/day; $M_{guesses} = 157.9 \pm 76.8$ [range = 31–300] min/day), sedentary screen time (recommendation = 60 min or less/day; $M_{guesses} = 49.3 \pm 57.1$ [range = 0–279] min/day), sleep (recommendation = 10–13 hr/day; $M_{guesses} = 11.2 \pm 1.7$ [range = 8–14] hr/day). Over half of parents (57.1%) guessed that physical activity recommendations were lower than 180 min/day; about a

quarter (23.3%) guessed that sedentary screen time recommendations were higher than 60 min/day. Most parents (78.6%) guessed within the correct range for sleep recommendations. Data collection is ongoing. Planned analyses will examine associations between movement behavior engagement and children's socioemotional functioning, as well as group differences based on guideline adherence, to better understand psychological implications of early movement behaviors and parental knowledge and perceptions.

Contextualizing mental health and physical activity: Exploring differences in affective responses by physical activity domain

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Latina women in the United States engage in disproportionately low levels of moderate-to-vigorous physical activity (MVPA) and higher subsequent levels of mental distress. Evidence suggests that Latinas tend to engage in more occupation-related MVPA compared to their White peers, and the domain in which physical activity occurs may impact mental health. This study examines acute affective responses to physical activity and how they differ when Latinas engage in occupation-related tasks, using Ecological Momentary Assessment (EMA) and accelerometry. For seven days, Latinas ($N=66$; $M_{\text{age}}=39$ years; 70% earning $< \$20,000/\text{year}$; 79% foreign-born) wore accelerometers and completed surveys delivered via EMA prompts three times/day. EMA items assessed positive and negative affect (5 items) and occupation-related tasks (single item) at each prompt. Accelerometers measured device-based MVPA in the 30 and 60 min before the EMA prompt. Multilevel models evaluated within-person associations between MVPA in the 30 and 60 min before the EMA prompt and subsequent affect, and the moderating role of completing occupation-related tasks. On occasions when Latinas engaged in more MVPA than was typical for them, they experienced high levels of positive affect (30 min before: $b=.04$, $p=.01$; 60 min before: $b=.03$, $p<.01$), regardless of completion of occupation-related tasks (30 min before: $b=-.01$, $p=.097$; 60 min before: $b=-.01$, $p=.81$). On occasions when Latinas reported completing occupation-related tasks and engaging in more MVPA than typical for them, they experienced higher levels of subsequent negative affect (30 min before: $b=.02$, $p=.06$; 60 min before: $b=.02$, $p<.01$). But this physical activity-negative affect association did not exist when Latinas reported not completing occupation-related tasks (30 min before: $b=-.01$, $p=.46$; 60 min before: $b=-.01$, $p=.16$). This offers preliminary evidence that momentary physical activity may enhance positive affect among Latinas across work and non-work contexts. Additionally, associations with subsequent negative affect may be context dependent, with activity in occupation-related settings linked to higher negative affect. These results underscore the value of EMA for capturing dynamic behavior-affect relationships and suggest the potential need for culturally tailored interventions that address work contexts to optimize mental health outcomes.

“Energy to give”: How thriving fuels sustained social support in women's high-performance sport – a teammate-researcher season-long investigation

Lauryl Bons, Brock University; Blair Evans, Western University; Dan Brown, University of Bath; Luc Martin, Queen's University

Despite the growing recognition that well-being and performance are connected, sport research has seldomly explored how athletes experience these processes simultaneously. Thriving, understood as the joint psychological experience of well-being and high subjective performance, is shaped not only by personal qualities (e.g., resilience), but also by

contextual factors (e.g., growth environments). In interdependent sport, teammates represent central figures in shaping those contextual factors. Notably, they are ideally suited to provide emotional, esteem, tangible, and informational support. Yet, how these everyday interactions contribute to athletes' lived experiences of thriving remains underexamined, especially in the realm of high-performance women's sport. Positioned as both researcher and teammate in this study, I explored how social support influenced athletes' experiences of thriving throughout a competitive season. This intensive qualitative study involved a combination of data collection approaches, including observations, informal conversations, and semi-structured interviews ($N=13$). My insider role provided access to nuanced interpersonal moments and allowed for an ongoing, relational understanding of support as it unfolded in real time. Reflexive thematic analysis was used to construct themes and connect patterns across field-work and interview accounts, enabling a rich interpretation of how social support and thriving were experienced and sustained. Findings highlight the central role of esteem support in enabling thriving, while also emphasizing the importance of capacity and emotional intelligence in effective support provision. Factors such as selection pressure/positional competition, injuries, and status were common themes that contributed to periods of languishing and reduced capacity to support others. Overall, this study offers important insights for researchers, coaches, and athletes by illustrating the dynamic, deeply relational, and cyclical nature of thriving in women's high-performance sport, and by emphasizing the shared responsibility in cultivating environments that promote both well-being and high-performance simultaneously. Funding source: SSHRC.

Mapping psychological skills in sport officiating: A scoping review

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Sport officials operate in fast-paced, high-pressure environments that place substantial psychological demands on performance. Although psychological skills are widely recognized as fundamental to effective officiating, the extent to which these skills are studied, trained, and applied within officiating contexts remains unclear. Therefore, the purpose of this scoping review was to systematically map the literature examining sport officials' education, acquisition, and use of psychological skills. Guided by the Arksey and O'Malley's (2005) framework, four databases were searched that yielded 30 eligible sources comprised of quantitative, qualitative, and mixed-methods designs across multiple sports and countries. Across the 2,623 participants included, only 9.2% identified as female, with most samples drawn from elite or high-performance officiating contexts and predominantly interactor (e.g., center soccer referee) officiating roles. Officials employed a wide range of psychological skills, including imagery, goal setting, self-talk, routines, attentional control and focus strategies, as well as arousal and anxiety regulation. Psychological skill use varied across sports, experience levels, contexts, and authors differed in how psychological skills were defined and conceptualized. Psychological skills were commonly used to proactively and reactively regulate emotions and stress, manage errors, maintain or regain concentration, and prepare for competition. Psychological skill acquisition occurred through both informal pathways (e.g., peer mentorship, prior athletic experience) and formal training programs, with officials and practitioners expressing a need for organization-led, sport-specific, and practitioner-facilitated delivery models. Evidence from intervention studies suggests that psychological skills training could enhance officiating performance and readiness. However, approaches varied in structure, theoretical grounding, and reporting quality. Future research should employ theory-driven, ecologically valid, and longitudinal designs to clarify how psychological skills are learned and enacted among sport officials, while addressing persistent demographic,

role-based, and sport-specific gaps to inform comprehensive, contextually grounded training.

The intersection of youth sport and mental health in the United States: A rapid review

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Research on the intersection of youth sport and mental health is growing, but there is a gap in understanding the impact of U.S.-based interventions makes it difficult to establish the state of the evidence and develop effective, data-driven interventions in the U.S. This rapid review aimed to identify and evaluate studies on mental health interventions in U.S. youth sport. The review's objectives were to: (a) identify studies representative of U.S. youth sport mental health interventions; (b) evaluate the quality and certainty of the evidence and compare correlational and experimental study results; and (c) synthesize and report on the mechanisms and conditions under which youth sport impacts mental health. A total of 17,478 studies and 37 grey literature sources were screened, with 30 included studies. The review followed PRISMA guidelines. Studies were required to be published in English between 2015 and 2025, focus on U.S. youth under 20, and include sport as part of an intervention measuring mental health outcomes. The Downs and Black Checklist and GRADE assessment tool were used to evaluate study quality and certainty of evidence. Consistent associations were found between sport participation and improved mental health, including reduced anxiety, depression, and suicidal thoughts, as well as increased self-esteem. However, the quality of the evidence was limited, with 4 studies rated as good, 12 studies rated as fair, and 14 rated as poor. The review found no significant differences in findings based on study quality or design. The benefits of sport were not uniform across all groups. The review identified several contextual factors associated with positive mental health outcomes, such as coach-athlete relationships and psychological need satisfaction. The review confirms a positive association between youth sport participation and mental health outcomes in the U.S. The findings highlight the importance of specific social and environmental factors in creating a beneficial experience. However, the lack of high-quality research, particularly for diverse populations, indicates a significant gap in the literature. Future research should prioritize high-quality, longitudinal, and equity-focused studies to inform the development of effective and equitable sport-based mental health interventions.

Supporting young athletes: Parental motivations and challenges in girls' youth sports

Diana Avans, Vanguard University of Southern California; Zoey Barajas, Vanguard University of Southern California; Jazmine Cervantes, Vanguard University of Southern California

Parental support is a central component of children's development, particularly in youth sports where participation and motivation are shaped by parents' attitudes and behaviors. This study examined parental motivation and involvement in their daughters' youth sport participation. Fifty-three parents of girls aged 8-16 completed the Parental Involvement in Sports Questionnaire (PISQ; Lee & MacLean, 1997), along with demographic, participation/motivation, and free response. The PISQ includes three subscales (1-5 Likert): Directive Behavior, Active Involvement, and Praise and Understanding. The sample included 41 female and 12 male parents, primarily ages 25-35 (37.7%), with similar proportions in the 36-45 and 46-55 groups (both 30.2%). Participants identified primarily as

Hispanic/Latino (52.8%), Native Hawaiian/Pacific Islander (26.4%), Caucasian (11.3%), Asian (5.7%), and American Indian /Alaska Native (3.8%). Parents expressed strong beliefs about the benefits of sport participation, identifying leadership development, confidence building, physical fitness, social relationships, college or career opportunities, and mental well-being as leading motivations. Parents also described feeling empowered through pride in their daughters' achievements, strengthened bonds, and increased advocacy for girls' sports. Descriptive results showed that sports contribute to future success ($M = 4.47 \pm .93$), enhance development ($M = 4.47 \pm .82$), and build character ($M = 4.53 \pm .72$). Positive changes observed included increased confidence, improved health, stronger leadership, expanded social networks, and better academic performance. Financial pressures and limited time/transportation were the most common barriers to participation. Independent *t*-tests revealed a significant gender difference on the Directive Behavior subscale, with men scoring higher than women ($p = .014$). Praise and Understanding had the highest overall mean ($M = 4.34 \pm .6$). A one-way ANOVA indicated ethnic differences on Directive and Praise, with Native Hawaiian/PI parents scoring higher than Caucasian parents on Directive ($p = .017$), and Caucasian, Hispanic, and Native Hawaiian/PI parents scoring higher on Praise ($p = .005$). Overall, findings highlight strong parental commitment to supporting girls in sport and the developmental benefits that accompany their involvement. Future research should draw from more diverse samples, incorporate daughters' perspectives, and further explore cultural influences on parental involvement.

Psychological recovery between training sessions: Applying the Recovery Experiences Framework to endurance training

Stuart G. Wilson, Queen's University; Andres Moran-MacDonald, Queen's University; Brendon J. Gurd, Queen's University

Recovery is the multidimensional process through which athletes restore performance capability between training sessions (Kellmann et al., 2018). While athletes describe psychological recovery experiences as highly important for their performance development (Eccles & Kazmier, 2019; Wilson & Young, 2023, 2024), sport psychology currently lacks ecologically valid approaches for representing the complexity of those experiences between real training sessions. In organizational psychology, recovery is conceptualized as a set of psychological experiences pursued during non-work time, including detachment from work, relaxation, and perceived control during leisure time (Sonnentag & Fritz, 2007). Within this framework, psychological recovery experiences are associated with improved performance and wellbeing (Steed et al., 2021), yet this approach has not been fully applied within sport training contexts. The purpose of this study was to explore correlates of psychological recovery experiences within a six-week endurance training intervention, including whether experiences following a workout were associated with (a) perceptions of recovery and stress before the next workout, (b) performance during that next workout, and (c) the type of training prescribed. 20 trained cyclists and triathletes (17 male, 3 female; ages 18-60) completed six weeks of high-intensity interval training (HIIT; three sessions/week), either alone (HIIT-only) or with an additional 240 min/week of Zone 2 cycling (HIIT-Z2). Before each HIIT session, participants completed the Short Recovery and Stress Scale (Kölling et al., 2020) and the Recovery Experiences Questionnaire (Sonnentag & Fritz, 2007). In-session performance was indexed using calories expended. Multilevel models (responses nested within participants) were used to test within-person associations (all $p < .05$). Within individuals, greater perceived control during leisure time was associated with higher perceived physical and psychological recovery ($\beta = .24-.36$) and lower psychological stress ($\beta = -.28$), but lower subsequent workout performance ($\beta = -.628$). Perceived control diminished over time in the HIIT-Z2 group ($\beta = -.013$). Detachment and relaxation

were largely unrelated to recovery, stress, performance, or intervention condition. Overall, perceived control emerged as a psychologically salient recovery experience that both shaped and was shaped by endurance training demands, underscoring the need for more refined conceptualization and measurement of psychological recovery in sport.

Help-seeking intentions among Canadian high-performance athletes: Preliminary findings from a national baseline study

Lauryn Bons, Brock University; Emma Shea, Brock University; Corliss Bean, Brock University; Paul Gorczynski, University of Greenwich; Stuart Vella, Wollongong University; Philip Sullivan, Brock University

Athletes experience elevated mental health challenges compared to the general population (Durand-Bush & Van Slingerland, 2021). However, the extent of this burden varies across high-performance (HP) contexts, with student-athletes often reporting mental health levels similar to their non-athletic peers (Saarinen et al., 2025), whereas national and Olympic/Paralympic athletes tend to exhibit heightened psychological distress (Poucher et al., 2021). Despite this variability, a consistent pattern emerges across all levels of sport: athletes are markedly reluctant to seek professional mental health support (Drew & Matthews, 2019). While research has shown that identity dimensions, including sex (Kew et al., 2025), as well as racial and sexual minority status (Anchuri et al., 2020; Gorczynski et al., 2017), are associated with disproportionate impacts of mental illness, it remains unclear whether these identity-based disparities extend to mental health assets such as help-seeking. This gap is particularly important within the context of The Mental Health Strategy for High Performance Sport in Canada, which emphasizes not only the reduction of mental illness, but also the importance of the proactive development of mental health assets (e.g., mental health literacy, perceptions of stigma, and help-seeking behaviors; Durand-Bush & Van Slingerland, 2021). Therefore, establishing a national baseline for these assets among Canadian HP athletes is critical for monitoring asset development over time and informing meaningful intervention efforts. To contribute to this need, the present study aimed to assess help-seeking intentions among Canadian HP athletes using the General Help-Seeking Questionnaire (Wilson et al. 2005). A preliminary analysis ($N = 125$; data collection ongoing) was conducted using a 2 (sex) by 3 (HP context) factorial MANCOVA, to assess help-seeking for emotional problems and suicidal thoughts, with age, and racial and sexual minority status included as covariates. The analysis revealed no significant main effects or interactions. This pattern contrasts with established disparities in mental illness prevalence and symptom severity among HP athletes. Help-seeking may function differently from other mental health indicators within HP sport. These results highlight the need for further investigation into the factors contributing to the consistency observed in mental health assets across diverse athlete identities. Funding source: SSHRC.

Building gatekeeper confidence in collegiate sport: An extension of the Talk Today program

Joshua Celebre, Brock University; Philip Sullivan, Brock University

The Talk Today program is a season-long, team-based mental health program developed by the Canadian Mental Health Association (Canadian Mental Health Association, 2019). A pillar of this program is the embedment of a Mental Health Champion (MHC) into the participating team(s). The MHC acts as a gatekeeper, connecting psychologically distressed athletes with community mental health resources. The present study examined Canadian collegiate sport mental health champions' (1) mental health help seeking attitudes (via Mental Help Seeking Attitudes

Scale; MHAS), (2) confidence in their skills to contact a suicidal collegiate athlete (via Gate Keeper Self-Efficacy Scale; GKSE), and (3) confidence in their skills to help a collegiate athlete deal with a mental health issue (via Referral Efficacy Scale; RES). Eleven mental health champions were assessed on these variables at baseline (September), 1 month (October), and 5 months (January). Following baseline assessment, MHCs completed a series of training modules focused on suicide symptom recognition and intervention, setting boundaries values and peer support, privacy and confidentiality, and trauma informed care. Separate repeated measures ANOVAs were conducted to examine changes in each outcome across the three timepoints. Results indicated a significant effect of time on gatekeeper self-efficacy, $F(1.18, 11.76) = 16.22$, $p = .001$, $\eta_p^2 = .62$. No significant effects were found of time on MHAS or RES. These findings suggest that the Talk Today program may be associated with improvements in MHCs' confidence to intervene with suicidal athletes, while help seeking attitudes and referral efficacy remained stable.

Scrolling, comparing, and checking: Social media's role in muscle dysmorphia risk

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Social media has become integral to daily life, shaping communication, engagement, and self-perception. Platforms that encourage constant self-presentation can negatively influence body image, particularly among men pressured to achieve an ideal muscular physique (Wilson, 2010). This study examined men's Appearance-Related Social Media Consciousness (ASMC), their awareness of how attractive they may appear to a social media audience, and its relationship to muscle dysmorphia symptoms. Thirty-four male participants aged 18–36 ($M = 26 \pm 4.6$) completed a survey including the ASMC scale, the Muscle Appearance Satisfaction Scale (MASS), and items assessing demographics, social media use, and exercise habits. Participants averaged 3.5 ± 1.75 hr of social media use per day and 5.2 ± 1.44 days per week. They exercised approximately $1.7 \pm .44$ hr per session, primarily through weight training. The total MASS score (52.91 ± 12.39) indicated moderate but below-median muscle dysmorphia symptoms. Subscale means for Bodybuilding Dependence ($3.05 \pm .75$), Injury Risk (3.03 ± 1.48), Muscle Checking ($2.63 \pm .90$), Muscle Satisfaction ($2.63 \pm .97$), and Substance Use ($2.54 \pm .99$) all fell below the 25th percentile. ASMC scores ($M = 36 \pm 17.4$) were also moderate but below the 50th percentile. Participants who reported comparing themselves to others on social media scored significantly higher on the MASS ($p = .042$) and ASMC ($p = .043$) than those who did not, and they reported more frequent muscle-checking behaviors ($p = .017$). A significant moderate correlation emerged between MASS and ASMC ($r = .52$, $p = .002$). Muscle checking ($r = .603$, $p = .0001$) and bodybuilding dependence ($r = .46$, $p = .006$) showed the strongest subscale associations with ASMC. In conclusion, although this sample did not demonstrate significant muscle dysmorphia symptoms, the results did show that men who engage in social comparison on social media exhibit higher appearance-related awareness and greater muscle dysmorphia symptoms. Future research should include larger and more diverse samples, assess longitudinal changes in ASMC and dysmorphia symptoms, and examine the psychological mechanisms that may mediate the influence of social media on men's body image.

Dancing through a lifetime: Motivators and stressors

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Dance is a lifelong activity; however, many think of it as an activity for young people. While some continue to dance throughout their lifetime recreationally, others simply start dancing later in life. Dance participation provides numerous health benefits from childhood development (Žošić et al., 2025) to slowing the aging process (Yan et al., 2018). Dance participation in young and adult populations is associated with thinner cortical gray matter than non-dancers, which indicates a greater cognitive reserve (Porat et al., 2016). Additionally, adults benefit from health improvements in gait, balance, posture, aerobic fitness, muscle strength, flexibility, body composition, and possibly metabolic and cardiovascular risk markers (Rodrigues-Krause et al., 2019). A Qualtrics survey was distributed to dance students aged 7+, through email and social media, to identify how often they experience specific stressors and motivators related to dance and to observe which change with age and which remain consistent across all ages. Responses from participants ($N=67$) between 9 and 80 years of age were analyzed. Chi-square analyses indicated significant differences ($\chi^2(78)=130.6$ $p<.002$) among the 4 age groups (youth(Y) 7-18 years, young adult(YA) 19-25 years, adult(A) 26-50 years, and older adult(OA) 51+ years) supporting the observation that dance students experience less stress with age. Responses from OA indicated they experience minimal stress and maximal enjoyment. The Y and YA had the greatest variability in their responses and indicated more stressors. Results indicate that the most frequently experienced stressors for the Y and YA were how their body looks in class, how their dance teacher and peers perceive their dance ability in class, and balancing school and dance. Adults of all ages are more motivated to dance due to the physical and mental health benefits, and the community, while the youth were motivated by achieving their goals and the physical health benefits. The most frequently experienced motivations for taking dance class were that they enjoy dancing, they get to challenge themselves, and they enjoy exercise. Overall, the consistent response from participants of every age was that they dance because they enjoy it.

In what ways are contextual and organizational factors related to coaching for life skill development? Results from the National Coach Survey

Samantha Bates, The Ohio State University; Dawn Anderson-Butcher, The Ohio State University; Kylee Ault-Baker, The Ohio State University; Emily Nothnagle, The Ohio State University; Sydney Mack, The Ohio State University

Youth sport is widely regarded as a context for promoting positive youth development and life skill development. Critically, relationships with coaches are thought to be powerful mechanisms through which the life skill development can occur (Ault et al., 2024; Pierce et al., 2024). However, inequities in youth sport systems may limit coaches' ability to foster these outcomes due to less access to resources, organizational support, and training. Guided by opportunity structure theories (Cloward & Ohlin, 1960; Roberts, 2009), this study examined how contextual and organizational factors were associated with coaches' perceived impact on athletes' life skill development. Using data ($N=4,964$) from the U.S. National Coach Survey (Anderson-Butcher & Bates, 2022), hierarchical linear regression analyses were conducted with predictors entered in three blocks: (1) structural context (school vs. community setting, urbanicity, proportion of athletes experiencing poverty, proportion of athletes identifying as Black, Latino, Asian/Pacific Islander, or multiracial), (2) access to community-level resources (e.g., equipment, transportation, safe practice spaces), and (3) organizational support and training (e.g., certification, life skills training, administrative support, peer support, paid vs. unpaid status, and having been evaluated last season). Context accounted for 5.4% of the variance in perceived impact. Access to community-level resources explained an additional 1.7% of the variance, with equipment availability

emerging as a significant predictor. Organizational support and training variables accounted for the largest increase in variance ($\Delta R^2=6.4\%$, $p<.001$), with training in life skill development, peer support, coaching certification, paid status, and prior evaluation associated with greater perceived impact ($\text{Adj } R^2=.133$). Findings suggest that coaches' capacity to support life skill development can be shaped by access to opportunity structures within youth sport organizations. Implications for future research, coach education and development, and systems change in youth sport will be discussed. Funding source: Susan Crown Exchange, Nike.

The influence of passion for a sports-based extracurricular activity at the end of high school on adolescents' performance and well-being at school during their first year in postsecondary education

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Participation in extracurricular activities (EAs) in high school is associated with positive educational outcomes, such as better academic performance and a higher likelihood of pursuing postsecondary education (Farb & Matjasko, 2012). Specifically, sports-based EAs have been associated with the development of life skills and higher cognitive functioning among adolescents (Biddle et al., 2019; Camiré et al., 2019). However, studies have shown that sports-based EAs are related both positively (Hamer et al., 2009) and negatively (Sabato et al., 2016) to adolescents' mental health. Moreover, participation in sports-based EAs in high school has students facing both academic (Conley et al., 2014; Gaudreau et al., 2023) and athletic (Alfermann & Stambulova, 2012; Hansen et al., 2019) challenges when they transition to postsecondary education. Grounded in the Dualistic Model of Passion (Vallerand et al., 2003), this study aimed to investigate how adolescents' passion for a sports-based EA in high school is associated with their performance and well-being at school during their first year of postsecondary education. Additionally, this study investigated the mediating role of basic psychological needs satisfaction and frustration in these relationships (Deci & Ryan, 2000). Therefore, 102 students (56% female, $M_{\text{age}}=17.71$ years, $SD_{\text{age}}=1.29$ years) partaking in their first year of postsecondary education and who practiced sports-based EAs during high school were surveyed. Results of path analyses showed that harmonious passion for sports-based EAs was positively associated with basic psychological needs satisfaction ($b=.31$; $p<.01$), while obsessive passion was positively associated with basic psychological needs frustration ($b=.35$; $p<.001$), during the transition. In turn, basic psychological needs satisfaction was positively related to academic performance ($b=.33$; $p<.01$) and subjective well-being ($b=.40$; $p<.001$), while obsessive passion was negatively associated with academic performance ($b=-.29$; $p<.01$) and subjective well-being ($b=-.29$; $p<.01$), during the first year of postsecondary education. These results suggest that, depending on its nature, adolescents' passion for sports-based EAs during high school impacts them beyond sports and beyond high school, and distinctively influences their thriving in postsecondary education.

A randomized study of coach education training in education-based athletics: Coaching beyond the Xs and Os

Samantha Bates, The Ohio State University; Dawn Anderson-Butcher, The Ohio State University; Kylee Ault-Baker, The Ohio State University; Emily Nothnagle, The Ohio State University

Building the competencies of coaches to support mental health and well-being is of increasing importance given escalating rates of mental health challenges in the general population and among student-athletes (Chandler et al., 2019). Coach Beyond is a program developed by LiFEsports at The Ohio State University in partnership with the Ohio High School Athletic

Association to equip coaches with knowledge, skills, and efficacy in psychosocial aspects of sport to coach “beyond the Xs and Os.” Coach Beyond programming includes a series of coach and student-athlete sessions designed to support student-athlete mental health, enhance mental performance skills, foster a positive team environment, and promote leadership and life skills. All trainings are grounded in experiential learning techniques, are gamified, and center around interaction and reflection (Bates et al., 2024a; Bates et al., 2024b), as recommended by Ciampolini and colleagues (2019). In partnership with the Ohio Department of Behavioral Health, this study used a randomized design to assess the effectiveness of the Coach Beyond programming. Three middle schools in one school district were randomized into three conditions, receiving: 1) both coach and student-athlete trainings, 2) coach trainings only, and 3) no training. Across the sites, 24 coaches participated in four hours of Coach Beyond training and 10 received no training. Repeated Measures ANOVAs assessed changes from pre- to post-intervention across the two groups on the Coach Beyond Behaviors Scale and specific training learning objectives (Bates et al., 2025). Repeated Measures ANOVAs revealed significant Time x Training interactions, indicating differential changes over time between the trained and untrained group. Specifically, a significant interaction was observed for overall coaching behaviors, $F(1, 32) = 8.52, p = .006$, with the trained group demonstrating greater improvement from pre- to post-intervention. Significant Time x Training interactions also were found for efficacy to link and refer student-athletes to mental health resources, $F(1, 32) = 5.62, p = .003$, and for awareness of mental skills to improve performance, $F(1, 32) = 5.39, p = .02$. This presentation will further highlight results of the study and future directions for research designed to improve coach competencies to support student-athletes. Funding source: Ohio Department of Behavioral Health.

The lasting effect of athlete aggressive behavior beyond sport

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The motivational climate fostered by coaches in youth sport contexts have been shown to influence athletes' immediate psychological and behavioral responses. Less, however, is known about the long-term impact on athlete aggressive behaviors and tendencies beyond sport. This retrospective mixed-methods study examined the relationship between the perceived motivational climate during participants' most impactful youth sport experience and their self-reported aggression in early adulthood. College students ($N = 114$; $M_{age} = 20.17$ years) completed the Buss-Perry Aggression Questionnaire (BPAQ; Buss & Perry, 1992), retrospective-adapted measures of the Perceived Motivational Climate in Sport Questionnaire (PMCSQ; Sefriz et al., 1992) and Caring Climate Scale (CCS; Newton et al., 2007) reflecting their most salient youth sporting experience. Multiple regression analyses were conducted to examine whether perceived motivational climate (i.e., caring, task-involving, and ego-involving) predicted aggression outcomes. Results indicated that perceiving a caring motivational climate during one's most impactful sporting experience significantly predicted lower levels of aggression in adulthood. Additionally, gender differences were observed, such that men reported significantly greater physical aggression compared to women. Qualitative analyses were conducted to further contextualize participants' experiences and interpretations of their past sport environments. Findings indicated that ego-involving climates (EICs) were commonly associated with maladaptive responses, including heightened frustration, and justification of aggressive behaviors in early adulthood. In contrast, caring, task-involving climates (CTICs) were linked to more favorable moral behaviors and adaptive responses that extended beyond the context of sport, such as emotional regulation techniques, conflict resolution, and personal growth.

Collectively, these findings suggest that motivational climates in youth sport may have enduring implications for aggression in early adulthood. Specifically, environments that emphasize effort, personal improvement, and foster a sense of belonging (i.e., CTICs) may function as protective factors by supporting emotional regulation and adaptive behavioral responses, whereas EICs appear to be associated with maladaptive outcomes, including increased aggressive behavioral engagement. These results highlight the importance of coach-created motivational climates in youth sport and underscore the need for longitudinal research examining how early sport environments shape psychosocial development across the lifespan.

Empowering or disempowering? Instructor climates, self-compassion, and their relation to music performance anxiety and motivation in college musicians

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College music students must manage constant performance and evaluative demands, including high instructor expectations, extensive practice requirements, and peer competition, which may influence motivation and music performance anxiety (MPA; Farley & Kelly, 2023). As leaders of college music studios, instructors facilitate the social and motivational climate in which these demands are experienced (Herrera et al., 2023). However, limited research has explored how students perceive these climates or how their perceptions relate to psychological outcomes. Drawing from Duda (2013)'s coach-created motivational climate framework, I investigated college music students' perceptions of the instructor-created motivational climate (ICMC) and whether distinct ICMC profiles emerged. I further examined whether these profiles differed in MPA and motivation, and whether the relationship between self-compassion and MPA varied across profiles. 221 college music students ($M_{age} = 20.44$) from 20 U.S. institutions completed surveys assessing perceptions of the ICMC, MPA, motivation, and self-compassion. Latent profile analysis indicated a two-profile solution. Profiles were characterized as “Empowered” ($n = 184, 83.3\%$), indicating high empowering and low disempowering climate perceptions, and “Disempowered and Isolated” ($n = 37; 16.7\%$), indicating low empowering and high disempowering perceptions. No significant differences in profile membership emerged based on demographic variables. Accounting for profile classification uncertainty using the BCH method, profiles were compared on MPA and motivation. Participants in the “Empowered” profile reported significantly higher intrinsic motivation than the “Disempowered” profile. Although the “Disempowered” profile showed higher mean levels of MPA and controlled motivation, the differences were not significant. Moderated regression analyses indicated that self-compassion was a significant negative predictor of MPA. However, self-compassion did not interact with profile membership. The findings indicate that a meaningful subset of college music students perceive disempowering and socially isolating studio environments. Further, the “Disempowered” profile was associated with lower intrinsic and higher controlled motivation. However, ICMC perceptions were not strong predictors of MPA. In contrast, self-compassion negatively predicted MPA across profiles, suggesting that coping strategies may impact musicians' anxiety experiences more than the instructional climate. Thus, these findings indicate ICMC as an important factor in music student motivation, while personal coping skills may be more critical for managing MPA.

“I will focus on my breathing during games and not just get mad”: SMART skills that high school athletes found helpful to implement

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Carolina at Greensboro; Grace Rogers, IMG Academy; Bryce Trail, Sharpen Skills Training; Margaret Njomo, Springfield College

Self-compassion has gained traction in sport psychology due to recent research showing its role in promoting athletes' psychological outcomes (see Cormier et al., 2025). However, evidence-based self-compassion interventions in sport settings are scarce, especially at the youth level. Implementing the Self-compassion, Mindfulness, & Athlete Resilience Training (SMART) Program, this study examined (1) session ratings of perceived helpfulness/enjoyability, (2) what athletes found helpful, (3) how and when they planned to implement skills learned, and (4) their suggestions to strengthen session effectiveness. Four groups of high school basketball and soccer athletes ($N=38$; 21 boys, 17 girls) in the southeast U.S. participated in the eight-week SMART Program, delivered virtually by mental performance consultants. SMART was modified from Germer and Neff's (2019) Mindful Self-Compassion Program with shorter sessions (45 min) and sport-specific components. After each session, the athletes indicated session helpfulness and enjoyability on a scale of 1–5 and completed open-ended prompts. Athletes viewed SMART as favorable, reporting small degrees of variation in perceived helpfulness ($M=4.14-4.44$) and enjoyability ($M=3.96-4.38$) across sessions. Thematic analysis (Braun & Clark, 2019) revealed the following patterns. Helpful Aspects: brief mindful/breathing resets and inner-coach self-talk were most frequently mentioned techniques due to their immediacy, portability, and "next-play" focus; breakout room peer sharing contributed to common humanity. Implementation (how and when): use resets and cue-words before games, at halftime/timeouts, and during game/bench moments for composure and refocus; practice skills in training and daily routines (e.g., brief breathing at home or school). Suggestions: shorter, more active segments; more peer interaction; personalization/1:1 options; inclusion of journaling prompts; coach involvement; clearer logistics; and continuity/follow-ups to sustain momentum. Findings support the acceptability of the SMART Program for high school athletes, although its delivery needs to be more activity-based, personalized, and structured in follow-ups to enhance its effectiveness. Athletes perceived in-the-moment skills as highly useful, strategically timed at pregame and in-game moments. SMART, and other self-compassion interventions alike, should emphasize brevity, peer interaction, and coach-aligned continuity to strengthen transfer to sport, school, and life.

The mediating role of psychological need satisfaction in the relation between passion for sport-based extracurricular activities and the functioning of high school students in school and sports

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Extracurricular activities (ECAs) often represent a passion for adolescents who engage in them. These adolescents are required to juggle the tasks and demands of both ECAs and school simultaneously. While related, those domains have specificities that are likely to foster or hinder adolescents' psychological needs. In line with the Dualistic Model of Passion and Self-Determination Theory, this research investigated how passion for an ECA influences adolescents' psychological need satisfaction (NS) and need frustration (NF) in ECAs and in school, as well as their functioning in both domains. Results of a study with 224 high school students ($M_{\text{age}} = 14.36$, $SD_{\text{age}} = 1.10$) showed that harmonious passion for ECA was positively related to NS in ECA ($\beta = .588$, $p < .001$) and school ($\beta = .460$, $p < .001$), while being negatively related to NF in ECA ($\beta = -.329$, $p < .001$) and school ($\beta = -.355$, $p < .001$). In contrast, obsessive passion for ECA was positively associated with NF in ECA ($\beta = .264$, $p < .001$) and school

($\beta = .229$, $p = .001$), while being negatively related to NS in school ($\beta = -.209$, $p = .002$). In turn, NS in ECA was positively associated with ECA subjective well-being ($\beta = .231$, $p < .001$), while NF in ECA was negatively related to it ($\beta = -.141$, $p = .011$). NF in ECA ($\beta = .327$, $p < .001$) and school ($\beta = .176$, $p = .009$) were positively related to ECA dropout intentions. School NS was positively associated with school subjective well-being ($\beta = .255$, $p < .001$), whereas school NF was negatively associated with it ($\beta = -.343$, $p < .001$). Finally, school NS was negatively associated with school dropout intentions ($\beta = -.229$, $p = .002$), while NF in ECA ($\beta = .160$, $p = .035$) and in school ($\beta = .162$, $p = .024$) were positively associated with them. Ultimately, the study reveals that passions for ECAs can either propel adolescents toward thriving or pull them into cycles that threaten both their well-being and their commitment to ECAs and to school. Funding source: Fonds de Recherche du Québec, Société et Culture.

Mapping the ride: A qualitative evaluation of character development in a youth mountain biking camp

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Out-of-school-time (OST) mountain biking has experienced substantial growth over the past five years; however, empirical research examining its specific impact on resilience, character development, and related psychosocial outcomes remains limited. This study investigated the experiences and character-related impacts of junior high and high school mountain biking athletes using a Positive Youth Development (PYD) framework (e.g., Shek et al., 2019). OST programs have been shown to encourage PYD while helping youth develop problem-solving competence and healthy social networks (Hill et al., 2020), including grit and resilience, by providing a supportive setting in which participants can practice handling challenges. Through semi-structured interviews, qualitative data were analyzed following Braun and Clarke's (2006) six-phase thematic framework to identify developmental patterns among athletes participating in the National Interscholastic Cycling Association (NICA) program. The analysis yielded three interrelated themes: (1) Family and Community Influence, (2) Personal Growth Through Challenge, and (3) Enjoyment and Identity as Riders. Participants emphasized that their engagement was heavily shaped by family traditions and encouragement from siblings or peers. One athlete shared, "My brother joined the team, and he really liked it, and he just kind of encouraged me to do it". Furthermore, athletes described how overcoming physical and technical challenges contributed to significant gains in confidence, responsibility, and perseverance. As one participant reflected, "It's helped me become like who I am today... it helps me keep my responsibility up". Finally, despite the sport's demanding nature, riders consistently highlighted mountain biking as a source of intrinsic motivation and joy, central to their emerging identity. One rider stated, "It's a sport that makes me happy. I can succeed... and still be excited to learn". Collectively, these findings highlight the developmental value of OST activities such as mountain biking in fostering competence, social development, and a resilient mindset, and provide a potential roadmap for parks and recreation agencies to leverage mountain biking as a scalable intervention to promote youth activity, resilience, character-building, and well-being in young people.

Exercise-induced muscle soreness reduces global sensorimotor processing efficiency without altering body representations

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Delayed-onset muscle soreness (DOMS), occurring 24–72 hours after unaccustomed or intense exercise, is characterized by muscle pain, stiffness, and neuromuscular fatigue. Beyond peripheral symptoms, emerging evidence suggests that DOMS primarily affects muscle spindle afferents rather than structural muscle damage. Such proprioceptive disruption could impair the central integration of sensory inputs and, in turn, reduce the efficiency of body-related motor imagery processes. The present study investigated whether DOMS alters implicit motor imagery performance and whether such effects are effector specific. In a first experiment, sport-science students were assigned either to a DOMS group, in which soreness was induced through whole-body eccentric exercises, or to a Control group. Forty-eight hours post-exercise, participants completed mental rotation tasks involving images of arms and legs, which require implicit motor imagery, as well as a rotated-number task serving as a non-body-related control. Response times and accuracy were recorded. Accuracy in motor imagery tasks was preserved in the DOMS group, indicating intact body representations. However, participants experiencing DOMS exhibited significantly longer response times for body-related stimuli. No such slowing was observed for the number rotation task. A second experiment examined whether the impact of DOMS was specific to the exercised effector. Two DOMS groups were formed: one following upper-body exercises and the other following lower-body exercises, each compared with a Control group. Results revealed that both DOMS groups displayed increased response times for mental rotation of limb images (arms and legs), irrespective of the exercised body segment, again without accuracy impairment. Together, these findings demonstrate that DOMS affects not only peripheral neuromuscular function but also cognitive aspects of motor control. Specifically, DOMS induces a transient, global slowing of body-related sensorimotor processing during implicit motor imagery, without compromising the accuracy of body representations. This central alteration in sensorimotor efficiency may hinder performance, delay recovery, and increase injury risk by impairing the timely proprioceptive updates required for effective movement planning and adjustment.

A VR-based cognitive-motor dual-task increases perceived mental fatigue and selectively impairs task-switching: A crossover comparison study of different task modalities

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Mental fatigue (MF) is a psychobiological state that results in increased perceived effort and potential impairments in cognitive and physical performance. Cognitive-only fatigue induction paradigms are well-studied but cognitive-physical dual-task and integrated cognitive-motor task modalities are underexplored. This study compared acute effects of three modalities on subjective MF, inhibitory control and cognitive flexibility using a within-subject four-condition (24-min each) counterbalanced crossover design. Twenty-eight participants (age: 23.3 ± 3.6) watched a neutral documentary (C), performed a cued color-shape task-switching paradigm seated (Cog), while walking on a treadmill (DT), and a VR-based cognitive-motor dual-task that demands set-shifting (VR-CMDT) on separate days. Subjective MF was assessed by a visual analog scale (VAS) immediately before and after each condition. A flanker task and a digit-based task-switching task were employed in pre- and post-tests to assess inhibitory control and cognitive flexibility. Interference-cost and switch-cost were calculated using balanced integration scores to control for speed-accuracy tradeoff. NASA-TLX mental and physical demands were collected immediately after each condition as manipulation checks. Condition $\times$ Time (4x2) ANOVAs were applied to VAS and cognitive performance scores, and one-way ANOVAs to NASA-TLX subscales with Bonferroni-adjusted post-hoc comparisons. Mental demand were higher in VR-CMDT, DT and Cog than C ($ps < .001$), and physical demand were higher in VR-CMDT and DT than Cog and C ($ps < .001$). A conditiontime

interaction was obtained in VAS ($p < .001$) showing increased MF from pre to post in VR-CMDT, DT and Cog ($ps < .01$), but not in C. The conditiontime interaction was also obtained for the switch-cost measure ($p = .003$). It increased after VR-CMDT ($p = .025$) and decreased after Cog ($p = .016$) with no change for DT and C. A main effect of time was found in interference-cost ($p < .001$), with lower post-test than pre-test scores. No condition*time interaction was found ($p = .688$). In conclusion, active conditions increased perceived MF and workload, but effects on executive control differed by modalities. Despite all active conditions employed a set-shifting based task, switch-cost was worsened after VR-CMDT and improved after Cog while inhibitory control showed practice-related effects without any interaction. These findings suggest that MF-induction modality may have an important role in subsequent cognitive performance.

Emotion regulation and leadership stability across a competitive season: A longitudinal network analysis

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Leadership within athletic teams is often treated as a stable individual attribute. However, recent network-based work has demonstrated that leadership structures are dynamic and can change substantially across a competitive season. In prior analyses presented at the International Network for Social Network Analysis (INSA), longitudinal leadership networks revealed meaningful instability, including the emergence and decline of perceived leaders over time. The present study builds on this structural evidence by examining emotion regulation as a candidate psychological mechanism associated with leadership stability under pressure. Participants were collegiate athletes from multiple Division I teams who completed preseason and postseason assessments and peer-based leadership evaluations. Leadership perceptions were quantified using weighted social network metrics, allowing identification of leadership stability and transition patterns across the season. These included stable leaders, emerging leaders, and fading leaders, which were athletes who were perceived as leaders early in the season but no longer held central leadership positions at postseason. Across athletes, cognitive reappraisal was positively associated with adaptive performance-related characteristics, including coping with adversity and peaking under pressure, whereas expressive suppression showed limited or negative associations with adaptive outcomes. Importantly, emotion regulation was not consistently associated with leadership rank at a single time point. Instead, preliminary longitudinal analyses indicated that higher cognitive reappraisal was associated with leadership retention across the season, whereas athletes lower in reappraisal were more likely to exhibit leadership decline. In contrast, personality traits such as extraversion were associated with early-season leadership visibility but did not reliably distinguish stable from fading leaders over time. These findings extend prior network-based evidence of leadership instability by identifying emotion regulation as a psychological process linked to leadership durability rather than momentary influence. Results highlight the importance of examining leadership as a dynamic, system-level phenomenon and suggest that regulation under pressure may play a critical role in maintaining leadership influence as team demands intensify across a season. This work underscores the need for longitudinal, context-sensitive approaches to understanding leadership and emotion regulation in sport.

NEKOMA Big Air Study: Predicting competitive snowboarding performance through pre-competition gait dynamics

Takahiro Hirao, National Institutes for Quantum Science and Technology; Daisuke Matsuyoshi, National Institutes for Quantum Science and Technology; Kazuma Mori, National Institutes for Quantum Science and

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Somatic and psychological conditioning is essential for elite athletes to maximize competitive performance; however, the precise relationship between these psychophysical states and competition outcomes remains elusive. This ambiguity stems from the fact that conventional questionnaires alone may not fully capture an athlete's psychological readiness around competition. Positing that gait kinematics serve as embodied indicators of these internal states, this study investigated whether gait dynamics during the pre-competition period could reflect readiness and predict performance. The study analyzed elite snowboarders participating in the "Nekoma Mountain Big Air" in Japan. Although fourteen athletes were initially recruited, data from 12 participants (aged 12–25, $M \pm SD$: 18.0 ± 4.0) were included in the final analysis due to sensor data loss. Instrumented footwear was distributed on the evening of February 9, and participants were instructed to wear the shoes for mobility within their hotel. Following retrieval on February 11, gait data recorded during the night before the competition and the competition morning were analyzed. To address high inter-variable correlations, nine kinematic variables (e.g., cadence and ground contact time) were reduced via principal component analysis into six interpretable gait features. Subsequently, a three-component Gaussian mixture model with a missing-data-aware framework was implemented to identify latent locomotor states. Finally, competition outcome was modeled using a hierarchical Bayesian ordered logistic regression, incorporating the six gait features and three latent locomotor states as predictors; top ranks indicate better performance. Model convergence was confirmed for all parameters ($R\text{-hat} < 1.01$, bulk ESS > 2500, no divergences). Model comparison revealed that gait data obtained on the morning of the competition provided a superior predictive fit compared to data from the previous night ($R^2 = .60$ vs. $-.63$). At the inter-individual level, athletes with top ranks tended to exhibit higher cadence (posterior mean = -1.49 , 94% HDI = $[-4.39, 1.85]$, $P(\beta < 0) = .84$: lower rank number indicates better performance), where negative coefficients correspond to better performance. Consistent with links between cadence and arousal, these findings suggest that gait analysis can capture performance-relevant factors, including psychophysical states, in elite athletes. Funding source: JST Moonshot R&D Grant (JPMJMS2295-01, JPMJMS2295-02, JPMJMS2295-05).

NEKOMA Big Air Study: Eye movements under sustained attention at rest are associated with world-class snowboarding competition rankings

Kazuma Mori, National Institutes for Quantum Science and Technology; Takahiro Hirao, National Institutes for Quantum Science and Technology; Daisuke Matsuyoshi, National Institutes for Quantum Science and Technology; Sorato Minami, NTT, Inc.; Seiji Matsumura, NTT, Inc.; Naoki Saijo, NTT, Inc.; Makio Kashino, NTT, Inc.; Makiko Yamada, National Institutes for Quantum Science and Technology

Eye movements support athletic performance, and many prior studies have examined eye movements during sports performance. Analogous to how high-performance athletes exhibit distinctive resting cardiovascular indices (e.g., lower heart rates), we hypothesized that oculomotor activities at rest may have stable patterns linked to elite performance. We therefore examined whether eye movements under sustained attention at rest are associated with rankings in Japan's premier Big Air competition. We measured eye movements using high-quality eye-tracking glasses (100 Hz sampling rate) in professional Big Air athletes ($n = 23$), with recordings collected several months after the competition. The athletes performed a 5-min sustained fixation task (fixating a central target) to capture oculomotor

activities at rest, not sports performance. We quantified eight metrics, including fixation, saccades, and pupil dilation, and examined their associations with competition rankings based on three judges' total scores. The correlation analysis showed significant associations between the competition rankings and four eye metrics (number of fixations: $\rho = .7$; duration of fixation: $\rho = -.67$; number of saccades: $\rho = .53$; and ratio of saccade duration to fixation duration: $\rho = .54$, $ps < .01$). Furthermore, using machine learning analysis (support vector regression with two-fold cross-validation), we found that the eight-eye metrics can accurately predict competition rankings ($\rho = .68$, $p < .001$). These results suggest that top-ranked Big Air athletes tend to show distinctive baseline eye movement patterns during sustained fixation, characterized by more frequent fixations and saccades and relatively longer saccade durations. Top-ranked Big Air athletes may rapidly sample the external environment through these eye movements under sustained attention. Although the present study is correlational and measured post-competition, eye-movement metrics at rest may serve as a candidate biomarker for elite performance and motivate future work on oculomotor-focused conditioning and training. Funding source: JST Moonshot R&D Grant (JPMJMS2295-01, JPMJMS2295-02, JPMJMS2295-05).

NEKOMA Big Air Study: Run-by-run mindset tracking relates to subjective performance ratings and objective scores during world-class snowboarding competition

Daisuke Matsuyoshi, National Institutes for Quantum Science and Technology; Kazuma Mori, National Institutes for Quantum Science and Technology; Takahiro Hirao, National Institutes for Quantum Science and Technology; Sorato Minami, NTT, Inc.; Seiji Matsumura, NTT, Inc.; Naoki Saijo, NTT, Inc.; Makio Kashino, NTT, Inc.; Makiko Yamada, National Institutes for Quantum Science and Technology

In world-class competition, performance unfolds under extreme environmental, physical, and social-evaluative constraints that are difficult to reproduce in laboratory settings. We conducted a multimodal field study during a real-world elite snowboard big air competition to capture these dynamics directly in situ. Here, as part of our broader study, we report a psychological aspect that characterizes run-by-run links between in-competition mindset and performance, focusing on metacognition and momentary psychological state. Elite snowboarders completed brief assessments immediately after each competitive run, including mental readiness, state flow, and subjective performance ratings. Objective performance was indexed using official judges' scores. We also collected personality measures as supplementary trait-level correlates. Our analyses revealed that metacognitive accuracy (calibration between self-evaluation and performance) and precision (run-by-run stability) were strongly associated with objective scores, such that higher metacognitive fidelity—more accurate and more stable self-evaluation—corresponded to higher scores. In a supplementary analysis, trait-level energetic extraversion was negatively associated with objective performance. Comparing training and competition contexts, athletes showed a modest increase in confidence and a higher propensity to enter flow during competition. Importantly, tension and flow exhibited dissociable relationships with performance indices: greater tension-related effort (i.e., effortful strain) was associated with higher subjective performance ratings, whereas flow aligned more closely with higher objective scores. These findings demonstrate that the in-competition mindset contains separable components that differentially relate to perceived versus scored performance, while metacognitive alignment provides an additional, strong marker of objective success. By quantifying these relationships in a world-class competitive environment, the study clarifies how metacognition and state dynamics jointly shape appraisal and achievement under pressure, informing training approaches that target in-competition mindset calibration. Looking ahead, our in-

competition measurement framework can be extended to integrate concurrent psychological, physiological, neural, and movement markers to support mechanistic modeling of performance under real-world competitive demands. Funding source: JST Moonshot R&D Grant (JPMJMS2295-01, JPMJMS2295-02, JPMJMS2295-05).

Designing tailored physical literacy resources to improve wellbeing in children: A co-creation study with community football coaches

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Physical literacy (PL) is recognised as a holistic framework for supporting children's physical, psychological, and social wellbeing. Community football coaches play a key role in shaping children's movement experiences; however, many report difficulties translating PL theory into accessible, context-specific practice. The aim was to co-create tailored PL education resources with grassroots football coaches to support children's wellbeing. Research questions were: (1) what type and format of coach education best supports coaches' understanding of PL and its relationship with child wellbeing? and (2) how can PL resources be designed to foster both PL development and wellbeing within football practice? Twelve male coaches were recruited using purposive sampling from clubs affiliated with the Warwickshire County Football Association (United Kingdom). All held a recognised coaching qualification and had at least two years' coaching experience. Data collection was completed through in-person co-creation workshops lasting 2–3 hours. Guided by principles of Participatory Action Research, workshops were structured using the Double Diamond Design Approach to support both divergent and convergent thinking. Coaches engaged in collaborative activities to explore barriers to PL implementation, evaluate existing PL materials, and co-design football-specific PL resources. The co-creation process resulted in the development of 12 resource plans centred on warm-ups, small-sided games, and football skill development activities, each mapped to recognised PL frameworks. Following the workshops, the lead researcher worked with a panel of expert stakeholders, including experienced football coaches and academics, to refine the resources and align them with wellbeing-focused coaching behaviors informed by Self-Determination Theory. This process emphasised autonomy-supportive, competence-enhancing, and relatedness-focused coaching practices. Data were collected via audio-recorded group discussions, participant worksheets, and researcher observation notes. Data were transcribed verbatim and analysed using reflexive thematic analysis. Coaches viewed PL as a useful framework for promoting inclusive, enjoyable, and confidence-building environments, and prioritised the development of football-specific resources that embedded PL principles within familiar coaching activities and session structures to create more awareness for supporting child wellbeing. Development of coach-led, co-created PL resources that support holistic child wellbeing in grassroots football contexts provides a tangible way for coaches to be guided in bringing PL theory into practice.

Head coaches' perceptions of the coach-athlete relationship in professional ice hockey

Adam Pilotte, McGill University; Marc Glaude, McGill University; Gordon A. Bloom, McGill University

High quality coach-athlete relationships in elite team sports contribute to numerous individual and team outcomes. Although research on the coach-athlete relationship is well established in youth and intercollegiate contexts, little attention has been given to this construct in professional sport. This represents an important gap, as developing strong coach-athlete

relationships in professional sport may be particularly challenging, with organizations often prioritizing winning over athletes' personal growth, development, and well-being. Therefore, the purpose of this study was to explore head coaches' perspectives on the coach-athlete relationship in the National Hockey League (NHL). We conducted 13 semi-structured interviews with current or former NHL head coaches, who averaged 24 years of coaching experience, including 11 years in the NHL and 5.5 years as NHL head coaches. Our reflexive thematic analysis revealed several challenges to the development of strong coach-athlete relationships in the NHL, including job instability that offered little time to forge strong relationships and the need to navigate multiple stakeholders (e.g., general managers, player agents). Despite these constraints, coaches described a recent shift in NHL coaching practices, moving from autocratic methods towards more collaborative, relationship-centered approaches. While some coaches favored a professional, task-oriented coaching style, most aimed to build close relationships by understanding players on a personal level, consistently emphasizing the importance of genuine care and authenticity. Once positive relationships were established, proactive and individualized communication methods were viewed as essential for maintaining them over time. When addressing performance – an inevitable component of the professional context – coaches described separating the person from the player by providing objective feedback on task-related behaviors as a way to protect the closeness of their relationship. This study is among the first to examine the coach-athlete relationship in professional ice hockey and illustrates how coaches in high-performance, results-driven environments build and sustain relationships with players. These findings provide practical coaching strategies for fostering strong coach-athlete relationships in performance-driven sport environments and also identify organizational implications for teams that value coach-athlete relationships, including the provision of sufficient stability to enable relationship building. Funding source: SSHRC.

Determining the internal structure of responses to a physical activity self-efficacy scale under a latent variable approach: A Monte Carlo study

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The objective of this study was to improve the validity of physical activity self-efficacy measurement. To accomplish this objective, a latent variable approach commonly advocated, but infrequently used in practice, for determining the internal structure of response patterns to a PASE scale was systematically studied. Specifically, experimental methods were used to investigate two research questions: (1) How many factors appeared to be warranted to explain the internal structure of response patterns to a PASE scale when the population model was one-dimensional? (2) How many factors appeared to be warranted to explain the internal structure of response patterns to a PASE scale when the population model was two-dimensional? Data were simulated using Monte Carlo methods under conditions commonly observed in practice (e.g., N; exploratory/confirmatory latent variable models) to maximize potential implications for data analysis in practice. Three sets of outcomes were monitored: convergence issues, model-data fit, and parameter estimates. A one-dimensional model outperformed two-dimensional exploratory and confirmatory models when the population model was one-dimensional. This finding should give self-efficacy researchers confidence that when the traditional unidimensional assumption is true then: (a) a one-dimensional latent variable approach is very likely to provide empirical evidence in support of this internal structure and (b) a two-dimensional latent variable approach, under either an exploratory or a confirmatory model, is very unlikely to provide a more compelling alternative when both conceptual and empirical evidence are considered. A two-dimensional exploratory model

outperformed a two-dimensional confirmatory model and a one-dimensional model when the population model was two-dimensional. This finding should give self-efficacy researchers confidence that when the traditional unidimensional assumption is false and a complex two-dimensional internal structure is true then: (a) a two-dimensional latent variable exploratory approach is very likely to provide empirical evidence in support of this internal structure and (b) neither a one-dimensional latent variable approach nor a two-dimensional latent variable confirmatory approach is very likely to provide a more compelling alternative when both conceptual and empirical evidence are considered. Findings from this study provide an empirical basis for fuller adherence to the long proposed, but rarely fully followed, validity guideline for measuring PASE.

When winning matters most: Performance pressure and coach mental health

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While the coaching profession is widely recognized as demanding, complex, and performance-driven, coaches' mental health experiences remain underexplored beyond accounts that focus on stressors, coping strategies, and burnout. In particular, limited research has examined how performance pressure intersects dynamically with coaches' identities, meanings, and mental health over the course of their careers. To address this gap, this study adopted a narrative inquiry approach using creative non-fiction to capture the lived experiences of Canadian university sport coaches with regards to their own mental health and related challenges. Eight Canadian university head coaches participated in individual semi-structured interviews. Data were analyzed using a narrative thematic analysis and presented through creative non-fiction vignettes. Findings indicated that, early in their careers, coaches experienced losses as deeply personal and emotionally taxing, often eliciting shame, guilt, irritability, sleep disturbances, diminished motivation, and, for some, contemplation of leaving coaching altogether. Across narratives, performance pressure was closely tied to fluctuations in mental health throughout competitive seasons. Many coaches described a turning point, often within the first five years of head coaching, where they began to redefine success beyond wins and losses. This shift toward values-based coaching, athlete development, and long-term program building reduced the emotional volatility associated with outcomes and supported greater fulfillment and sustainability. Reflective processes that clarified personal values and sources of self-worth were key in cultivating identities less dependent on performance outcomes. These findings position the performance narrative as a dynamic process through which coaches' mental health fluctuates over time, rather than as a fixed risk factor. This work has practical implications for coach education, mental health literacy, and reflective practice, offering narrative tools that may reduce stigma, normalize shared struggles, and support healthier, more sustainable coaching careers. Funding source: Fonds de Recherche du Québec—Société et Culture.

Relationships between quality participation in exercise and happiness among people with disabilities: An explanatory mixed methods study

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The Quality Participation Framework (QPF; Evans et al., 2018) is an evidence-informed framework designed to guide research and practices to optimize exercise participation for people with disabilities. Quality participation experiences may help to explain how exercise influences happiness (cf., Côté-Leclerc et al., 2017; Giacobbi et al., 2008). The objective of this explanatory mixed-methods study was to explore whether quality

exercise participation experiences are related to happiness among individuals participating in an exercise program for adults with physical disabilities. Participants were adults with spinal cord injury, multiple sclerosis or cerebral palsy ($n = 19$; $M_{age} = 47.1 \pm 15.6$ years; 7 women, 12 men) enrolled in an exercise program delivered in 24-session blocks. After each session, participants rated their quality participation using the Measure of Experiential Aspects of Participation (Caron et al., 2019) and happiness using the Subjective Happiness Scale (Lyubomirsky & Lepper, 1999). Interviews were carried out with a subset of eight participants. N-of-1 analyses were conducted to test contemporaneous correlations and predictive relationships between quality participation and happiness. Content analysis was undertaken to provide context on survey responses. Participants completed between 2 and 177 exercise sessions ($M = 48.3 \pm 53.6$ sessions). Contemporaneous correlations between quality participation and happiness differed across individuals in both the direction and strength of the correlations ($r = -.38$ to $.63$, $p < .001$ to $p = .96$). Quality participation did not predict happiness in future exercise sessions. Participants highlighted different QP elements (e.g., belongingness, challenge, meaning) as most important to their experience and had different explanations for their happiness scores. Some participants viewed themselves as happy people, while others viewed quality exercise experiences as a necessary component for their happiness. Overall, results suggest that quality participation experiences are related to happiness, but in a highly individual manner. Thus, exercise programs should be individualized to provide the quality-participation experiences that a participant deems most important. We encourage continued research on quality participation using idiographic/N-of-1 methods to shed light on how exercise experiences influence happiness and other aspects of subjective well-being on a case-by-case basis.

The meaning of sedentary behavior for people with spinal cord injury: A reflexive thematic analysis

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People with spinal cord injury (SCI) accumulate 10-13 hr of sedentary behavior a day. It is unclear what people with SCI know about sedentary behavior and its potential impact. Uncovering their understanding of sedentary behavior is an important first step to understanding and modifying the behavior. The purpose of this study was to examine how people with SCI interpret sedentary behavior, particularly sitting, and media messaging about sedentary behavior and sitting. We used semi-structured interviews to explore the meaning of sedentary behavior among people with SCI. Data analysis followed Braun and Clarke's six-phase reflexive thematic analysis approach, guided by a constructivist paradigm. We recruited 12 participants who were medically stable, living in the community, had a chronic SCI, and primarily used a wheelchair when outside of the home ($M_{age} = 49.2$ years, $SD = 12.7$; $M = 17.3$ years since injury, $SD = 8.6$). The five themes generated were: (i) understanding of sedentary behavior is constructed through personal experiences and social interactions, (ii) sedentary time exacerbates symptoms and symptoms increase sedentary time, (iii) activity is intentional, interruptions of sedentary behavior are incidental, (iv) life as a sitter: sedentary behavior becomes part of the identity, and (v) 'that's not for me,' media messages exclude people with SCI. People with SCI have a general understanding about sedentary behavior and its negative effects. Their experience with sedentary behavior is centred around their negative symptoms which are often unique to SCI. Sitting is perceived as unavoidable, but also as restorative and necessary. Media messages related to sedentary behavior are ableist and exclude individuals who use wheelchairs. Funding source: UBC

Public Scholars Initiative funded by UBC Faculty of Graduate and Postdoctoral Studies; UBC Pritchard Graduate Student Grant funded by the UBCO Vice Principal Research.

“I only have so much to give”: University students with disabilities’ experiences of coaching and enacting behavior change techniques

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Self-enactable behavior change techniques (BCTs; e.g., goal setting, self-talk) are designed to support behavior change, but how they are enacted to support physical activity participation remains understudied. To improve self-enactable BCTs, a BCT-focused coaching intervention was developed that shifted from guiding university students with disabilities through BCTs to coaching them on how to enact BCTs in daily life (e.g., teaching the mechanics of goal setting rather than helping a participant set a goal). However, little is known about students’ experiences of being coached and their lived experiences towards enacting BCTs in daily life. The purpose of this qualitative study was to explore how university students with disabilities experienced enacting BCTs while engaging in the intervention. Eleven university students (age range: 21–56 years) with physical disabilities (4) or chronic conditions (7) participated in the 12-week intervention and completed semi-structured interviews following the program. Interview data were analysed using reflexive thematic analysis, informed by a constructivist understanding of how participants made meaning of their experiences. Three themes captured participants’ experiences. The first theme, “I Only Have So Much to Give”: Behavior Change with Limited Bandwidth, reflected how pain, symptom unpredictability, fatigue, and competing demands constrained participants’ capacity to engage in BCT enactment (e.g., goal-setting). The second theme, “The Battle Was Mostly Internal”: Emotions, Mindset, and Mental Blocks, highlighted how fear, guilt, shame, and perfectionistic tendencies shaped enactment, while reflective conversations and strategies (e.g., positive-reframing) supported emotional relief and an open mindset. The third theme, “It Wasn’t Just About Exercise”: BCTs as Life Skills, captured how participants transferred BCTs beyond physical activity (e.g., problem-solving) and applying them to academic work, emotional regulation, and daily decision-making on managing disability flare-up. The findings suggest that coaching BCTs for enactment may support both physical activity behavior change and broader self-regulatory and emotional skills, advancing understanding of how future interventions can better support behavior change. Funding source: This research was supported by the Social Sciences and Humanities Research Council and Natara Ng was supported by a Social Sciences and Humanities Research Council Canada Graduate Scholarship (CGS-M). Shane N. Sweet was also supported by a Canada Research Chair.

Can algorithms build athletes? A scoping review of machine learning applications in athlete development

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The topic of athlete development encompasses a range of dimensions such as physical, psychological, skill and career development, each playing a critical role in guiding athletes through sport. In recent years, artificial intelligence (AI) and machine learning (ML) have emerged as tools for predicting and streamlining processes for evaluation in areas such as talent identification, performance outcomes, training load, and injury prediction. Despite the growing interest, the extent and nature of ML applications in athlete development remain unclear. This scoping review aimed to map the current research landscape regarding how ML has been used in athlete

development, identifying key focus areas, methodological trends, and gaps for future research to address. Following Arksey and O’Malley’s (2005) and Colquhoun et al’s (2014) six-step framework, our search identified 26 studies using ML in athlete development. Extracted data included ML techniques, athlete population and the developmental area of focus. The most prominent development areas include a) performance prediction, b) training load and recovery monitoring, c) skill evaluation and tactical analysis and d) talent identification and athlete selection. In terms of methodological approaches, supervised learning was most common and typically used for classification and prediction. Current research also reflected a strong focus on athlete performance and physical measures, rather than examining aspects such as psychological and career development. Few studies addressed how to interpret the ML models and their findings, which may be a critical limitation if results are meant to be readily adopted by coaches and other interest-holders. While current ML applications in athlete development show promise for enhancing athlete performance and balancing training load, the narrow focus on the physical and technical domains highlights the need for holistic models incorporating psychological, social, and career factors. Future research should explore ways to prioritize knowledge translation and explainability to ensure applicability of these ML models in supporting sustainable athlete development.

Using knowledge holders’ perspectives to build a physical activity-based intervention for the self-management of cancer-related cognitive impairment

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Individuals living with and beyond cancer (“patients” henceforth) often experience cancer-related cognitive impairment (CRCI)—i.e., issues with thinking, remembering, and focusing that can severely impair quality of life. Growing evidence suggests physical activity (PA) may be a promising form of behavioral self-management (SM) for cognitive health. However, many patients report low PA after a cancer diagnosis, highlighting the need to increase SM support. To design effective PA-based SM support interventions for CRCI, understanding and utilizing patients’ perspectives and needs is essential. Within a larger research program, the aims of this qualitative study were: (1) explore patients’ knowledge/perceptions of CRCI guidelines, treatment gaps, and SM strategies (with emphasis on PA), and (2) identify their barriers, facilitators, and needs for receiving PA-based CRCI SM support. Fifteen Ontario-based patients ($M_{\text{age}} = 58 \pm 14$; 87% female) were recruited purposively until data saturation to complete an online survey and semi-structured interview guided by the Theoretical Domains Framework (TDF). Descriptive statistics were computed for survey data; interviews were analyzed via hybrid deductive-inductive content analysis to map findings to TDF domains and the Capability, Opportunity, Motivation-Behavior Model (COM-B), as well as to identify emergent themes. Seven TDF domains across all COM-B components captured determinants influencing patients’ management of CRCI using PA and non-PA means. Key barriers were: (1) CRCI knowledge gaps, consistent with low-moderate survey scores ($M_{\text{score}} = 54 \pm 16\%$, range = 13–75%; knowledge), (2) pervasive cognitive difficulties (memory, attention, and decision processes), (3) distress and fear (emotions), (4) social invalidation/trivialization of CRCI (social influences), and (5) lacking/inaccessible resources and guidance (environment context/resources). Facilitators included: (1) patient initiative/activation (skills) and (2) valuing PA’s health benefits (belief about consequences). Patients voiced needs for greater CRCI and PA education (knowledge), emotional support (emotions), and personalized, evidence-based resources provided by informed and competent facilitators (environmental context/resources); preference was indicated for blended delivery formats incorporating

various learning techniques (e.g., hands-on activities, self-monitoring tools, discussion opportunities). By highlighting the content, delivery format, and actionable strategies necessary to strengthen patients' Capability, Opportunity, and Motivation to manage CRCI using PA, results can inform the development of future PA-based CRCI SM supports.

Negative peer relationships, athlete social identity, and maladaptive sport motivation in youth athletes

Justin T. Worley, The Ohio State University; Alan L. Smith, Utah State University

Positive peer relationships reflected in friendship quality and peer acceptance contribute to adaptive sport motivation in youth athletes, in part through their contribution to athlete social identity (Worley & Smith, 2025). Although negative interpersonal relationships are a naturally occurring feature in youth sport (Fraser-Thomas & Côté, 2009), less peer-focused research has examined challenges in peer relationships when studying motivational processes. Examination of how negative features of peer relationships are tied to athletes' social identity and sport motivation is warranted to better understand the social context of youth sport. The purpose of this study was (a) to examine the associations of negative peer relationships (friendship conflict, friendship victimization, peer rejection) with maladaptive sport motivation (constrained commitment, controlled motivation, athlete burnout), and (b) to examine whether athlete social identity (cognitive centrality, ingroup affect) mediated these associations. Participants from youth sport teams ($N=160$, male $n=105$, $M_{\text{age}}=15.1 \pm 1.9$ years) completed valid and reliable measures of the study constructs. Observed variable path analysis was used to test the hypothesized model and bootstrapped confidence intervals were used to assess indirect effects. While friendship conflict and friendship victimization were not directly associated with maladaptive sport motivation, peer rejection was positively associated with controlled motivation ($\beta=.26$, $p=.001$) and athlete burnout ($\beta=.20$, $p=.013$). Peer rejection was negatively associated with ingroup affect ($\beta=-.30$, $p<.001$). In turn, cognitive centrality was positively associated with constrained commitment ($\beta=.27$, $p=.005$) and controlled motivation ($\beta=.39$, $p<.001$) and ingroup affect was negatively associated with constrained commitment ($\beta=-.43$, $p<.001$), controlled motivation ($\beta=-.34$, $p<.001$), and athlete burnout ($\beta=-.38$, $p<.001$). Indirect effects showed that peer rejection was positively associated with constrained commitment ($\beta=.13$), controlled motivation ($\beta=.10$), and athlete burnout ($\beta=.11$) by way of lower ingroup affect. Results suggest that feeling frequently rejected by one's teammates may be tied to maladaptive sport motivation by diminishing the positive feelings associated with group membership. Funding source: This research received funding from the Utah State University College of Education & Human Services Graduate Student Research Award.

Social network analysis of perceived physical ability and peer exposure over time among adolescents in school-based physical activity

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Extensive research demonstrates that maintaining a physically active lifestyle confers lifelong health benefits such as physical fitness, cardiometabolic health, and enhanced cognitive and mental outcomes. Adolescence is a pivotal developmental stage when engagement in physical activity (PA), along with other enduring lifestyle choices and behavioral patterns, begins to take shape. In school-based PA settings, adolescents' perceived physical ability (PPA) and their exposure to peers' PPA may

shape PA behaviors and represent interrelated, modifiable factors. This study examined the longitudinal relationship between PPA and exposure to peers' PPA among Korean adolescents in school-based PA. A total of 112 South Korean middle school students participated in the study, with Time 1 (T1) conducted at the end of the spring term in 2019 and Time 2 (T2) at the end of the fall term in 2019. Social network analysis was applied to the longitudinal data because it offers a robust framework for examining the structure and dynamics of interpersonal-level relationships (e.g., exposure to peers) while incorporating intrapersonal-level attributes (e.g., peers' PPA). Participants self-reported PPA and nominated peers, enabling the calculation of exposure to peers' PPA. Exposure networks were analyzed using UCINET and NetDraw. A just-identified, two-time-point, two-variable cross-lagged panel model was fit in Stata 15. Gender was controlled for as a covariate in the model. The Type I error rate was set at .05. Results showed that exposure to peers' PPA increased over time, as indicated by higher mean out-degree, stronger tie strength, and more cohesive network visualizations. PPA at T1 significantly predicted exposure to peers' PPA at T2, whereas exposure at T1 did not predict PPA at T2. Collectively, these findings indicate that school-based PA networks can be more cohesive over time, reducing social isolation and enhancing peer connectivity. More importantly, this study suggests that adolescents' PPA is an important modifiable predictor of subsequent exposure to peers and may be more internally driven (e.g., mastery experiences) than socially constructed. These findings underscore the need to consider both intrapersonal and interpersonal factors, including their direct and indirect effects on PA, when designing effective school-based PA interventions.

Independent and combined associations of perceived stress, psychological resilience, and seeking social support with student-athlete mental health

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Conceptual models of resilience (e.g., Fletcher & Sarkar, 2012) highlight psychological resilience and adaptive coping strategies as key mechanisms that may buffer from potentially maladaptive effects of stress on sport performance. An emerging priority for sport psychology researchers is to better understand how the interplay of perceived stress, psychological resilience, and coping strategies may be tied to athlete mental health experiences (Bryan et al., 2023). The purpose of this study was to examine the independent and combined associations of perceived stress, psychological resilience, and seeking social support with college student-athlete mental ill-health (i.e., anxiety, depression) and well-being (i.e., hope). We hypothesized: (a) perceived stress would be positively associated with anxiety and depression, and negatively associated with hope, (b) psychological resilience and seeking social support would be negatively associated with anxiety and depression, and positively associated with hope, and (c) psychological resilience and seeking social support would weaken the associations between perceived stress with anxiety and depression, and strengthen the association between perceived stress and hope. Participants were Division I college student-athletes ($N=209$) at a large U.S. Midwestern university. Hierarchical regression analyses showed: (a) perceived stress was positively associated with anxiety ($b=.73$, $p<.001$) and depression ($b=.50$, $p<.001$), and negatively associated with hope ($b=-.18$, $p=.029$), (b) psychological resilience ($b=.69$, $p<.001$) and seeking social support ($b=.24$, $p<.001$) were positively associated with hope, and (c) there were two-way interactions between perceived stress and psychological resilience with anxiety and

depression, along with perceived stress and seeking social support with anxiety. Positive associations between perceived stress with anxiety and depression were progressively weaker at higher levels of psychological resilience. However, the positive association between perceived stress with anxiety was progressively stronger at higher levels of seeking social support. In total, 59%, 54%, and 51% of variance was explained for anxiety, depression, and hope, respectively. Results suggest psychological resilience can partly attenuate deleterious effects of perceived stress on mental ill-health. Future research is needed to address a broader array of student-athlete mental health indicators. Funding source: This research was supported by funding from The Ohio State University Jeffrey Schottenstein Program for Resilience.

Strokes vs. steps: A mixed-methods comparison of guided mindfulness on perceived exertion, attention, and self-compassion for rowing and walking

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Mindfulness-based interventions are increasingly utilized to enhance physical activity experiences (Bakken-Gay et al., 2025; Cox et al., 2020; Ullrich-French et al., 2024), yet the influence of movement modality on the efficacy of these interventions remains under-explored. The purpose of this study was to examine the impact of a guided mindfulness intervention during two distinct physical tasks, walking and rowing, on state mindfulness, attention, and psychophysiological responses. Participants ($N=22$; 17 female, 5 male; $M_{\text{age}}=20.50$) completed two experimental sessions. In each session, participants engaged in a self-selected intensity 20-min activity (walking or rowing) while listening to a matched guided mindfulness script, recorded by a female researcher. Measures included heart rate, rating of perceived exertion (RPE), and attentional focus collected at 5-min intervals, as well as post-session measures of state mindfulness, exercise enjoyment, and body image self-compassion. Participants also completed open-ended items regarding task experience. Results indicated a significant main effect for time ($p < .01$) for heart rate, RPE, and attention, reflecting expected increase in effort and more associative attention as exercise duration progresses, however, no significant condition main effects were observed for these metrics. Post-session analyses revealed significantly higher levels of session attention, indicating more associative attentional focus ($p = .011$), and mindfulness of the body ($p = .012$) during the rowing condition compared to walking, with additional trends emerging for mindfulness of the mind and self-kindness. Quantitative findings were corroborated by qualitative themes, where participants described rowing as a novel task requiring conscious focus on mechanics, contrasting the automaticity of walking. Exercise enjoyment analysis revealed no significant difference between modalities. The novelty or biomechanical complexity of rowing may facilitate greater somatic awareness compared to the automated nature of walking. These findings suggest that characteristics of the movement task itself may meaningfully influence mindful engagement during exercise, with implications for how mindfulness-based physical activity experiences are structured and delivered. Funding source: Texas Christian University Harris College Student Research and Creative Scholarship Grant.

Testing the displacement hypothesis: Does physical activity mediate the association between screen-based sedentary behaviors and health-related quality of life?

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Screen-based sedentary behaviors are often conceptualized as a risk factor for lower health-related quality of life (HRQoL) in young people. The displacement hypothesis posits that physical activity may mitigate these effects by displacing time spent in sedentary behaviors. This study examined whether physical activity mediates the association of three screen-based sedentary behaviors (i.e., watching videos, playing video games, and using social media and chat applications) with HRQoL. Participants ($n=330$, $M_{\text{age}}=18.15 \pm 2$ years) were predominantly male ($n=182$, 55%), with the majority (~60%) currently enrolled as full-time university or college students. Screen-based sedentary behaviors in min per day were assessed using an adapted version of the questionnaire for the screen time of adolescents (QueST), while physical activity (weekly volume was measured using an adapted version of the Self-Administered Physical Activity Checklist (SAPAC). HRQoL was assessed using the KIDSCREEN-10 index. Causal mediation analyses using nonparametric bootstrapping with 1,000 simulations were conducted to estimate Average Causal Mediation Effects (ACME) and Average Direct Effects (ADE). Results indicated that physical activity did not significantly mediate the relationship between any screen-based sedentary behavior and HRQoL (all ACME $p > .05$). However, significant direct effects were observed for specific screen-based sedentary behaviors. Time spent on social media showed a significant direct association with HRQoL ($b = .010$, 95% CI [.002, .018], $p = .014$), as did time spent watching videos ($b = .008$, 95% CI [.001, .015], $p = .032$). Conversely, video gaming was not significantly associated with HRQoL ($b = .006$, 95% CI [-.006, .016], $p = .298$). These findings do not support the displacement hypothesis in this context, as physical activity did not serve as a mechanistic pathway linking screen-based sedentary behavior to HRQoL. Consequently, interventions targeting screen-based sedentary behaviors and physical activity may need to treat these as distinct contributors to HRQoL rather than assuming a compensatory relationship. Funding source: Social Sciences and Humanities Research Council.

Relationships between quality of physical activity participation, physical activity levels, and subjective well-being in people with physical disabilities: A latent class analysis

Kenneth S. Noguchi, University of British Columbia; Amy E. Latimer-Cheung, Queen's University; Kathleen A. Martin Ginis, University of British Columbia

Physical-activity participation levels are critically low in people with disabilities. To improve subjective well-being, it is not enough for people with disabilities to simply perform physical activity, they must also experience full and effective (i.e., quality) participation. Quality physical-activity participation is positively associated with subjective well-being but to date has explained little variability in physical-activity levels. It may be that person-centred approaches are needed to delineate these complex relationships. The objectives of this study were to (1) identify subgroups of scores on a measure of quality physical-activity participation among people with physical disabilities, and (2) examine differences in physical-activity participation levels and subjective well-being between the subgroups. This study was a secondary analysis of a national cross-sectional study in people with physical disabilities in Canada. Participants were eligible if they reported participating in exercise or sport. Quality of physical-activity participation was measured using the 12-item Measure of Experiential Aspects of Participation (MeEAP), which assesses levels of autonomy, belongingness, mastery, challenge, engagement, and meaning. Latent class analysis was performed to generate subgroups based on MeEAP scores and compared across distal outcomes of interest, including physical-activity levels, life satisfaction, self-reported health, and perceived stigma.

Participants ($n = 504$, age: 50.1 ± 15.4 years, 46% women) belonged to one of four subgroups: (1) Unfulfilled (low scores on all items), (2) Fulfilled (highest scores on all items), (3) Fulfilled-Independent (high scores on most items, low scores on social items), and (4) Unfulfilled-Autonomous (low scores on all items, except autonomy). The highest physical-activity levels were observed in the Fulfilled (159.8 min/week [95% confidence interval, CI: 17.4 - 302.2]) and Fulfilled-Independent subgroups (178.5 min/week [95%CI: 59.2 - 297.8]). The Fulfilled subgroup had the highest life satisfaction, self-reported health, and lowest perceived stigma, whereas the Unfulfilled subgroup had the least favourable scores. Person-level subgroups of physical activity experiences may exist among people with physical disabilities. The results from this study suggest that, while high scores across every element may not be necessary for physical-activity participation, autonomy alone is not sufficient. Moreover, those social elements (e.g., belongingness) are likely to play a key role in subjective well-being in people with physical disabilities. Funding source: Supported by a partnership grant from the Social Sciences and Humanities Research Council of Canada (grant no. 895-2013-1021) for the Canadian Disability Participation Project (www.cdpp.ca).

Awareness of physical activity guidelines and their relevance to Canadians with disabilities: A nationwide survey

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There is little reported evidence on awareness of the Canadian 24-Hour Movement Guidelines (hereafter, 'the Guidelines') and the relevance of its messaging in Canadians with disability. This study had three research questions: (1) What is the prevalence of awareness about the Guidelines in a representative sample of Canadians with disability? (2) What is the association between awareness and adherence to the Guidelines? and (3) Are inclusive adaptations of the Guidelines' messaging more relevant than the original messages? This study was a cross-sectional survey sent to Canadians with disability conducted by ParticipACTION. Awareness of the Guidelines was measured using a single-item measure with four response options. The Leisure Time Physical Activity Questionnaire for People with Disabilities was used to assess Guideline adherence levels in the last seven days. Four inclusive physical activity (PA) messages were created by replacing terms such as "step" or "sit" from the original Guidelines with alternate wording, such as "wheel" or "being active in your own way". Relevance of the PA guideline messages was rated using a seven-point Likert scale. Of the 1,001 respondents (51.6 years, 58% women), 4.8% (95% confidence interval [CI]: 2.7-8.1%) were very aware of the Guidelines. Men, younger adults, and persons with physical disability were most likely to be aware of the Guidelines. Having at least heard the name of the Guidelines was associated with greater odds of meeting all components of the Guidelines ($OR = 5.13$, 95% CI: 2.43, 10.8, $p < .001$), and both moderate to vigorous PA ($OR = 2.05$, 95% CI: 1.38, 3.05, $p < .001$) and resistance training ($OR = 2.19$, 95% CI: 1.46, 3.28, $p < .001$) guideline components. Inclusive adaptations of the Guidelines were perceived as more relevant to Canadians with disability than the generic versions for two messages ($M_{\text{differences}} = .37$ to $.49$, $p = .002$ to $.006$). Overall, awareness of the Guidelines is low among Canadians with disability, which was associated with low PA adherence. More inclusive messages, particularly those that encourage diverse movements (e.g., moving or wheeling) rather than walking or standing, were perceived as more relevant. Targeted PA campaigns that use inclusive messaging could improve guideline relevance, awareness and, distally, PA levels in Canadians with disability.

Affective valence across exercise intensity: An individual participant data meta-analysis

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A growing body of evidence indicates a robust link between affective responses to exercise, commonly assessed using the Feeling Scale (FS), and long-term exercise adherence. Understanding the factors that shape affective responses during exercise is therefore of central importance. Dual-Mode Theory (DMT) posits a curvilinear relationship between exercise intensity and affect, characterized by pronounced inter-individual variability. Although numerous studies have examined affective responses to exercise, this proposed intensity-affect relationship has not yet been systematically investigated at scale using individual participant data (IPD), which are required to adequately capture its average form and the variability across individuals. To address this gap, we conducted a systematic review and IPD meta-analysis examining the relationship between exercise intensity and affective valence during aerobic exercise, using age-predicted percentage of maximal heart rate (%HRmax) as a practically accessible proxy for intensity. Compared with aggregate-data approaches, IPD meta-analysis allows for more precise estimation of nonlinear effects and the direct modeling of inter-individual differences. A total of 173 studies assessing FS responses and heart rate during aerobic exercise bouts of at least 10 min were identified. Of these, 43 studies provided usable IPD, comprising 1,707 participants ($M_{\text{age}} = 26.88 \pm 11.21$ years; 47.1% female) and 17,924 observations. Multilevel regression models were used to predict FS responses as a function of age-predicted %HRmax (Tanaka formula) and its quadratic term. Random effects accounted for the hierarchical structure of observations nested within participants, studies, and study conditions. The intercept was centered at 50% HRmax. Results supported the curvilinear relationship proposed by DMT, indicated by the linear and quadratic fixed effects (intercept = 3.02 [95% CI: 2.71 , 3.33]; $b = -.0167$ [95% CI: $-.0257$, $-.0078$]; $b_2 = -.00089$ [95% CI: $-.00112$, $-.00066$]). Importantly, substantial between-person variability was observed, underscoring that affective responses differ markedly between individuals. Overall, this IPD meta-analysis provides a more precise and differentiated characterization of the intensity-affect relationship than previously available. By leveraging participant-level data across multiple studies, the findings strengthen the empirical foundation of DMT and highlight the importance of accounting for individual differences when using affective responses to inform exercise intensity prescription and future theory-driven research.

The moderated mediation model of mindfulness, competitive anxiety, and self-efficacy in Taekwondo athletes with and without injury

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Combat sport athletes experience a relatively high risk of injury, which is frequently associated with elevated competitive anxiety and reduced pre-event self-efficacy. Mindfulness has been proposed as a potential psychological buffer against these adverse effects. Accordingly, the present study investigated the association between athletic mindfulness and pre-event self-efficacy through competitive anxiety; and further examined whether this relationship was moderated by injury status among taekwondo athletes. Collegiate taekwondo athletes ($N = 105$) who competed in a national competition were assessed for athletic mindfulness, competitive anxiety, pre-event self-efficacy, and injury status. A moderated mediation model was tested using PROCESS Models 4 and 59. Athletic mindfulness

significantly predicted pre-event self-efficacy through cognitive anxiety. Injury status did not moderate the overall mediation pathway, instead, it moderated the relationship between athletic mindfulness and cognitive anxiety. These findings suggest a pathway through which mindfulness enhances pre-event self-efficacy by reducing cognitive anxiety among combat sport athletes. Although injury status did not significantly moderate the overall mediation effect, the association between mindfulness and cognitive anxiety appeared to be stronger among injured athletes. Funding source: National Taiwan Normal University (NTNU) within the framework of the Higher Education Sprout Project (HESP) by the Ministry of Education (MOE) in Taiwan.

Visuomotor skills do not differ by sex in highly trained baseball athletes

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Performance in interceptive sports like baseball requires the continuous integration of complex visual perceptual information by the athletes to either catch or strike the ball. While it has been shown that males often outperform females in basic visuospatial tasks within non-athletic samples, recent evidence suggests these differences are reduced when individuals possess similar sport-specific expertise. Therefore, this study aimed to compare sex differences in underlying visuomotor skills and baseball batting performances. Seventy highly trained adolescent and young adult baseball athletes (42 males; $16.40 \pm .89$ years; range: 15-19, 28 females; 16.93 ± 1.89 years; range: 15-23) participated in the study. Athletes completed a standardized computer-based testing battery (Senaptec Sensory Station, OR, USA) which assessed six visuomotor skills: visual acuity, contrast sensitivity, accommodation, multiple-object tracking, working memory, and reaction time. Batting performance data, including maximal and average ball exit speed and distance, and average launch angle were collected using a Trackman B1 unit (Vedbæk, Denmark). Independent samples *t*-tests were performed to compare male and female athletes across all variables. The independent samples *t*-test results revealed no significant differences between male and female baseball athletes across any of the visuomotor variables (all $p > .05$). Specifically, no sex-based differences were found for binocular visual acuity ($p = .581$), contrast sensitivity ($p = .235$), accommodation ($p = .734$), working memory ($p = .989$), multiple-object tracking ($p = .514$) and reaction time ($p = .140$). In contrast, significant sex differences were observed for all batting performance indicators ($p < .01$), with male athletes showing higher maximal and average ball exit speed, greater average and maximal batted ball distances, and higher average launch angles. These findings support the hypothesis that visuomotor skills do not differ by sex among highly trained baseball athletes practicing the same sport with similar level of expertise. While more physical-related batting performance remains distinct between sexes, the underlying information-processing and visual systems appear to adapt similarly to the demands of sport-specific high-level training and expertise. Funding source: M.T. doctoral studies are supported by the Natural Sciences and Engineering Research Council of Canada (NSERC), [ES D - 589592 - 2024].

Aging well and moving more: The development and evaluation of a physical activity education program for older adults in a rural community

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Physical activity (PA) is a modifiable behavior that can benefit the health and quality of life of older adults. However, adherence to recommended PA guidelines remains low among older adults, particularly those living in rural communities, where PA levels are lower and health outcomes are poorer. The Appalachian Institute for Health and Wellness is a University-managed health clinic that supports the local aging population of rural western North Carolina through the Aging Well program's comprehensive health assessment. While this assessment provides general PA recommendations, individualized guidance and follow-up support for PA are not provided. Additional evidence-based programs targeting the unique PA-related challenges experienced by older adults in rural communities are needed to help this population incorporate regular PA into their lifestyle. To address this critical need, a team of kinesiology professionals with diverse expertise in PA and aging developed the Aging Well and Moving More program for adults 45 years and older. First, the team conducted focus group interviews with previous participants of the Aging Well health assessment to ascertain their PA needs, preferences, and barriers. This feedback informed the program's aims: to increase participants' knowledge of PA health benefits, build competence for safe and effective PA participation, promote autonomy through self-selected PA goals and plans, and foster relatedness through supportive social connections. Next, the team met biweekly over four months to systematically design the program structure, define specific topics, develop participant materials, and plan evaluations of program effectiveness. As a result of these efforts, the complete program consists of group education sessions paired with small group workshops. Evaluations of program effectiveness are integrated into the program structure, including learning checks after each session and post-program focus group interviews to collect participant feedback. Additionally, pre- and post-program assessments of motivation, cardiovascular health, gait and balance, and cognitive function are obtained, along with the administration of PA and health surveys at regular intervals for one-year following program completion. This data will inform iterative program refinement, and the Aging Well and Moving More program can serve as a blueprint for the development of PA education programs for older adults in other communities.

Systematic observation and analysis of Taekwondo coaching behavior in competition

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Since access to an expert coach is critical for athletic excellence, understanding the qualities of expert coaches warrants investigation. However, previous investigations have typically occurred in practice settings, with considerably less work examining competition contexts. The competition context provides an important venue for understanding how combat sport coaches communicate with their athletes. This study developed and tested a methodology for assessing combat sport coaching and provided a preliminary baseline of Taekwondo-specific coaching behaviors. A mixed methods approach was used to examine the behaviors used by high-performance Taekwondo coaches during elite competition to explore the coaching strategies underpinning different behaviors, and how these behaviors and strategies may differ depending on athlete experience level (i.e., Cadet (12–14 years), Junior (15–17 years), and Senior (17+ years)). Five Canadian National Team coaches were audio and video recorded during the National Championships. Recordings were analyzed using Behavioral Observation Research Interactive Software (BORIS; Friard & Gamba, 2016), and behaviors were coded using pre-established definitions and decision trees. Additionally, qualitative

interviews using video-stimulated recall were conducted after the competition to provide insight into the coaches' decision-making processes. Quantitative analysis showed no significant differences in behavior frequency between matches with a win or loss outcome. In the age group analysis, significant differences were found in management behavior, $F(2, 52) = 7.10, p = .002, \eta^2 = .21$, and performance-state silence, $F(2, 51) = 7.64, p < .001, \eta^2 = .23$, with higher frequencies of management behavior and longer durations of performance-state silence at the Senior level of competition compared to Cadets and Juniors. No other statistically significant differences were found. Qualitative analysis provided context for these quantitative findings by highlighting how expert coaches effectively use behaviors that directly influence match score, adapt behavior according to athlete individual differences, and balance the desire for development with the desire for results. Future research should explore larger samples, address gaps identified in the current coaching behavior observation scale, expand investigation into practice settings across the pre-competition training cycle, and include coaches from multiple countries to examine cross-national differences in the effects of coaching behaviors.

Are we preparing for the next chapter? Social norms in academic support and career services use among NCAA student-athletes

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Failure to prepare for a career beyond college sports can result in mental health challenges, financial instability, and difficulties securing post-college employment among student-athletes. Although most institutions provide academic and career support services, many student-athletes do not use them due to stigma. Social norms, which refer to individuals' perceptions of how others think or act, may help mitigate stigma and promote services use; if student-athletes believe others commonly use and approve of these services, they may be more likely to use these services themselves. Given that prior studies have shown that social norms influence student-athletes' behaviors across various domains (e.g., help-seeking, alcohol consumption), social norms may relate to student-athletes' services use. This study examined the prevalence of NCAA student-athletes' academic support and career services use and its relationship with their social norm perceptions. NCAA student-athletes ($N = 116$; $M_{age} = 19.6$ years; 64.7% female, 35.3% male) across three divisions from various sports completed an online survey assessing their perceived availability, familiarity, and frequency of use of academic support and career services. They also reported their perceptions of how often (1) their teammates, (2) other student-athletes at their institution, and (3) NCAA student-athletes nationwide use the services (i.e., social norm perceptions). Ordinal logistic regression was conducted to examine the relationships between student-athletes' social norm perceptions from the three referent groups and their own services use. Most participants reported that their institution offers academic support services ($n = 88$; 75.9%) and career services ($n = 101$; 87.1%). Among student-athletes who reported having academic support services (75.9%), perceptions of their teammates' use of services were positively associated with athletes' own use ($OR = 1.04, \beta = .044, p = .005$), whereas perceptions of how often other student-athletes at their institution and NCAA student-athletes nationwide use the services were not statistically significant correlates. Among those who reported that their institutions offer career services (87.1%), social norm perceptions from all referent groups were not significantly associated with student-athletes' own use. Findings highlight the importance of proximal peer influences, which could inform future interventions promoting services use to support student-athletes' academic and career development.

Effects of a pregame speech intervention on coach pregame speech efficacy and coaching efficacy

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Pregame speech is not a coaching skill often taught formally. The purpose of this study was to investigate the effectiveness of a self-efficacy theory-driven intervention created to educate and build coach confidence in their ability to provide effective pregame speech. Participants ($n = 37$) were recruited from attendees at a coaches' association clinic. Using a 2 x 2 randomized experimental design (group by pre-/post-test), participants were randomly assigned into intervention ($n = 20$) and control ($n = 17$) groups. The 23-min intervention was designed based on Bandura's four sources of self-efficacy to provide participants with opportunities to engage with material regarding what pregame speech is, how it has been previously used, and how pregame speech can be used effectively in their own sport contexts. All participants completed measures of pregame speech self-efficacy (Coach Precompetitive Communication Questionnaire-Self Efficacy; CPCQ-SE) and coaching efficacy (Coach Efficacy Scale II-High School Teams; CESII-HST) prior to and following the intervention. Data were analyzed in Mplus via two regression models to estimate mean differences for the intervention group as compared to the control group on post-intervention scores (e.g., CPCQ-SE or CESII-HST), while controlling for scores at baseline. One-tailed hypothesis tests were used based on positive results from a pilot trial. Both statistical significance ($\alpha = .05$) and effect size (Cohen's $d \geq .20$) were considered. Results of the regression models showed significant positive effects between the intervention group and post-intervention mean scores for tactical content ($B = 1.70, p = .04, d = .74$), emotional content ($B = 1.07, p = .04, d = .77$), and delivery ($B = 1.38, p = .02, d = .87$) of the CPCQ-SE and motivation ($B = 1.70, p = .01, d = .89$) of the CESII-HST. Other CESII-HST subscales did not have significant associations with the intervention group (game strategy, $B = .69, p = .16, d = .45$; character building, $B = .26, p = .33, d = .18$). The results of this study provide evidence for the effectiveness of the intervention on improving coach pregame speech efficacy and coaching efficacy compared to a control group.

Leadership development in men's university ice hockey: From first-year player to team captain

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University sport provides a compelling context for examining athlete leadership development as athletes typically spend four to five years within the same program, creating a natural developmental trajectory. As athletes progress from first-year players to senior leaders, they encounter evolving challenges, responsibilities, and mentorship opportunities that may contribute to the development of leadership skills. Yet, there is limited empirical research examining how leadership skills evolve across an athlete's university career, particularly from the perspective of athlete leaders themselves. Therefore, the purpose of this study was to explore the experiences that contributed to the development of leadership skills among men's university ice hockey captains. Participants were six NCAA Division I men's ice hockey captains (five current and one who was captain the previous season). All completed individual virtual semi-structured interviews, which were analyzed using reflexive thematic analysis. Findings demonstrated that leadership development in men's university ice hockey is largely affected by experiential, relational, and cultural factors.

Experiential development occurred as athletes learned through exposure to challenges, observing veteran players and coaches to understand expectations and standards, and taking on increased responsibilities over time. Athletes described “earning the right to lead” through accumulated experiences and consistent work ethic. Relational development was grounded in mentorship and authentic relationships with teammates and veteran leaders who provided guidance, feedback, and space to develop their voices as leaders. Honest communication with coaches and trustworthy relationships with teammates were highlighted as critical to leadership growth. Cultural development reflected the influence of established team norms and the presence of strong athlete leaders during participants’ early university years, which shaped how they understood, internalized, and eventually embodied the program’s standards and expectations as leaders themselves. Overall, these findings highlight the dynamic and socially embedded nature of athlete leadership development in university sport. The results can inform coaches and sport organizations seeking to design intentional leadership development pathways that leverage experiential learning, relational support, and team culture to support athlete leadership growth and enhance team functioning. Funding source: The Social Sciences and Humanities Research Council (SSHRC).

Domain-specific physical activity and mental well-being in emerging adults with physically demanding roles

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Many emerging adults struggle to meet recommended levels of physical activity (PA), which poses significant implications for mental well-being. While previous research has explored total PA in relation to aspects of mental well-being, little attention has been given to the role of domain-specific PA. Evidence suggests occupational PA may adversely affect mental health, but such research often overlooks certain physically demanding roles such as military personnel and athletes. The present study examines associations between occupational, leisure-time, and transportation-related PA with indicators of mental well-being in emerging adults. Using data from a cross-sectional study of emerging adults (18–29 years) identifying as collegiate student-athletes, military college students, or recent military school alumni ($N = 258$, 66% men), we analyzed differential impacts of PA domains on mental well-being. PA was assessed using the International Physical Activity Questionnaire Long-Form and operationalized as self-reported min of domain-specific moderate- to vigorous-intensity PA (MVPA) per week. Participants reported PA associated with military or student-athlete training under the occupational domain. Mental well-being indicators were assessed using subscales from the PROMIS Global Health measure (i.e., mental health-related quality of life) and the Short Form Health Survey (i.e., emotional well-being, energy). Linear regressions revealed distinct associations between PA domains and mental well-being. Transportation-related MVPA was negatively associated with emotional well-being ($\beta = -.006$, $p = .035$), whereas leisure-time MVPA showed positive associations across indicators (energy: $\beta = .010$, $p = .004$; mental health-related quality of life: $\beta = .003$, $p = .011$). Occupational MVPA was non-significant across all indicators. By delineating the unique contributions of PA domains on mental well-being, this study enhances our understanding of how PA context shapes one’s overall health. These findings emphasize the importance of promoting leisure-time PA to support mental well-being in emerging adults, while recognizing the potential adverse or limited effects of transportation-related and occupational MVPA, respectively. This research informs targeted interventions

aimed at optimizing PA engagement to improve mental well-being in this specialized, physically active population. Activities undertaken out of obligation, such as commuting or work-related tasks, may involve stressors and/or lack of choice—which undermine psychological benefits typically linked to PA. Funding source: School of Health and Human Sciences at the University of North Carolina at Greensboro and the Virginia Military Institute Department of Human Performance and Wellness.

Associations between overall fitness and individual fitness components with cognitive performance

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Existing studies report inconsistent results on the relationship between fitness and cognition in healthy young adults and often examine only one fitness component (e.g., Gajewski et al., 2023; Hayes et al., 2014). Furthermore, it is unclear whether an individual fitness component or overall fitness is more strongly associated with cognitive performance. Therefore, we investigated the associations between individual fitness components and an overall fitness composite (COMP) with information processing and inhibitory control in 34 healthy young adults (18–29 years). To evaluate information processing, participants completed two eight-trial blocks of a simple reaction task using EMG to fractionate responses into premotor and motor components. One, 120-trial block of the Flanker task, consisting of congruent (e.g., <<<<<), incongruent (e.g., <<<><), and neutral (e.g., ##>##) conditions, assessed inhibitory control. Next, participants completed four fitness tasks: three maximal countermovement jumps (CMJ) on a force plate (Bertec FP5060) to assess lower-body power; three maximal grips (GRIP) with each hand using a dynamometer (Takei KiKi Kogyo) to assess upper-body strength; three one-min trials attempting to maintain parallel on a mediolateral stabilometer (Lafayette Instrument Company, USA) to assess balance; and the three-min YMCA step test and one-min recovery heart rate to evaluate cardiorespiratory fitness. We averaged individual fitness outcomes and used z-scores to create a composite (COMP) representing overall fitness. Pearson’s correlation analyses demonstrated inverse relationships between individual and overall fitness with cognitive performance. Specifically, COMP related to faster Flanker (congruent: $r = -.386$, $p = .032$; incongruent: $r = -.404$, $p = .024$; neutral: $r = -.419$, $p = .019$) and reactions (total: $r = -.508$, $p = .002$; premotor: $r = -.448$, $p = .008$). CMJ related to faster reactions (total: $r = -.524$, $p = .001$; premotor: $r = -.441$, $p = .009$), and GRIP related to faster incongruent Flanker ($r = -.417$, $p = .020$) and reactions (total: $r = -.430$, $p = .011$). We did not detect any other significant associations. Our findings suggest that overall fitness is a broader correlate of central processing speed and inhibitory control compared to individual fitness components in healthy young adults.

The crux of safely teaching group decisions under competition: Initial insights from a case study of the German Alpine Club’s expedition squad

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Although groups provide their members with numerous benefits, they can also entail drawbacks such as skewed and inaccurate decision-making (Janis, 1972; Myers & Lamm, 1976). A physical activity context in which both groups and decisions are prevalent – and where inaccurate decisions may entail severe consequences – is mountaineering (Burnette et al., 2011). Although we are starting to better understand the psychology of mountaineering in general (Jackman et al., 2020) and know a bit about

group-decisions in the lab or individual decision-making in the mountains (Crust et al., 2016; van Knippenberg et al., 2000), we know little about the ways in which groups make decisions in these dynamic and uncertain environments and which factors discriminate between success and failure – or survival and death. To advance our understanding of the phenomenology, consequences, and influences of group decision-making at the (aspiring) elite level of mountaineering, we conducted an instrumental case study (Hodge & Sharp, 2017) of the male expedition training squad of the German Alpine Club (3 coaches, 9 athletes). Operating from a relativist-subjectivist position (Thorpe & Olive, 2017), leaning on the conceptual framework for sport teams (Eys et al., 2020) and IGLOO model of ecological nesting (Nielsen et al., 2018), we implemented pre- and post-event focus groups and interviews, a pre-event questionnaire as well as daily debriefs, questionnaires, and observations across a five-day course of big wall climbing in the Dolomites. Our ongoing reflexive thematic analysis (Braun & Clarke, 2022) suggests that group decision-making in this context can be captured by themes of (a) What Counts as Decisions – Which Decisions Count, (b) Competing Feelings at the End of the Day, (c) Different People and Dynamics, (d) Teaching Standards and Challenges, and (e) The Limits of Legacy. Although these insights pertain to our unique case, they highlight that group decision-making in mountaineering may be an everyday process that requires the coordination of different individuals and individual differences and that it may be challenging to define decision quality and teach decision-making skills safely. Funding source: Albrecht von Dewitz foundation.

“I can’t stop thinking about my body”: When rumination shapes emotional pathways linking body evaluation to mental health

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Sport and physical activity (PA) contexts frequently prompt individuals to evaluate their bodies, yet these evaluations may have unintended consequences for mental health (MH). Importantly, individuals differ in the extent to which they dwell on body-related evaluations, and this body-related rumination may intensify emotional responses to these evaluations. However, the emotional mechanisms through which body evaluation influences MH, and the conditions under which these mechanisms operate, remain insufficiently understood. The present study examined whether body-related emotions mediate the relationship between body evaluation and MH, and the influence of body-related rumination on these pathways. Data were collected from Canadian adults ($N = 1938$; 51.9% female) who reported engaging in PA at least once per week or expressed a desire to be physically active. Moderated mediation models were performed separately for each emotion. Shame significantly mediated the association between body evaluation and MH, with the indirect effect strengthening as rumination increased (index = $-.05$, 95% CI $[-.08, -.01]$). Indirect effects were significant at moderate ($b = -.08$, 95% CI $[-.11, -.05]$) and high ($b = -.12$, 95% CI $[-.17, -.08]$) levels of rumination, but not at low levels. A similar pattern emerged for embarrassment, which also demonstrated significant moderated mediation (index = $-.06$, 95% CI $[-.10, -.03]$) only at moderate ($b = -.09$, 95% CI $[-.12, -.06]$) and high ($b = -.15$, 95% CI $[-.20, -.11]$) levels of rumination. Conversely, guilt ($b = -.07$, 95% CI $[-.09, -.05]$) and envy ($b = -.03$, 95% CI $[-.05, -.01]$) demonstrated significant indirect effects of body evaluation on MH that did not vary by rumination, indicating mediation without moderation. Authentic pride fully mediated ($b = -.13$, 95% CI $[-.16, -.10]$, direct effect $p = .16$), whereas hubristic pride partially mediated ($b = -.06$, 95% CI $[-.08, -.04]$, direct effect $p < .001$), the association between body evaluation and MH with no evidence of moderation by rumination. Findings indicate that body evaluation influences MH primarily through emotional responses, with body-related rumination selectively shaping the strength of some emotional pathways, but not others. Future

research should examine how these distinct emotional pathways unfold over time and across activity contexts, and whether targeting rumination alters emotion-specific mechanisms linking body evaluation to MH.

Understanding the meaning of psychological readiness to return to competition following injury: Perspectives of adolescent athletes and support providers

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Despite an increased awareness of the importance of psychological readiness to return to competition following injury, little consensus exists on what it means for athletes to be psychologically ready. As adolescents may be prone to psychological distress following injury (e.g., catastrophizing, pain sensitivity, athletic identity issues), it seems reasonable that such disturbances may undermine their readiness to resume competitive sport. To gain insight into the notion of psychological readiness, we conducted 28 semi-structured interviews with 7 adolescent athletes experiencing a current, severe sport-injury (minimum 4-week sport absence) and 21 support providers (parents, sport medicine physicians, sports psychology professionals, physio/athletic therapists). Data were analyzed using Braun and Clarke's (2006) six-step thematic analysis. Results indicate that athletes felt psychologically ready to return to competition when they believed they were a) physically recovered, b) able to safely perform technical (sport-specific) skills commensurate with their competitive division/level of play, and c) felt a sense of belonging in their sport/team. Additionally, support providers specified that athletes' beliefs about their physical abilities need to coincide with clinical assessments and performance indicators to be psychologically ready. Both athletes and support providers also identified the ability to focus on and react to appropriate cues or relevant tactical information as a component of readiness. Furthermore, positively-valenced emotions such as excitement, joy, and anticipation for their return indicated an approach (versus avoidant) orientation to competitive sport, a key feature of being ready. Conversely, athletes were “not” ready when pain or anxiety about reinjury resulted in hesitant behaviors such as a) looking away from coaches or teammates, b) holding back during practices, c) no longer engaging in sport with pre-injury tenacity, and d) guarding or bracing the formerly injured body part. Finally, support providers detailed how individual differences, including trait anxiety, self-awareness, perseverance and extraversion/introversion, could influence the outward expression or manifestation of readiness. This study emphasizes the complex nature of psychological readiness, highlighting dimensions that support providers can consider when making return to competition decisions. Findings can be used to develop scale measures or other assessments for determining adolescents' psychological readiness to return to competition.

COM-B-ing for Change: What moves you—comparative study on AI-vs. human-generated messaging

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Motivation is a central driver of behavior change, yet scalable strategies to enhance motivation for physical activity remain limited. Digital tools to improve physical activity may be instrumental for influencing change. This exploratory study assessed whether messaging grounded in the COM-B

framework (Capability, Opportunity, Motivation) influences intention to be physically active and whether message source (i.e., human- versus AI-generated) moderates this effect. RM-ANOVA was used to explore data from 373 Canadian adults recruited online via Prolific. Participants were randomized to one of seven conditions: Capability–Human, Opportunity–Human, Motivation–Human, Capability–AI, Opportunity–AI, Motivation–AI, Standard messages. Each participant viewed 20 messages tailored to their assigned condition. Intentions to be physically active were measured before and after exposure. Human-generated messages were created by physical activity behavior scientists; AI-generated messages were produced with ChatGPT 4.1 using specific prompts to produce COM-B centered messaging. The three-way interaction among time (pre-post), messaging type (COM), and message source (human/AI) was statistically significant, $F(2, 288) = 6.96, p < .01, \eta^2 = .05$. For increasing intention to be physically active, human-generated capability messages pre-test $M = 5.55, 95\%CI[5.16-5.94]$; post-test $M = 6.03, 95\%CI[5.69, 6.38]$, and AI-generated motivational messages were more effective, pre-test $M = 5.69, 95\%CI[5.29, 6.09]$; post-test $M = 6.00, 95\%CI[5.64-6.35]$. For a subset of participants not meeting physical activity guidelines, the three-way interaction was statistically significant, $F(2, 111) = 3.56, p = .3, \eta^2 = .06$. For increasing intention to be physically active, AI-generated motivational messages, pre-test $M = 4.54, 95\%CI[3.55-5.54]$; post-test $M = 4.99, 95\%CI[4.07-5.91]$, and human-generated capability messages were more effective, pre-test $M = 4.33, 95\%CI[3.58-5.08]$; post-test $M = 5.00, 95\%CI[4.31-5.70]$. AI- and human-generated opportunity messages had similar effects. No difference in intentions were found across demographic variables. Inconsistent preferences for human- compared to AI-generated messages suggests that a focus on careful language training of physical activity-promoting tools is needed. AI-generated motivational prompts appeared to be effective, offering a scalable and innovative solution for physical activity promotion. Future research should examine behavioral outcomes, extending these findings and exploring the role of AI in advancing personalized, theory-based strategies for behavior change.

A qualitative comparison of the exerciser experience of rowing with digital mindfulness

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The integration of digital mindfulness tools into physical activity is a growing area of interest, yet the mechanisms through which these interventions shape the subjective exercise experience remain underexplored. This study examined the experiences of active individuals engaging in a four-week rowing program accompanied by two distinct video interventions. Participants ($N = 30, M = 21.00 \pm 3.77$) were assigned to either a mindfulness intervention group, which watched episodes of the Netflix series Headspace Guide to Meditation, or an active control group that viewed the nature documentary series Wild Babies. Following the four-week protocol, participants underwent semi-structured interviews. Interviews were analyzed via reflexive thematic analysis for each condition, followed by a comparative analysis. The comparative analysis revealed that while both groups reported heightened feelings of accomplishment, gratitude, and resilience, the psychological mechanisms underlying these states differed. Participants in the mindfulness group utilized the content as an internal cognitive reset, serving as a transition tool to clear daily stressors before exercise. These participants successfully engaged in skill transfer, applying nonjudgmental body awareness to manage physical discomfort during rowing and utilizing mindfulness to regulate stress in real-world scenarios. Conversely, engagement in the nature group was driven by narrative transportation and dissociation. While these participants experienced a calm state, it was achieved through distraction rather

than engagement. Furthermore, without explicit education, the nature group demonstrated conceptual confusion, as they frequently conflated mindfulness with associative strategies or engaging in performance monitoring rather than nonjudgmental acceptance. Critically, the results highlighted the limitations of the digital delivery method. While the pre-exercise videos supported state mindfulness, the gap between viewing and rowing made it difficult for novices to maintain an internal focus under physical stress without real-time prompts. These findings suggest that while passive digital delivery can induce calm through distraction, active engagement in mindfulness education is essential to facilitate the purposeful transition into exercise and true skill acquisition. Implications for practice suggest that digital interventions should not be treated as stand-alone solutions but must be supplemented with explicit instruction to help exercisers distinguish between standard performance focus and mindful awareness.

A non-parametric test-retest analysis of individual item scores from a novel self-report balance confidence inventory

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Balance confidence is a proven factor of health in geriatric and neurological populations (e.g., fall risk, functional mobility, quality of life). However, in healthy, younger-to-middle-aged adults, the measurement (and subsequently the understanding) of balance confidence is scant. The purpose of this study was to investigate the test-retest properties of individual item scores from a novel, 14-item self-report balance confidence inventory. Using various recruitment techniques, a sample of non-pathological adults completed an electronic Qualtrics survey which included the newly developed balance confidence inventory ($N = 1,259$; 18-62 years $M_{age} = 22.8, SD = 5.5$; 70% women). Thereafter, 27% of the sample completed the inventory a second time (i.e., retest; $N = 345$; 18-52 years, $M_{age} = 21.3, SD = 4.4$; 74% women). Participants were required to wait at least seven (but no more than 31) days to complete the inventory a second time. To examine the test-retest reliability of the scores from the proposed 14-item balance confidence inventory, Svensson's non-parametric methods for paired ordinal data were used. Per Svensson's methodology, the calculated statistics were relative position, relative concentration, relative rank variance, internal rank variance, measure of disorder, augmented rank-order agreement coefficient, and coefficient of monotonic agreement. All analyses were completed in Excel and/or RStudio. Scores from 13 of the 14 items presented with consistent and stout test-retest properties. Scores for one item (i.e., parkour/freerunning) presented with mixed (but adequate) test-retest properties. Limitations of this study included the use of convenient sampling as well as skewed biological sex and age representations within the sample (i.e., roughly a 3:1 women-to-men ratio; majority of participants were in early adulthood). Further, these results cannot be generalized to all individuals, samples, and populations. Potential and future adopters should have strong confidence in the item-level test-retest properties of scores from the newly introduced self-report balance confidence inventory. Additional research in line with the current investigation is advised (e.g., replication, different populations, extended test-retest intervals).

A causal pathway linking anxiety to reduced physical activity

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Although regular physical activity (PA) benefits mental health, evidence on the inhibitory effects of anxiety on sustained PA remains limited. Anxiety's suppressive influence on PA may partially explain the modest impact of exercise interventions on mental health outcomes. Clarifying this

bidirectional relationship, particularly how anxiety inhibits PA, can help define the optimal conditions for effective interventions. This study used secondary, open-access data from the four-month, multi-modal longitudinal LifeSnaps Study (Yfantidou et al., 2022) to examine how anxiety influences daily PA and whether this relationship differs by sex. The analytic sample included $N = 53$ participants (men = 35, women = 18) from Sweden, Italy, Greece, and Cyprus. Causal inference analyses were conducted separately for women and men, controlling for age and sleep duration as confounders. In women, Bayesian hierarchical models suggested a positive but non-significant association between anxiety and physical activity ($\beta = .04$, 95% HDI $[-.06, .15]$). In men, causal forest analysis revealed that higher anxiety levels were associated with lower physical activity (mean change = -16.96 MET-min/day, 95% CI $[-18.21, -15.71]$), showing an inverted-U pattern with the strongest negative effect around an anxiety score of 52.9. A Bayesian additive regression tree (BART) model further confirmed that participants with high anxiety (anxiety ≥ 52.9) had significantly lower daily activity levels than those with low anxiety (posterior mean difference = -1.11 , 95% CI $[-2.13, -.06]$), with strong posterior evidence supporting a negative effect. Multiple causal and sensitivity analyses confirmed the robustness of the BART results. These findings identify a potential causal pathway through which anxiety impairs PA and suggest a threshold effect that may guide tailored exercise interventions for individuals with anxiety symptoms. Elucidating anxiety's inhibitory role within the anxiety-activity feedback loop is essential for designing targeted, evidence-based mental health interventions.

Examining the moderating role of industriousness on associations between mental fatigue, message framing, and physical activity motivation

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Many adults fail to achieve recommended physical activity (PA) guidelines, highlighting the need for effective strategies supporting PA. Mental fatigue reduces the likelihood of choosing to engage in PA. Presenting PA-related messages framed to emphasize either benefits (gain-framed) or costs (loss-framed) may help enhance PA motivation. Individual differences in dispositional tendencies toward effort and tolerance for challenge may moderate these effects. For instance, learned industriousness reflects how effort becomes reinforcing through repeated associations with reward. This study examined whether dispositional industriousness moderates associations between mental fatigue and message framing on PA motivation. Undergraduate students ($N = 320$) completed a measure assessing industriousness and were then randomized to be exposed to a series of gain-framed or loss-framed PA messages. Next, they completed an online questionnaire assessing their motivation to engage in light-, moderate-, and vigorous-intensity PA across progressively longer durations (10, 20, 30, 40, 50, and 60 min) using an effort discounting paradigm. Mental fatigue was assessed throughout the questionnaire. Results showed lower motivation for PA at higher intensities and durations ($p < .001$). The interaction between message framing and industriousness ($p < .001$) demonstrated the effects of framed messages on PA motivation were dependent on dispositional industriousness. Specifically, individuals higher in industriousness showed greater PA motivation in response to gain-framed messages, whereas PA motivation decreased under loss-framed messages as industriousness increased. Finally, a significant interaction between mental fatigue and industriousness ($p = .003$) indicated participants with higher industriousness (75th percentile) reported lower PA motivation as mental fatigue increased ($p < .001$), whereas participants lower in industriousness (25th percentile) showed no statistically significant change in motivation with increasing fatigue ($p = .90$). Findings suggest that highly industrious

people may be more responsive to gain-framed than loss-framed PA messages, and more affected by mental fatigue than less industrious people, possibly due to heightened awareness of effort costs. Together, results support industriousness as a theoretically meaningful moderator of associations between message framing, mental fatigue, and PA motivation. Future work should examine the effectiveness of messaging interventions tailored to situational (i.e., fatigue) and dispositional (i.e., industriousness) traits on PA motivation and behavior.

The impact of physical exercise on positive coping strategies in early adolescents: A sequential mediation analysis and intervention study

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This study examined the relationship between physical exercise and positive coping strategies among middle school students, focusing on the sequential mediating roles of perceived social support and core self-evaluation. It further assessed whether an eight-week exercise intervention could improve these psychological constructs to inform adolescent mental health practices. In Study 1, 783 students completed scales measuring exercise, coping, social support, and core self-evaluation. Data were analyzed using SPSS 26.0 and PROCESS 4.2, including tests for common method bias, t -tests, ANOVA, correlations, and sequential mediation analysis. Study 2 randomly assigned 70 eighth-grade students to an intervention group (8-week structured exercise, 30 min/week) or a control group. Pre- and post-intervention data were analyzed with SPSS 26.0. Study One revealed significant positive correlations among physical exercise, positive coping, perceived social support, and core self-evaluation ($p < .05$). Physical exercise positively predicted positive coping ($p < .01$), perceived social support ($p < .01$), and core self-evaluation ($p < .01$), while the latter two also positively predicted positive coping ($p < .001$). Students with moderate exercise levels scored higher in positive coping than those with high or low levels. Perceived social support (mediation effect = .049) and core self-evaluation (mediation effect = .045) partially mediated the relationship between exercise and coping, with a significant combined sequential mediation effect of .023. Study Two demonstrated that the intervention group showed significant post-test improvements in positive coping, perceived social support, and core self-evaluation ($p < .05$), whereas the control group did not ($p > .05$). Furthermore, the post-intervention scores of the intervention group were significantly higher than those of the control group ($p < .05$). Moderate exercise yields the highest positive coping scores. Exercise correlates positively with coping, social support, and self-evaluation, with the latter two serving as sequential mediators in the pathway: Exercise $\rightarrow$ Support $\rightarrow$ Coping; Exercise $\rightarrow$ Self-Evaluation $\rightarrow$ Coping; Exercise $\rightarrow$ Support $\rightarrow$ Self-Evaluation $\rightarrow$ Coping. An eight-week exercise program effectively enhances these psychological constructs.

Intentional movement reorganizes auditory-motor synchronization and groove experience by shifting dominant coordination timescale

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Groove—the pleasurable urge to move to music—is commonly theorized to arise from embodied engagement with rhythmic structure, particularly through entrained bodily movement. However, it remains unclear how different movement contexts shape the relationship between groove experience and auditory-motor synchronization, and whether stronger synchronization uniformly enhances groove. The present study aimed to clarify how musical syncopation, bodily movement context, and timing-

specific synchronization jointly organize the experience of groove. Participants listened to musical rhythms with three levels of syncopation (low, medium, high) under three movement conditions: Free (movement neither instructed nor restricted), Static (movement restricted), and Dynamic (intentional rhythmic movement at the neck required). After each stimulus, participants rated urge-to-move and pleasure. Auditory-motor synchronization was quantified using phase-locking values (PLV) between musical rhythms and head movement at beat-related frequencies (1 Hz and 2 Hz). To characterize the functional organization of groove across movement contexts, analyses focused on structural equation modeling, complemented by supplementary linear mixed-effects models to assess trial-by-trial associations. Across all movement conditions, urge-to-move robustly increased synchronization at 2 Hz, indicating that motivational drive toward movement facilitates temporal alignment with musical rhythms even when overt movement is constrained. Critically, however, synchronization was not uniformly beneficial for groove. In Free and Dynamic conditions, increased PLV exerted negative effects on groove-related ratings, suggesting that overly rigid temporal alignment may constrain the subjective groove experience. Moreover, the functional role of synchronization differed across movement contexts: groove-related effects were primarily associated with beat-level (1 Hz) synchronization in the Free condition, whereas sub-beat-level (2 Hz) synchronization played a dominant role in the Dynamic condition, indicating context-dependent temporal scaling of auditory-motor coupling. These findings indicate that groove is not simply amplified by stronger synchronization. Instead, groove emerges from a context-dependent balance between entrained bodily engagement with music and the degree to which movement becomes temporally stabilized to rhythmic structure. We propose an Embodied Groove-Synchrony Model, which distinguishes entrained bodily engagement from phase-based synchronization and shows that excessive temporal stabilization can, depending on movement context, attenuate rather than enhance the groove experience.

Psychosocial influences on talent development and relative age effects in esports

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Relative age effects (RAEs), defined as the over-representation of individuals born earlier in a selection year, are well documented in traditional sport but remain relatively underexplored in non-traditional sports, particularly esports. Recent analysis of date-of-birth data provided by Esports Wales identified a clear RAE in a sample of approximately 1,600 Welsh esports athletes, demonstrating that age-related participation biases can emerge even within non-physical, predominantly open-age competitive systems. Whilst these findings challenged assumptions of age neutrality in esports the research did not examine the developmental mechanisms underpinning such effects. The aim of the present study was to investigate whether psychosocial characteristics and early developmental experiences differ systematically across relative-age quartiles in competitive esports players to better understand potential mechanisms for the existence of the RAE bias. Competitive participants aged 18 years and over completed an online questionnaire assessing date of birth, developmental background, and competitive history, alongside validated measures of personality, motivation, confidence, and resilience. Relative age was operationalised using country-specific cut-off dates and birth-quartile classification. Quantitative analyses examined associations between relative age and psychosocial variables, whilst qualitative thematic analysis of retrospective developmental reflections provided contextual insight into early experiences of reinforcement, challenge, and support. Results revealed both a RAE and differences in athlete psychological characteristics and early developmental experiences across birth quartiles. This research adopted a cumulative advantage framework to examine whether psychosocial factors

commonly implicated in talent development may help explain the emergence and persistence of RAEs in esports. By shifting focus from prevalence-based detection toward examination of underlying developmental processes, the study extends relative age theory into emerging performance domains. The findings aim to inform more developmentally sensitive talent identification and support practices within esports and contribute to a broader understanding of how early developmental biases shape long-term participation trajectories in cognitively demanding competitive environments.

Identifying dose-response dynamics for collision avoidance during an overground virtual pursuit task

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Traditional research has demonstrated that high doses of stressors like task complexity inhibit our performance. Put simply, when a task becomes too challenging, we cannot perform to our potential. However, these insights are often based on 2x2 research paradigms that compare relatively low doses of a stressor with high doses of a stressor, which constrains our ability to resolve or interpret more nuanced dose-dependent effects. Although the literature has also documented that small to moderate doses of task complexity can facilitate performance or learning (e.g., challenge point), very few studies systematically vary task complexity to identify at what doses stimulative or inhibiting effects occur. Therefore, we designed an overground target interception task in virtual reality (VR) during which athletes ($n=24$) had to run to a target while avoiding collisions with an increasing number of virtual defenders. The target's speed was standardized to each participant using the same task without any defenders. Next, the participants completed two blocks varied from two to five defenders (with two different configurations of three defenders) in a counterbalanced manner yielding 48 trials in total. This included four trials without any defenders in each block and excluded for the present analyses. Note that the defenders were not static obstacles, but dynamic virtual defenders programmed to impede the participant's path. For each trial, we assessed the completion time and the number of collisions with the virtual defenders, both averaged for each defender configuration (i.e., dose). A repeated-measures ANOVA revealed that completion time significantly differed across the five doses, $F(4, 92)=14.10$, $p<.001$, $\eta_p^2=.38$, with three defenders yielding the fastest times. Interestingly, the speed-accuracy tradeoff did not occur. The number of collisions also differed significantly, $F(4, 92)=9.34$, $p<.001$, $\eta_p^2=.29$, with the lowest number occurring at the same condition as the fastest completion times. Taken together, the results suggest that performance benefits emerge at intermediate levels of task complexity, highlighting the importance of dose-specific task design. In this way, the present approach offers a useful framework for calibrating task demands in VR-based performance assessment. Funding source: NIH.

Social norms, self-efficacy, and engagement in lifestyle physical activity among people with overweight or obesity

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Although individuals with overweight/obesity understand the health benefits of physical activity (PA), they tend to have low levels of PA.

Specifically, individuals with overweight/obesity often experience negative affective responses to exercise—purposeful PA completed for health benefits—and report low exercise self-efficacy (i.e., confidence to exercise), which represent key barriers to their exercise participation. Lifestyle PA (LSPA), which represents activities that are naturally incorporated into daily lives (e.g., taking stairs), may be more feasible than exercise for this population because LSPA involves sporadic lower-intensity movements as opposed to structured, higher-intensity exercise (e.g., workouts). Although LSPA provides health benefits comparable to exercise and individuals with overweight/obesity adhere to LSPA better than exercise, exercise remains the dominant theme of PA messaging. Thus, individuals with overweight/obesity may be more likely to conceptualize PA as exercise rather than LSPA. At the same time, those who realize that others commonly engage in LSPA (i.e., positive social norm perceptions) may be more likely to feel confident that they can engage in LSPA themselves (i.e., “if they can do it, so can I”). This study examined the perceptions of individuals with overweight/obesity related to exercise and LSPA. An online survey was completed by 357 participants with overweight/obesity ($M_{\text{age}} = 41.9$) twice over one week. Participants were asked to describe what comes to their mind when they hear “physical activity”, and subsequently indicate how confident they feel with engaging in, how often they engage in, and how often they believe others with overweight/obesity engage in, exercise and LSPA. Findings demonstrated that 6.3% of the participants’ descriptions of PA pertained to LSPA, whereas 93.7% pertained to exercise/sports. Participants reported greater self-efficacy to engage in LSPA than exercise ($d = .27, p < .001$). Polyserial correlations revealed that those who held stronger perceptions that others engaged in various forms of LSPA reported higher confidence and engagement in the same forms of LSPA themselves ($p = .22-.55, p < .001$). The results suggest that promoting positive social norms surrounding LSPA may be effective for helping people with overweight/obesity increase their PA.

Experience exercise differently: A qualitative study of the affect-oriented exercise program FEEL

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Interventions for promoting exercise have predominantly relied on cognitive-behavioral approaches, often neglecting affective processes during and after exercise, despite growing evidence that these processes play a central role in exercise adoption and maintenance. The FEEL exercise program addresses this gap by aiming to promote affect regulation competence, positive affective exercise experiences, well-being, and, consequently, exercise behavior. FEEL is a holistic, experience-based intervention that integrates key topics, including affect-guided intensity regulation, attentional strategies, and motive-matched activity types. The purpose of this qualitative study was to explore participants’ subjective experiences with the FEEL program. We aimed to understand how physically inactive young adults perceived the processes of initiating and re-engaging in exercise within an affect-oriented intervention. The study was embedded within a larger randomized controlled trial examining the efficacy of the FEEL program. Twenty-two participants ($M_{\text{age}} = 27.6$ years; 65.2% female) from the FEEL program took part in individual semi-structured interviews. The sample was purposively selected to reflect variation in attendance, gender, and age within the predefined inclusion criteria (18–35 years). All participants were previously physically inactive. Data were analyzed using inductive reflexive thematic analysis. Preliminary analyses identified three overarching themes. Theme 1, “Affect orientation as a relieving alternative,” highlights that participants experienced the FEEL program as a contrast to prior exercise experiences centered on performance and comparison. Emphasizing affect was

described as relieving, allowing participants to reframe their understanding of exercise. Theme 2, “A non-judgmental and safe setting,” demonstrates that participants felt welcomed and experienced the instructors and group setting as psychologically safe, particularly because abilities and fitness levels were not evaluated. Exercising with others who shared similar starting points reduced fear and fostered a sense of not being alone. Theme 3, “Affective learning: experiencing exercise differently,” indicates that participants learned to experience exercise as pleasant and enjoyable, in contrast to previous negative experiences, and to adapt exercise to their personal preferences. In summary, the findings suggest that participants benefited from reflective and practical engagement with affect in exercise, contributing to a reframing of exercise and changes in how they approached exercise behavior. Funding source: The Swiss Federal Office of Sports (FOSPO) fully funds the Bern study site. The intramural project funding for postdocs of the University of Tübingen partially supports the Tübingen study site. The funding association is not involved in the study design, the collection, analysis, and interpretation of data, in the writing of the report, nor in the decision to submit an article for publication.

Affective forecasting among Toronto Blue Jays fans throughout the 2025 World Series

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People need to predict how certain choices or events will make them feel in order to help make decisions and navigate the social world. Previous research with sports fans has shown that, like people in other contexts experiencing negative events, they tend to exhibit impact bias and overestimate how badly they will feel after a team loss. However, as sports fans know, some losses are more painful than others. This was particularly true for fans of the Toronto Blue Jays during the 2025 World Series, which the Blue Jays lost in heartbreaking fashion to the Los Angeles Dodgers. Our aim in this research was to test if Blue Jays fans were accurate in their forecasts of how they felt after the Series ended. We assessed two affective outcomes: negative affect (i.e., how badly the fans felt), and savoring (i.e., the extent to which fans looked on the bright side and appreciated the successful season despite the World Series loss). We also focused on both absolute (i.e., overestimation or underestimation) and relative (i.e., rank-order) assessments of affective forecasting accuracy. We recruited 427 Blue Jays fans who completed online surveys before and after the World Series. The pre-series survey included scales assessing forecasted negative affect and savoring in the event of a Blue Jays loss, and the post-series survey included scales assessing experienced negative affect and savoring. The results showed that fans’ absolute affective forecasting was generally inaccurate, and they tended to underestimate their negative affect after the series ended ($d = .17, p < .001$). However, relative affective forecasting accuracy was quite strong for both negative affect and savoring (each $d = 1.5, p < .001$). These results mean that, although fans did not accurately estimate how they would feel after the loss and tended to feel greater negative affect than expected, they were accurate in predicting how their feelings would rank relative to others. The results also demonstrate that there are times when fans feel worse, not better, than expected after a team loss, supporting early theorizing on the importance of event construal. Funding source: Social Sciences and Humanities Research Council of Canada.

Who is coaching the coaches? A qualitative exploration of the learning and mentoring experiences of Special Olympics Canada coaches

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The Special Olympics is the largest sporting organization for athletes with intellectual and developmental disabilities. Coach learning and development in the Special Olympics has received scarce empirical attention compared to coaching in generic sport contexts, leaving a limited understanding of how coaches acquire the knowledge and skillset to be effective in this domain. Previous research has identified how coaching in the Special Olympics context requires patience, creativity, and adaptability to enhance learning and how coach mentorship would help them better prepare for the demands of training and competition (Alexander-Urquhart et al., 2025; Cybulski et al., 2016; Turgeon et al., 2023). The purpose of this study was to explore the learning and mentoring experiences of Canadian Special Olympics coaches. Data were collected using semi-structured individual interviews (Smith & Sparkes, 2016) with eight Special Olympics coaches of varying levels of experience, sport type, and geographic locations across Canada, and analyzed using a reflexive thematic analysis (Braun & Clarke, 2019). Among the findings, coaches described how their initial exposure to the Special Olympics was considered overwhelming at times (e.g., first practices, major events), and that coaches used reflection and trial and error when to enhance their coaching knowledge and behaviors as it helped them acquire a more refined, confident approach to their coaching practices. Coaches appreciated learning experientially from other coaches, acquiring validation in their ability to coach, brainstorming ways to progress the quality of Special Olympics, and observing innovative or alternative practices from coaches in other sports or locations. All coaches spoke of being mentored or mentoring other coaches within their careers, advocating for mentorship as a beneficial opportunity to learn in a practical, hands-on manner and provided their thoughts on what would make an effective formal coach mentorship program. Efforts towards providing experiential learning through mentorship can allow Special Olympics coaches to acquire the confidence, motivation, and knowledge needed to provide high-quality coaching to their athletes and foster safe, inclusive, and positive sport experiences for all. Funding source: Special Olympics Canada.

“I feel connected and present”: Pilot research exploring embodied physical activity among college students

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Physical activity may shape embodied experiences such as body connection, agency, comfort, and self-attunement; however, empirical work specifying how engagement in physical activity aligns with distinct domains of embodiment remains limited. Guided by the Developmental Theory of Embodiment, this pilot study examined college students' embodied physical activity (EPA) experiences using quantitative indicators and qualitative self-reports. Participants ($n = 105$; $M_{\text{age}} = 20.83$) completed the Experiences of Embodiment Scale (EES) and the Embodied Physical Activity Questionnaire (EPAQ). A subset of participants ($n = 45$) responded to three open-ended items about EPA experiences. Correlational analyses showed that more frequent EPA experiences were associated with Positive Body Connection and Comfort (PBCC; $r = .69$), Agency and Functionality (AF; $r = .50$), Attuned Self-Care (ACS; $r = .41$), and Body Unencumbered Adjustment (BUA; $r = .38$), with all associations significant at $p < .001$. Resisting Objectification was not significantly associated (RO; $r = \text{ns}$). Qualitative responses indicated that EPA was most strongly experienced through positive body connection and agency. Participants described embodiment as “a deep connection between my mind and body, where I feel present, alive, and fully aware of each movement and sensation,” and as an opportunity to “honor what my body is capable of—the arms that hold me up, the legs that help me move, and my muscles that make me

feel free.” Experiences reflecting self-care were also common, including mental and emotional regulation, e.g., “My mind completely gets a reset... nothing compares to running.” In contrast, unencumbered experience and resistance to objectification were rarely mentioned. Enhanced motivation for physical activity emerged as a prominent outcome of EPA, e.g., “Experiencing embodiment enhances my engagement... encouraging consistent participation and a deeper appreciation for how my body feels and functions.” Participants also described environmental conditions, physical health, and psychological states as shaping their embodied experiences. Collectively, these findings suggest that EPA among college students is characterized primarily by experiential processes of body connection, agency, and self-care. These results extend the Developmental Theory of Embodiment by highlighting EPA as a context-sensitive, motivating process through which specific dimensions of embodiment may be cultivated.

Domain-specific changes in physical activity across the educational transition in adolescence: A one-year longitudinal study

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Research among adolescents has predominantly focused on overall physical activity levels, despite growing evidence that different physical activity domains (e.g., household activities, sport and exercise) are associated with distinct health outcomes. Moreover, these domains may respond differently to social and environmental changes associated with life events, such as the transition from lower to upper secondary education. However, domain-specific evidence on physical activity changes during this transition remains limited. Therefore, this study examined changes across multiple physical activity domains during the educational transition in adolescence. In a one-year longitudinal study, 393 adolescents undergoing an educational transition were included ($N_{t1} = 1156$; $M_{\text{age}, t1} = 15.27$ years, $SD_{\text{age}, t1} = .68$; 53.7% female; 72.8% Swiss nationality). Physical activity was assessed six months before ($t1$) and six months after ($t2$) the transition using a validated questionnaire (Fuchs et al., 2015) capturing four domains: sport and exercise, active transport to school/work by foot, active transport to school/work by bicycle, and household activities. For exploratory purposes, adolescents reported other critical life events and changes experienced during the study period (Siegrist & Geyer, 2014). Changes in domain participation were analyzed using McNemar tests. Significant changes were observed in three domains. Participation in sport and exercise declined from 85.0% ($t1$) to 71.2% ($t2$) ($\chi^2(1) = 33.44$, $p < .001$). Active transport by walking decreased from 49.6% to 38.7% ($\chi^2(1) = 8.13$, $p = .004$), and engagement in household activities decreased from 83.4% to 71.8% ($\chi^2(1) = 9.38$, $p = .002$). No change was observed for active transport by bicycle (46.1% vs. 41.7%; $\chi^2(1) = 1.99$, $p = .158$). Overall, the findings indicate that educational transitions are associated with selective declines in specific physical activity domains rather than uniform reductions in overall activity. These domain-specific changes may be linked to alterations in adolescents' environments. For example, 49.6% of adolescents reported a significantly longer commute to school or work, suggesting that increased commuting distances to new schools or workplaces may limit walking while still allowing cycling. A domain-specific perspective therefore provides a foundation for designing targeted interventions across the educational transition. Funding source: Suzanne and Hans Biäsch Foundation for the promotion of applied psychology.

Does being kind to oneself help to support others? Fitness professionals' self-compassion as a predictor of participant-perceived social support

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Physical activity (PA) can offer opportunities for social support (SS), helping adults initiate and sustain PA behavior, reduce isolation, and connect with others. Fitness professionals play a crucial role in providing and facilitating SS, but little research exists examining personal skills and resources they can develop and draw upon to be effective supporters. Self-compassion, defined as being kind and non-judgmental towards oneself, may be a relevant personal resource for fitness professionals to help navigate the emotional and interpersonal demands of providing SS. This study examined whether fitness professionals' self-reported self-compassion and provision of SS, and participants' self-reported self-compassion, predict participant-perceived SS provided by fitness professionals. 31 fitness professionals ($M_{\text{age}} = 43.2$ years, $SD = 14.0$) completed a questionnaire assessing self-compassion (self-kindness, mindfulness, common humanity, self-judgment, over-identification, isolation) and SS provision (autonomy support and thwarting, competence support, relatedness support and thwarting). 189 adult PA participants ($M_{\text{age}} = 60.7$ years, $SD = 14.3$) instructed by those fitness professionals, completed the same questionnaires assessing their own self-compassion and perception of their instructor's SS. Structural equation models were estimated using robust maximum likelihood with cluster-robust standard errors to account for participants being nested within classes taught by the same fitness professional. Separate models were run for each participant-perceived SS outcome with fitness professional self-compassion and SS, and participant self-compassion as predictors. Fitness professional self-kindness ($\beta = .34$), mindfulness ($\beta = -.41$), and over-identification ($\beta = -.41$) predicted participant-perceived autonomy support. Fitness professional self-kindness ($\beta = -.41$), self-judgment ($\beta = -.35$), mindfulness ($\beta = .39$), and over-identification ($\beta = .38$) predicted participant-perceived autonomy thwarting. Participant isolation ($\beta = -.23$) predicted participant-perceived competence support. Fitness professional self-kindness ($\beta = .57$) and isolation ($\beta = .28$) predicted participant-perceived relatedness support. Participant over-identification ($\beta = -.20$) and isolation ($\beta = .25$) predicted participant-perceived relatedness thwarting. High self-kindness and low over-identification in fitness professionals consistently predicted more supportive and less thwarting participant-perceived behaviors; however, some self-compassion subscales (e.g., mindfulness) showed unexpected associations. Participants' own self-compassion also may influence how they interpret fitness professionals' behavior. Findings highlight the importance of cultivating self-compassion in fitness professionals to foster supportive PA spaces. Funding source: Alberta Graduate Excellence Scholarships, Doctoral Research; Graduate Faculty Council Scholarship, Doctoral; Faculty of Graduate Studies Doctoral Scholarship.

Lessons learned from high school sport vs. extracurriculars: A dynamic topic modeling comparison

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Evidence has consistently shown that school sport can facilitate development for high school students (Camiré & Kendellen, 2016); however, a common critique of this notion is that life lessons may be developed in other contexts and is not unique to sport. This begs the question as to what unique contributions sport may provide for young people in comparison to other key developmental opportunities (i.e., marching band, volunteering, honors societies), referred to as extracurriculars, in the high school context. Therefore, the purpose of this investigation was to explore if, and if so how, students describe lessons learned from sport differently from those learned in other extracurriculars. This study employed Latent Dirichlet

Allocation (LDA) topic modeling (Blei et al., 2003), a natural language processing method, within R. The data included open-ended self-descriptive narratives from 6,392 high school students from two prompts, totaling a corpus of 12,784 essays. The first prompt asked student-athletes to describe lessons learned from high school sport participation, referred to as the sport essays, while the second asked for lessons learned from extracurricular participation, referred to as the non-sport essays. A model for each essay type was fit, resulting in $k = 12$ topics and $k = 13$ topics for the sport and non-sport essays, respectively. Then, for both models, researchers divided the topics thematically into (1) Context for Lessons Learned and (2) Life Skills and Lessons Learned. Topics from the sport context theme include individual competitions, exposure to adversity, and skill success, while the sport life skills and lessons theme included sportsmanship, overcoming adversity and communication. Themes from the non-sport context included assisting younger kids, service through honors societies, and volunteering, while life skills and lessons learned included growing from discomfort, general life preparation, and earnings. Key similarities between sport and non-sport context themes included time management, leadership, and growth. Uniquely, sport essays described differences in building meaningful relationships and developing work ethic without a direct reward, suggesting these life lessons may be more specific to sport contexts. Moving forward, it is important to continue exploring this nuance to understand the unique contribution of sport-based relationships and work ethic.

A qualitative analysis: Body appreciation, perceived stress, and body composition across training phases in competitive female powerlifters

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Body appreciation, perceived stress, and body composition have been linked as influential factors in an individual's body image perception. Understanding the interrelationships of these factors in strength-based athletes, such as powerlifters, is important as the sport has a dual emphasis on aesthetics and performance. The purpose of this study was to qualitatively examine body appreciation, perceived stress, and body composition across a competitive season in female powerlifters. Fifteen female powerlifters were recruited (mean age 22.13 ± 5.60 years). Four assessments were conducted across 12 weeks of periodized training, around a competition. Semi-structured interviews using thematic analysis were utilized to understand the individual's lived experiences surrounding changes in body appreciation, perceived stress, and body composition. Four major themes emerged from the interviews: 1) Perceived stability of the body and stress, 2) Stress linked to external demands, 3) Body size, femininity, and appreciation, and 4) Redefining body appreciation through function. In the first theme, participants perceived minimal visible changes in their body composition and stress levels. Stability was a consistent characteristic across experiences of body composition, body appreciation, and perceived stress; however, stability was interpreted differently (e.g., balance, stagnation, demotivating). The second theme highlighted the stressful influence of external demands (e.g., relationships, academics). Stress emerged as a motivator and a constraint that undermined motivation and enjoyment in training and their personal lives. Within the third theme, participants reflected on how societal standards of femininity influenced their perceptions of body size and appreciation. Additionally, these ideals conflicted with the muscular physique and strength that are central to powerlifting. The fourth theme emphasized a shift in how participants evaluated their bodies – from appearance and aesthetics to appreciation of functionality and capability. This study highlights the complex relationship between body appreciation, perceived stress, and body composition in female powerlifters. These athletes continue to change the notion of what it means to be a female athlete, but they are heavily pressured by traditional beauty standards.

Increased levels of body appreciation countered the stress of those standards by shifting their perspective from appearance to function. However, longitudinal data is needed to examine this relationship further.

Physical activity and sleep are associated with early information processing in young adults

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Sleep and physical activity are two key lifestyle behaviors that together reflect the rest-activity cycle and have broad implications for cognitive function and brain health. Disruptions in early sensory processing have been etiologically linked to psychiatric disorders, particularly schizophrenia. Notably, these disorders most commonly emerge during late adolescence and early adulthood, highlighting young adulthood as a critical developmental window in which lifestyle behaviors may confer neuroprotective benefits relevant to risk and vulnerability. Although sleep and physical activity have been associated with higher-order cognitive functions such as attention, executive function, and memory, their associations with pre-attentive, bottom-up processes that support early information processing remain largely unexamined. The present pilot study examined associations between physical activity, sleep, and early sensory processing in 27 healthy young adults. Early information processing was indexed using the mismatch negativity (MMN), an event-related potential captured during an auditory oddball task that reflects pre-attentive information processing. More moderate-to-vigorous physical activity (IPAQ; $r = -.39, p < .05$) and better sleep quality (PSQI; $r = -.60, p < .01$) were associated with more negative MMN amplitudes, indicating enhanced early information processing. When both lifestyle behaviors were examined concurrently, only moderate-to-vigorous physical activity remained significantly associated with MMN amplitude after controlling for sleep quality (partial $r = -.43, p < .05$), whereas the association between sleep quality and MMN was attenuated. These findings suggest that physical activity and sleep are both associated with early sensory processing. However, physical activity may demonstrate a more robust, independent association with early stages of information processing. Although limited to a nonclinical sample, the present study highlights physical activity and sleep as modifiable lifestyle behaviors that may confer protective benefits for neural processes implicated in serious mental illness. The next stage of this study will extend and compare these findings to patients with schizophrenia to probe potential mechanisms through which these lifestyle behaviors influence perturb early sensory processing.

Differential associations between body- and appearance-based self-conscious emotions and physical activity engagement

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Despite established public health guidelines and broad cardiometabolic, physical and mental health benefits, physical activity levels remain insufficient worldwide. These deficiencies indicate that knowledge of health benefits alone fails to motivate sustained engagement, highlighting the need to identify other factors that influence physical activity behavior. Affective and self-evaluative processes related to body image may play an important role in shaping motivation for physical activity and maintaining engagement. Body- and appearance-based self-conscious emotions (e.g., guilt, shame, and pride) reflect evaluative responses to physical activity and appearance-related standards and may differentially support or undermine engagement. Understanding the role of these self-conscious emotions may provide important insight into motivational pathways of engagement, and in-turn, inform strategies to maintain physical activity participation. To examine the contribution of body image-related self-conscious emotions to physical activity engagement, this observational study investigated associations between body- and appearance-related self-conscious emotions and physical activity levels among 36 healthy young adults (15 females). Physical activity was assessed using the International Physical Activity Questionnaire (IPAQ), and self-conscious emotions were measured using the Body and Appearance Self-Conscious Emotions Scale (BASES). Multivariable regression models adjusted for sex and body mass index (BMI) were used to examine emotion-specific associations with physical activity. In adjusted analyses, lower body-related guilt was associated with greater physical activity ($\beta = -.97, p = .001$), whereas higher hubristic pride was positively associated with physical activity ($\beta = .65, p = .01$). Interestingly, shame and authentic pride were not significantly associated with physical activity. Together, these results demonstrate that specific self-evaluative emotions may reflect distinct motivational pathways underlying physical activity engagement. Findings may suggest that lower guilt may support physical activity engagement by reducing negative self-evaluation, whereas higher hubristic pride may positively reinforce participation. These findings highlight affective and self-evaluative mechanisms as relevant targets for understanding and promoting physical activity engagement, suggesting that targeting body image-related emotional processes may support sustained participation.